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BASIC EMERGENCY OBSTETRIC & NEWBORN CARE (BEmONC)

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Training Manual



Federal Democratic Republic of Ethiopia Ministry of Health

January 2018

**BASIC EMERGENCY
OBSTETRIC
&
NEWBORN CARE
(BEmONC)**

Training Manual



**Federal Democratic Republic of Ethiopia
Ministry of Health**

January 2018

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Acronyms

AMDD	Averting Maternal Death and Disability
AMTSL	Active Management of Third Stage of Labor
ANC	Ante Natal Care
APH	Ante Partum Hemorrhage
ART	Anti retroviral Therapy
ARV	Anti RetroViral
BEmONC	Basic Emergency Obstetric & Newborn Care
CBT	Competency-based training
CCT	Controlled Cord Traction
CEmONC	Comprehensive Emergency Obstetric & Newborn
CareCPD	Cephalo Pelvic Disproportion
C/S	Cesarean Section
EDD	Expected Date of Delivery
EDHS	Ethiopian Demographic Health Survey
EOC	Emergency Obstetric Care
EmONC	Emergency Obstetric & Newborn Care
FANC	Focused antenatal care
FGC	Female Genital Cutting
FHB	Fetal Heart Beat
FP	Family Planning
GA	Gestational Age
GBV	Gender Based Violence
HELLP	Haemolysis Elevated Liver enzymes
and Low Platelets	HLD High Level Disinfection
HTC	HIV Testing and Counseling
ICPD	International Conference on Population and Development
IM	Intra Muscular
IMPAC	Integrated Management of Pregnancy and Childbirth
IMNCI	Integrated Management of Newborn
and Childhood Illnesses	IP Infection Prevention

ITN	Insecticide Treated (bed)Nets
IUCD	Intra Uterine ContraceptiveDevice
IUGR	Intra Uterine GrowthRestriction
IV	IntraVenous
KMC	Kangaroo MotherCare
LAM	Lactational AmenorrhoeaMethod
LBW	Low BirthWeight
LNMP	Last Normal MonthlyPeriod
LRP	Learning ResourcePackage
MDG	Millennium DevelopmentGoal
MVA	Manual VacuumAspiration
PID	Pelvic InflammatoryDisease
POC	Products OfConception
PPH	Post PartumHemorrhage
PS	PatientSafety
SBA	Skilled BirthAttendant
STD	Sexually TransmittedDisease
STI	Sexually TransmittedInfection
TOT	Training OfTrainers
TT	TetanusToxoid
UNFPA	United Nations PopulationFund
UNICEF	United Nations Children'sFund
VAW	Violence AgainstWomen
WHO	World HealthOrganization

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PREFACE

Most pregnancies and births are uneventful with good maternal and perinatal outcome, however, approximately 15% of all pregnant women develop potentially life-threatening complications that call for skilled care and some will require a major obstetrical intervention to survive. This training manual is a component of the **Basic Emergency Obstetric and Newborn Care (BEmONC)** learning resource package (LRP) prepared for use in Ethiopia in in-service training of doctors, midwives, health officers and/or nurses with midwifery skills who, as team members, will provide Basic Emergency Obstetric and Newborn Care (BEmONC) at health centers and district hospitals to avert maternal and newborn mortality & morbidity. Although the learning resource package is developed primarily for use in district hospital and health centers, it can also be used by the same mid-level health care providers in higher level facilities to provide initial care and until consultation.

Components of the learning resource package are: *Training manual, Facilitator's guide, Participant's Guide, power point presentations, technical videos and other relevant resources.* There are eight modules in the package and each module describes the learning objectives, learning outlines, learning materials and assessment tools. Module one is introduction to maternal and newborn health, module two on rapid initial assessment and emergency management; module three on care during pregnancy; module four on hypertensive disorders of pregnancy, module five on care during labor and child birth, module six on obstetric hemorrhage, module seven on postpartum maternal and newborn care and module eight on infection in obstetrics.

The training manual contains updated and summarized essential technical information from the relevant references and is intended to be used as the reference manual by both the facilitators and participants. The facilitator's guide part contains the course outline, learning guides and checklists, exercises, role plays and answer keys for each module and will guide how to facilitate the training. The participant handout is intended primarily to serve for the participant and contains, learning guides and checklists, exercises, role plays and answer keys of each module.

This training is intended to be completed over a period with 5 days' classroom theoretical sessions & practice on model and 7 days of clinical practice in selected health facilities.

MODULE -1

INTRODUCTION TO MATERNAL & NEWBORN HEALTH

Time: 140 min

Participant learning objective: after completing this module, participants will be able to describe the global situation of maternal and newborn morbidities & mortality best practices in maternal and newborn care and emergency management principles

ENABLING OBJECTIVES

1. Describe the magnitude of maternal and neonatal mortality & morbidity in the developing world generally and in our countryspecifically.
2. Describe the current approach to reduction of maternal and neonatalmortality.
3. Describe the principles of woman friendly care
4. Describe Infection Prevention practices

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UNIT-1: INTRODUCTION TO MATERNAL AND NEWBORN HEALTH

The health of mothers and children is central to global and national concerns, and improvements in maternal and child survival are always given priorities. Apart from the obvious linkages between health programmes, mother and child health is intimately bound up with economic development, education, gender issues and rights. Although most pregnancies and births are uneventful, approximately 15% of all pregnant women develop a potentially life-threatening complication that calls for skilled care and some will require a major obstetrical intervention to survive. For an individual woman, the risk of maternal death is influenced both by the risk associated with pregnancy and by the number of times she becomes pregnant. Each time a woman becomes pregnant, she runs the risk of maternal death again, and the risk adds up over her lifetime. In developing countries, where both mortality and fertility tend to be high, the lifetime risk of maternal death can be astoundingly high.

A **maternal mortality** is defined as 'the death of a woman while pregnant or within 42 days of termination of pregnancy, irrespective of the duration or site of pregnancy, from any cause related to or aggravated by the pregnancy or its management but not from accidental or incidental causes. And, a late **maternal mortality** is defined as the death of a woman or girl from direct or indirect causes more than 42 days but less than one year after termination of pregnancy. There are three main ways in which maternal mortality is measured:

- **Maternal Mortality Rate:** Number of maternal deaths per year for every 100,000 women aged 15-49. This measure reflects both the risk of death among pregnant and recently pregnant women, and the proportion of all women who become pregnant in a given year. But because this denominator is difficult to determine precisely, most widely used is the maternal mortality ratio.
- **“Maternal Mortality Ratio:** Number of maternal deaths per 100,000 live births during a given time period. This measure indicates the risk of maternal death among pregnant and recently pregnant women. It is a measure of obstetric risk and a reflection of a woman's basic health status, her access to health care, and the quality of service that she receives.
- **Lifetime Risk of maternal death:** This measure reflects the probability of maternal death faced by an average woman cumulated over her entire reproductive life-span. Like the maternal mortality rate, it reflects both a woman's risk of dying from maternal death, as well as her risk of becoming pregnant in the course of a reproductive lifetime.

A maternal death could be due to a direct or an indirect cause. A direct maternal death is an obstetric death resulting from obstetric complications of the pregnancy state, labor, or puerperium. And, an indirect maternal death is an obstetric death resulting from a disease previously existing or developing during the pregnancy, labor, or puerperium; death is not directly due to obstetric causes but may be aggravated by the physiologic effects of pregnancy.

In addition to the women who die, many more suffer from serious but not fatal health problems as a result of pregnancy or childbirth. Most women who have obstetric complications recover, but some suffer long-term disabilities including sterility and obstetric fistula. Obstetric urinary fistula is a condition in which prolonged obstructed labor produces a hole between the vagina and the urinary system, resulting in chronic incontinence. This is not only painful, but if left untreated (as is usually the case in developing countries) it can lead to social stigmatization and isolation. Sterility commonly results from untreated or recurrent pelvic infection. Beyond frustration and disappointment, sterility can have profound social and economic consequences for women in societies where women's value is largely determined by the children they bear. There is little reliable information on the prevalence of maternal morbidity, but the number of women affected is sure to be several times greater than the number of those who die. Fortunately, interventions that reduce maternal deaths will also reduce maternal morbidity.

Perinatal mortality tends to follow the same geographical pattern as for maternal deaths. Stillbirths, neonatal deaths, and maternal morbidity and mortality fit together as public health priorities. A very large proportion of maternal and perinatal deaths are avoidable. Most deaths occur due to poor service provision, as well as lack of access to and use of these services. Interventions that can prevent mortality from the major causes of death are known, and can be made available even in resource-poor settings. These include focusing on adequate care and preparation in the household, assuring quality services close to where women live and systematically detecting and managing complications at an early stage.

The purpose of this module is to introduce the participant to the global situation of maternal and neonatal mortality & morbidities, overview of best practices in maternal and neonatal care and emergency management principles.

UNIT-2: CURRENT APPROACH TO REDUCTION OF MATERNAL AND NEONATAL MORTALITY

2.1. UNIT LEARNING OBJECTIVES

By the end of this unit participants will be able to:

- Recognize the situation of maternal and neonatal morbidity & mortality globally and in Ethiopia specifically.
- Describe factors affecting maternal and perinatal mortality and morbidity.
- Review current interventions to reduce maternal and neonatal mortality

2.2. GLOBAL AND NATIONAL SITUATION OF MATERNAL AND PERINATAL MORBIDITY AND MORTALITY

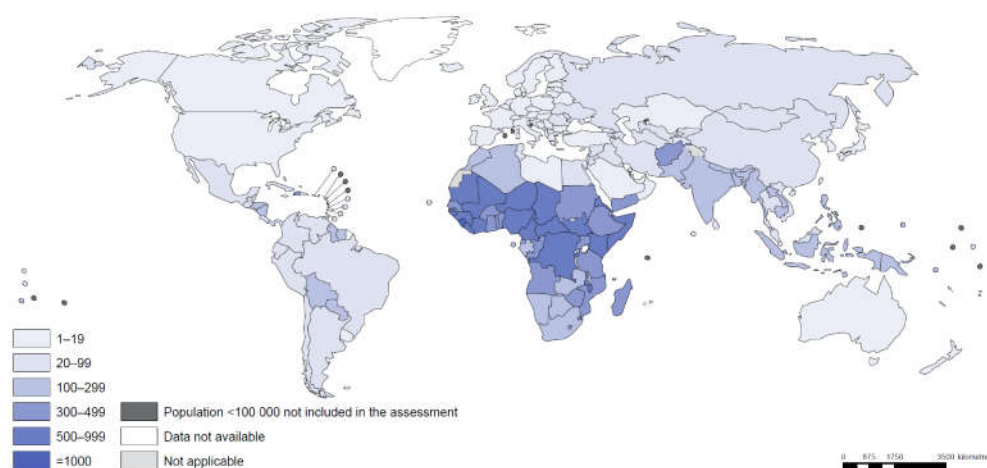
The global maternal mortality is unacceptably high. According to World Health Statistics 2015 released by the World Health Organization: every year some 303,000 women die of complications during pregnancy or childbirth globally. About 830 women die from pregnancy- or childbirth-related complications around the world every day.

The overall MMR in developing regions is 239, which is roughly 20 times higher than that of developed regions, where it is just 12. Sub-Saharan Africa has a very high MMR with a point-estimate of 546. Developing regions account for approximately 99% (302 000) of the estimated global maternal deaths in 2015, with sub-Saharan Africa alone accounting for roughly 66% (201 000), followed by Southern Asia (66 000). The lifetime risk of maternal mortality is estimated at 1 in 36 in sub-Saharan Africa, contrasting sharply with approximately 1 in 4900 in developed countries.

Nigeria and India account for over one third of all estimated global maternal deaths in 2015, with an approximate 58 000 maternal deaths (19%) and 45 000 maternal deaths (15%), respectively. Ten countries account for nearly 59% of global maternal deaths. In addition to Nigeria and India, they are (in descending order of numbers of maternal deaths): Democratic Republic of the Congo (22 000), Ethiopia (11 000), Pakistan (9700), United Republic of Tanzania (8200), Kenya (8000), Indonesia (6400), Uganda (5700) and Bangladesh (5500).

The disparity between developed and developing countries is greater for maternal mortality than for any other commonly-used index of health. Whereas levels of infant mortality are, on average, 10 times higher in developing than in developed countries, maternal mortality in developing countries is more than 100 times higher than in industrialized countries.

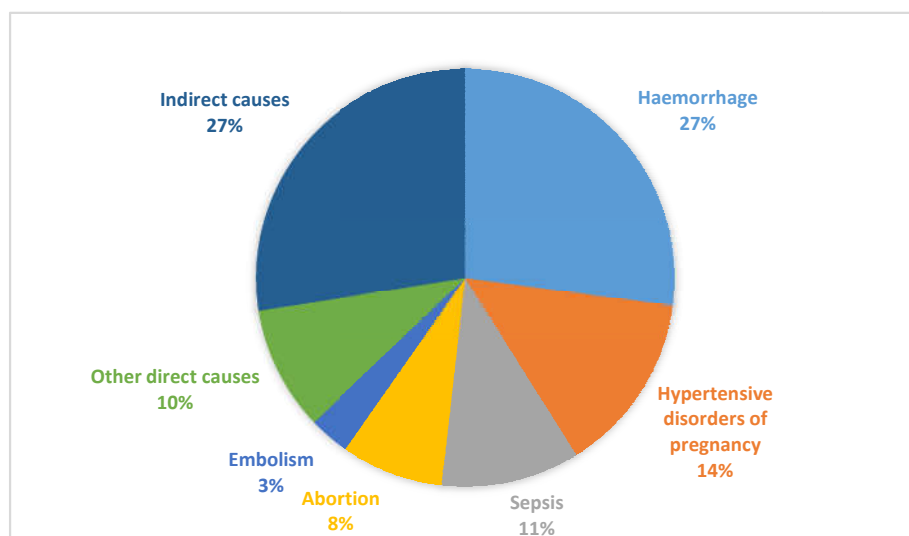
Figure 1-1: Maternal mortality ratio (MMR, maternal deaths per 100 000 live births), WHO - 2015



Women die from a wide range of complications in pregnancy, childbirth or the postpartum period. Most of these complications develop because of their pregnant status and some because pregnancy aggravated an existing disease. The five major global causes of maternal death are: severe bleeding (mostly bleeding postpartum), infections (also mostly soon after delivery), unsafe induced abortion, hypertensive disorders in pregnancy (eclampsia) and obstructed labor (*See Figure 1-2*). Globally, about 80% of maternal deaths are due to these causes. Hemorrhage alone accounts for one third of all maternal deaths in Africa, yet many of these deaths are preventable. Among the indirect causes (20%) of maternal death are diseases that complicate pregnancy or are aggravated by pregnancy, such as malaria, anemia and heartdisease.

Most of the maternal deaths around the world could have been prevented by improving women's access to good-quality reproductive health care and effective interventions. Many women die because of poor health at conception and a lack of adequate care needed for the healthy outcome of the pregnancy for themselves and their babies. The 303,000 maternal deaths are the tip of the iceberg, and many more women are estimated to suffer pregnancy-related illnesses, near-miss events, and other potentially devastating consequences after birth. Obstetric fistula resulting from obstructed labor is a long term complication suffered by as many as two million women. Generally, about 15 percent of all pregnant women have childbirth complications that require emergency obstetric care (EmOC), yet few are able to access such services.

Figure 1-2: Causes of maternal deaths, global.



Source: Global causes of maternal death: a WHO systematic analysis, June 2014

Mortality rate in the perinatal period is another important indicator of the quality of care during pregnancy, labor & delivery and the post-partum period. It shows the obvious disparity between developed and developing countries and is used to evaluate the outcome of pregnancy and monitor the quality of prenatal and newborn care. The perinatal mortality rate includes both fetal and early neonatal mortality. Fetal death is defined as death of the fetus occurring from 28 weeks of gestation but prior to delivery; while neonatal death is defined as a live born infant who dies before 28 days of age. Neonatal deaths can occur early, within the first seven days after birth, or late, on days eight through 28. Approximately 98% of the 5.7 million perinatal deaths suffered globally occur in developing countries. According to WHO data, 2.7 million babies are born dead each year and another 3 million do not survive beyond the first week of life.

Newborn health and survival are closely linked to care the mother receives before and during pregnancy, childbirth, and the postnatal period. Most of the perinatal deaths are avoidable. Throughout the continuum of care, the period with the highest risk of death and disability for both mothers and newborns is labor, birth, and the first few hours after birth. Complications and lack of care at this crucial time has consequences for mothers and babies. About one-third of perinatal deaths in developing countries are related to intrapartum complications leading to birth asphyxia (See the global causes of newborn causes in **Figure 1-3**). Preterm birth, malformations, and infections related to pregnancy and birth contribute to the remainder of the early neonatal b[7] deaths. Representing a substantial portion of overall child deaths, early neonatal deaths account

for 38% of all infant mortality and 29% of 'under-five' mortality in developing countries. Late neonatal deaths are to some extent due to perinatal conditions, but mostly to infections acquired after birth, many of which are associated with poor hygiene, lack of information on adequate newborn care and/or poor neonatal feeding practices. Often the death of mothers is closely connected with newborn deaths, as maternal mortality and morbidity have a direct negative impact on the survival chances of the newborn.

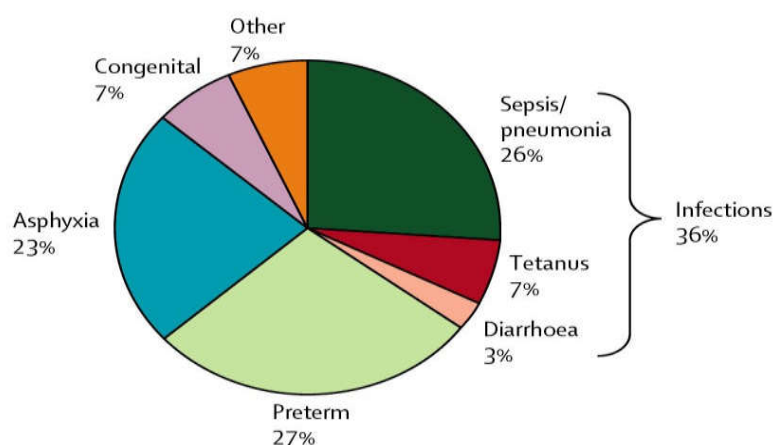


Figure 1-3: Causes of Newborn Death, global.

Skilled care at birth and immediately thereafter would save the lives of many mothers and babies and prevent countless complications. Yet almost 50 percent of African women give birth without a skilled attendant, the average coverage of births with a skilled attendant on the continent has not increased significantly. Two in three women who need emergency obstetric care do not receive it. Scaling up skilled attendance and emergency obstetric care is fundamental to reaching to reducing maternal and neonatal mortality.

The situation in Ethiopia is similar to the situation in many developing countries. According to the EDHS 2016 only 28% were attended by a skilled health provider. Although the countries' maternal mortality ratio has decreased from 871/100 000 live births in 2000 to 412/100 000 live births in 2016 (DHS-2016), it is still very high and access to emergency obstetrical care is still limited.

A facility based National Baseline Assessment for **Emergency Obstetric & Newborn Care done in 2016** has revealed that 90% of women with obstetric complications who visited the facilities had direct complications and 10% had indirect complications (*see Table 1.1*). According to the report; APH and PPH/retained placenta were responsible for 12% of all maternal deaths and severe pre-eclampsia /eclampsia for 10%.

Table 1-1: The leading causes of maternal mortality in Ethiopia (a facility based study, 2016).

	Women with Complications (%)	Maternal Deaths (%)
Total DIRECT complications/causes	90	43
• APH	4	2
• PPH/Retainedplacenta	7	10
• Ruptured uterus	1	2
• Obstructed/ Prolonged labor/	18	4
• Postpartumsepsis	1	2
• Severe pre-eclampsia /eclampsia	6	10
• Severe complications ofabortion	2	1
• Others:		
oAbortions with less severe complications	25	NA
o Direct complications from other causes	26	12
Total INDIRECT complications/causes	10	5
• Malaria	0	0
• HIV/AIDS -related	7	0
• Anemia	1	2
• Other indirectcauses	2	2
• Undefinedcause	NA	52
TOTAL	100%	100%

2.3. FACTORS AFFECTING MATERNAL AND PERINATAL MORBIDITY AND MORTALITY

Multiple factors are responsible for the tragic high maternal and perinatal morbidity and mortality in the developing world including our country. These factors lead to delays in appropriate interventions at different levels. The ability of families and communities to recognize and access care quickly in case of an emergency determines the survival and health of both mother and fetus/newborn. For some obstetric complications, particularly haemorrhage, the window of opportunity to respond and save the life of the mother may be measured in hours. For the fetus or the newborn death can come even more quickly. Any delay may have fatal consequences (Box 1- 1).

BOX 1-1: Deadly delays

Three delays in care seeking affect the survival of both mothers and newborns.

1. Delays in recognizing problems and deciding to seekcare
 - Complications not recognized as serious
 - Family members delay care seeking
 - Spiritual or cultural beliefs may reinforce delays or result in other treatments
2. Delays in transportation to reach appropriate care
 - Lack of transport and/or funds
 - Distance and travel time to reach health facilities
3. Delays in receiving appropriate care at the health facility
 - Lack of appropriately trained staff and negative attitudes of health workers
 - Lack of essential equipment, drugs and supplies

The first two delays reveal questions about seeking care at the family and community level. Are families equipped to make healthy choices? Can the family and community support women when transportation and emergency costs are necessary? In many cultures, a woman must receive permission and money from her husband or other family members to seek care when complications take place. Long distance, high cost, and poor quality of care also contribute to the first and second delays. The third delay is related to health care providers, the facility, and the health system. In South Africa, data collected for the national perinatal problem identification programme, which now covers over one third of South Africa's births, show that while the majority of avoidable factors for stillbirths and neonatal deaths are related to poor maternal care during labor and the immediate postnatal period, about one third are due to delays at home and in transportation.

2.4. CURRENT APPROACHES TO REDUCTION OF MATERNAL AND NEONATAL MORTALITY

After reviewing the results of the millennium development goal and seeing that it is possible to accelerate the decline in maternal and neonatal mortality, countries have now united behind a new target to reduce maternal mortality even further. One target under Sustainable Development Goal 3 is to reduce the global maternal mortality ratio to less than 70 per 100 000 births, with no country having a maternal mortality rate of more than twice the global average. To achieve this goal, countries throughout the world are investing more energy and resources into providing equitable, adequate maternal health services.

Progress in many countries has led to a growing consensus in the maternal health field that reducing maternal and newborn deaths and disability can be achieved by ensuring

- 1) Access to family planning,
- 2) A skilled health professional present at every delivery and
- 3) Access to emergency obstetric and newborn care (EmONC),

Neonatal deaths are more common than maternal deaths and can be reduced through a range of approaches: institutional or community-based, ante partum, peripartum, and postpartum. Within this spectrum, skilled birth attendance is particularly advantageous for both maternal and neonatal survival. Associations between place of birth (the presence of a skilled attendant) and neonatal deaths are similar to those for maternal deaths; 90% coverage of facility-based clinical care alone could reduce neonatal mortality by 23–50%. The three biggest causes of neonatal death are preterm delivery, complications of presumptive birth asphyxia, and infection. The first two of these are manifest at the time of birth and about three-quarters of neonatal deaths occur in the first week, most of them in the first 2 days. If we can achieve high coverage of intrapartum care based in health centers, a qualitative change in labor monitoring and in early care for preterm newborn babies is likely to translate into a fall in early neonatal mortality.

In childbearing, women need a continuum of care to ensure the best possible health outcome for them and their newborns. The successful provision of the continuum of care requires a functioning health care system with the necessary infrastructure in place, including transport between the primary level of health care and referral clinics and hospitals. It also needs effective, efficient and proactive collaboration between all those involved in the provision of care to pregnant women and newborns. The skilled attendant is at the centre of the continuum of care. In 1999, a joint WHO/UNFPA/ UNICEF/World Bank statement¹ called on countries to “ensure that all women and newborns have skilled care during pregnancy, childbirth and the immediate postnatal period”. Skilled care refers to the care provided to a woman and her newborn during pregnancy, childbirth and immediately after birth by an accredited and competent health care provider who has at her/ his disposal the necessary equipment and the support of a functioning health system, including transport and referral facilities for emergency obstetric care. Since

skilled care as defined above can be provided by a range of health professionals, whose titles may vary according to specific country contexts, it has been agreed to refer to this health care provider as the “skilled attendant” or, “skilled birth attendant”, so as to avoid confusion over titles. Thus: a skilled attendant is an accredited health professional — a midwife, doctor, health officer or nurse—who has been educated and trained to proficiency in the skills needed to manage normal (uncomplicated) pregnancies, childbirth and the immediate postnatal period, and in the identification, management and referral of complications in women and newborns.

One way of reducing maternal mortality is by improving the availability, accessibility, quality and use of services for the treatment of complications that arise during pregnancy and childbirth. These services are collectively known as Emergency Obstetric and Newborn Care (EmONC) (See **Table 1-2** for classification and Criteria for EmONC services). Newborn deaths cannot be substantially reduced without efforts to reduce maternal deaths and improve maternal health. However, care during pregnancy and delivery must be accompanied by appropriate care of newborns and measures to reduce newborn deaths due to postnatal causes such as infections (tetanus, sepsis), hypothermia and asphyxia. Most postnatal deaths are caused by preventable and/or treatable diseases. Preventive interventions are simple, inexpensive, available and cost-effective. Most direct obstetric complications can be treated by a package of interventions identified by the World Health Organization (WHO), the United Nations Children's Fund (UNICEF) and the United Nations Population Fund (UNFPA) that, taken together, are known as emergency obstetric and newborn care (EmONC).

Table 1-2: Criteria for Basic and Comprehensive EmONC services.

In Primary Health Care Facility	In District Hospital
Basic EmONC	Comprehensive EmONC
• Parenteral antibiotics • Parenteral oxytocic drugs • Parenteral anticonvulsants • Manual removal of placenta • Removal of retained products of conception • Assisted vaginal delivery (vacuum extraction) • Newborn care	All Basic EmONC + • Surgery (caesarean section) • Blood transfusion

UNIT-3: WOMEN FRIENDLYCARE

3.1. UNIT LEARNINGOBJECTIVES

By the end of this unit participants will be able to:

- Define woman friendlycare.
- Describe cares which are womanfriendly.
- Describe examples of cares which are not womanfriendly.

3.2. WOMEN FRIENDLYCARE

Women-friendly care is life-saving as studies have shown that women may refuse to seek care from a provider who “abuses” them or does not treat them well, even if the provider is skilled in preventing and managing of complications.

Woman friendly care is a care that: -

- **Provides services that are acceptable to thewoman:**
 - Respects beliefs, traditions, and culture
 - Includes family, partner, or other support person in care
 - Provides relevant and feasibl eadvice
- **Empowers woman and her family to become active participants incare**
- **Considers the rights of the woman:**
 - Right to information about her health
 - Right to be informed about what to expect duringvisit
 - Obtains permission/consent prior to exams andprocedures
- **Ensures that all healthcare staff use good interpersonal communication skills**
- **Considers the emotional, psychological, and social well-being of the woman**

Some examples of care that is women-friendly: -

- Individualizes care to woman’sneeds
- Recognizes the richness and spiritual significance of community andculture
 - Is aware of traditional beliefs regarding pregnancy and childbirth

- Cooperates and liaises with traditional healthcare system when possible
- Provides culturally sensitive care
- Respects and supports the mother-newborn dyad:
 - Encourages bonding
 - Keeps newborn with mother
 - Places newborn on mother's abdomen (at breast) immediately after birth
- Speaks to the woman in her own language
- Observes rules and norms of her culture as appropriate
- Is aware of who makes decisions in her life and involves that person in discussions and decisions
- Works with traditional birth attendants when possible
- Learns about traditional practices:
 - Promotes/builds on positive traditional practices
 - Offers alternatives to those that are harmful

Some examples of care that is not women-friendly: -

- Does not respect woman or her culture or background
- Rude, offensive, demeaning language by health personnel
- Physically restrains, pushes or hits the woman
- Insists on routine procedures that are convenient for the healthcare provider but may be shameful or disgusting to the woman, eg lithotomy position only, routine episiotomy, frequent vaginal exams, assembly-line fashion of care
- Excludes partner or companion from care
- Separates mother and newborn

UNIT 4: INFECTION PREVENTION

4.1 Unit Learning Objectives

- Describe Infection Prevention and Patient safety
- Demonstrate key Infection Prevention and Patient Safety practices
- Discuss appropriate hand hygiene practice, Personal protective equipment and antisepsis,
- Discuss healthcare waste management, Instrument processing, health risk management, safe handling of sharps and House-keeping practices

4.2 Infection prevention practices

Infections are one of the main causes of maternal and neonatal mortality/morbidity. Infections are caused by pathogens, which in most cases are transferred from health care workers' hands, either from another patient or from contaminated equipment and surfaces. These infections can be prevented.

Follow these precautions to protect the woman, her baby, and the health provider, from infections with bacteria and viruses, including HIV.

Wash hands (*See the picture below for technique of hand washing*)

- Wash hands with soap and water:
- Before and after caring for a woman or newborn, and before any treatment procedure.
- Whenever the hands (or any other skin area) are contaminated with blood or other body fluids.
- After removing the gloves, because they may have holes.
- After changing soiled bedsheets or clothing.
- Keep nails short.

Wear gloves

- Wear sterile gloves when performing vaginal examination, delivery, cord cutting, repair of episiotomy or tear, blood drawing.
- Wear long sterile gloves for manual removal of placenta.
- Wear clean gloves when:
 - Handling and cleaning instruments
 - Handling contaminated waste
 - Cleaning blood and body fluid spills.
 - Drawing blood.

Figure 1-4. Technique of hand washing



Protect yourself from blood and other body fluids during deliveries

- Wear gloves; cover any cuts, abrasions or broken skin with a waterproof bandage; take care when handling any sharp instruments (use good light); and practice safe sharps disposal.
- Wear a long apron made from plastic or other fluid resistant material, and shoes.
- Protect your eyes and mouth from splashes of blood.

Practice safe sharps disposal

- Keep a puncture resistant container nearby.
- Use each needle and syringe only once.
- Do not recap, bend or break needles after giving an injection.
- Drop all used (disposable) needles, plastic syringes and blades directly into this container,

without recapping, and without passing to another person.

- Empty or send for incineration when the container is three-quarters full.

Practice safe waste disposal

Dispose of placenta or blood, or body fluid contaminated items, in leak-proof containers.

- Burn or bury contaminated solid waste.
- Wash hands, gloves and containers after disposal of infectious waste.
- Pour liquid waste down a drain or flushable toilet. Deal with contaminated laundry
- Collect clothing or sheets stained with blood or body fluids and keep them separately from other laundry, wearing gloves or use a plastic bag. DO NOT touch them directly.
- Rinse off blood or other body fluids before washing with soap.

Sterilize and clean contaminated equipment

- Make sure that instruments which penetrate the skin (such as needles) are adequately sterilized, or that single-use instruments are disposed of after one use.
- Thoroughly clean or disinfect any equipment which comes into contact with intact skin
- Use bleach for cleaning bowls and buckets, and for blood or body fluid spills.

MODULE 2

INITIAL RAPID ASSESSMENT AND MANAGING EMERGENCIES

b

Time:90min

Participant learning objective: After completing this module, participants will be able to provide care to obstetric emergencies in an organized and effective approach

Enabling objectives:

- Describe how to organize the facility to respond to an obstetric emergency.
- Describe key steps in rapid initial assessment of a woman with emergency problems.
- Demonstrate key emergency management steps for specific obstetric emergency problems.
- Describe pre-referral management to a recognized emergency situation
- Demonstrate steps in detection and management of “shock”.

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UNIT-1 INTRODUCTION

RAPID INITIAL ASSESMENT AND MANAGING EMERGENCIES

UNIT- INTRODUCTION

Responding to an emergency promptly and effectively requires that members of the clinical team know their roles and how the team should function to respond most effectively to emergencies. Rapid initiation of treatment requires immediate recognition of the specific problem and quick action. The skilled provider should perform a rapid initial Assessment to determine what is needed for immediate stabilization, management, and referral.

It is the responsibility of the skilled provider (nurse, Midwife, or physician) to make sure that all staff at the health institution (whether a clerk, guard, doorkeeper, cleaner, etc.) know how to respond to an emergency. The purpose of this module is to introduce participants to an organized and effective approach in providing care to obstetric emergencies.

Participant learning objective: After completing this module, participants will be able to provide care to obstetric emergencies in an organized and effective approach

Enabling objectives:

1. Describe how to organize the facility to respond to an obstetric emergency.
2. Describe key steps in rapid initial assessment of a woman with emergency problems.
3. Outline key emergency management steps for specific obstetric emergency problems.
4. Describe pre-referral management to a recognized emergency situation

UNIT-2: RAPID INITIAL ASSESMENT ANDMANAGINGEMERGENCIES

2.1. UNIT LEARNINGOBJECTIVES

By the end of this unit participants will be able to:

- Describe key steps in rapid initial assessment of a sickwoman.
- Outline key emergency management steps for specific emergencyproblems
- Describe important elements in assessment and management ofshock.
- Describe the emergency management of difficulty of breathing

2.2. BEING PREPARED AND RESPONDING TO ANEMERGENCY

Responding to an emergency promptly and effectively requires that members of the clinical team know their roles and how the team should function to respond most effectively to emergencies. Rapid initiation of treatment requires immediate recognition of the specific problem and quick action. The provider and staff should work together to plan for a way to respond to emergencies and all instruments and supplies needed for emergency care should be ready for use all the time.

Every staff at the health facility must be ready to help if a woman arrives with an obstetric emergency or signs and symptoms of advanced labor i.e. a staff person should be able to identify a woman with danger signs of pregnancy or in advanced labor, take her to the treatment area, and call for the skilled provider. To make this possible the skilled provider should give staff the information and training they need to be able to perform their jobs if there is an emergency.

IMPLEMENTING A RAPID INITIAL ASSESSMENT SCHEME

Rapid initiation of treatment requires immediate recognition of the specific problem and quick action. This can be done by:

- Training all staff—including clerks, guards, door-keepers or switchboard operators—to react in an agreed upon fashion (“sound the alarm”, call for help) when a woman arrives at the facility with an obstetric emergency or pregnancy complication or when the facility is notified that a woman is being referred;
- Clinical or emergency drills with staff to ensure their readiness at alllevels;
- Ensuring that access is not blocked (keys are available) and equipment is in working order (daily checks) and staff are properly trained to useit;
- Having norms and protocols (and knowing how to use them) to recognize a genuine

emergency and know how to react immediately;

- Clearly identifying which women in the waiting room—even those waiting for routine consultations—warrant prompt or immediate attention from the health worker and should therefore pass to the front of the queue (agreeing that women in labor or pregnant women who have any of the problems noted in table 1-8 should immediately be seen by a health worker);
 - Agreeing on schemes by which women with emergencies can be exempted from payment, at least temporarily (local insurance schemes, health committee emergency funds).

All staff at the health facility should be able to perform a Quick Check

QUICK CHECK

- **Look at the woman:**
 - Did someone carry her into the health institution? (possible sign of shock)
 - Is there blood on her clothing or on the floor beneath her? (sign of bleeding in pregnancy.)
 - Is she grunting, moaning, or bearing down? (possible signs of advanced labor)
- **Ask the woman or someone who is with her whether she has now or has recently had:**
 - Vaginal bleeding
 - Severe headache/blurred vision
 - Convulsions or loss of consciousness
 - Difficulty breathing
 - Fever
 - Severe abdominal pain
 - Labor pains
- **If the woman has or recently had ANY of these danger signs, or signs and symptoms of advanced labor, immediately:**
 - Shout for help.
 - Stay calm. Focus on the woman.
 - Do not leave the woman alone.
 - Notify the skilled provider.

2.3. RAPID INITIAL ASSESMENT

When a woman of childbearing age presents with a problem, perform a rapid initial assessment to determine what is needed for immediate stabilization, management, and referral. Rapidly assess her condition with a quick examination including blood pressure, pulse, respiration, temperature, and skin color to determine her degree of illness. See table 2-1 for specific steps.

TABLE 2-1:- Rapid initial assessment

Assess	Danger Signs	Consider
Airway and breathing	LOOK FOR: • cyanosis (blueness) • respiratory distress EXAMINE: • skin: pallor • lungs: wheezing or rales	• severe anemia • heart failure • pneumonia • asthma See Difficulty in breathing section 2.6
Circulation (signs of shock)	EXAMINE: • skin: cool and clammy • pulse: fast (110 or more) and weak • blood pressure: low (systolic less than 90 mm Hg)	Shock,
Vaginal bleeding (early or late pregnancy or after childbirth)	ASK IF: • pregnant, length of gestation • recently given birth • placenta delivered EXAMINE: • vulva: amount of bleeding, placenta retained, obvious tears • uterus: atony • bladder: full DO NOT DO A VAGINAL EXAM AT THIS STAGE	• abortion • ectopic pregnancy • molar pregnancy (See Vaginal bleeding in early pregnancy in obstetric hemorrhage module) • abruptio placentae • ruptured uterus • placenta previa (See Vaginal bleeding in later pregnancy and labor,) • atonic uterus • tears of cervix and vagina • retained placenta • inverted uterus (See Vaginal bleeding after childbirth,)
Unconscious or convulsing	ASK IF: • pregnant, length of gestation EXAMINE: • blood pressure: high (diastolic 90 mm Hg or More) • temperature: 38°C or more	• eclampsia • malaria • epilepsy • tetanus (Convulsions or loss of consciousness,)

^a This list does not include all the possible problems that a woman may face in a pregnancy or the puerperal period.

It is meant to identify those problems that put the woman at greater risk of maternal morbidity and mortality.

TABLE 2-2:-Cont. Rapid initial assessment

Assess	Danger Signs	Consider
Dangerous fever	ASK IF: • weak, lethargic • frequent, painful urination EXAMINE: • temperature: 38°C or more • unconscious • neck: stiffness • lungs: shallow breathing, consolidation • abdomen: severe tenderness • vulva: purulent discharge • breasts: tender	• urinary tract infection • malaria (See Fever during pregnancy and labor,) • metritis • pelvic abscess • peritonitis • breast infection (See Fever after childbirth) • complications of abortion (See Vaginal bleeding in early pregnancy, • pneumonia) (See Difficulty in breathing,)
Abdominal pain	ASK IF: • pregnant, length of gestation EXAMINE: • blood pressure: low (systolic less than 90 mm Hg) • pulse: fast (110 or more) • temperature: 38°C or more • uterus: state of pregnancy	• ovarian cyst • appendicitis • ectopic pregnancy (See Abdominal pain in early pregnancy,) • possible term or preterm labor • amnionitis • abruptio placentae • ruptured uterus (See Abdominal pain in later pregnancy and after childbirth,)

The woman also needs **prompt attention** if she has any of the following signs: blood-stained mucus discharge (show) with palpable contractions; ruptured membranes; pallor; weakness; fainting; severe headaches; blurred vision; vomiting; fever or respiratory distress. The woman should be sent to the front of the queue and promptly treated.

2.4. EMERGENCY MANAGEMENT PRINCIPLES

Emergencies can happen suddenly, as with a convulsion, or they can develop as a result of a complication that is not properly managed or monitored.

PREVENTING EMERGENCIES

Most emergencies can be prevented by:

- careful planning
- following clinical guidelines;
- Close monitoring of the woman.

RESPONDING TO AN EMERGENCY

Responding to an emergency promptly and effectively requires that members of the clinical team know their roles and how the team should function to respond most effectively to emergencies. Team members should also know:

- clinical situations and their diagnoses and treatments;
- drugs and their use, administration and side effects;
- emergency equipment and how it functions.

INITIAL MANAGEMENT

In managing an emergency:

- Stay calm. Think logically and focus on the needs of the woman.
- Do not leave the woman unattended.
- Take charge. Avoid confusion by having one person in charge.
- **SHOUT FOR HELP.** Have one person go for help and have another person gather emergency equipment and supplies (e.g. oxygen cylinder, emergency kit).
- If the **woman is unconscious**, assess the airway, breathing and circulation.
- If **shock is suspected**, immediately begin treatment. Even if signs of shock are not present, keep shock in mind as you evaluate the woman further because her status may worsen rapidly. If **shock develops**, it is important to begin treatment immediately.
- Position the woman lying down on her left side with her feet elevated. Loosen tight clothing.
- Talk to the woman and help her to stay calm. Ask what happened and what symptoms she is experiencing.
- Perform a quick examination including vital signs (blood pressure, pulse, respiration, temperature) and skin color. Estimate the amount of blood lost and assess symptoms and signs.

REFERRING THE WOMAN FOR CARE

- After emergency management, discuss decision to refer with woman and family.
- Quickly organize transport and possible financial aid.
- Inform the referral center by radio or phone.
- Give the woman a referral slip containing the following information:
 - Name, age, address
 - Obstetric history (parity, gestational age, antenatal complications)
 - Relevant past obstetric complications
 - Specific problem for which she is referred
 - Treatment applied thus far and results
- Send with the woman:
 - A health worker trained in childbirth care
 - Essential emergency drugs and supplies
 - A family member who can support and attend her

- If there is a newborn, send with the mother if there is a family member who can go with the mother to care for the neonate.
- During journey:
- Maintain IV infusion.
 - Keep the woman (and newborn, if born) warm but do not overheat.
 - If journey is long, give appropriate treatment on the way.
 - Keep record of all IV fluids, medications given, time of administration, and woman's condition.

2.5. MANAGING A PATIENT IN “SHOCK”

Shock is characterized by failure of the circulatory system to maintain adequate perfusion of the vital organs. Shock is a **life-threatening condition** that requires **immediate and intensive treatment**.

Suspect or anticipate shock if at least one of the following is present:

- Bleeding in early pregnancy (e.g. abortion, ectopic or molar pregnancy);
- Bleeding in late pregnancy or labor (e.g. placenta praevia, abruptio placentae, ruptured uterus);
- Bleeding after childbirth (e.g. ruptured uterus, uterine atony, tears of genital tract, retained placenta or placental fragments);
- Infection (e.g. unsafe or septic abortion, amnionitis, metritis, pyelonephritis);
- Trauma (e.g. injury to uterus or bowel during abortion, ruptured uterus, tears of genital tract).

SYMPTOMS AND SIGNS

Diagnose shock if the following symptoms and signs are present:

- Fast, weak pulse (110 per minute or more);
- Low blood pressure (systolic less than 90 mmHg).

Other symptoms and signs of shock include:

- Pallor (especially of inner eyelid, palms or around mouth);
- Sweatiness or cold clammy skin;
- Rapid breathing (rate of 30 breaths per minute or more);
- Anxiousness, confusion or unconsciousness;

- Scanty urine output (less than 30 mL per hour).

MANAGEMENT

IMMEDIATE MANAGEMENT

- **SHOUT FOR HELP.** Urgently mobilize all available personnel.
- Monitor vital signs (pulse, blood pressure, respiration, temperature).
- Turn the woman onto her side to minimize the risk of aspiration if she vomits and to ensure that an airway is open.
- Keep the woman warm but do not overheat her as this will increase peripheral circulation and reduce blood supply to the vital centers.
- Elevate the legs to increase return of blood to the heart (if possible, raise the foot end of the bed).

SPECIFIC MANAGEMENT

- Start an IV infusion (two if possible) using a large-bore (16-gauge or largest available) cannula or needle. Collect blood for estimation of haemoglobin, immediate cross-match and bedside clotting (see below), just before infusion of fluids:
 - Rapidly infuse IV fluids (normal saline or Ringer's lactate) initially at the rate of 1 L in 15–20 minutes;
 - Give at least 2 L of these fluids in the first hour. This is over and above fluid replacement for ongoing losses.

Note: A more rapid rate of infusion is required in the management of shock resulting from bleeding. Aim to replace 2–3 times the estimated blood loss.

Do not give fluids by mouth to a woman in shock.

- If a **peripheral vein cannot be cannulated**, perform a venous cut-down by skilled person
- Continue to monitor vital signs (every 15 minutes) and blood loss.
- Catheterize the bladder and monitor fluid intake and urine output.
- Give oxygen at 6–8 L per minute by mask or nasal cannulae.

BEDSIDE CLOTTING TEST:

- Assess clotting status using this **bedside clotting test**:
 - Take 2 mL of venous blood into a small, dry, clean, plain glass test tube (approximately 10 mm x 75mm);
 - Hold the tube in your closed fist to keep it warm (+37°C);
 - After 4 minutes, tip the tube slowly to see if a clot is forming. Then tip it again every minute until the blood clots and the tube can be turned upside down;
 - Failure of a clot to form after 7 minutes or a soft clot that breaks down easily suggests coagulopathy.

DETERMINING AND MANAGING THE CAUSE OF SHOCK

Determine the cause of shock after the woman is stabilized.

1. If **heavy bleeding** is suspected as cause of shock:
 - Take steps simultaneously to stop bleeding (e.g. oxytocics, uterine massage, bimanual compression, aortic compression, preparations for surgical intervention);
 - Determine the cause of bleeding and manage:
 - If **bleeding occurs during first 28 weeks of pregnancy**, suspect abortion, ectopic or molar pregnancy;
 - If **bleeding occurs after 28 weeks or during labor but before delivery**, suspect placenta praevia, abruptio placentae or ruptured uterus;
 - If **bleeding occurs after childbirth**, suspect ruptured uterus, uterine atony, tears of genital tract, retained placenta or placental fragments.
 - Reassess the woman's condition for signs of improvement.
 - Consider consultation/referral for transfusion as soon as possible to replace blood loss;
2. If **infection is suspected** as the cause of shock:
 - i. Collect appropriate samples (blood, urine, pus) for microbial culture before starting antibiotics, if facilities are available;
 - ii. Give the woman a combination of antibiotics to cover aerobic and anaerobic infections and continue until she is fever-free for 48 hours:
 1. Penicillin G 2 million units OR ampicillin 2 g IV every 6 hours;
 2. PLUS gentamicin 5 mg/kg body weight IV every 24 hours (~80 mg tid);
 3. PLUS metronidazole 500 mg IV every 8 hours.

Do not give antibiotics by mouth to a woman in shock.

- iii. Reassess the woman's condition for signs of improvement.
3. If **trauma is suspected** as the cause of shock, prepare for surgical intervention.

REASSESSMENT

- Reassess the woman's response to fluids within 30 minutes to determine if her condition is improving. Signs of improvement include:
 - Stabilizing pulse (rate of 90 per minute or less);
 - Increasing blood pressure (systolic 100 mm Hg or more);
 - Improving mental status (less confusion or anxiety);
 - Increasing urine output (30 mL per hour or more).
- If the woman's **condition improves**:
 - Adjust the rate of infusion of IV fluids to 1 L in 6 hours;
 - Continue management for the underlying cause of shock.
- If the woman's condition **fails to improve or stabilize**, she requires further management.

FURTHER MANAGEMENT

- Continue to infuse IV fluids, adjusting the rate of infusion to 1 L in 6 hours and maintain oxygen at 6–8 L per minute.
- Closely monitor the woman's condition.
- Perform laboratory tests including hematocrit, blood grouping and Rh typing and consider referral/consultation for transfusion if needed.

2.6. 1 MANAGING A PATIENT WITH DIFFICULTY IN BREATHING PROBLEM

- A woman is short of breath during pregnancy, labor or after delivery.

GENERAL MANAGEMENT

- Make a rapid evaluation of the general condition of the woman including vital signs (pulse, blood pressure, respiration, temperature).
- Prop up the woman on her left side.
- Start an IV infusion and infuse IV fluids.
- Give oxygen at 4–6 L per minute by mask or nasal cannulae.
- Obtain haemoglobin estimates.

DIAGNOSIS

See **Table 2.3** below for diagnostic approach

Table 2.3. Diagnosis of difficulty in breathing

Presenting Symptom and Other Symptoms and Signs Typically Present	Symptoms and Signs Sometimes Present	Probable Diagnosis
•Difficulty in breathing •Pallor of conjunctiva, tongue, nail beds and/or palms •Haemoglobin 7g per dL or less •Hematocrit 20% or less	•Lethargy and fatigue •Flat or concave nails	Severe anemia,
•Symptoms and signs of severe anemia	•Oedema •Cough •Rales •Swelling of legs •Enlarged liver •Prominent neck veins	Heart failure due to anemia,
•Difficulty in breathing •Diastolic murmur and/or •Harsh systolic murmur with palpable thrill	•Irregular heart beat •Enlarged heart •Rales •Cyanosis (blueness) •Cough •Swelling of legs •Enlarged liver •Prominent neck veins	Heart failure due to heart disease,
•Difficulty in breathing •Fever •Cough with expectoration •Chest pain	•Consolidation •Congested throat •Rapid breathing •Rhonchi/rales	Pneumonia,
•Difficulty in breathing •Wheezing	•Cough with expectoration •Rhonchi/rales	Bronchial asthma,
•Difficulty in breathing •Hypertension •Proteinuria	•Rales •Frothy cough	Pulmonary oedema associated with pre-eclampsia^a

^a Withhold fluids and give frusemide 40 mg IV once .

6.2.3 SPECIFIC MANAGEMENT

A. SEVERE ANEMIA

- If *Plasmodium falciparum* malaria is suspected, manage as severe malaria.
- Where **hookworm is endemic** (prevalence of 20% or more), give one of the following anthelmintic treatments:
 - Albendazole 400 mg by mouth once;
 - OR mebendazole 500 mg by mouth once or 100 mg two times per day for 3 days;
 - OR levamisole 2.5 mg/kg body weight by mouth once daily for 3 days;
 - OR pyrantel 10 mg/kg body weight by mouth once daily for 3 days.
- If **hookworm is highly endemic** (prevalence of 50% or more), repeat the anthelmintic treatment 12 weeks after the first dose.
- Consult/refer for transfusion as necessary:

B. HEART FAILURE

HEART FAILURE DUE TO ANEMIA

- Consult/refer for transfusion of packed or sedimented cells (almost always necessary in heart failure due to anemia):

HEART FAILURE DUE TO HEART DISEASE

- Treat acute heart failure. Drugs used may include:
 - Morphine 10 mg IM as a single dose;
 - OR furosemide 40 mg IV, repeated as necessary;
 - OR digoxin 0.5 mg IM as a single dose;
 - OR nitroglycerine 0.3 mg under the tongue, repeated in 15 minutes, if necessary.
- Refer to a higher level if needed.

MANAGEMENT OF HEART FAILURE DURING LABOR

- Such a case should be referred immediately to a higher facility; but use the following steps if she comes late in labor or in all cases until referral.
- Prop up the woman on her left side.
- Limit infusion of IV fluids to decrease the risk of circulatory overload and maintain a strict fluid balance chart.
- Ensure adequate analgesia.

Note: Do not give ergometrine

- Have the woman avoid sustained bearing down efforts during the expulsive stage, if possible.
- If **necessary to decrease the woman's workload during delivery**, perform an episiotomy and assist delivery by vacuum extraction or forceps.
- Ensure active management of third stage.

C. PNEUMONIA

Inflammation in pneumonia affects the lung parenchyma and involves respiratory bronchioles and alveoli. There is loss of lung capacity that is less tolerated by pregnant women.

- A radiograph of the chest may be required to confirm the diagnosis of pneumonia.
- Give erythromycin base 500 mg by mouth four times per day for 7 days.
- Give steam inhalation.

Consider the possibility of tuberculosis in areas where it is prevalent.

D. BRONCHIAL ASTHMA

Bronchial asthma complicates 3–4% of pregnancies. Pregnancy is associated with worsening of the symptoms in one-third of affected women.

- If **bronchospasm occurs**, give bronchodilators (e.g. salbutamol 4 mg by mouth every 4 hours or 250 mcg aerosol every 15 minutes for 3 doses).
- If there is **no response to bronchodilators**, give corticosteroids such as hydrocortisone IV 2 mg/kg body weight every 4 hours as needed.
- If **there are signs of infection** (bronchitis), give ampicillin 2 g IV every 6 hours.
- Avoid the use of prostaglandins. For prevention and treatment of postpartum haemorrhage, give oxytocin 10 units IM or give ergometrine 0.2 mg IM.
- After acute exacerbation has been managed, continue treatment with inhaled bronchodilators and inhaled corticosteroids to prevent recurrent acute episodes.

MODULE - 3

BASIC PREGNANCY CARE(ANTENATAL CARE)

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UNIT-1: INTRODUCTION

Good care during pregnancy is important for the health of the mother and the development of the fetus. Pregnancy is a crucial time to promote healthy behavior and parenting skills. Good Antenatal Care links the woman and her family with the formal health system, increases the chance of using a skilled attendant at birth and contributes to good health through the life cycle. Inadequate care during this time breaks a critical link in the continuum of care, and affects both women and babies:

It has been estimated that 25 percent of maternal deaths occur during pregnancy, with variability between countries depending on the prevalence of unsafe abortion, violence, and disease in the area. Between a third and a half of maternal deaths are due to causes such as hypertension (pre-eclampsia and eclampsia) and ante partum hemorrhage, which are directly related to inadequate care during pregnancy. Certain pre-existing conditions become more severe during pregnancy. Malaria, HIV/AIDS, anemia and malnutrition are associated with increased maternal and newborn complications as well as death where the prevalence of these conditions is high.

In sub-Saharan Africa, an estimated 900,000 babies die as stillbirths during the last twelve weeks of pregnancy. According to the 2016 EmONC report, the institutional stillbirth rate was 15 per 1,000 deliveries, and the neonatal death rate 3 per 1,000 live births in Ethiopia. It is estimated that antepartum stillbirths (fetal deaths before onset of labor), account for two-thirds of all stillbirths in countries where the mortality rate is greater than 22 per 1,000 births – nearly all African countries. Ante partum stillbirths have a number of causes, including maternal infections – notably syphilis – and pregnancy complications, but systematic global estimates for causes of ante partum stillbirths are not available. Newborns are affected by problems during pregnancy including preterm birth and restricted fetal growth, as well as other factors affecting the child's development such as congenital infections (e.g HIV, Toxoplasmosis, Syphilis).

Despite the Ethiopian FMOH efforts to increase access to ANC from skilled provider, the antenatal care coverage is still low. The 2016 EDHS results show that 62 percent of women who gave birth in the five years preceding the survey received antenatal care from a skilled provider at least once for their last birth.

Participant learning objective: after completing this module, participants will be able to understand the evidence based approaches in the care of women during pregnancy which will enhance the knowledge, attitudes and practices of skilled health attendants on quality antenatal care.

Enabling objectives:

1. Describe focused antenatal care.
2. Provide focused antenatal care to the pregnant woman.
3. Describe antenatal care provision for PMTCT
4. Recognize an emergency situation during pregnancy which requires immediate treatment

and urgent referral to a higher level health facility.

UNIT-2: FOCUSED ANTENATAL CARE

2.1. LEARNING OBJECTIVES

- To enhance the knowledge, attitudes and practices of skilled health attendants on quality antenatal care.

At the end of this unit, participants will be able to:

- Define focused ANC.
- Appreciate the value of focused ANC
- Describe the basic components of focused ANC

2.2. INTRODUCTION

Preventing maternal and perinatal mortality and morbidity depends on an operational continuum of care with accessible, high quality care before and during pregnancy, childbirth, and the postnatal period.

ANC is defined as the complex of interventions that a pregnant woman receives from organized health care facilities with the objective of assuring every pregnancy to culminate in the delivery of a healthy child without impairing the health of the mother. The goal of the ANC package is to prepare for birth and parenthood as well as prevent, detect, alleviate, or manage the three types of health problems during pregnancy that affect mothers and babies:

- Complications of pregnancy itself,
- Pre-existing conditions that worsen during pregnancy,
- Effects of unhealthy lifestyles.

ANC also provides women and their families with appropriate information and advice for a healthy pregnancy, safe childbirth, and postnatal recovery, including care of the newborn, promotion of early, exclusive breastfeeding, and assistance with deciding on future pregnancies in order to improve pregnancy outcomes. The care should be appropriate, cost-effective and based on individual needs of the mother. An effective ANC package depends on competent health care providers in a functioning health system with referral services and adequate supplies and laboratory support. ANC improves the survival and health of babies directly by reducing stillbirths and neonatal deaths and indirectly by providing an entry point for health contacts with the woman at a key point in the continuum of care.

ANC indirectly saves the lives of mothers and babies by promoting and establishing good health before childbirth and the early postnatal period – the time periods of highest risk. ANC often

presents the first contact opportunity for a woman to connect with health services, thus offering an entry point for integrated care, promoting healthy home practices, influencing care seeking behavior, and linking women with pregnancy complications to a referral system. Women are more likely to give birth with a skilled attendant if they have had at least one ANC visit.

While research has demonstrated the benefits of ANC through improved health of mothers and babies, the exact components of ANC and what to do at what time have been matters of debate. In recent years, there has been a shift in thinking from the high risk approach to focused ANC. The high risk approach intended to classify pregnant women as “low risk” or “high risk” based on predetermined criteria and involved many ANC visits. This approach was hard to implement effectively since many women had at least one risk factor, and not all developed complications; at the same time, some low risk women did develop complications, particularly during childbirth. The 2002 WHO ANC model suggested that essential interventions can be provided over four goal –oriented visits at specified intervals, at least for healthy women with no underlying medical problems. This model has been further defined by what is done in each visit, and is often called *focused antenatal care*. Focused ANC is an approach to ANC that emphasizes on: -

- ☐ Individualized & client centered care,
- ☐ Fewer but comprehensive visits,
- ☐ Disease detection not risk classification and
- ☐ Care by a skilled provider.

2.3. FOCUSED ANTENATALCARE

2.3.1 OVERVIEW

Focused or goal oriented ANC service emphasizes the quality of care rather than the quantity, provides specific evidence-based interventions for all women, carried out at certain critical times in the pregnancy. For normal pregnancies WHO recommends only four antenatal visits. The major goal of focused antenatal care is to help women maintain normal pregnancies through:

- ☐ Early detection and treatment of problems and complications
- ☐ Prevention of complications and disease
- ☐ Birth preparedness and complication readiness
- ☐ Health promotion

For many of the essential interventions in ANC, it is crucial to have early identification of underlying conditions – for example, prevention of congenital syphilis, control of anemia, and prevention of malaria complications. Hence the first ANC visit should be as early as possible in pregnancy, preferably in the first trimester. The last visit should be at around 37 weeks or near the expected date of birth to ensure that appropriate advice and care have been provided to prevent and manage problems such as multiple births (e.g. twins), post maturity (which carries an increased risk of fetal death), and fetal malpresentations.

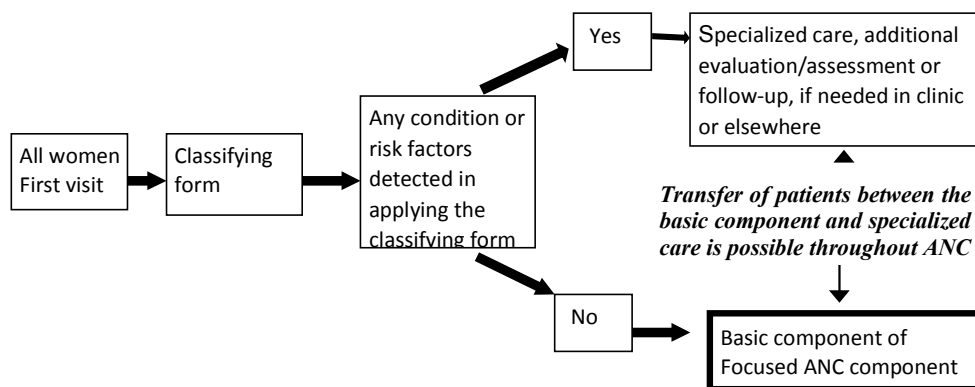
At the outset, focused antenatal care segregates pregnant women into two groups based on their specific health conditions or risk factors (**Figure 3-1**):

- Those eligible to receive routine ANC (called the **basic component**);and
- Those who need **specialized care**

Pre-set criteria are used to determine the eligibility of women for the basic component. The women selected to follow the basic component are considered not to require any further assessment or special care at the time of the first visit regardless of the gestational age at which they start the programme. The remaining women are given care corresponding to their detected condition or risk factor. The women who need special care will represent, on average, approximately 25% of all pregnant women initiating antenatal care.

A **classifying checklist** (**Figure 3-2**) is used at the first antenatal visit to the clinic to decide which women will follow the basic component of the new WHO model and which will require special care. The form contains 19 checklist questions that require binary responses (yes/no). They cover the patient's obstetric history, their current pregnancy and general medical conditions. Women who answer 'yes' to any of the 19 questions would not be eligible for the basic component of the new WHO antenatal care model; they should receive care corresponding to the detected condition.

Figure 3-1: The Focused ANC Model



Transfer of patients between the basic component and specialized care, however, is possible throughout ANC. A woman in the basic care component category can be transferred any time during ANC follow up to a specialized care category. It is also possible that a woman who is initially referred to a higher level of care because of a condition identified in the classifying form is subsequently considered suitable to follow the basic component of the focused ANC model. In such a situation, the woman would have to undergo all the activities included in the basic component that correspond to her fetus's gestational age. In addition, she would have to undergo

all activities that she missed owing to her late entry into the basic component that were not performed during her visit(s) to the higher level of care. The activities included in the basic component fall within three general areas:

- **Problem identification: Screening** for health and socio-economic conditions likely to increase the possibility of specific adverse outcomes;
- **Care provision:** Providing **therapeutic interventions** known to be beneficial; and
- **Health promotion:** health education and educating pregnant women about planning for safe birth, emergencies during pregnancy and how to deal with them (birth planning & emergency preparedness).

The activities distributed over the four visits are presented in the **basic component checklist (Figure 3-3)**. This checklist should be used to record tests and interventions performed at each ANC visit and should be incorporated into the medical records for each patient. Results of tests or treatments recommended should be recorded in the clinic's medical records as is normally done. The checklist is not intended to replace the clinic's medical records. Rather, it is designed to serve as a reminder of the activities that have been and must be performed.

Ideally, the first visit should occur in the first trimester, around, or preferably before, week 12 of pregnancy. However, regardless of the gestational age at first enrolment, all pregnant women coming to the clinic for ANC will be enrolled and examined according to the norms for the first, and subsequent, visits. As the basic component includes only four visits, sufficient time must be made during each visit for discussion of the pregnancy and related issues with the pregnant woman. The first visit can be expected to take 30–40 minutes. Individual interaction between the patient and health care provider is an essential element of focused ANC model. For details of activities see learning guide 3-1 in the Participant's handout, and the national management protocol on selected obstetric topics.

N;B One ultrasound scan before 24 weeks of gestation (early ultrasound) is recommended for pregnant women to estimate gestational age, improve detection of fetal anomalies and multiple pregnancies, reduce induction of labor for post-term pregnancy, and improve a woman's pregnancy experience (WHO 2016).

Figure 3-2: -Focused ANC classifying form;

Federal Ministry of Health Integrated Antenatal, Labor, Delivery, Newborn and Postnatal Care Card			
Date: _____ ANC Reg.No: _____ Medical RecordNumber(MRN): _____			
NameofClient: _____ Nameof Facility _____			
Woreda: _____ Kebele: _____ HouseNo: _____			
Age(Years) _____ LMP ____/____/____ EDD ____/____/____			
Gravida ____ Para ____ Number ofchildren alive _____ Marital Status _____			
INSTRUCTIONS to Fill Classifying form: Answer all of the following questions by placing a cross mark in the corresponding box.			
OBSTETRIC HISTORY		No	Yes
1. Previous stillbirth or neonatal loss?		☐	☐
2. History of 3 or more consecutive spontaneous abortions?		☐	☐
3. Birth weight of last baby < 2500g		☐	☐
4. Birth weight of last baby > 4000g		☐	☐
5. Last pregnancy: hospital admission for hypertension or pre-eclampsia/eclampsia?		☐	☐
6. Previous surgery on reproductive tract?(Myomectomy, removal of septum, fistula repair, cone biopsy, CS, repaired rapture, cervical circlage)		☐	☐
CURRENT PREGNANCY		No	Yes
7. Diagnosed or suspected multiple pregnancy?		☐	☐
8. Age less than 16 years?		☐	☐
9. Age more than 40 years?		☐	☐
10. Isoimmunization Rh (-) in current or in previous pregnancy		☐	☐
11. Vaginal bleeding		☐	☐
12. Pelvic mass		☐	☐
13. Diastolic blood pressure 90mm Hg or more at booking?		☐	☐
GENERAL MEDICAL		No	Yes
14. Diabetes mellitus		☐	☐
15. Renal disease		☐	☐

16. Cardiac disease					
17. Chronic Hypertension					
18. Known 'substance' abuse (including heavy alcohol drinking, Smoking)					
19. Any other severe medical disease or condition TB, HIV, Ca, DVT..					

A "Yes" to any ONE of the above questions (i.e. ONE shaded box marked with a cross) means that the woman is not eligible for the basic component of the new antenatal care mode and require more close follow up or referral to specialty care. If she needs more frequent ANC visits use and attach additional recording sheets

Figure 3-3:- New WHO antenatal care model basic component checklist, Integrated client card

II. Initial Evaluation plus Promotive and Preventive Care				Counseling /Testing, HIV+ Care and follow up	
General Exam		Gyn Exam			
General		Vulvar Ulcer	☒	Danger signs in pregnancy & delivery advised Birth	☒
			☒	Preparedness advised	☒
Pallor	☒	Vaginal discharge	☒	MOTHER	
	☒		☒	HIV test accepted	☒
Jaundice		PelvicMass		HIV test result	☒
			☒		
ChestAbn.	☒		☒	PARTNER	
			☒	Partner HIV test result	
Heart abnormality	☒	Uterine size (Wks)			
	☒	Cervical Lesion	☒		
III. Present Pregnancy: Follow Up					
	1st visit (better before 16 wks)	2nd visit (better 24 - 28 wks)	3rd visit (better 30 -32 wks)	4th visit (better 36-40w ks)	
Date of visit					
Gestation age (LMP)					
BP					
Weight (Kg)					
Pallor					
Uterine height (Wks)					
Fetal heart beat					
Presentation					
Urine test for infection					
Urine test for protein					
Rapid syphilis test					
Hemoglobin					
Blood Group and Rh					
Ultrasound					
TT (dose)					
Iron/Folic Acid					
Mebendazole					
Use of ITN/ITP					
ARV Px (type)					
Remarks					

Firstvisit Secondvisit ThirdVisit FourthVisit

Danger signs identified and Investigation:				
Action, Advice, counseling				
Appointment for next follow-up				
Name and Sign of Health care Provider				

2.3.2. PROVIDE PREVENTIVE MEASURES

☐ **i) iron and folic acid supplements: -**

- ☐ Daily oral iron and folic acid supplementation is recommended for pregnant women to prevent maternal anaemia, puerperal sepsis, low birth weight, and preterm birth.
- ☐ → 60mg elemental iron and 400 µg folic acid once daily in pregnancy and until 3 months after delivery or abortion.

→ Twice daily as treatment for anemia (double dose for 3 months).

Note: 1 Iron Folate tablet = ~60mg Iron, folic acid = 400µg

- Check woman's supply of iron and folic acid at each visit and dispense 3 month supply.
- To enhance the absorption of iron, instruct mothers to take iron when eating meat or vitamin-rich foods (fruits and vegetables). Avoid tea, coffee, and milk at the same time when taking iron; it interferes with the body's absorption of iron. Iron can also be taken between meals.
- Advise to store iron safely:
 - Where children cannot get it
 - In a dry place.

N;B Intermittent oral iron and folic acid supplementation with 120 mg of elemental iron and 2800 µg (2.8 mg) of folic acid once weekly is recommended for pregnant women to improve maternal and neonatal outcomes if daily iron is not acceptable due to side-effects, and in populations with an anaemia prevalence among pregnant women of less than 20%.

ii) Prevention of Malaria

- ☐ Protection against malaria with insecticide-treated bed nets is recommended.

iii) Immunization against tetanus:-

■ Check tetanus toxoid (TT) immunization status.

- if the woman has not previously been vaccinated, or if her immunization status is unknown, give two doses of TT/Td one month apart before delivery (last dose at least 2 weeks before delivery), and further doses as per **Table 3.1**;
- if the woman has had 1–4 doses of tetanus toxoid in the past, give one dose of TT/Td before delivery
- if the woman can show written proof of vaccination in childhood or adolescence with tetanus-containing vaccine (e.g. DTP, DT, Td, TT) administer 1 dose of TT at present contact/pregnancy and a second dose a year later.

- Give tetanus toxoid if has not been vaccinated. For TT administration schedule see table 3-1 below.

■

Table 3-1:-Schedule for Tetanus Toxoid administration.

Dose	Time for administration	Duration of protection
TT1	At first contact	No protection
TT2	4 weeks after TT1	Three years
TT3	At least 6 months after TT2	Five years
TT4	At least one year TT3	Ten years
TT5	At least one year after TT4	For thirty years (throughout a woman's reproductivelife

iv) Preventive anti-helmenthic treatment (De-worming):-

- ☐ Studies have shown that eradication of hookworms in endemic areas could prevent 41 to 56% of moderate to severe anemia.
- ☐ Give mebendazole 500 mg once in second or third trimester to every woman in hookworm endemic areas.

Careprovision based on screeningtest results

v) Antibiotics for asymptomatic bacteriuria (ASB)

A seven-day antibiotic regimen with amoxyciline or cephalaxine is recommended for all pregnant women with asymptomatic bacteriuria (ASB) to prevent persistent bacteriuria, preterm birth and low birth weight. The best criteria to diagnose ASB is: a single urinary bacterial isolate of 10^5 CFU in mid-stream urine culture. But in low resource settings a urine dip stick test positive for either leucocytes or nitrates can be used.

vi) treat Latent (asymptomatic) syphilis

- In a pregnant woman with positive VDRL/ RPR laboratory tests, consider a possible syphilis & give benzathine penicillin 2.4miu im every week for 3 weeks. In case of penicillin allergy, give erythromycin. Encourage woman to bring her sexual partner for treatment. Paln to treat the newborn after delivery (Neonatal blood tests & treat with ampiciline/ benzyi penicillin if positive for VDRL/RPR).

vii) RH (Anti-D) prophylaxis

Anti_D (Rhogam, 300microgram) should be administered as early as possible after delivery (preferably within 72hrs) in RH negative women if antibody screening test (coomb's test) is negative (i.e mother is unsensitized).

N;B an antenatal administration of anti-D at GA of 28wks can be considered after counseling of the woman & the family including a discussion on the economic implications of anti-D & the very low risk of antenatal sensitization (WHO concluded that it is only recommended in the context of rigorous research.

2.3.3 HEALTH PROMOTION

Health promotion refers to any activity that aims to achieve better health in a community or a country. It includes the health education of individuals to enable them to control and change their lifestyles so that their health is improved. Health promotion activities go far beyond changing individual behavior, and include a wide range of social and environmental interventions that increase health and wellbeing in populations as well as individuals. Health promotion also includes disease prevention — actions taken to prevent a disease from developing, and health screening — the routine testing of individuals to see if they are at risk of developing a health problem

Counseling about important issues affecting a woman's health and the health of the newborn is a critical component of focused ANC. Counsel the woman and provide the services as necessary:

□ **i) Nutrition**

- Counselling about healthy eating and keeping physically active during pregnancy is recommended for pregnant women to stay healthy and to prevent excessive weight gain during pregnancy. In undernourished populations, nutrition education on increasing daily energy and protein intake is recommended for pregnant women to reduce the risk of low-birth-weight neonates.
- For pregnant women with high daily caffeine intake (more than 300 mg per day, equivalent of 5 cups of coffee), lowering daily caffeine intake during pregnancy is recommended to reduce the risk of pregnancy loss and low-birth-weight neonates (WHO 2016).

ii) Advice & counsel on healthy life styles

Advise the woman to:

- Rest and avoid lifting heavy objects. Have adequate sleep
- NOT to take medication unless prescribed at the health centre/hospital.
- Avoid substance abuse (alcohol, tobacco... use during pregnancy.)
- Avoid exposure to second-hand smoke.
- Do not take any drugs or Nicotine Replacement Therapy for tobacco cessation.

iii) Hygiene & infection prevention

During pregnancy, women should be especially careful about personal hygiene. Pregnant women sweat more and have more vaginal discharge than non-pregnant women (due to hormonal changes), and they may be more vulnerable to infection by germs in the environment. Keeping the body clean helps prevent infection. Hand washing with soap is the most important hygiene action she can take, especially before preparing food and after going to the toilet. If possible, a pregnant woman should wash her body every day with clean water — especially her genital area.

Dental hygiene is especially important during pregnancy because increased oestrogen levels can cause swelling and increased sensitivity in gum tissues. Whether she cleans her teeth with a dental stick or a toothbrush and toothpaste, the pregnant woman should do so regularly.

iv) Advise on danger signs:

Teach the pregnant woman and her family to report any of the following conditions immediately:

- Vaginal bleeding
- Sudden gush of fluid or leaking of fluid from vagina
- Severe headache not relieved by simple analgesics (e.g paracetamol)
- Dizziness and blurring of vision
- Sustained vomiting
- Swelling (hands, face, etc.)
- Decreased or Loss of fetal movements
- Convulsions
- Premature onset of contractions (before 37 weeks)
- Severe or unusual abdominal pain
- Chills or fever

V) Counseling On Family Planning:

Counsel on the importance of family planning

- ☐ If appropriate, ask the woman if she would like her partner or another family member to be included in the counseling session.
- ☐ Explain that after birth, if she has sex and is not exclusively breastfeeding, she can become pregnant as soon as 4 weeks after delivery. Therefore, it is important to start thinking early about what family planning method they will use. For contraceptive options see tables 3-2 and 3-3.
 - If she (and her partner) want more children, advise that waiting at least 2-3 years between pregnancies is healthier for the mother and child.
 - Information on when to start a method after delivery will vary depending on whether a woman is breastfeeding or not.
 - Make arrangements for the woman to see a family planning counselor, or counsel her directly (see the *Decision-making tool for family planning providers and clients* for information on methods and on the counseling process).
- ☐ Advise the correct and consistent use of condoms for dual protection from sexually transmitted infection (STI) or HIV and pregnancy. Promote their use, especially if at risk for sexually transmitted infection (STI) or HIV.
- ☐ For HIV-positive women, advise dual protection.
- ☐ Her partner can decide to have a vasectomy (male sterilization) at anytime.

Special considerations for family planning counseling during pregnancy:

Counseling should be given during the third trimester of pregnancy.

- ☐ If the woman chooses female sterilization:

- can be performed immediately postpartum if no sign of infection (ideally within 2 days, or delay for 6weeks).
- plan for delivery in health facility where they are trained to carry out the procedure.
- ensure counseling and informed consent prior to labor and delivery.
- If the woman chooses an intrauterine device(IUD):
 - can be inserted immediately postpartum if no sign of infection (up to 48 hours, or delay 4weeks)

Table 3-2:-Method options for the non-breastfeeding woman

Can be used immediately postpartum	• Condoms • Progestogen-only oral contraceptives • Progestogen-only injectables • Implant • Spermicide • Female sterilization (within 2 days or delay 6weeks) • IUD (within 48 hours or delay 4weeks)
Delay 3 weeks	• Combined oral contraceptives • Combined injectables • Diaphragm • Fertility awareness methods

Table 3-3:- Method options for the breastfeeding woman

Can be used immediately postpartum	• Lactational amenorrhoea method(LAM) • Condoms • Spermicide • Female sterilization (within 2 days or delay 6weeks) • IUD (within 48 hours or delay 4weeks) • Progestogen-only oral contraceptives • Implants
Delay 6 weeks	• Diaphragm
Delay 6 months	• Combined oral contraceptives • Combined injectables • Fertility awareness methods

2.3.4 EARLY DETECTION and responding to COMPLICATIONS during pregnancy (including Referral)

Recognizing an emergency situation during pregnancy which requires immediate treatment and urgent referral to a higher level facility is important to prevent maternal morbidity and mortality. During each visit health providers should perform a thorough evaluation of the pregnant woman for pre-existing or new-onset disease conditions that may affect the outcome of pregnancy, require

immediate treatment or require a higher level of care and follow-up care over the course of pregnancy.

Table3:4 Symptom signs of Common pregnancy complications

Symptoms	Signs	Possible Diagnosis
BEFORE 20WKS OF GA		
Persistent nausea, vomiting, weight loss	Dehydration, weight loss of 5 kg and above, urine analysis shows ketones 2+ or more	Hyperemesis gravidarum
- Vaginal bleeding with or without passage of clots and fleshy material, - crampy lower abdominal pain	Tachycardia and or hypotension Cervix may open	Abortion (All types of spontaneous abortions except missed abortion)
Pregnancy symptoms disappear, Minimal dark vaginal bleeding	No increase in Uterine size, absent FHB, cervix closed	Missed Abortion
Vaginal bleeding (menstrual-like), lower abdominal pain, missed or irregular period	Hypotension, pallor, abdominal tenderness	Ectopic pregnancy
Vaginal bleeding (fresh), passage of tissues which look like an ice spoiled with blood (grape-like tissues),	fast abdominal growth	Molar pregnancy
AFTER 20WKS OF GA (N:B for viability GA of 28wks should be considered)		
Headache, RUQ/epigastric pain, blurred vision,	SBP > 140 and/or DBP > 90mmHg Proteinuria > 1+	Hypertensive disorders of pregnancy
Sudden gush or intermittent leakage of watery vaginal fluid	Wet vulva, pad test, leakage on speculum examination	Prelabor Rupture of Membranes
Leakage of fluid, abdominal pain, fever	Fever 38oC. Fetal tachychardia Foul-smelling discharge. Uterine tenderness	Uterine and fetal infection (chorio-amnionitis)
No change in abdominal growth, +/- fetal kick felt less than 10 times in 12 hours.	Small uterine size for gestational age (no increase in symphysis fundal height)	Intrauterine fetal growth restriction (IUGR)
Absent fetal kick for more than 6 hours	No FHB (Fetoscope)	Intrauterine fetal death (IUFD)
Blood-stained mucus or watery vaginal discharge (show) lower abdominal (pushing) pain before 37wks	- Cervical dilatation and effacement - Contractions	Possible preterm labor,
Bleeding, with or without abdominal pain	Blood at introitus, uterine contractions, malpresentation,..FHB abnormalities	Antepartum haemorrhage (placenta previa or Abruptio placenta)
THROUGHOUT PREGNANCY		

vaginal discharge No history of leakage of fluid Itching	Abnormal vaginal discharge (foul smelling, yellowish, curdy) on inspection or digital examination	Vaginitis (candidiasis, bacterial vaginosis..) cervicitis(Gonorrhea, chlamydia,)
Painful urination, urgency, pus or bloody urine + Pain in the sides (flanks), fever, vomiting,	- Urinalysis: Leukocytes (>5-10/HPF) + - Flank tenderness; WBC casts (U/A)	Urinary tract infections (UTIs, cystitis, urethritis) Pyelonephritis
Fever, headache, chills, rigor, sweating, feels thirsty, anorexia, generalized aching pain,	Fever, tachycardia, enlarged spleen Suggestive Blood tests results	Malaria, typhoid or typhus fever,
loss of appetite, RUQ pain, exhaustion, nausea and vomiting	Yellowish discoloration of the eyes, tender liver	Hepatitis
Easy tiredness, breathlessness, palpitation, head ache, vertigo	- Pallor + Haemoglobin 7-11 g/dl. - tachycardia, tachypnea (>30 breaths per minute) + Haemoglobin <7 g/dl.	Moderate Anemia Sever anemia

For emergency treatment of common pregnancy complications refer to Module 2 (rapid initial assessment and management). For a detailed discussion of the obstetric complications refer to the respective modules.

Box 3.1 contents of Referral note

- Date of the referral and time
- Name of the health facility you are sending the patient to
- Name, date of birth, ID number (if known) and address of the patient
- Relevant medical history of the patient
- findings from physical examinations and tests
- suspected diagnosis
- Any treatment you have given to the patient
- reason for referring the patient
- name, date and signature.

2.3.5 DEVELOP A BIRTH PREPAREDNESS AND COMPLICATION READINESS PLAN

Approximately 15% of women will develop a life-threatening complication during pregnancy, childbirth, and the postpartum/newborn period. Therefore; women and newborns need timely access to skilled care during these periods. Too often, however, their access to care is impeded by delays—delays in deciding to seek care, delays in reaching care, and delays in receiving care. These delays have many causes, including logistical and financial concerns, unsupportive policies, and gaps in services, as well as inadequate community and family awareness and knowledge about maternal and newborn health issues. The causes of these delays are common and predictable. However, in order to address them, women and families—and the communities, providers, and facilities that surround them—must be prepared in advance and ready for rapid emergency action. Birth Preparedness and Complication Readiness is the process of planning for

normal birth and anticipating the actions needed in case of an emergency. So, every woman and her family should have a plan for the following:

- A skilled attendant at birth
- The place of birth and how to get there including how to access emergency transportation if needed
- Items needed for the birth
- Money saved to pay for transportation, the skilled provider and for any needed medications and supplies that may not be provided for free
- Support during and after the birth (e.g., family, friends)

Assisting the woman to prepare for birth including:

- Items needed for clean birth
- Identification of skilled attendant for the birth
- Plan for reaching skilled attendant at the time of labor and delivery
- Identification of support people to help with transportation, care of children/household, and accompaniment to health facility
- Complication Readiness Plan in case of emergency: emergency funds, transportation, and decision-making
- Counseling/educating the woman and family on danger signs, nutrition, family planning, breastfeeding, HIV/AIDS
- Informing woman and family of existence of emergency funds if available
- Referring to higher levels of care when appropriate

Readiness plan

- Identifies transportation systems and where to go in case of emergency, support persons to accompany and stay with family
- Speaks out and acts on behalf of her and her child's health, safety and survival
- Knows that community and facility emergency funds are available
- Ensures personal savings and how to access it in case of need
- Chooses skilled attendant and place of birth in antenatal period.
- Recognizes normal labor and complications

Facility delivery:

Explain why birth in a facility is recommended:-

- Any complication can develop during delivery - they are not always predictable.

- A facility has staff, equipment, supplies and drugs available to provide best care if needed, and a referral system.

Advise how to prepare

- Review the arrangements for delivery: -

Advise when to go:-

- If the woman lives near the facility, she should go at the first signs of labor.
- If living far from the facility, she should go 2-3 weeks before expected date of delivery and stay either at the maternity waiting home or with family or friends near the facility.
- Advise to ask for help from the community, if needed.

Advise what to bring: -

- Home-based maternal record.
- Clean clothes for washing, drying and wrapping the newborn.
- Additional clean clothes to use as sanitary pads after birth.
- Clothes for mother and newborn.
- Food and water for woman and support person.

2.4 ANC FOR PMTCT

Provider initiated routine HIV testing and counseling (HTC) using the opt out approach is the recommended approach for all clients seen within the context of maternal care (i.e. at ANC, labor, immediate postpartum). The client is given pre test information in a group or on individual basis on HIV/AIDS and PMTCT and will be told that her routine antenatal laboratory tests will include an HIV test. The provider also must inform the client that, she has the right to say “no” (to opt out), and this by no means affects the services that she is going to get from the health facility. Rapid HIV testing must be used so results can be provided on the same day. (Table 3-5).

Additional Antenatal Care Needs for HIV-positive Women

HIV-positive women need focused antenatal care as described in the previous section, but need *extra* care, including prevention and early treatment of opportunistic infections. This can reduce risk of adverse pregnancy outcomes and the likelihood of mother-to-child HIV transmission. All HIV-positive pregnant women should be eligible to HAART according to the new PMTCT recommendation to implement option B+. CD4 determination is done for follow up either by sending blood samples or referring the client directly to a centre where CD4 testing is available. At each antenatal clinic appointment, HIV-positive mothers should be routinely reassessed for OI prophylaxis. HIV infected women and their families should be enrolled in HIV care and treatment services.

Additional history and clinical examination for HIV-positive pregnant women:

- Past history of HIV-related illness and HAART
- Duration of known HIV-positive status
- Assessment for symptoms of AIDS and HIV as per WHO Clinical Staging System for HIV status of other children and partner
- HIV and health status of other children and partner
- Partner disclosure and referral
- Any medications for HIV-related illness taken since beginning of pregnancy (e.g. TB or malaria medications, antiretroviral drugs, antibiotics for opportunistic infections)

Additional laboratory assessment of HIV-positive pregnant women:

- CD4 count/percentage is routinely recommended, where the service is available for follow up. Additional laboratory investigations are recommended as relevant and indicated to diagnose opportunistic infections.

Antiretroviral therapy:

- All HIV-positive women should be on HAART (option B+), at point of care or at HIV care/ART clinic.
- Link to ART clinic for care and support.

Tuberculosis (TB):

- Screen all pregnant HIV-positive pregnant women for TB

Malaria:

- All pregnant women should be advised to use insecticide treated bed nets to prevent malaria.

Infant Care:

- Counsel pregnant women about infant feeding with emphasis on exclusive breastfeeding for the first six months of life, with introduction of appropriate complementary feedings at six months with continued breastfeeding until 12–18 months. Educate mothers on the importance of infant follow-up, cotrimoxazole preventive therapy and early infant diagnosis.

Counseling on signs and symptoms of HIV/AIDS disease progression:

- Provide information and instructions on seeking care for symptoms of HIV disease progression. HAART is provided at ANC however women should be referred to ART sites for appropriate management of complications, consultation or ongoing care when indicated.

Partners and family:

- Help women through the process of disclosure, involving partner and/or couple counseling, and on effective involvement of the family in care and support.

Support:

- Counsel and refer to community care and support organizations and ensure feedback.

Prevention:

- Counsel: primary prevention, condoms, postpartum contraception including counseling for family planning and provision of contraceptive methods.

Table 3-5:- Counsel all pregnant women for HIV at first visit. Check status during each visit.

ASK, CHECK RECORD	LOOK, LISTEN , FEEL	SIGNS	CLASS IFY	TREAT AND ADVISE
Have you ever been tested for HIV? If yes, do you know the result? (Explain to the woman that she has the right not to disclose the result.) Has the partner been tested?		* Known HIV-positive.	HIV-POSITIVE	Ensure that she has received the necessary information about MTCT prevention. Enquire about the ARV treatment and inform her to start HAART right away. Enquire how she will be supplied with the drugs. Enquire about the infant feeding option chosen. Advise on additional care during pregnancy, delivery and postpartum. Advise on correct and consistent use of condoms. Counsel on benefits of involving and testing the partner.
		No HIV test results or not willing to disclose	UNKNOWN HIV STATUS	Provide key information on HIV. Inform her about VCT to determine HIV status. Advise on correct and consistent use of condoms. Counsel on benefits of involving and testing the partner.
		Known HIV-negative	HIV-NEGATIVE	Provide key information on HIV. Assess for signs suggesting advanced HIV infection Counsel on benefits of involving and testing her partner. Counsel on the importance of staying negative by correct and consistent use of condoms.

For more information refer to the national PMTCT guideline (2017).

MODULE 4

HYPERTENSIVE DISORDERS IN PREGNANCY

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UNIT 1. INTRODUCTION

Preeclampsia is a potentially serious complication of pregnancy with increasing significance worldwide. Preeclampsia is the cause of 9%–26% of global maternal mortality and a significant proportion of preterm delivery, and maternal and neonatal morbidity [WHO, MCPC 2017]. Incidence is increasing in keeping with the increase in obesity, maternal age, and women with medical comorbidities entering pregnancy. However, the impact of the disease is disproportionately higher in developing countries.

The incidence of eclampsia reported in the limited studies in Ethiopia varies from 0.3% to 2.3%. Only estimated 3.8% of eclamptic women attended emergency obstetric services in Ethiopia. According to the 2016 EMONC assessment severe pre-eclampsia/eclampsia was the 3rd leading cause of death in Ethiopia accounting for 10% of all direct causes of maternal mortality.

New tools for early detection, prevention, and management of preeclampsia have the potential to revolutionize practice in the coming years. There are no reliable tests or symptoms for predicting preeclampsia. But Eclampsia can be prevented by early detection and management of preeclampsia. Institutional care with close maternal monitoring; prevention and control of seizures through the use of anticonvulsants; treatment of severe hypertension; and timely termination of pregnancy reduce maternal mortality and serious morbidity.

Participant Learning Objectives: After completing this module, participants will be able to understand the principles in the diagnosis of the different spectrum of hypertensive disorders of pregnancy and be able to provide comprehensive preeclampsia/eclampsia care.

Enabling objectives

By the end of this module, the participants will be able to:

- Understand the variable clinical presentation of hypertension in pregnancy
- diagnose preeclampsia/eclampsia
- provide clinical care for women with preeclampsia/eclampsia
- provide appropriate referral of women with severe preeclampsia & complications
- understand recent updates in the prevention of preeclampsia/eclampsia

UNIT 2. CLASSIFICATION & DIAGNOSIS OF HYPERTENSIVE DISORDERS IN PREGNANCY

2.1 Classification & definitions

The hypertensive disorders of pregnancy include (see also table):

- *chronic hypertension* (elevation of the blood pressure noted before 20 weeks of gestation or persisting beyond 12 weeks postpartum);
- *Gestational hypertension*; an elevated blood pressure (SBP $\geq$ 140mmhg and/or DBP $\geq$ 90hg) after 20 weeks of gestational age and subsides before 12 weeks of postpartum period with no symptom signs of preeclampsia.
- *Mild pre-eclampsia*; gestational hypertension plus proteinuria
- *Severe pre-eclampsia*; hypertension with severe features (symptom/signs, blood pressure, and/or laboratory abnormalities see table 4.1)
- *Eclampsia*; Convulsion and/or coma in a preeclampsia woman with no other identifiable cause
- *Chronic hypertension with superimposed pre-eclampsia*: The development of features of preeclampsia in a patient with chronic hypertension

2.2 Diagnosis

The diagnosis of the different forms of hypertensive disorders of pregnancy depends on findings from a thorough clinical assessment (symptom/signs) and some specific laboratory tests (mainly urinalysis). It is important to remember that the clinical manifestation of Preeclampsia varies widely.

i) symptoms/signs

The common presentation is that of pregnant woman or a woman who recently gave birth (i.e. less than six weeks postpartum):

- Is found unconscious or having convulsions (seizures);
- Has elevated blood pressure;
- Complains of: severe headache and/or blurred vision; epigastric or right upper quadrant pain; difficulty of breathing

NB Headaches, blurred vision, convulsions and loss of consciousness are often associated with hypertension in pregnancy, but are not necessarily specific to it. Other conditions that may cause convulsions or coma include epilepsy, complicated malaria, head injury, meningitis, encephalitis, etc. See **Table 4.1**, for more information on diagnosis.

ii) Blood pressure

The hallmark of hypertensive disorders of pregnancy is elevated blood pressure. Clinical criteria for diagnosis of hypertension in pregnancy are:

- * Systolic blood pressure (SBP) greater than or equal to 140 mmHg and/or
- * Diastolic blood pressure (DBP) greater than or equal to 90 mmHg (two consecutive readings four hours or more apart)

iii) Laboratory tests

- Urinalysis (proteinuria): the most feasible simple tests in many health facilities is performing U/A for the detection of proteinuria.

Diagnostic criteria for proteinuria include:

- 0.3 g protein in a 24-hour urine specimen or
- 1+ on dipstick (specific gravity < 1030) or
- 2+ protein on dipstick (twice taken 6hrs apart)
- PROTEIN/CREATININE RATIO OF 0.3

N;B Because vaginal secretions or amniotic fluid may contaminate urine specimen, only clean-catch midstream specimens should be used. Catheterization for this purpose is not justified due to the risk of urinary tract infection. Other conditions, such as urinary infection, severe anemia, heart failure and difficult labor, can cause proteinuria and false positive results are possible.

- Blood chemistry tests

In areas with laboratory capacities, these tests are very important especially in the diagnosis of severe preeclampsia (preeclampsia with end organ damage).

- Liver enzymes; if elevated 2-3 times above the normal range (normal value < 42 IU/L)
- Renal function Tests (Serum creatinine): if > 1.2 mg/dl
- Platelets count: if < 100,000/microlitre

For specific features (diagnostic characteristics) of the different forms of hypertensive disorders in pregnancy see the table below

TABLE 4.1. Differential diagnosis of elevated blood pressure, headache, blurred vision, convulsions or loss of consciousness

	Symptoms, Signs and Laboratory Findings Sometimes Present	Probable Diagnosis
•Systolic blood pressure (SBP) 140 mmHg or higher and/or diastolic blood pressure (DBP) 90 mmHg or higher before the first 20 weeks of gestation		Chronic hypertension
• SBP 140 mmHg or higher and/or DBP 90 mmHg or higher before 20 weeks of gestation		Chronic hypertension with superimposed pre-eclampsia
• After 20 weeks: Proteinuria 2+ on dipstick Presence of any pre-eclampsia features		
•Two readings of SBP 140 mmHg or higher but lower than 160 mmHg and/or DBP 90 mmHg or higher but lower than 110 mmHg four hours apart after 20 weeks of gestation		Gestational hypertension
• No proteinuria		
• No features of preeclampsia		
• Two readings of SBP 140 mmHg or higher but lower than 160 mmHg and/or DBP 90 mmHg or higher but lower than 110 mmHg four hours apart after 20 weeks of gestation		Mild preeclampsia,
• Proteinuria 2+ on dipstick		

• SBP 160 mmHg or higher and/or DBP 110 mmHg or higher after 20 weeks of gestation • Proteinuria 2+ on dipstick	• Headache (increasing frequency, unrelieved by regular analgesics) • Vision changes (e.g. blurred vision) • Oliguria (passing less than 400 mL urine in 24 hours) • Upper abdominal pain (epigastric pain or pain in right upper quadrant) • Difficulty breathing (rales on auscultation of lungs due to fluid in lungs) • Nausea and vomiting • Hyperreflexia or clonus	Severe preeclampsia,^{a,b}
	In facilities with laboratory capacity: • Liver enzymes (transaminases) more than twice the normal range • Serum creatinine higher than 1.2 mg/dL or a doubling, or higher, of the baseline serum creatinine concentration in the absence of other renal disease • Platelets less than 100,000 cells/mcL	
• Convulsions • SBP 140 mmHg or higher or DBP 90 mmHg or higher after 20 weeks of gestation	• Coma (unconscious) • Other symptoms and signs of severe pre-eclampsia	Eclampsia
• Trismus (difficulty opening mouth and chewing)	• Spasms of face, neck, trunk • Arched back • Board-like abdomen • Spontaneous violent spasms	Tetanus
• Convulsions • Past history of convulsion • Normal blood pressure	Epilepsy,	
• Fever • Chills/rigors • Headache • Muscle/joint pain	• Enlarged spleen	Malaria
• Symptoms and signs of uncomplicated malaria • Coma • Anaemia	* Convulsions * Jaundice	Severe malaria
• Headache • Stiff neck • Photophobia	* Convulsions * Confusion * Drowsiness * Coma	Meningitis^{c,d} or encephalitis^{c,d}

* head ache	* vomiting	Migraine
* blurring of vision		

^a If a woman has any one of the symptoms or signs listed for severe pre-eclampsia (with the exception of proteinuria 2+ on the dipstick), diagnose severe pre-eclampsia.

he HELLP syndrome is a severe form of pre-eclampsia; the acronym stands for “haemolysis, elevated liver enzymes and low platelets.” ^c If a diagnosis of eclampsia cannot be ruled out, continue treatment for eclampsia. ^d Examine cerebrospinal fluid and give appropriate treatment for meningitis or encephalitis. ^e Give analgesics (e.g. paracetamol 500 mg by mouth as needed).

Remember:

- Mild pre-eclampsia often has no symptoms.
- Pre-eclampsia, including mild pre-eclampsia, may not have symptoms, so it is important to be vigilant with any woman with hypertension in pregnancy and to watch for the subtle or overt onset of symptoms that suggests worsening of disease.
- Edema of the feet and lower extremities is not considered a reliable sign of pre-eclampsia
- Increasing proteinuria is a sign of worsening pre-eclampsia.
- Mild pre-eclampsia may progress rapidly to severe pre-eclampsia. The risk of complications, including eclampsia, increases greatly in severe pre-eclampsia.
- The development of grand mal seizures or coma in a woman with preeclampsia indicates eclampsia. These convulsions:
 - Can occur regardless of the severity of hypertension; Around 20% of women with eclampsia will have normal blood pressure.
 - Treat all women with convulsions as if they have eclampsia until another diagnosis is confirmed.
 - Are difficult to predict and typically occur in the absence of hyperreflexia, headache or visual changes;
 - Occur after childbirth in about 20% of cases;
 - Are tonic-clonic and resemble grand mal convulsions of epilepsy;
 - may recur in rapid sequence, as in status epilepticus, and may end in death; will not be observed if the woman is alone;
 - May be followed by coma that lasts minutes or hours depending on the frequency of convulsions.

UNIT 3. MANAGEMENT OF HYPERTENSIVE DISORDERS IN PREGNANCY

3.1 Gestational Hypertension (*Hypertension Without Proteinuria*):

Manage on an outpatient basis:

- Monitor blood pressure, urine (for proteinuria) and fetal condition weekly.
- If **blood pressure worsens or the woman develops features of pre-eclampsia**, manage as pre-eclampsia.
- If there are **signs of severe fetal growth restriction or fetal compromise**, admit the woman to the hospital for assessment and possible expedited birth.
- Counsel the woman and her family about danger signs indicating severe pre-eclampsia or eclampsia.
- If all **observations remain stable**, allow to proceed with spontaneous labor and childbirth.
- In women with gestational hypertension, if **spontaneous labor has not occurred before term**, induce labor at term (around 40wks of GA).

3.2 Mild Pre-Eclampsia

I) Gestational Age Less Than 37 + 0/7 Weeks

As long as the well-being of the mother and fetus remains stable, the goal is for the woman to reach 37 + 0/7 weeks of gestation while monitoring of maternal and fetal status continues. However, it is important to remain vigilant because mild pre-eclampsia may progress rapidly to severe pre-eclampsia. The risk of complications, including eclampsia, increases greatly once pre-eclampsia becomes severe. Close monitoring and a high suspicion for worsening disease are important.

If blood pressure **and signs of mild pre-eclampsia remain unchanged or normalized**, follow the woman as an outpatient twice a week:

- Monitor blood pressure, reflexes and fetal condition.
- Monitor for danger signs associated with features of severe pre-eclampsia
- Counsel the woman and her family about danger signs associated with severe pre-eclampsia or eclampsia.
- Encourage the woman to eat a normal diet.
- Do not give anticonvulsants or antihypertensives unless clinically indicated (see severe pre-eclampsia and eclampsia,
- Do not give sedatives or tranquilizers.

If **follow-up as an outpatient is not possible**, admit the woman to the hospital:

- Provide a normal diet.
- Monitor blood pressure (four to six times daily) and urine for daily output.

- Do not give anticonvulsants unless blood pressure increases or other signs of severe pre-eclampsia appear (see severe pre-eclampsia and eclampsia).

- Do not give sedatives or tranquilizers.

Do not administer diuretics. Diuretics are harmful and only indicated for use in women with pre-eclampsia who have indications for a diuretic (such as pulmonary oedema).

- Monitor for danger signs associated with severe pre-eclampsia

If blood pressure decreases to normal levels or her condition remains stable, tell the woman that she can go home:

- Advise her to watch for symptoms and signs of severe pre-eclampsia (Table 4.1).
- See her twice weekly to monitor blood pressure and fetal well-being and to assess for symptoms and signs of severe pre-eclampsia.

* If systolic blood pressure is 160 mmHg or higher and/or diastolic blood pressure is 110 mmHg or higher, or if signs of severe preeclampsia appear, even if her blood pressure is normal, admit the woman and follow recommendations for management of severe preeclampsia and eclampsia below.

ii) Gestational Age at or More Than 37 + 0/7 Weeks

In women with mild pre-eclampsia is at term (37 + 0/7 weeks or more), induction of labor is recommended. If at a health center, the patient needs to be referred to a hospital. If at a higher facility (center with comprehensive emergency Obstetric Care services), prepare for delivery. Assess the cervix and induce or augment labor

3.3 Severe Pre-Eclampsia and Eclampsia

Severe pre-eclampsia and eclampsia are managed similarly, except that birth must occur within 12 hours of onset of convulsions in eclampsia.

- * General measures - supporting the specific treatments
- Prevent convulsion with magnesium sulfate (or Diazepam if Magnesium is not available)-
- Control hypertension
- Delivery as soon as possible

All women with severe preeclampsia/eclampsia seen at lower health facilities (e.g health center) should be referred and be managed at a higher center with availability of comprehensive emergencyobstetric care

i) GENERAL MANAGEMENT

- Secure IV line and infuse IV fluids (N/S maximum 2.4l/24hrs if no obvious loss of intravascular volume)
- If convulsing: DO NOT RUSH TO ADMINISTER ANTICONVULSANTS!! (the administration of anticonvulsants will not abort the ongoing seizure immediately)
 - Gather equipment (airway, suction, mask and bag, oxygen) and

- Give oxygen at 4–6 L per minute.
- Protect the woman from injury but do not actively restrain her.
- Protect the woman from tongue biting.
- Place the woman on her left side to reduce risk of aspiration of secretions, vomit or blood.

- After the convulsion, clear the mouth and throat as necessary.

* Administer magnesium sulfate (see below))

- Monitor vital signs (pulse, blood pressure, respiration and pulse oximetry), reflexes and fetal heart rate hourly.

- *If systolic blood pressure remains at 160 mmHg or higher and/or if diastolic blood pressure remains at 110 mmHg or higher, give antihypertensive drugs.(see below)*

Note: An important principle is to maintain blood pressures above the lower limits of normal.

- Catheterize the bladder to monitor urine output.
- Maintain a strict fluid balance chart (monitor the amount of fluids administered and urine output) to prevent fluid overload.

- If urine output is less than 30 mL per hour:

* Withhold magnesium sulfate and infuse IV fluids (normal saline or Ringer's lactate) at 1 L in eight hours.

– Monitor for the development of pulmonary oedema (increased respiratory rate and/or work of breathing, rales on auscultation of lungs).

- Never leave the woman alone. A convulsion followed by aspiration of vomit may cause death of the woman and fetus.

- Auscultate the lung bases hourly for rales indicating pulmonary oedema.

- If rales are heard, withhold fluids and administer furosemide 40 mg IV once.

- Assess clotting status with a bedside clotting test. Failure of a clot to form after seven minutes or a soft clot that breaks down easily suggests coagulopathy.

Anticonvulsive Therapy For Severe Pre-Eclampsia Or Eclampsia

• Magnesium Sulfate

A key factor in anticonvulsive therapy is timely and adequate administration of anticonvulsive drugs. Convulsions in hospitalized women are most frequently caused by under-treatment. Magnesium sulfate is the drug of choice for preventing and treating convulsions in severe preeclampsia and eclampsia. An intramuscular or intravenous regimen can be used (Box 4.1).

BOX 4.1. Magnesium sulfate regimens for severe pre-eclampsia and eclampsia

Intramuscular Regimen (Ethiopian national protocol)

Loading dose (IV and IM):

Give 4 g of 20% magnesium sulfate solution IV over five minutes.

- Follow promptly with 10 g of 50% magnesium sulfate solution: Give 5 g in each buttock as a deep IM injection with 1 mL of 2% lidocaine in the same syringe. Ensure aseptic technique when giving magnesium sulfate deep IM injection. Warn the woman that she will have a feeling of warmth when the magnesium sulfate is given.

Maintenance dose (IM):

Give 5 g of 50% magnesium sulfate solution with 1 mL of 2% lidocaine in the same syringe by deep IM injection into alternate buttocks every four hours. Continue treatment for 24 hours after birth or the last convulsion, whichever occurs last.

<u>Intravenous Regimen</u> (may be considered at tertiary centers using an infusion pump or continuous iv drip with close monitoring)
<u>Loading dose:</u> Give 4g of 50% magnesium sulfate solution IV.
• If convulsions recur after 15 minutes, give 2 g of 50% magnesium sulfate solution IV over five minutes.
<u>Maintenance dose (IV):</u>
• Give intravenous infusion 1g/ hour. Continue treatment for 24 hours after childbirth or the last convulsion, whichever occurs last.

Although magnesium toxicity is rare, a key component of monitoring women with severe pre-eclampsia and eclampsia is assessing for signs of magnesium toxicity. ***Before repeat administration, ensure that:***

- respiratory rate is at least 16 per minute;
 - patellar reflexes are present;
 - Urine output is at least 30 mL per hour over four hours.
- **If there are signs of toxicity**, delay the next IM dose or withhold the IV infusion of magnesium sulfate (**Box 4.2**).
 - If there are **no signs of toxicity**, give the next IM dose or continue the IV infusion of magnesium sulfate.

BOX 4.2 Signs indicating the need to withhold or delay maintenance dose of magnesium sulfate

<u>Closely monitor the woman for signs of magnesium toxicity.</u> To prevent magnesium intoxication, it is important to evaluate respiratory rate, deep tendon reflexes and urinary output before administering an additional dose.
<u>Withhold or delay drug if:</u>
• <u>respiratory rate falls below 16 breaths per minute;</u> • <u>patellar reflexes are absent;</u> • <u>Urinary output falls below 30 mL per hour over preceding four hours.</u>
<u>Keep antidote ready. In case of respiratory arrest (respiratory rate below 12 breaths per minute):</u>
• <u>assist ventilation (mask and bag, anesthesia apparatus, intubation);</u> • <u>give calcium gluconate 1 g (10 mL of 10% solution) IV slowly over three minutes, until respiration begins to counteract the effect of magnesium sulfate (usually one vial of calcium is enough).</u>

N;B Magnesium can be restarted with half of the standard dose if respiratory rate returns to normal and if the urine out put is adequate with normal serum creatinine level.

- **DIAZEPAM**

Give diazepam if convulsions occur in early pregnancy (because eclampsia is rare before 20wks of GA), if magnesium sulphate toxicity occurs or magnesium sulphate is not available. There is a greater risk for neonatal respiratory depression because diazepam passes the placenta freely. A single dose of diazepam to abort a convulsion seldom causes neonatal respiratory depression. Long-term continuous IV administration increases the risk of respiratory depression..

Loading dose IV: Give diazepam 10 mg IV slowly over 2 minutes.

If convulsions recur, repeat 10 mg.

Maintenance dose: Give diazepam 40 mg in 500 ml IV fluids (normal saline or Ringer's lactate) titrated over 6-8 hours to keep the woman sedated but arousable.

Stop the maintenance dose if breathing <16 breaths/minute.

Assist ventilation if necessary with mask and bag. Do not give more than 100 mg in 24 hours.

If IV access is not possible (e.g. during convulsion), give diazepam rectally.

Loading dose: rectally give 20 mg (4 ml) in a 10 ml syringe (or urinary catheter):

→Remove the needle, lubricate the barrel and insert the syringe into the rectum to half its length.

→Discharge the contents and leave the syringe in place, holding the buttocks together for 10 minutes to prevent expulsion of the drug. If convulsions recur, repeat 10 mg.

Maintenance dose: Give additional 10 mg (2 ml) every hour during transport.

- ii) **Antihypertensive Medications**

Antihypertensive medications should be started if the **systolic blood pressure is 160 mmHg or higher and/or the diastolic blood pressure is 110 mmHg or higher.**

Note: An important principle is to maintain blood pressures above the lower limits of normal.

The choice and route of administration of an antihypertensive drug for severe hypertension during pregnancy should be based primarily on the prescribing clinician's experience with that particular drug and its cost and local availability, while ensuring that the medication has no adverse fetal effects. If antihypertensive medication for acute treatment of severe hypertension cannot be given intravenously, oral treatment can be given (**Table 4.2.**).

TABLE 4.2. Antihypertensive medications and dosing options for acute treatment of severe hypertension in pregnancy

Antihypertensive Options	Dosing
Hydralazine	Intravenous treatment: • Administer 5 mg IV, <i>slowly</i> (risk of maternal hypotension; closely monitor blood pressure). • Repeat every 20 minutes until BP goal is achieved (DBP 90-105mmhg and SBP 140-155mmhg). • If BP is not controlled after administering a total of 20mg, another agent should be used (Nifedipine)
Labetalol	Oral treatment: • Administer 200 mg. • Repeat dose after one hour until the treatment goal is achieved. • The maximum dose is 1200 mg in 24 hours. Intravenous treatment: • Administer 10 mg IV. • If response is inadequate after 10 minutes, administer additional 20 mg IV. • The dose can be doubled to 40 mg and then 80 mg with 10-minute intervals between each increased dose until blood pressure is lowered below threshold. • The maximum total dose is 300 mg; then switch to oral treatment. Note: Women with congestive heart failure, hypovolemic shock or predisposition to bronchospasm (asthma) should not receive labetalol.
Nifedipine immediate-release capsule	Oral treatment: • Administer 5–10 mg under the tongue. Or oral • Repeat dose after 30 minutes if response is inadequate until optimal blood pressure is reached. • The maximum total dose is 30 mg in the acute treatment setting.^a
Alpha methyl dopa	Oral treatment: • Administer 750 mg orally. • Repeat dose after three hours until the treatment goal is achieved. • The maximum dose is 3 g in 24 hours.

^a Other treatment options should be considered if blood pressure is not lowered within the acute treatment phase of 90 minutes with 30 mg immediate-release nifedipine.

Once blood pressure is reduced to non-severe levels (lower than 160/110 mmHg), ongoing treatment should be continued using oral medication the target being diastolic between 90 -105 HmHg and systolic between 140 -155 HmHg(**Table 4.3**).

TABLE 4.3. Oral antihypertensive medications for non-severe hypertension

Antihypertensive Options	Dosing
Alpha methyl dopa	- Administer 250 mg every six to eight hours. - The maximum dose is 2000 mg per 24 hours.
Nifedipine immediate-release capsule ^a	- Administer 10–20 mg every 12 hours. - The maximum dose is 120 mg per 24 hours.
Labetalol	- Administer 200 mg every six to 12 hours. - The maximum dose is 1200 mg per 24 hours.

Note: Women with congestive heart failure, hypovolemic shock or predisposition to bronchospasm (asthma) should not receive labetalol.

^a **Note:** In some settings, nifedipine is available in more than one formulation (e.g. immediate-release, intermediate-release and sustained-release). Only immediate-release nifedipine is included on WHO's Essential Medicines List (WHO, 2015) and hence in **Tables 3 & 4**. To avoid medication errors, it is important to specify and confirm the nifedipine formulation before administering it.

iii) *Delivery & Optimal Timing For Birth*

Birth should be considered as soon as the woman's condition has stabilized. The decision about the optimal timing of childbirth should be made on an individual basis, taking into account, among other factors, gestational age, maternal and fetal status and well-being, cervical favorability, and urgency.

a) *Eclampsia*

Following an eclamptic convulsion, birth of the baby should occur within 12 hours of the onset of convulsions. Mode of delivery is determined based on maternal & fetal condition, cervical status & GA. Induction can be considered if the cervix is favorable, in the absence of contraindications to vaginal delivery

b) *Sever preeclampsia*

- **Gestation Less than 28 (Pre-Viable Fetus)**

Termination of pregnancy (Induction of labor) is recommended for women with severe pre-eclampsia if the fetus is not viable or is unlikely to achieve viability within one or two weeks.

- Assess the cervix and induce labor as per medical management of inevitable abortion if the gestational age is less than 28wks; **or offer dilatation and evacuation for expedited birth.**

Gestation between 28 and 34 Weeks

In women with severe pre-eclampsia and a viable fetus before 34 weeks but above 28 weeks of gestation, expectant management is recommended until GA of 34wks provided that uncontrolled maternal hypertension, maternal danger signs (e.g. severe headache, visual changes and abdominal pain) and fetal distress are absent and can be monitored. When laboratory services are available, it is advisable to monitor the maternal laboratory values outlined in **Table 4.1**.

(creatinine, liver transaminases, LDH and platelets).

If it is not possible to monitor maternal and fetal well-being, transfer to a tertiary care hospital (or to the nearest higher center) is recommended. If referral to a tertiary hospital is not possible immediately, manage severe pre-eclampsia as eclampsia and refer as soon as possible

- Give antenatal corticosteroids to accelerate fetal lung maturation.

Antenatal corticosteroid therapy is recommended for women with pregnancies at a gestational age of 28–34 weeks for whom preterm birth is considered imminent (due to severe pre-eclampsia or eclampsia), if the following conditions are met:

 - Gestational age assessment can be accurately undertaken.
 - There is no clinical evidence of maternal infection.
 - Adequate childbirth care is available (including the capacity to recognize and safely manage preterm labor and birth), and the preterm newborn can receive adequate care if needed (including resuscitation, thermal care, feeding support, infection treatment and safe oxygen use).

- Corticosteroids dosing and timing of birth:

 - ▶ Give two doses of betamethasone 12 mg IM, 24 hours apart, or four doses of dexamethasone 6 mg IM, 12 hours apart.
 - ▶ If the **maternal or fetal condition is rapidly deteriorating**, expedite birth after the first dose of antenatal corticosteroids. Do not wait to complete the course of antenatal corticosteroids before the woman gives birth.
- Monitor the woman and fetus.
- Monitor fetal growth by symphysis-fundal height (weekly) or ultrasound, if available.
- Give magnesium sulfate.
 - If **the clinical picture is not worsening, premonitory signs of eclampsia** (e.g. increased reflexes associated with clonus, severe headache or visual disturbance) **are absent** and signs of **severe pre-eclampsia** (diastolic BP is 110 mmHg or higher, systolic BP is 160 mmHg or higher, elevated liver transaminases, elevated serum creatinine, and low platelets) **are not persistent, stop magnesium sulfate** after 24 hours.
- Give Antihypertensive

Monitor blood pressure at least every four hours and give antihypertensive medications if **systolic blood pressure is 160 mmHg or higher and/or the diastolic blood pressure is 110 mmHg or higher.**
- Delivery

If **severe pre-eclampsia or eclampsia persists** (e.g. uncontrolled hypertension despite antihypertensive medications, deterioration in blood tests or non-reassuring fetal status), or GA of 34wks is achieved, **expedite delivery.**

- **Gestational age of 34 to 36 6/7 Weeks**

Note: After 34 completed weeks of gestation, corticosteroids are not recommended for the indication of fetal lung maturation.

In women with severe pre-eclampsia and a viable fetus that is between 34 and 36 + 6/7 weeks of gestation, a policy of expectant management may be recommended, **provided that** uncontrolled maternal hypertension, worsening maternal status and fetal distress are absent and can be closely monitored. But generally termination of pregnancy is recommended

If any features of worsening severe pre-eclampsia or eclampsia are present, or if close monitoring of the woman and fetus is not feasible, transfer to a higher level facility. If transfer is not possible, the birth should be expedited.

- Monitor & Management is the same as detailed above.
- If **severe pre-eclampsia or eclampsia persists** (e.g. uncontrolled hypertension despite antihypertensive medications, deterioration in blood tests or non-reassuring fetal status), **expedite delivery**.

- **Gestational age after 37 Completed Weeks**

For women with pre-eclampsia at term (37 + 0/7 weeks), regardless of pre-eclampsia severity, giving birth is recommended.

- Assess the cervix and induce labor.
- If **vaginal birth is not anticipated** within 12 hours (eclampsia) or 24 hours (severe pre-eclampsia), cesarean section is recommended.
- If there are **fetal heart rate abnormalities** (less than 100 or more than 180 beats per minute), CS is recommended
- If **safe anaesthesia is not available for caesarean** or if the **fetus is dead**:
 - Aim for vaginal birth.
 - If the **cervix is unfavourable** (firm, thick, closed), ripen the cervix with misoprostol or intracervical foley catheter..

N:B the second stage of labor in women with sever preeclampsia/eclampsia should be shortened with vaccum or forceps delivery

Do not give ergometrine to women with pre-eclampsia, eclampsia or high blood pressure because it increases the risk of convulsions and cerebrovascular accidents.

iv) *Referral To a Tertiary Level Care*

Consider referral of women who have:

- HELLP-syndrome (haemolysis, elevated liver enzymes and low platelets) or coagulopathy;
- Persistent coma lasting more than 24 hours after convulsion;
- Severe pre-eclampsia and maternal and fetal well-being cannot be adequately monitored;
- Uncontrolled hypertension despite treatment with antihypertensive;
- Oliguria that persists for 48 hours after giving birth.

Elements of Appropriate referral

Refer the clients after giving loading dose of MgSO₄, appropriate nursing care with Ambulance

Inform referral center by radio or phone

Send the clients with skilled birth attendants, delivery set, essential emergency

Drugs, resuscitation materials and supplies

Keep the clients in left lateral position during transportation

Keep record of all IV fluids, Medications given, time of administration, women's conditions and basic laboratory results

3.4 Chronic Hypertension

- High levels of blood pressure maintain renal and placental perfusion in chronic hypertension; reducing blood pressure will result in diminished perfusion.

Note: Blood pressure should not be lowered below its pre-pregnancy level.
--

- If the **woman was on an antihypertensive medication before pregnancy** and her **blood pressure is well-controlled**, continue the same medication if acceptable in pregnancy or change to medication to one safely used in pregnancy.
- If the **systolic blood pressure is 160 mmHg or more or the diastolic blood pressure is 110 mmHg or more**, treat with antihypertensive medications.
- If **proteinuria or other signs and symptoms of pre-eclampsia are present**, consider superimposed pre-eclampsia and manage as pre-eclampsia.
- Monitor fetal growth and condition.
- If **fetal growth restriction is severe** and **pregnancy dating is accurate**, assess the cervix and induce labor.

Note: Assessment of gestation by ultrasound in late pregnancy is not accurate. Use a first trimester ultrasound scan, if available, to date the pregnancy.

- Observe for complications, including abruptio placentae and superimposed pre-eclampsia.
- If there are **no complications**, induce labor at term (>37wks).

- If there are **fetal heart rate abnormalities** (less than 100 or more than 180 beats per minute), suspect fetal distress, CS is indicated.

3.5 Management after Childbirth

- In women being treated with magnesium sulfate:
 - If magnesium sulfate was started before the birth, continue anticonvulsive therapy for 24 hours after childbirth or the last convulsion, whichever occurs last.
 - If magnesium sulfate was started in the postpartum period, continue anticonvulsive therapy for 24 hours total or for 24 hours after the last convulsion.
- Monitor blood pressure in all postpartum women.
 - Continue antihypertensive therapy postpartum, if it was started before the woman gave birth. - Blood pressures sometimes increase three to seven days after giving birth, so it is important to re-evaluate women one to two weeks after discharge.
 - Start antihypertensive therapy if the woman's systolic blood pressure is 160 mmHg or higher, or if the diastolic pressure is 110 mmHg or higher, during the postpartum period.
- Monitor urine output to ensure that it normalizes if it has been low.
 - If oliguria persists for 48 hours after giving birth, transfer to a tertiary care facility.
- Advise the woman and her family to watch for danger signs of severe pre-eclampsia and seek care immediately if they occur.
 - It is important to continue to monitor blood pressure and remain vigilant for new pre-eclampsia danger signs or worsening pre-eclampsia in the postpartum period because:
 - While most women with pre-eclampsia and eclampsia recover within the first 24 hours of giving birth, a minority will remain unstable or deteriorate.
 - Some women (20%) will present with pre-eclampsia or eclampsia for the first time after giving birth.

COUNSELLING TO REDUCE RISK OF PRE-ECLAMPSIA IN FUTURE PREGNANCIES

- *Counsel all women treated for pre-eclampsia or eclampsia about the risk of recurrence in future pregnancies and the interventions that may reduce risk of future disease.*
- *Counsel women on the importance of initiating early antenatal care in all future pregnancies and avoiding unwanted pregnancies.*

POSTPARTUM FAMILY PLANNING SERVICES

- *Due to the increased risk of recurrence of pre-eclampsia and eclampsia in future pregnancies, it is very important that women are appropriately counselled and offered an effective postpartum contraceptive method.*
- *Women treated for severe pre-eclampsia or eclampsia should have access to family planning particularly long-acting reversible (implants and intrauterine devices) and permanent (tubal ligation and vasectomy) contraceptive methods, which can be given postpartum prior to discharge.*

- Provide women with the contraceptive method of their choice (page) or link them with follow-up contraception services before discharge.

COUNSELLING TO REDUCE LIFETIME RISK OF CARDIOVASCULAR COMPLICATIONS

- Counsel women treated for pre-eclampsia and eclampsia about their increased risk of future cardiovascular disease (e.g. hypertension, stroke).
- Assess and address the woman's risk factors for cardiovascular disease (e.g. smoking, obesity, lack of physical activity, hyperlipidemia) before she is discharged from the facility.
- Emphasize the importance of regular medical follow-up and link women to follow-up primary care services before discharge.

UNIT 4. PREVENTION OF PREECLAMPSIA/ECLAMPSIA

To date there are no reliable clinical and biochemical tests for the prediction of preeclampsia. But recently evidences are suggesting that calcium and low dose aspirin have a role in the prevention of preeclampsia (see box 4.2, WHO 2016).

Eclampsia can be prevented by early detection, effective treatment of preeclampsia and timely delivery.

BOX 4.3 Prevention of pre-eclampsia and eclampsia in subsequent pregnancies

- In areas where dietary calcium intake is low, calcium supplementation during pregnancy (at doses of 1.5–2.0 g elemental calcium/day) is recommended for the prevention of pre-eclampsia for all women, but particularly those at high risk of pre-eclampsia.
- Low-dose acetylsalicylic acid (aspirin, 75 mg or 81mg) should be initiated before 20wks (and, if possible, as early as at 12) weeks of gestation for women at high risk of developing pre-eclampsia if they have one or more of the following risk factors: previous severe pre-eclampsia, diabetes, chronic hypertension, obesity, renal disease, autoimmune disease and multiple pregnancies. This list can be complemented based on local epidemiology.

SEVERE/COMPLICATED MALARIA

Women in the second and third trimesters of pregnancy are more likely to have severe malaria than other adults. Severe malaria in pregnancy may be misdiagnosed as eclampsia. **If a pregnant woman living in a malarial area has fever, headaches or convulsions, and malaria cannot be excluded, treat the woman for both malaria and eclampsia.** (Refer Module 8)

MODULE-5

CARE DURING LABOR, CHILD BIRTH & IMMEDIATE POSTPARTUM PERIOD

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	2.5 Supportive care of a woman in labor	
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UNIT-1 INTRODUCTION

Of the more than 130 million births occurring each year, an estimated 303 000 result in the mother's death, 2.6 million in stillbirth and another 2.7 million in a newborn death within the first 28 days of birth [WHO 2015]. Experiences from around the world suggest that about 15 percent of all pregnant women will develop obstetric complications, and that not all of these complications can be predicted through risk screening. Unless emergency care is available, the woman and the fetus/neonate could either die or develop severe disabilities. The availability and quality of skilled care at birth and immediately after birth is a major determinant of the survival and health of both mothers and babies. With essential preventive care, proper management of labor, and timely management of complications, we can prevent or successfully manage many obstetric and newborn complications as well as intrapartum stillbirths. Using a partograph to monitor labor will help to identify slow progress in labor, and providing early care and referral (or early consultation when appropriate) can prevent prolonged labor.

Only 42 percent of pregnant women in sub-Saharan Africa give birth with a skilled attendant present. According to the 2016 national EmONC assessment, the institutional birth rate was 66% while it was lower in the EDHS (2016) data which was 26%. Eighty percent of births to urban mothers were assisted by a skilled provider and 79 % were delivered in a health facility, as compared with 21 % and 20 %, respectively, of births to rural women.

Module objective: after completing this module, participants will be able to describe the continuum of care, care during child birth (labor, delivery and immediate postpartum) to save the lives of mothers and babies and promote overall health. It introduces the basic components in the provision of care to the mother and the newborn and the additional cares required for selected common problems to decrease maternal & perinatal morbidity and mortality

Enabling objectives

By the end of this module, the participants will be able to:

1. Provide basic care to the woman and the fetus during labor, delivery and early postpartum.
2. Manage birth asphyxia (neonatal resuscitation)
3. Provide basic care to the woman and newborn in the immediate postpartum period.
4. Detect and provide care for complications during labor and delivery.
5. Detect and provide care for complications to the woman in the immediate postpartum period.
6. Recognize an emergency situation during labor, delivery and immediate postpartum period which requires immediate treatment and, in most cases, urgent referral to a higher level health facility.
7. Provide pre-referral management to a recognized emergency situation.

UNIT-2: BASIC CARE DURING LABOR AND DELIVERY

2.1 LEARNING OBJECTIVES

At the end of this unit, participants will be able to:

- Describe basic care to the woman and fetus during labor.
- Describe basic care to the woman and fetus during delivery.
- Describe basic care to the woman and newborn during the immediate postpartum period.

Introduction

Every woman has a right to the highest attainable standard of health, the right to dignified, respectful health care throughout pregnancy and childbirth, and the right to be free from violence and discrimination. Abuse, neglect or disrespect during labor and/or childbirth is a violation of a woman's fundamental human rights, as described in internationally adopted human rights standards and principles.

Labor and child birth is a normal physiologic process that most women experience without complications. Labor is considered normal when the following conditions are fulfilled.

- Mother has no risk factors (eg. Pre-eclampsia, Previous scar),
- Labor should start spontaneously and at term,
- Fetal presentation must be vertex,
- Birth should be by spontaneous vertex delivery,
- Duration of all stages of labor should be normal
- The neonate is alive, normal and the woman has uncomplicated puerperium.

2.2 INITIAL ASSESSMENT AND DIAGNOSIS OF LABOR

2.2.1 RAPID ASSESSMENT

To ensure proper and effective delivery of care for a woman in labor first do Rapid assessment and management

- ☐ * Ask the woman if she has had any danger signs (vaginal bleeding, gush of fluid or any problems such as fever, difficulty breathing, cough, leg pain, etc.) or has any concerns.
- ☐ * Perform a rapid initial assessment of the woman (airway, breathing, circulation, vaginal bleeding, level of consciousness, convulsions, temperature higher than 38°C, abdominal pain).
- ☐ * Assess fetal condition:
- ☐ - Listen to the fetal heart rate immediately after a contraction.

- - Count the fetal heart rate for a full minute.
- - If there are fetal heart rate abnormalities (less than 100 or more than 180 beats per minute), suspect fetal distress.
- * Respond immediately to any abnormalities in vital signs or other problems identified during the rapid initial assessment.

2.2.2 HISTORY

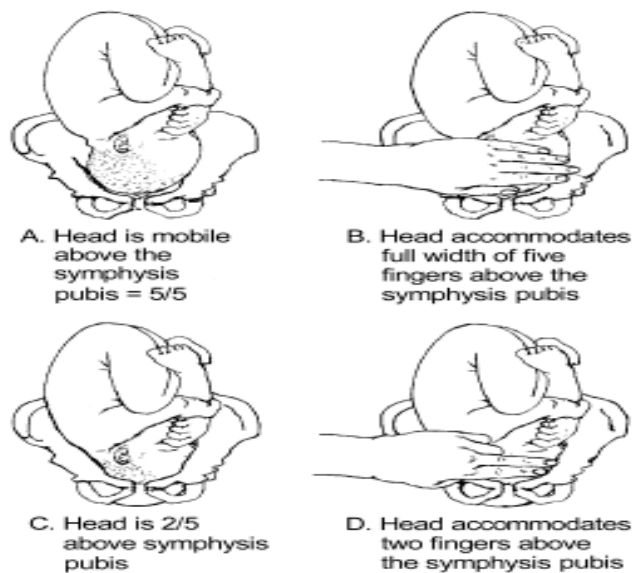
- Ask the woman about the history of this labor: approximate time contractions began and membranes ruptured.
- Ask the woman about or check her record (if she has ANC visits) for:
 - past obstetric, medical and surgical history, especially abdominal or pelvic surgeries or past caesarean(s);
 - problems during the current pregnancy (e.g. high blood pressure, pre-eclampsia, anemia, malaria, sexually transmitted infections) and management;
 - hemoglobin;
 - tetanus vaccination status; and
 - Syphilis, HIV, hepatitis and tuberculosis status.
- Check the woman's record for GA (by reliable LMP or early US [done before 20wks]). If she has no record:
 - Ask her if she remembers her LMP and calculate expected birth date.
 - Determine the gestational age. If preterm (before 37 weeks of pregnancy are completed), manage as preterm labor.

2.2.3 ABDOMINAL EXAMINATION

- Check the woman's abdomen for:
 - a caesarean scar;
 - a horizontal ridge across lower abdomen (if present, ask the woman to empty her bladder and observe her again).
- Assess uterine contractions:
 - Observe the woman's response to contractions:
 - Is she coping well or is she distressed?
 - Is she pushing or grunting? If so, consider possibility of second stage of labor.
 - Assess the number of uterine contractions in 10 minutes and duration in seconds.
- Assess fetal lie—longitudinal or transverse—and fetal presentation.
 - Fetal lie other than longitudinal and presentations in labor other than cephalic are considered malposition or malpresentations.
- Assess **descent** in terms of fifths of fetal head palpable above the symphysis pubis:

- A head that is entirely above the symphysis pubis is five fifths (5/5) palpable .
 - A head that is entirely below the symphysis pubis is zero fifths (0/5) palpable.
- . Assess FHR (count for full minute between and immediately after contractions)

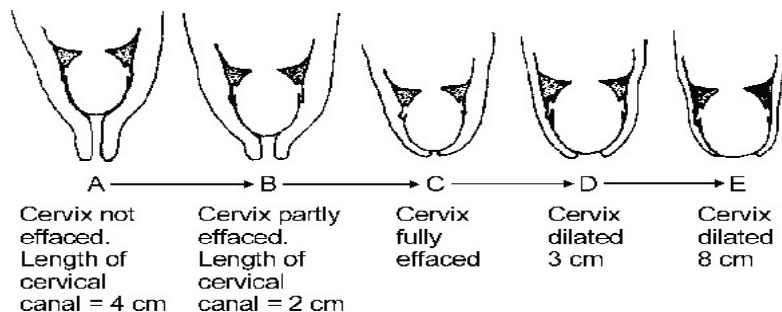
Figure 5.1 ABDOMINAL PALPATION FOR DESCENT OF FETAL HEAD



2.2.4 GENITAL EXAMINATION

- Inspect the external genitalia for female genital mutilation, edema. Wart, blood etc.
- Digital vaginal examination
 - * Assess cervical dilatation and effacement:

FIGURE 5.2 Effacement and dilatation of the cervix



- Note the status of membranes. If the **membranes have ruptured**, note the color of the amniotic fluid:
 - The presence of thick meconium indicates the need for close monitoring and possible intervention for management of fetal distress.

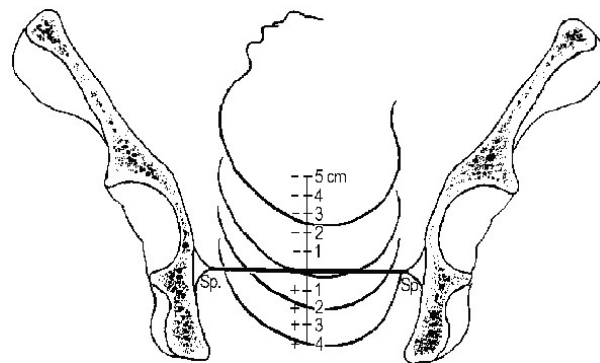
Routine antibiotics are **not recommended** in cases of meconiumstained amniotic fluid in the absence of other indications. Personnel experienced in neonatal resuscitation should attend the birth of infants if thick meconium liquor is noted, as meconium aspiration is clinically more significant in cases of thick meconium.

- The presence of foul-smelling watery discharge indicates the need to manage as amnionitis.

Antibiotics should be administered when characteristics of the liquor suggest infection.

- An absence of fluid draining after rupture of the membranes is an indication of reduced volume of amniotic fluid, which may be associated with fetal distress.
- If necessary, a vaginal examination may be used to assess **descent** by relating the level of the fetal presenting part to the ischial spines of the maternal pelvis.

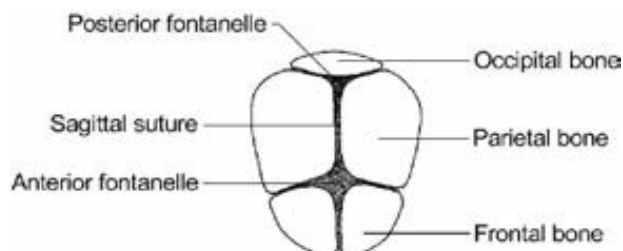
FIGURE 5.3: Assessing station of the fetal head by vaginal examination; 0 station is at the level of the ischial spine (Sp)



Note: When there is a **significant degree of caput or moulding**, assessment by abdominal palpation using fifths of head palpable is more useful than assessment by vaginal exam.

- Determine **presenting part** during vaginal examination:
 - The most common presenting part is the vertex of the fetal head. If the **vertex is not the presenting part**, manage as a malpresentation.
 - If the **vertex is the presenting part**, use landmarks on the fetal skull to determine the position of the fetal head in relation to the maternal pelvis.

FIGURE 5.4 Landmarks of the fetal skull



- Determine the **position of the fetal head** during vaginal examination:
 - The fetal head normally engages in the maternal pelvis in an **occiput transverse position**, with the fetal occiput transverse in the maternal pelvis (**Fig 5.5**).

FIGURE 5.5 Occiput transverse positions

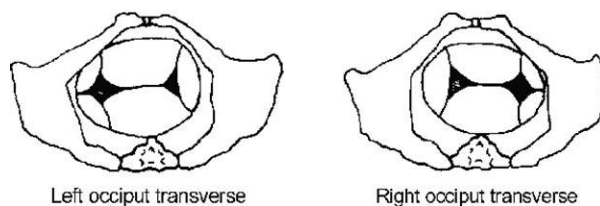
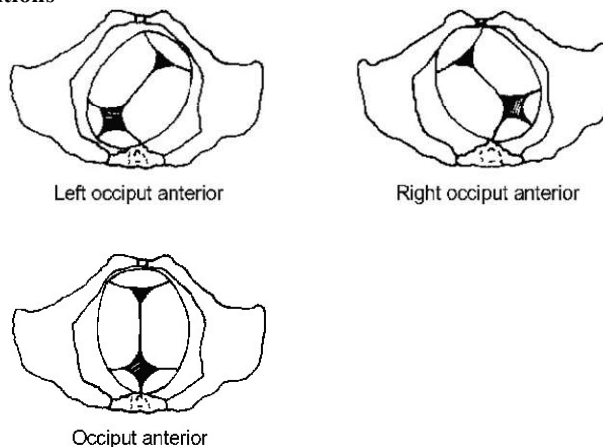


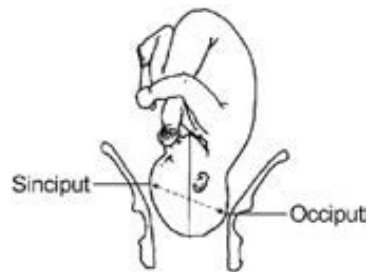
FIGURE 5.6 Occiput anterior positions



- With descent, the fetal head rotates so that the fetal occiput is anterior in the maternal pelvis (**occiput anterior position; Fig 5.6**). Failure of an occiput transverse position to rotate to an occiput anterior position should be managed as an occiput posterior position.

- An additional feature of a normal presentation is a **well-flexed vertex** (Fig. 5.7), with the occiput lower in the vagina than the sinciput.

FIGURE 5.7: Well-flexed vertex



NOTE: If there are positions or presentations in labor other than occiput anterior with a well-flexed vertex, manage as malpositions or malpresentations.

2.2.5 CONFIRMATION OF LABOR AND DIAGNOSIS OF STAGES OF LABOR

Based on initial clinical assessment of the women, the confirmation of the onset of labor and diagnosis of the stage of labor is important for subsequent intrapartum management & follow up.

Diagnosis and Confirmation of Labor

- Suspect or anticipate labor if the woman has:
 - Intermittent abdominal pain after 28 weeks of gestation;
 - Pain often associated with blood-stained mucus discharge (show);
 - Watery vaginal discharge or a sudden gush of water.
- Confirm the onset of labor if there is:
 - Pain full regular contractions of at least 2 contractions in 10 minutes
 - Cervical effacement—the progressive shortening and thinning of the cervix during labor; and Cervical dilatation—the increase in diameter of the cervical opening measured in centimeters.

(See table 5.1 for differentiating characteristics of false and True labor)

Table 5.1 Differences between characteristics of true and false labor

Characteristics	True labour	False labour
Uterine contractions	Contractions occur at regular intervals, but the interval between each contraction gradually becomes shorter	Contractions occur at irregular intervals
	Duration of each contraction gradually increases	Duration remains unchanged — either long or short
	Intensity of contractions becomes stronger and stronger	Intensity remains unchanged
Cervical dilation	Cervix progressively dilates	Cervix does not dilate, remains less than 2 cm
Pain	Discomfort at the back in the abdomen, cannot be stopped by strong anti-pain medication	Discomfort is non-specific (has no particular location) and is usually relieved by strong anti-pain medication or by walking

Diagnosis of the Stage and phases of Labor

The diagnosis of the stages and phases of labor is primarily depend on cervical dilatation and effacement (see table 5.2)

Table 5.2 Diagnosis of stage and phase of labor^a

Symptoms and Signs	Stage	Phase
• Cervix not dilated	False labor/ Not in labor	
• Cervix dilated less than 4cm	First	Latent
• Cervix dilated 4–9cm • Rate of dilatation typically 1 cm per hour or more • Fetal descent begins	First	Active
• Cervix fully dilated (10cm) • Fetal descent continues • No urge to push	Second	Early (non-expulsive)
• Cervix fully dilated (10cm) • Presenting part of fetus reaches pelvic floor • Woman has the urge to push	Second	Late (expulsive)

Summary

Review findings from the woman's history, physical examination and assessment, and make plans to care for the woman based on her individual needs and the stage and phase of labor.

Communicate with the woman and her companion about the plan:

- Administer intrapartum antibiotic (ampicillin or penicillin G) to all women with documented Group B streptococcus colonization, except where there are contraindications.
- Provide counselling and testing for HIV and syphilis if this was not done during the third trimester of pregnancy.
- As needed, provide prophylactic treatments and ongoing management of obstetric or medical problems per protocols.
- Plan to assess the progress of maternal condition, fetal condition and labor based on the phase and stage of labor, and any identified problems.
- Provide supportive care.
- If the woman is in the latent phase of the first stage of labor and able to be receptive to health education counselling, provide information regarding infant feeding and postpartum family planning.

2.3. DETECT AND RESPOND TO OBSTETRICAL PROBLEMS ON ADMISSION

Team approach is important in caring for laboring mothers, and all abnormal information should reach to the most senior personnel in charge of the labor ward activity. (*Use table 5.3 to assess for and respond to obstetric problems at admission*).

Table 5.3: chart to assess maternal and fetal status at admission and plan treatment of problems.

SIGNS	CLASSIFY	TREAT AND ADVISE
■ Transversellie. ■ Continuous contractions. ■ Constant pain between contractions. ■ Sudden and severe abdominal pain. ■ Horizontal ridge across lower abdomen. ■ Labor >24 hours.	OBSTRUCTED LABOR	■ If distressed, insert an IV line and give fluids. ■ If in labor >24 hours, give appropriate IM/IV antibiotics. ■ Consult or refer urgently to hospital.
■ Rupture of membranes and any of: → Fever >38°C → Foul-smelling vaginal discharge.	UTERINE AND FETAL INFECTION	■ Give appropriate IM/IV antibiotics. ■ If late labor, deliver and refer to hospital after delivery. ■ Plan to treat newborn.
■ Rupture of membranes at <8 months of pregnancy.	RISK OF UTERINE AND FETAL INFECTION	■ Give appropriate IM/IV antibiotics. ■ If late labor, deliver. ■ Discontinue antibiotic for mother after delivery if no signs of infection. ■ Plan to treat newborn.
■ Systolic blood pressure >90 mmHg - And/or Diastolic BP > 140 mmHg	PRE-ECLAMPSIA	■ Assess further and manage as hypertensive disorder
■ Severe palmar and conjunctival pallor and/or hemoglobin <7g/dl.	SEVERE ANEMIA	■ Manage as severe anemia;
■ Breech or other malpresentations. ■ Multiple pregnancy. ■ Fetal distress. ■ Prolapsed cord	OBSTETRICAL COMPLICATION	■ Follow specific instructions in respective sections.
■ Warts, keloid tissue that may interfere with delivery. ■ Prior third degree tear.	RISK OF OBSTETRICAL COMPLICATION	■ Do an episiotomy and carefully control delivery of the head. ■ If late labor, deliver. ■ Have help available during delivery.

■ Labor before 8 completed months of pregnancy (more than one month before estimated date of delivery).	PRETERM LABOR	■ Reassess fetal presentation (breech more common). ■ If woman is lying, encourage her to lie on her left side. ■ Call for help during delivery. ■ Conduct delivery very carefully as small fetus may pop out suddenly. In particular, control delivery of the head. ■ Prepare equipment for resuscitation of newborn.
■ Fetal heart rate <100 or >180 beats per minute.	POSSIBLE FETAL DISTRESS	■ Manage as fetal distress
■ Rupture of membranes at term and before labor.	TERM PRELABOR RUPTURE OF MEMBRANES	■ Give appropriate IM/IV antibiotics if rupture of membrane >12 hours. ■ Plan to treat the newborn.
■ If two or more of the following signs: → thirsty → sunken eyes → dry mouth → skin pinch goes back slowly.	DEHYDRATION	■ Give oral fluids. ■ If not able to drink, give 1 liter IV fluids over 3 hours and then more fluids as necessary.
■ HIV test positive. ■ Counselor on HAART treatment and infant feeding.	HIV-POSITIVE	■ Ensure that the woman takes ARV drugs as soon as possible according to the national guideline. ■ Support her choice of infant feeding.
■ No fetal movement, and ■ No fetal heart beat on repeated examination	POSSIBLE FETAL DEATH	■ Explain to the parents that the baby is not doing well ■ Us for confirmation. If no FHB, counsel on waiting for labor for 4wks or immediate induction

2.4 HIV testing and counselling during labor:

As up to two thirds of pregnant women attend health facilities for the first time when in labor, HIV counselling and testing should be offered routinely for all mothers admitted for delivery. Active identification of women in labor with unknown HIV status and offering of HIV counselling and testing shall be part of standard of care. HIV-positive women identified through this means shall receive prophylactic antiretroviral treatment and be linked to care for themselves and their infants. The right of women to decline HIV testing must always be respected.

The initial step is to check the woman's record to see if she has been tested for HIV and if the test result is recorded on the mother's card. The guiding principles of provider-initiated testing and counselling for HIV - confidentiality, informed consent, and post-test support and services - are tailored to the special needs of a woman in labor but not compromised when providing rapid testing in early labor. They still follow the principles of **3 C's**: Confidentiality, Informed Consent, and Counseling. Because the woman is in labor, there is an additional **C** that is important: Comfort of the woman.

Confidentiality: Protecting the confidentiality of the pregnant woman who receives HIV testing during labor is very important. However, in the busy and complex labor and childbirth unit, maintaining confidentiality requires that staff members be knowledgeable and vigilant.

Informed Consent for screening in early labor: The second of the guiding principles for testing and counselling for HIV, requires that the patient receives clear and accurate information about HIV testing and that the healthcare worker providing the information respects the individual's right to decide whether to be tested.

Counselling and post-test support and services: The result of HIV testing should always be offered in person. Along with the result, appropriate post-test information, counselling, and linkages with HIV care services should also be offered.

Comfort: The health worker should assess the woman's stage of labor, comfort level, and need for analgesics. Providers need to show empathy while presenting information about rapid HIV testing. The content covered should be short and to the point and should be explained **between contractions**. The health worker should ask the woman to signal for a pause when a contraction is starting. The health worker should always consider the woman's language and culture and, as needed, must adjust the terminology used. The health worker should make sure that the woman being counselled understands the content being covered by checking after each point is made and before beginning the next point to be sure she understands.

For more information refer to the national PMTCT guideline and PMTCT training manual (2007).

2.5 SUPPORTIVE CARE OF A WOMAN DURING LABOR AND DELIVERY

Communication

- Warm and friendly acceptance at arrival and admission for the woman and her family.
- Ensure good communication and support by staff.
- Explain all procedures, seek permission, and discuss findings with the woman.
- Keep her informed about the progress of labor.
- Praise her, encourage and reassure her that things are going well.
- Ensure and respect privacy during examinations and discussions.

If known HIV positive, find out what she has told the companion. Respect her wishes.

Cleanliness

- Maintain cleanliness of the woman and her environment:
 - Encourage the woman to wash herself or bath or shower at the onset of labor;
 - Wash the vulva and perineal areas before each examination;
 - Wash your hands with soap before and after each examination;
 - Ensure cleanliness of laboring and birthing area(s);
 - Clean up all spills immediately

■ **Note: DO NOT** routinely give an enema to women in labor.

Mobility

□ Ensure mobility:

- Encourage the woman to move about freely; during the first stage of labor.
- Support the woman's choice of position (left lateral, squatting, kneeling, standing supported by the companion) for each stage of labor and delivery.

Figure: 5.8 Positions that a woman may adopt during labor





Urination

- ☐ Encourage the woman to empty her bladder regularly/ frequently. Remind her every 2 hours.

Eating, drinking

- ☐ In general encourage oral intake of liquid diet (tea juice) throughout labor but not hard foods.
- ☐ Consider fluid diet as a source of water and energy for those mothers staying longer before delivery (e.g. small sips of sweetened tea or water)
- ☐ If the woman has visible severe wasting or tires during labor, make sure she drinks.

Breathing technique

- ☐ Teach her to notice her normal breathing.
- ☐ Encourage her to breathe out more slowly, making a sighing noise, and to relax with each breath.
- ☐ If she feels dizzy, unwell, is feeling pins-and-needles (tingling) in her face, hands and feet, encourage her to breathe more slowly.
- ☐ To prevent pushing at the end of first stage of labor, teach her to pant, to breathe with an open mouth, to take in 2 short breaths followed by a long breathout.
- ☐ During delivery of the head, ask her not to push but to breathe steadily or to pant.

Pain and discomfort relief

- ☐ Suggest change of position.
- ☐ Encourage mobility, as comfortable for her.
- ☐ Encourage companion to:
 - massage the woman's back if she finds this helpful.
 - hold the woman's hand and sponge her face between contractions.
- ☐ Encourage her to use the breathing technique.
- ☐ Encourage warm bath or shower, if available.
- ☐ If necessary, give pethidine 1 mg/kg body weight (but not more than 100 mg) IM or IV slowly or give morphine 0.1 mg/kg body weight IM.

- o Pethidine injection: 50 mg IM initially. Assess after ½ hour and if not adequate and side effects not troublesome, repeat 50 mg. Onset of action within 10 -20 min and lasts for 2-4 hours (it is better be avoided in late labor because of risk of neonatal depression).

- ☐ Help the woman in labor who is anxious, fearful or in pain
 - o Investigate the cause
 - o Give her praise, encouragement and reassurance;
 - o Give her information on the process and progress of her labor;
 - o Listen to the woman and be sensitive to her feelings

If pain is constant (persisting between contractions) and very severe or sudden in onset, consider abruptio placenta, uterine rupture or chorioamnionitis and manage accordingly (table 5.3).

Personal support (birth companion)

- ☐ Encourage the woman to have personal support from a person of her choice throughout labor and birth:
 - ☐ Encourage support from the chosen birth companion;
 - ☐ Arrange seating for the companion next to the woman;
 - ☐ Encourage the companion to give adequate support to the woman during labor and childbirth
- ☐ Describe to the birth companion what she or he should do:
 - ☐ Always be with the woman.
 - ☐ Encourage her.
 - ☐ Help her to breathe and relax.
 - ☐ Rub her back, wipe her brow with a wet cloth, do other supportive actions.
 - ☐ Give support using local practices which do not disturb labor or delivery.
 - ☐

Encourage woman to move around freely as she wishes and to adopt the position of her choice.

- ☐ Encourage her to drink fluids and eat as she wishes.
- ☐ Assist her to the toilet when needed.
- ☐ Ask the birth companion to call for help if:
 - ☐ The woman is bearing down with contractions.
 - ☐ There is vaginal bleeding.
 - ☐ She is suddenly in much more pain.
 - ☐ She loses consciousness or has fits.

- ☐ There is any other concern.
- ☐ Tell the birth companion what she or he **SHOULD NOT DO** and explain why:
 - ☐ **DO NOT** encourage woman to push.
 - ☐ **DO NOT** give advice other than that given by the healthworker.
 - ☐ **DO NOT** keep woman in bed if she wants to move around.

2.6 PARTOGRAPHIC FOLLOW UP OF LABOR

2.6.1 THEPARTOGRAPH and ITS COMPONENTS

The partograph is the graphic recording of the progress of labor and the salient condition of the mother and the fetus. It serves as an “early warning system” and assists in early decision to transfer/refer, augment and terminate labor. Partograph has been modified to make it simpler and easier to use. The latent phase has been removed and plotting on the partograph begins in the active phase when the cervix is 4 cm dilated. A sample partograph is included (**figure 5.9**). Note that the partograph should be enlarged to full size before use. Record the following on the partograph:

Patient information: Fill out name, gravida, para, hospital number, date and time of admission and time of ruptured membranes as HH/MM or Duration of ROM (in hrs).

Fetal heart rate: Record every half hour.

Amniotic fluid: Record the color of amniotic fluid at every vaginal examination:

- I: membranes intact;
- C: membranes ruptured, clear fluid;
- M: meconium-stained fluid;
- B: blood-stained fluid.

Moulding:

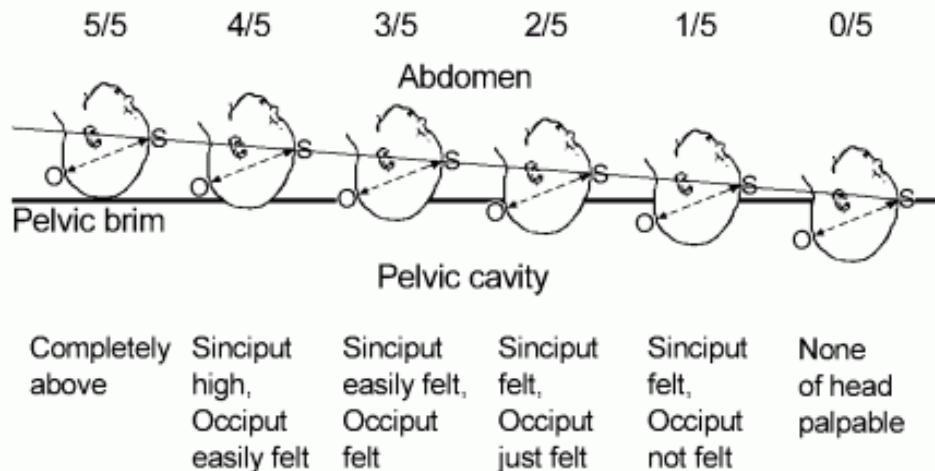
- 1: sutures apposed;
- 2: sutures overlapped but reducible;
- 3: sutures overlapped and not reducible.

Cervical dilatation: Assessed at every vaginal examination and marked with a cross (X). Begin plotting on the partograph at 4 cm.

Alert line: A line starts at 4 cm of cervical dilatation to the point of expected full dilatation at the rate of 1 cm per hour.

Action line: Parallel and 4 hours to the right of the alert line.

Descent assessed by abdominal palpation: Refers to the part of the head (divided into 5 parts) palpable above the symphysis pubis; recorded as a circle (O) at every vaginal examination. At 0/5, the sinciput (S) is at the level of the symphysis pubis.



Hours: Refers to the time elapsed since onset of active phase of labor (observed or extrapolated).

Time: Record actual time.

Contractions: Chart every half hour; palpate the number of contractions in 10 minutes and their duration in seconds.

- Less than 20seconds: ■■■
- Between 20 and 40seconds: ▨
- More than 40seconds: ▩

Oxytocin: Record the amount of oxytocin per volume IV fluids in drops per minute every 30 minutes when used in hospitals.

Drugs given: Record any additional drugs given.

Pulse: Record every 30 minutes and mark with a dot (●).

Blood pressure: Record every 4 hours and mark with arrows.

Temperature: Record every 2 hours.

Protein, acetone and volume: Record every time urine is passed.

Figure 5-9: -The Partograph

Name _____	Gravida _____	Para _____	Hospital number _____
Date of admission _____	Time of admission _____	Ruptured membranes _____	hours _____

Fetal heart rate

Amniotic fluid Moulding

Cervix (cm) [Plot X]

Descent of head [Plot O]

Hours

Time

Contractions per 10 mins

Oxytocin U/L drops/min

Drugs given and IV fluids

Pulse and BP

Temp °C

Urine { protein, acetone, volume

2.6.2 FOLLOW-UP

ASSESSMENT OF MATERNAL CONDITION

- Monitor the woman's condition:
 - During the **latent phase** of the first stage of labor: Check maternal mood and behavior (distressed, anxious) at least once every hour; check blood pressure, pulse and temperature at least once every four hours.
 - During the **active phase** of first stage of labor: Check maternal mood and behavior (distressed, anxious) at least once every 30 minutes; check blood pressure at least once every four hours, temperature at least once every two hours and pulse once every 30 minutes.
 - During **second stage**: Check maternal mood and behavior (distressed, anxious) at least once every five minutes.
- Evaluate the woman for emergency signs and for signs of distress, and respond appropriately:
 - If the **woman's pulse is increasing**, she may be dehydrated or in pain or she may be developing a fever.

Ensure adequate hydration via oral or IV routes (NS 1000l/4hrs).
Provide adequate analgesia (pethidine).
 - If the **woman's temperature is higher than 38°C**, manage the cause of the fever and monitor temperature at least every two hours.
 - If the **woman's blood pressure decreases**, suspect occult or frank hemorrhage.
 - If **acetone is present in the woman's urine**, suspect poor nutrition or dehydration and encourage her to eat or drink; otherwise, give dextrose IV.

ASSESSMENT OF FETAL CONDITION

- Count the fetal heart rate for a full minute at least once every hour during the latent phase, once every 30 minutes during the active phase and every five minutes during the second stage
- If there are **fetal heart rate abnormalities** (less than 100 or more than 180 beats per minute), suspect fetal distress.

ASSESSMENT OF PROGRESS OF LABOR

- Once diagnosed, progress of labor is assessed by:
 - measuring changes in cervical effacement and dilatation during the latent phase;
 - measuring the rate of cervical dilatation and fetal descent during the active phase; and
 - assessing further fetal descent during the second stage.
- If unsatisfactory progress of labor or prolonged labor is suspected, manage the cause slow progress.

Progress of the first stage of labor should be plotted on a partograph once the woman enters the active phase of labor. A sample partograph is shown in fig 5.9. Alternatively, plot a simple graph of cervical dilatation (centimeters) on the vertical axis against time (hours) on the horizontal axis.

UTERINE CONTRACTIONS

- Count the number of contractions in a 10-minute time period and their duration in seconds at least once every hour during the latent phase and once every 30 minutes during the active phase.
- Contractions are considered to be efficient if they result in progressive dilatation of the cervix.
- In general, contractions are considered to be efficient during the active phase if there are three to five contractions in 10 minutes, each lasting 40 to 60 seconds.
- The uterus should relax between contractions.

VAGINAL EXAMINATIONS

During the first stage of labor and after rupture of the membranes, vaginal examinations should be carried out once every four hours, unless there is a clinical indication to perform them more frequently. Plot the findings on a partograph if the woman is in the active phase of labor.

Multiple vaginal examinations increase a woman's risk of infection, especially in the presence of other infection risk factors (e.g. prolonged rupture of membranes and/or labor.) Do not perform vaginal exams more frequently than every four hours unless there is a clear indication. Always maintain hygienic technique when performing vaginal exams

- Inform the woman adequately about the necessity of the vaginal examination, what to expect and information to be gained by performing the exam; address the woman's fears and anxieties, and give her the opportunity to ask questions.
- The woman should not be assisted in the removal of her clothing unless it has been clarified that assistance is required.
- Obtain the woman's informed consent and permission before conducting examination.
- Do not force the woman's legs apart. Talk her gently through the exam and wait until she is ready for you to begin.
- If available, use a lubricant when performing a vaginal examination; if a lubricant is not available, wet the glove with clean water or saline solution.
- Assist the woman in assuming the position of her choice, and maintain eye contact during the examination to decrease any feelings of vulnerability.
- After the examination, provide information on your findings and how they affect the plan of care.
- At each vaginal examination, record the following:
 - status of membranes;
 - color of amniotic fluid (liquor), if membranes are ruptured;

- cervical dilatation and effacement; and
- descent (can also be assessed abdominally).
- If the cervix is not dilated on first examination, it might not be possible to diagnose labor.
- If contractions persist, re-examine the woman for cervical changes after four hours. At this stage, if there is effacement and dilatation, the woman is in labor; if there is no change, the diagnosis is false labor.
- In the second stage of labor, perform vaginal examinations once every hour.

PROGRESS OF FIRST STAGE OF LABOR

- Findings suggestive of **satisfactory progress** in first stage of labor are:
 - Regular contractions of progressively increasing frequency and duration;
 - Rate of cervical dilatation at least 1 cm per hour during the active phase of labor (cervical dilatation on or to the left of alertline);
 - Cervix well applied to the presenting part.
- Findings suggestive of **unsatisfactory progress** in first stage of labor are:
 - Irregular and infrequent contractions after the latent phase;
 - OR rate of cervical dilatation slower than 1 cm per hour during the active phase of labor (cervical dilatation to the right of alertline);
 - OR cervix poorly applied to the presenting part.

Unsatisfactory progress in labor can lead to prolonged labor. If **unsatisfactory progress of labor** or **prolonged labor** is suspected, manage the cause of slow progress.

PROGRESS OF SECOND STAGE OF LABOR

- Findings suggestive of **satisfactory progress** in second stage of labor are:
 - Steady descent of fetus through birth canal;
 - Onset of expulsive (pushing) phase.
- Findings suggestive of **unsatisfactory progress** in second stage of labor are:
 - Lack of descent of fetus through birth canal;
 - Failure of expulsion during the late (expulsive) phase.
 - Diagnosis of prolonged 2nd stage
 - Nullipara - 2 hrs without or 3hrs with epidural anesthesia
 - Multipara - 1 hrs without or 2hrs with epidural anesthesia

Note: - If second stage is prolonged without visible steady descent of the head, reevaluate and deliver using vacuum extractor or **refer urgently to hospital**.

SAMPLE PARTOGRAPHIC FOLLOW-UP OF NORMALLY PROGRESSING LABOR

Figure 5-10, is a sample partograph for normal labor:

- A primigravida was admitted in the latent phase of labor at 5AM:
 - Fetal head 4/5 palpable;
 - Cervix dilated 2cm;
 - 3 contractions in 10 minutes, each lasting 20seconds;
 - Normal maternal and fetal condition.

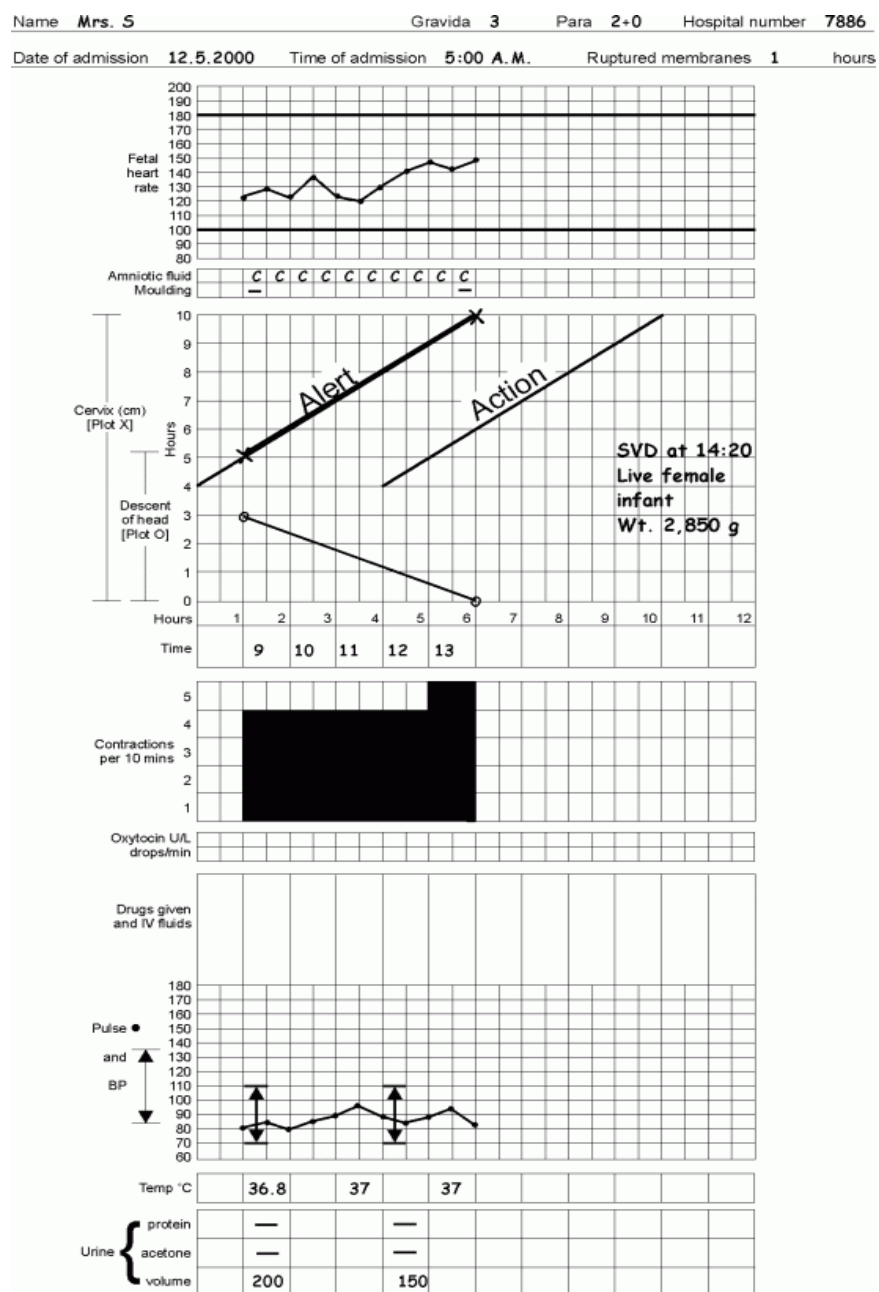
Note: This information is not plotted on the partograph.

- At 9AM:
 - Fetal head is 3/5 palpable;
 - Cervix dilated 5cm;

Note: The woman was in the active phase of labor and this information is plotted on the partograph. Cervical dilatation is plotted on the alert line.

- 4 contractions in 10 minutes, each lasting 40seconds;
 - Cervical dilatation progressed at the rate of 1 cm per hour.
- At 2PM:
 - Fetal head is 0/5 palpable;
 - Cervix is fully dilated;
 - 5 contractions in 10 minutes each lasting 40seconds;
 - Spontaneous vaginal delivery occurred at 2:20PM.

Figure 5-10:- Sample partograph for normal labor



2.7 INTRAPARTUM CARE FOR PMTCT

2.7.1 INTRODUCTION

Majority of infant born from an HIV positive mothers acquire infection during labor and delivery as compared to that of ANC and postpartum. Therefore, understanding the standardized approach to attend labor and delivery as well as interventions to PMTCT is very important to have an effective program to avert infection amongst the new generation.

Proper diagnosis and follow up of labor is of paramount importance to minimize maternal morbidity and mortality. In addition to such a benefit it will also help prevent mother to child transmission of HIV during labor and delivery. a very important way of making sure labor is being followed is the use of partograph.

Factors that increase the risk of MTCT of HIV:

There are certain factors that are believed to increase the risk of MTCT of HIV. Some of the factors have strong evidence associated with an increased risk while others have only weaker evidence linking them to an increased risk as outlined in the table 4-3 below

Table 5-4: Factors increasing mother-to-child transmission of HIV during labor and delivery

Strong evidence	Weaker evidence
• High viral load (from recent infection or advanced disease/clinicalAIDS) • Prematurity • Vaginal delivery • Rupture of membranes for more than 4 hours • Prolonged labor • Instrumental delivery (forceps or vacuum extraction)	• Chorioamnionitis • Suctioning newborn unless necessary for thick meconium or to resuscitate • Invasive obstetrical procedures – Episiotomy – Artificial rupture of membranes – Frequent vaginal examinations

2.7.2 INTERVENTIONS FOR PMTCT:

- Administer ARV treatment or prophylaxis during labor in accordance with national guideline.
- Continue ARV treatment/prophylaxis or give ARV prophylaxis at labor (single-dose NVP+AZT+ 3TC), to reduce maternal viral load and provide protection to the infant.

Use good infection prevention practices for all patient care

Use standard precautions for infection prevention, which include use of protective gear, safe use and disposal of sharps, sterilization of equipment and safe disposal of contaminated materials.

Minimize cervical examinations

- Perform cervical examination with appropriate clean techniques and only when absolutely necessary.

Avoid prolonged labor

- Consider use of oxytocin to shorten labor when appropriate.
- Use non-invasive fetal monitoring to direct need for early intervention.

Avoid routine rupture of membranes

- Use a partograph to measure the progress of labor.
- Avoid artificial rupture of membranes, unless necessary.

Avoid unnecessary trauma during delivery

- Avoid invasive procedures including use of scalp electrodes or scalp sampling.
- Avoid routine episiotomy.
- Minimize the use of forceps or vacuum extractors. If instrumental delivery is a must, forceps is preferable.

Considerations regarding mode of delivery

Caesarean section, when performed before the onset of labor or rupture of membrane, has been associated with reduced MTCT. But in women on HAART with viral load of < 1000/microlit, the risk of PMTCT is very low and CS does not seem to have benefit. In such cases, the use of Caesarean section for PMTCT should be given due consideration provided that the limited capacity of most of the health facilities. Thorough counselling of the Family should be done on the benefits and risks of vaginal delivery versus elective caesarean section, including the safety of blood supply and the risk of peri-operative complications before decision making. Nevertheless C-section for other emergency obstetrical indications should proceed emergently regardless of time lag or if ARVs not able to be given.

For further reading refer to the national PMTCT guideline (2007).

2.8 CARE DURING 2ND STAGE AND ASSISTING NORMAL CHILDBIRTH

Notify nursing staff that delivery is imminent. Ensure all delivery equipment and supplies, including newborn resuscitation equipment in a pre-warmed neonatal corner, are available, and place of delivery is clean and warm (25°C). In addition:

- Ensure bladder is empty.
- Assist the woman into a comfortable position of her choice, as upright as possible (*figure 5.11*). Avoid supine position till head is visible.
- Stay with her and offer her emotional and physical support.
- Allow her to push as she wishes with contractions avoiding early push; it should start spontaneously.
- Wait until head visible and perineum distending.
- Wash hands with clean water and soap. Put on gloves just before delivery.
- Clean the vulva and perineum with antiseptic (downward and away from the introitus). If pieces of faeces get expelled, wipe them downward.
- See Universal precautions during labor and delivery.
 - Attendant should be dressed and gloved appropriately (gloves, gowns, apron, masks, caps, eyeprotection)
 - Sterile draping in such a way that only the immediate area about the vulva is exposed.

Figure 5-11: -Positions that a woman may adopt during childbirth



Note: - Episiotomy is no longer recommended as a routine procedure. There is no evidence that routine episiotomy decreases perineal damage, future vaginal prolapse or urinary incontinence. In fact, routine episiotomy is associated with an increase of third and fourth degree tears and subsequent anal sphincter muscle dysfunction.

- Timing of episiotomy – performed when fetal head has distended the vulva 2-3cms unless early delivery is indicated.
- Use analgesia/anesthesia for making episiotomy and repair.

ASSIST DELIVERY OF THE BABY:-

General methods of supportive care during labor are most useful in helping the woman tolerate labor pains

- Once the **cervix is fully dilated** and the woman is **in the expulsive phase of the second stage**, encourage the woman to assume the position she prefers and encourage her topush.

Delivery of the head:

- Ask the woman to pant or give only small pushes with contractions as the fetal head delivers.
- To control birth of the head, place the fingers of one hand against the fetal head to keep it flexed (bent).
- Continue to gently support the perineum as the fetal head delivers. If necessary; assist extension of the head with a hand protected with sterile towel placed on the perineum and the fetal chin palpated and pressed up ward gently effecting extension.
- Once the fetus's head delivers, ask the woman not to push.
- After delivery of the head, wipe the mouth, oro-pharynx first (routine suctioning not recommended even in neonates with meconium stained liquor).
- Feel around the fetal neck for the umbilical cord:
 - If the **cord is around the neck but is loose**, slip it over the fetus's head;
 - If the **cord is tight around the neck**, doubly clamp and cut it before unwinding it from around the neck.

Completion of delivery:

- Allow the fetal head to turn spontaneously.
- After the head turns, place a hand on each side of the fetus's head. Tell the woman to push gently with the next contraction.
- Reduce tears by delivering one shoulder at a time. Move the fetus's head posteriorly to deliver the shoulder that is anterior.

Note: If there is **difficulty delivering the shoulders**, suspect shoulder dystocia.

- Lift the fetal head anteriorly to deliver the shoulder that is posterior.
- Support the rest of the fetus's body with one hand as it slides out.
- Place the newborn on the mother's abdomen. Note the time of birth.

Note: If placing newborn on abdomen is not acceptable, or the mother cannot hold the newborn, place the newborn in a clean, warm, safe place close to the mother.

- Thoroughly dry the baby and assess breathing.

Note: Most babies begin crying or breathing spontaneously within 30 seconds of birth:

- Note: For neonates born through meconium-stained amniotic fluid who start breathing on their own, suctioning of the mouth or nose or tracheal suctioning should not be performed.
- Take cord blood if indicated
- Palpate the mother's abdomen to rule out the presence of additional baby (ies) and proceed with active management of the third stage.

2.9 ACTIVE MANAGEMENT OF THE THIRD STAGE OF LABOR

The third stage is the time period from delivery of the baby to the expulsion of the placenta. The average duration of the 3rd stage is 15 min and it is said to be prolonged if the placenta is not delivered 30min after birth of the baby.

Post-partum hemorrhage is the most important single cause of maternal death in the world. The majority of these deaths (88%) occur within 4 hours of delivery, indicating that they are a consequence of events in the third stage of labor. Postpartum haemorrhage is a complication which occurs at the transition between labor and the postpartum period. The major causes of hemorrhage are uterine atony and retained placenta (80%). Vaginal or cervical lacerations and (occasionally) uterine rupture or inversion also play a role. Most PPH occur in the first 2 hours after delivery and this critical period is sometimes referred as the 4th stage of labor.

Active management of the third stage (active delivery of the placenta) helps prevent postpartum haemorrhage. AMTSL is associated with decreased bleeding in 3rd stage, low rate of retained placenta and reduced risk of maternal anemia after childbirth. It includes administration of a uterotonic medication within one minute after the birth of the baby. It may also include:

- controlled cord traction; and
- Verification of uterine tone and, if the uterus is not well contracted, uterine massage (WHO, MCPC, 2017).

The most important intervention to reduce postpartum hemorrhage is the immediate postpartum administration of a uterotonic within one minute of birth. Controlled cord traction is optional and should only be performed by a skilled provider.

2.9.1 IMMEDIATE POSTPARTUM UTEROTONIC

- Immediately after the birth of the baby, palpate the abdomen to rule out the presence of an additional baby(s).
- In the absence of an additional baby(s), **give oxytocin 10 units IM.**

Make sure there is no additional baby(s) before giving an injectable uterotonic medication IM or giving large doses of misoprostol orally.

Note: Oxytocin is preferred because it is effective two to three minutes after injection, has minimal side effects and can be used in all women.

-If oxytocin is not available, give:

- oral misoprostol 600 mcg;
- OR ergometrine (0.2 mg IM) or methylergometrine;
- OR the fixed drug combination of oxytocin and ergometrine (1 mL = 5 IU oxytocin + 0.5 mg ergometrine).

Do not give ergometrine to women with pre-eclampsia, eclampsia or high blood pressure because it increases the risk of convulsions and cerebrovascular accidents.

2.9.2 CONTROLLED CORD TRACTION

The placenta may be allowed to deliver physiologically or can be delivered via controlled cord traction by a skilled provider. Controlled cord traction is contraindicated in settings without a skilled birth attendant.

- After cutting the cord, clamp the cord close to the perineum using sponge forceps. Hold the clamped cord and the end of the forceps with one hand.
- Place the other hand just above the woman's pubic bone and stabilize the uterus by applying counter-traction during controlled cord traction. This helps prevent inversion of the uterus.

Never apply cord traction (pull) without applying counter-traction (push) above the pubic bone with the other hand.

- Keep slight tension on the cord and await a strong uterine contraction (two to three minutes).
- When the **uterus becomes rounded (globular) or the cord lengthens**, very gently pull downward on the cord to deliver the placenta. Do not wait for a gush of blood before applying traction on the cord. Continue to apply counter-traction to the uterus with the other hand.
- If the **placenta does not descend** (i.e. there are no signs of placental separation) during 30 to 40 seconds of controlled cord traction, do not continue to pull on the cord. Instead:
 - Gently hold the cord and wait until the uterus is well contracted again. If necessary, use sponge forceps to clamp the cord closer to the perineum as it lengthens.
 - With the next contraction, repeat controlled cord traction with counter-traction.
- As the placenta delivers, the thin membranes can tear off. Hold the placenta in two hands and gently turn it several times in a clockwise fashion until the membranes are twisted.
- Slowly pull to complete the delivery.
- If the **membranes tear**, gently examine the upper vagina and cervix while wearing sterile gloves. Use some sponge forceps to remove any pieces of membrane that are present.
- Examine the placenta carefully to be sure none of it is missing. If a **portion of the maternal surface is missing or if there are torn membranes with vessels**, suspect retained placental fragments).
- **If uterine inversion occurs**, reposition the uterus.
- If the **cord is pulled off**, manual removal of the placenta may be necessary.
- Dispose of the placenta in a correct, safe and culturally appropriate manner.

2.9.3 VERIFICATION OF UTERINE TONE

- Immediately assess uterine tone. If **the uterus is soft**, massage the fundus of the uterus through the woman's abdomen until the uterus is contracted.

Note: Sustained uterine **massage** is not recommended as an intervention to prevent postpartum hemorrhage in women who have received a prophylactic uterotonic.

- Ensure that the uterus does not become relaxed (soft) after you stop uterine massage.
- Teach the woman how to assess uterine tone and massage her own uterus should it become soft.
- Estimate and record blood loss.

THE WHO SAFE CHILD BIRTH LIST

WHO developed the *WHO Safe Childbirth Checklist* (WHO, 2015) as a tool to support the delivery of essential maternal and perinatal care practices and hence improve the quality of care for women and babies at the time of childbirth. The Checklist is an organized list of evidence-based essential birth practices targeting major causes of maternal deaths (haemorrhage, infection, obstructed labor and hypertensive disorders), intrapartum-related stillbirths and neonatal deaths (birth asphyxia, infection and complications related to prematurity) that occur in facilities around the world. It was developed following a rigorous methodology and tested for usability in 10 countries across Africa and Asia.

SEE TABLE BELOW (5.5)

1

On Admission

Does mother need referral?

- ☐ No
- ☐ Yes, organized

Check your facility's criteria

Partograph started?

- ☐ No, will start when ≥ 4 cm
- ☐ Yes

Start plotting when cervix ≥ 4 cm, then cervix should dilate ≥ 1 cm/hr

- Every 30 min: plot HR, contractions, fetal HR
- Every 2 hr: plot temperature
- Every 4 hr: plot BP

Does mother need to start:

Antibiotics?

- ☐ No
- ☐ Yes, given

Ask for allergies before administration of any medication

Give antibiotics to mother if any of:

- Mother's temperature $\geq 38^{\circ}\text{C}$
- History of foul-smelling vaginal discharge
- Rupture of membranes > 18 hrs

Magnesium sulfate and antihypertensive treatment?

- ☐ No
- ☐ Yes, magnesium sulfate given
- ☐ Yes, antihypertensive medication given

Give magnesium sulfate to mother if any of:

- Diastolic BP ≥ 110 mmHg and 2+ proteinuria
- Diastolic BP ≥ 90 mmHg, 2+ proteinuria, and any: severe headache, visual disturbance, epigastric pain

Give antihypertensive medication to mother if systolic BP > 160 mmHg

- Goals: keep BP $< 159/100$ mmHg

- ☐ Confirm supplies are available to clean hands and wear gloves for each vaginal exam.

- ☐ Encourage birth companion to be present at birth.

- ☐ Confirm that mother or companion will call for help during labour if needed

Call for help if any of:

- Bleeding
- Severe abdominal pain
- Severe headache or visual disturbance
- Unable to urinate
- Urge to push

2

Just Before Pushing (Or Before Caesarean)

Does mother need to start:

Antibiotics?

- ☐ No
- ☐ Yes, given

Ask for allergies before administration of any medicine

Give antibiotics to mother if any of:

- Mother's temperature $\geq 38^{\circ}\text{C}$
- History of foul-smelling vaginal discharge
- Rupture of membranes >18 hrs
- Caesarian section

Magnesium sulfate and antihypertensive treatment?

- ☐ No
- ☐ Yes, magnesium sulfate given
- ☐ Yes, antihypertensive medication given

Give magnesium sulfate to mother if any of:

- Diastolic BP ≥ 110 mmHg and 3+ proteinuria
- Diastolic BP ≥ 90 mmHg, 2+ proteinuria, and any: severe headache, visual disturbance, ep

Give antihypertensive medication to mother if any of:

- Goal: keep BP $<150/100$ mmHg

Confirm essential supplies are at bedside and prepare for delivery:

For mother

- ☐ Gloves
- ☐ Alcohol-based handrub or soap and clean water
- ☐ Oxytocin 10 units in syringe

Prepare to care for mother immediately after birth:

Confirm single baby only (not multiple birth)

1. Give oxygen within 1 minute after birth

2. Deliver placenta 1-3 minutes after birth

3. Massage uterus after placenta is delivered

4. Confirm uterus is contracted

For baby

- ☐ Clean towel
- ☐ Tie or cord clamp
- ☐ Sterile blade to cut cord
- ☐ Suction device
- ☐ Bag-and-mask

Prepare to care for baby immediately after birth:

1. Dry baby, keep warm

2. If not breathing, stimulate and clear airway

3. If still not breathing:

- clamp and cut cord
- clear airway if necessary
- ventilate with bag-and-mask
- shout for help

- ☐ Assistant identified and ready to help at birth if needed.

3

Soon After Birth (Within 1 Hour)

Is mother bleeding abnormally?

- ☐ No
- ☐ Yes, shout for help

If bleeding abnormally:

- Massage uterus
- Consider more uterotonic
- Start IV fluids and keep mother warm
- Treat cause: uterine atony, retained placenta/fragments, vaginal tear, uterine rupture

Does mother need to start:

Antibiotics?

- ☐ No
- ☐ Yes, given

Ask for allergies before administration of any medication

Give antibiotic to mother if placenta manually removed or if mother's temperature $\geq 38^{\circ}\text{C}$ and any of:

- Chills
- Foul-smelling vaginal discharge

If the mother has a third or fourth degree of perineal tear give antibiotic to prevent infection

Magnesium sulfate and antihypertensive treatment?

- ☐ No
- ☐ Yes, magnesium sulfate given
- ☐ Yes, antihypertensive medication given

Give magnesium sulfate to mother if any of:

- Diastolic BP ≥ 110 mmHg and 3+ proteinuria
- Diastolic BP ≥ 90 mmHg, 2+ proteinuria, and any: severe headache, visual disturbance, epigastric pain

Give antihypertensive medication to mother if systolic BP > 160 mmHg

- Goal: keep BP $< 150/100$ mmHg

Does baby need:

Referral?

- ☐ No
- ☐ Yes, organized

Check your facility's criteria.

Antibiotics?

- ☐ No
- ☐ Yes, given

Give baby antibiotics if antibiotics given to mother for treatment of maternal infection during childbirth or if baby has any of:

- Respiratory rate $> 60/\text{min}$ or $< 30/\text{min}$
- Chest in-drawing, grunting, or convulsions
- Poor movement on stimulation
- Baby's temperature $< 35^{\circ}\text{C}$ (and not rising after warming) or baby's temperature $\geq 38^{\circ}\text{C}$

Special care and monitoring?

- ☐ No
- ☐ Yes, organized

Arrange special care/monitoring for baby if any:

- More than 1 month early
- Birth weight < 2500 grams
- Needs antibiotics
- Required resuscitation

☐ Started breastfeeding and skin-to-skin contact (if mother and baby are well).

☐ Confirm mother / companion will call for help if danger signs present.

4

Before Discharge

☐ Confirm stay at facility for 24 hours after delivery.

Does mother need to start antibiotics?

- ☐ No
☐ Yes, given and delay discharge

Ask for allergies before administration of any medication.
 Give antibiotics to mother if any of:
 • Mother's temperature $\geq 38^{\circ}\text{C}$
 • Foul-smelling vaginal discharge

Is mother's blood pressure normal?

- ☐ No, treat and delay discharge
☐ Yes

Give magnesium sulfate to mother if any of:
 • Diastolic BP ≥ 110 mmHg and 3+ proteinuria
 • Diastolic BP ≥ 90 mmHg, 2+ proteinuria, and any: severe headache, visual disturbances, epigastric pain
 Give antihypertensive medication to mother if systolic BP > 160 mmHg
 • Goal: keep BP $< 150/100$ mmHg

Is mother bleeding abnormally?

- ☐ No
☐ Yes, treat and delay discharge

If pulse > 110 beats per minute and blood pressure < 90 mmHg:
 • Start IV and keep mother warm
 • Treat cause (hypovolemic shock)

Does baby need to start antibiotics?

- ☐ No
☐ Yes, give antibiotics, delay discharge, give special care

Give antibiotics to baby if any of:
 • Respiratory rate $> 60/\text{min}$ or $< 30/\text{min}$
 • Chest in-drawing, grunting, or convulsions
 • Poor movement on stimulation
 • Baby's temperature $< 35^{\circ}\text{C}$ (and not rising after warming) or baby's temperature $\geq 38^{\circ}\text{C}$
 • Stopped breastfeeding well
 • Umbilical redness extending to skin or draining pus

Is baby feeding well?

- ☐ No, establish good breastfeeding practices and delay discharge
☐ Yes

☐ Discuss and offer family planning options to mother.

☐ Arrange follow-up and confirm mother / companion will seek help if danger signs appear after discharge.

Danger Signs

Mother has any of:

- Bleeding
- Severe abdominal pain
- Severe headache or visual disturbance
- Breathing difficulty
- Fever or chills
- Difficulty emptying bladder
- Epigastric pain

Baby has any of:

- Fast/difficult breathing
- Fever
- Unusually cold
- Stops feeding well
- Less activity than normal
- Whole body becomes yellow

2.10 IMMEDIATE (essential) NEW BORN CARE

Definition and Components

Definition: Essential Newborn Care is a package of basic care provided to newborns to support their survival and wellbeing. The definition of essential newborn care can be broadened to describe a “comprehensive strategy designed to improve the health of newborns through interventions before conception, during pregnancy, at and soon after birth, and in the postnatal period”.

The care we give for most of the babies immediately after birth is simple but very important to improve their survival and health. However, the standardized procedure for providing Essential Newborn Care is not commonly practiced. This results in serious consequences of unacceptably high neonatal morbidity and mortality. About 50% and 75% of neonatal mortalities occur in the first 24 hours after birth and within the first week of life of the newborn, respectively mainly due to asphyxia, hypothermia, hypoglycemia, infection, and complications related to prematurity etc.

The initial steps in the care of the baby at birth, such as drying, wrapping, and evaluation of breathing, are similar for all babies. Subsequent care, however, may be different if there are problems such as birth asphyxia, prematurity or infection. The box below summarizes the immediate care package that should be provided to all babies immediately after birth. The order of the care given may vary depending on the specific needs of the baby.

- Step 1: Dry and stimulate the baby by delivering on mother's abdomen.
- Step 2: Assess breathing. Make sure the baby is breathing well.
- Step 3: If the baby does not breathe, clamp/tie and cut the cord immediately and start resuscitation. If the baby does cry/breathes well, clamp/tie and cut the cord after pulsations stop or after 2-3 minutes.
- Step 4: Place the newborn in skin-to-skin contact on the mother's chest and cover both with clean linen and blanket as required. Carry out all the steps noted below up to step #9, preferably with the baby on the mother's chest.
- Step 5: Initiate breastfeeding within the first hour.
- Step 6: Administer tetracycline eye ointment.
- Step 7: Apply 4% chlorhexidine gel on the cord
- Step 8: Administer vitamin K1.
- Step 9: Place the baby identification bands on the wrist and ankle.
- Step 10: Weigh the infant when he/she is stable.
- Note: Record observations and treatment provided in the registers/appropriate chart/cards.
Defer the bath for at least 24 hours. Clean the HIV exposed newborn with clean towel and give HIV prophylaxis for the newborn based on national guideline.

Step 1: Dry and Stimulate

While the baby is on the mother's abdomen dry and stimulate.

Dry the baby thoroughly at birth. Drying helps keep the baby warm and stimulates breathing. Dry the baby's body, arms, legs, and head by gently rubbing with a cloth; as you dry stimulate breathing. Wipe the face clean of blood and maternal faeces. A newly born baby wet with amniotic fluid can become cold even in a warm room.

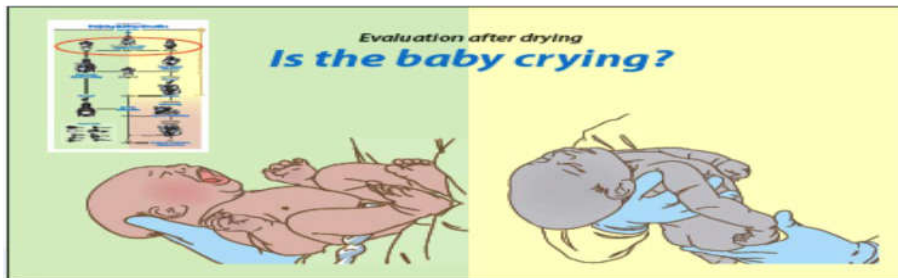
Drying often provides sufficient stimulation for breathing to start in mildly depressed newborn babies. Do your best not to remove the vernix (the creamy, white substance which may be on the skin) as it protects the skin and may help prevent infection.

Remove the wet cloth. Note the time at birth; you need to record the time after you complete the essential care. Wrap with another dry towel and cover the head while the baby is on mother's abdomen. Then wrap the baby with another dry cloth and cover the head.

- Do not let the baby remain wet, as this will cool the body and make him/her hypothermic.
- In neonates born through meconium-stained amniotic fluid who start breathing on their own, tracheal suctioning should not be performed and routine suctioning of the mouth or nose is not recommended. In neonates born through meconium-stained amniotic fluid who do not start breathing on their own, suctioning of the mouth and nose should be done before initiating positive-pressure ventilation.



Step 2: Assess Breathing and color; if not breathing or gasping, then resuscitate.



* Check if the baby is crying while drying it.

Crying is possible when large amount of air moves in and out of the lungs. The crying baby usually moves his or her arms and legs and has good muscle tone. After crying for some seconds, a baby may stop crying and begin to breathe quietly and regularly. A baby also may continue to cry for some time. If the baby does not cry, check the baby's breathing. Call for help & **Start resuscitation** if any of the following signs are seen

- 1) breathing less than 30/min
- 2) having trouble breathing
- 3) not breathing at all

This step is used for assessing and classifying the newborn for birth asphyxia. If the baby needs

resuscitation quickly clamp or tie and cut the cord leaving a stump at least 10 cm long for now, stop the subsequent steps of essential newborn care and start resuscitation. Functional resuscitation equipment should always be ready and close to the delivery area since you must start resuscitation

- The assistant can provide basic care for the mother while you provide the more specialized care for the baby who is not breathing. (see section on Neonatal resuscitation)
- Do not do suction of the mouth and nose *as a routine*. Do it only if there is meconium, thick mucus, or blood.

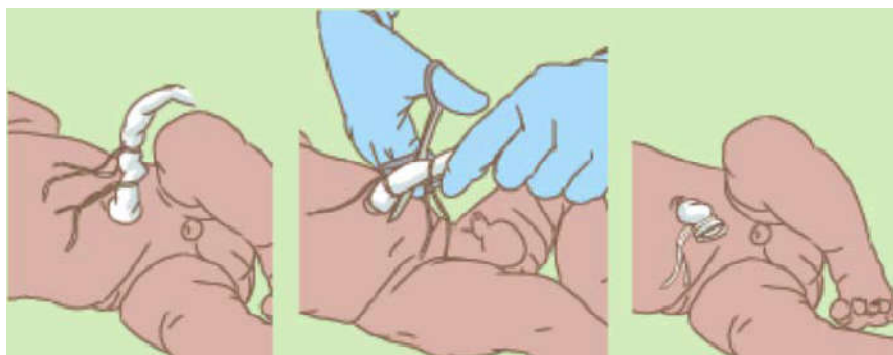
Step 3: If the baby does cry or breaths well clamp/tie and cut the cord

Optimal cord care consists of the following:

Clamping the cord: If the baby does not need resuscitation, wait for cord pulsations to cease or approximately 2-3 minutes after birth, whichever first, and then place one clamp/cord tie 2 fingerbreadths from the baby's abdomen and the second clamp/tie another 5 fingerbreadths from the abdomen. Make sure that it is tied tightly. Cutting the cord soon after birth can decrease the amount of blood that is transfused to the baby from the placenta and, in preterm babies; it is likely to result in subsequent anemia and increased chances of needing a blood transfusion. Check for bleeding/oozing. If bleeding or oozing occurs, place a second clamp or tie between the first one and the baby's skin and retie if necessary.



comes



Cutting the cord: Cut the cord between the two clamps or ties with sterile scissors or surgical blade, under a piece of gauze in order to avoid splashing of blood. At every delivery, a clean separate pair of scissors or blade should be designated for this purpose. Everything that touched the umbilical cord should be clean to avoid infection. Use clean gloves when clamping or tying and cutting the cord. Be careful not to cut or injure the baby. Either cut away from the baby or place your hand between the cutting instrument and the baby.

Step 4: Skin to Skin Care

Place the infant in skin-to-skin contact on the mother's chest and cover both with clean linen and blanket as required.

Position the baby skin-to-skin on the mother's abdomen. The warmth from the mother's body is one of the best ways to keep the baby warm. Keep the baby warm by placing it in skin-to-skin contact on the mother's chest.

- Cover the baby's body and head with warm, dry and clean cloth and cap or other head covering. Otherwise, cover the baby with part of the mother's clothing. If the room is cool ($<25^{\circ}\text{C}$), use a blanket to cover the baby over the mother.
- * The first skin-to-skin contact should last uninterrupted for at least 1 hour after birth or until after the first breast feed.
- * The newborn should not be bathed at birth because a bath can cool him dangerously. After 24 hours, he can have the first sponge bath, if his temperature is stabilized



Figure 5.12: Keeping the baby warm after delivery

Step 5: Initiate breastfeeding immediately within 1 hour

Initiate breastfeeding within the first hour. Select the appropriate method of feeding for the HIV-infected mother, based on informed choice.

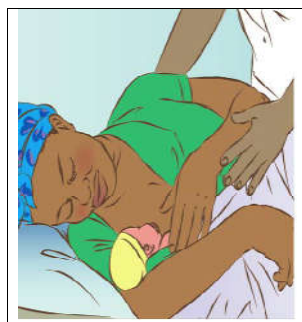
Breast milk and colostrum provide nutrition that is easy to digest and contain antibodies that protect against infection. Babies should receive only breast milk for the first 6 months. Babies who receive other food or liquids before 6 months of age are more likely to develop diarrhea. Advise women about breastfeeding during antenatal visits and discuss it again before birth occurs.

Starting breastfeeding within the first hour after birth helps mothers to provide enough milk later. It also helps the uterus contract and reduces maternal bleeding.

Starting breastfeeding within the first hour after birth helps mothers to provide enough milk later. It also helps the uterus contract and reduces maternal bleeding. Some babies may not breastfeed well soon after birth, but it is important to encourage breastfeeding during this time. To encourage early breastfeeding, keep mother and baby together unless a problem separates them. Babies are often alert immediately after birth will move toward the mother's breast but may not suck.

Signs of readiness to feed include:

- 1) Licking movements



and

- 2) Eyes open
- 3) The baby's head slightly back
- 4) Tongue down and forward
- 5) Mouth open

Teach mothers how to recognize these signs and initiate breastfeeding early.

Skin to skin contact & early initiation of breast feeding are the 2 most important measures to prevent hypothermia after delivery.

Step 6: Give eye care (while the baby is held by his mother) by applying Tetracycline eye ointment once on both eyes.

Within 90 minutes after birth administration of eye drops/eye ointment

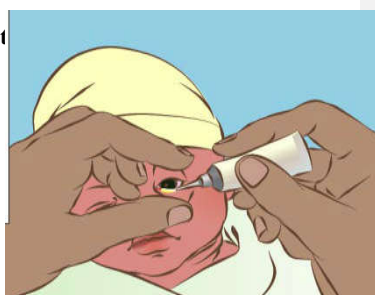
Infections can pass from the mother to the newborn during birth.

Infections of the eye can result in blindness.

Applying tetracycline eye ointment/drops to the inside of the lower both eyes soon after birth can prevent these infections which can blindness. Eye treatment can be delayed until the baby has breastfed, but provide eye care within the first 90 minutes after

Steps for giving eye care:

- Wash your hands with soap and water
- Clean eyes immediately after birth with swab soaked in sterile water, using separate swab for each eye.
- Clean from medial to lateral side.
- Pull down the lower lid of the eye.
- Place a portion of the ointment or drops inside the lower lid for both eyes.
- Hold one eye open and apply a rice grain size of tetracycline 1% eye ointment along the inside of the lower eyelid. Make sure not to let the medicine dropper or tube touch the baby's eye or anything else.
- Apply the ointment from medial to lateral
- Repeat this step to put medication into the other eye.
- Don't put anything else in baby's eyes as it can cause infection.
- Watch out for discharge from the eyes, especially with redness and swelling around the eyes.
- Do not rinse out the eye medication.



Step 7. Apply Chlorhexidine gel (4%) on the cord

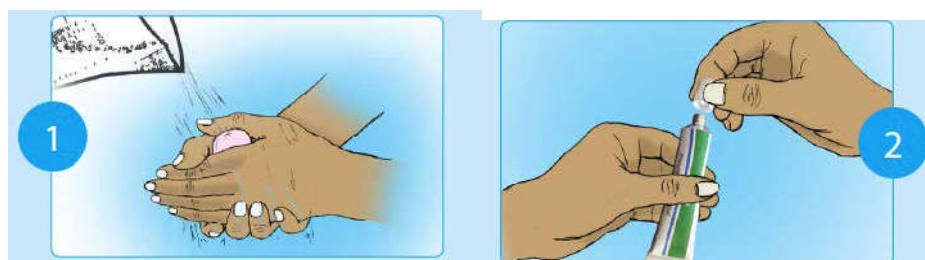
Within 30 minutes after birth apply Chlorhexidine to the cord

Application of Chlorhexidine on the cord prevents babies from getting infection. Mothers/caretakers need to be able to properly apply the Chlorhexidine before they leave the health facility. Once the mother/caretaker returns home from the health facility she should continue applying the Chlorhexidine Gel on the cord once per day for **seven consecutive days** (for six additional days if the first dose is given at the health facility in the first 30 minutes after birth).

The technique for applying Chlorhexidine Gel on the cord is as follows:

- Explain the procedure to the mother.
- Wash your hands thoroughly with soap and water, air-dry or dry with clean towel.
- Open the lid of the Chlorhexidine Gel. When you open the package of the gel for the first time use the sharp end of the back of the lid to open the cover
- Gently press the package of the gel and put the ointment on the cord
- Use the tip of your finger to apply the Chlorhexidine Gel on the cut end of the cord and around the cord
- Leave the cord area open for 3 minutes after the application of the Chlorhexidine Gel. Dress the baby well and wrap him/her after 3 minutes to keep the baby warm
- Close the lid of the Chlorhexidine Gel firmly and put in a place where children cannot reach. **DO NOT USE THE CHLORHEXIDINE GEL TO TREAT ANY OTHER WOUND OR ILLNESS ESPECIALLY ON THE EYE, EAR OR MOUTH.**

Fig 5.13 Application of chlorhexidine gel



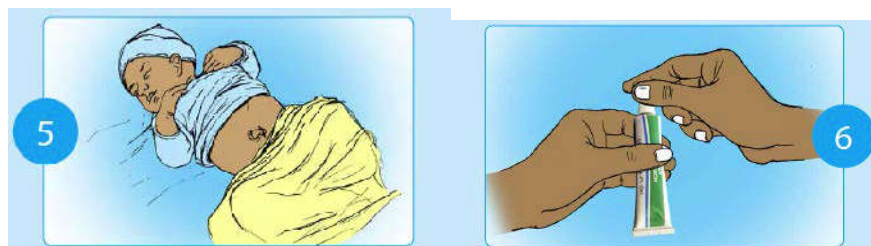
Wash your hands thoroughly with soap and or dry with clean towel.

Open the lid of the Chlorhexidine Gel. When water, air-dry you open the package of the gel for the first time use the sharp end of the back of the lid to open the cover



Gently press the package of the gel and cord

Use the tip of your finger to apply the put the ointment on the Chlorhexidine Gel on the cut end of the cord and around the cord



Leave the cord area open for 3 minutes after the application of the Chlorhexidine Gel. Dress the baby well and wrap him/her after 3 minutes to keep the baby warm

Close the lid of the Chlorhexidine Gel firmly and put in a place away from the reach of children.

Step 8: Give Vitamin K, 1mg IM on anterior mid-lateral thigh (while baby held by his mother).

Within 90 minutes after birth administer vitamin K

Vitamin K protects babies from serious bleeding that may result in death or brain damage. Every newborn should be given vitamin K. ***Because this treatment is painful, it should not be given during the first hour after birth, a when the mother and baby should not be disturbed.***

The dose of vitamin K is 1 mg (0.5 mg for babies <1500 grams), ***Check the preparation of this dose carefully as more than one concentration may be available.***



time

The technique for giving an intramuscular injection in the newborn is as follows:

- Explain the procedure to the mother.
- Wash your hands thoroughly with soap and water, air-dry or dry with clean paper towel, put on clean gloves.
- Gather the necessary equipment: disposable syringe, needle, medication, alcohol/antiseptic solution, and clean, preferably sterile gauze/cotton.
- Examine carefully the medication's label to verify the name, expiration date, instructions for dilution, if any, or any other special notes.
- Calculate the amount to be given where required.
- Draw out the medication:
- Clean the rubber stopper with alcohol swab/cut the ampoule at its neck.
- Push the needle into the bottle/ampoule.
- Draw the calculated amount and pull the needle out.
- Remove the air while holding the syringe with the needle pointing up and tapping on the syringe barrel.
- Expose the baby's thigh and gently hold the knee so the baby is unable to kick.
- Grasp the muscle of the antero-lateral part of the upper thigh, clean the skin with the alcohol/antiseptic, and let it dry for a few seconds.
- In one quick movement put the needle in the muscle straight in, pull back on the plunger a little bit to make sure that the tip of the needle is not in a blood vessel.
- If blood comes to the syringe, take the needle out and apply pressure at the site to prevent bleeding. Re-inject in a fresh spot.
- Inject the drug slowly, remove the needle, and apply gentle pressure for a short while and ensure that there is no oozing of blood upon removal of the swab.
- Discard the needle and syringe immediately in a "sharps" disposal container.

Step 9: Place the baby identification bands on the wrist and ankle

Within 90 minutes after birth, place the baby identification bands on the wrist and ankle

Place the identification tag /label on the wrist and ankle. If a ready-made disposable identification is not available, prepare one locally using sticking plaster and gauze strips. Note that, at a minimum, the

names of the mother and, if available, the father, and the date and time of birth should be written on the identification bands.

Putting the identification bands on the hands and ankle will save you from misshaping babies in busy delivery rooms.

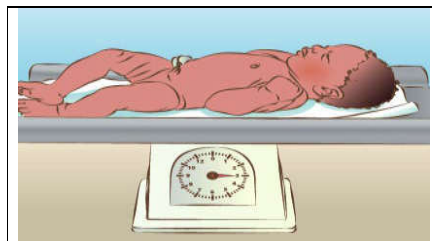
Step 10: Weigh the Newborn

Within 90 minutes after birth weigh the newborn when he/she is stable

- Birth weight helps identify babies at higher risk, provides a baseline for monitoring growth and may also be necessary for calculating drug doses.
- Babies should be weighed within 90 minutes of birth. However, weighing should be deferred if an infant is cold unless needed for calculating antibiotic doses. Use scales designed for weighing babies. Zero the scales before each use to test that they function properly. Clean the scales with dilute bleach solution or other safe cleaning product before each use to prevent infection.
- Babies with birth weights under 2500 grams may require special care to prevent low body temperature. Babies with birth weight under 2000 grams should receive prolonged skin-to-skin care. These babies may need alternative feeding methods and more frequent assessment to identify problems and Danger Signs. Babies with birth weights under 1500 grams should be referred for advanced care when possible.
- Always document birth weights. Use established regional or national forms and guidelines for documenting birth weight, for example on partograph/maternal/ newborn charts and delivery room register.

Weighing the newborn when he/she is stable

- Place a clean linen or paper on the pan of the weighing scale.
- Adjust the pointer to zero on the scale *with* the linen/paper on the pan.
- **Place the naked baby on the paper/linen. If the linen is large, cover the baby with the cloth.**
- Note the weight of the baby when the scale stops moving.
- Never leave the baby unattended on the scale.
- Record the baby's weight in partograph/maternal/ newborn charts and delivery room register and inform the mother



Additionally, to the ENC steps above, give BCG and OPV0 before discharge.

Recording

Within 90 minutes after birth record observations and treatment provided in the registers/appropriate chart/cards

Recording complete information about the health status of the mother and the baby in the partograph/maternal/baby charts and in the delivery room registers is very important to provide comprehensive follow up care for the mother and the baby.

Complete information about the mother and babies is also useful for the health facility to know the status of mothers and babies that receive care in the facility and to improve quality of maternal and newborn health care services.

Secondary Full Examination

Within 90 minutes after birth examine the baby to tell if a baby is well or has a problem

A complete exam should be performed within 90 minutes of birth or whenever a baby appears unwell. During the exam, evaluate a baby by looking, listening and feeling. Focus on the following features:

Breathing: A baby should breathe easily between 40-60 times per minute. Count a baby's breathing rate for one minute.

Movement and tone: When active, well babies have spontaneous movements of arms and legs that are equal on both sides. Limbs are flexed at rest. The tone should be neither floppy nor rigid.

Skin color: The normal skin color of a newborn is pink, but hands and feet may still look pale or blue soon after delivery. Pink color may be difficult to detect in dark-skinned babies. The inside of the mouth should be pink in all babies. Babies with jaundice may have yellow skin. Recognizing jaundice is important because severe jaundice may require advanced care.

Cord appearance: On the initial exam, there should be no drainage or bleeding from the cord.

Other features of a general exam: Inspect the baby's entire body for abnormalities.

Document the results of this exam even if all findings are normal. Perform the exam in front of the parents and communicate the findings to them.

Measure Temperature

Within 90 minutes after birth measure temperature to identify babies who require special care

Keeping body temperature normal helps a baby stay healthy. Low temperature can cause death.

It is better to prevent low temperature than to warm a baby who is cold. Monitor temperature in the first hours after birth. Low temperature is common among premature and low-weight babies. Prevent or correct low temperature with changes in care.

The normal temperature range is 36.5-37.5 °C. A temperature 35.5 °C-36.4 °C requires improved thermal care. A temperature below 35.5 °C is a Danger Sign. A temperature above 37.5 °C not due to over-warming (for example being placed in direct sunlight) is a Danger Sign.

Feeling the skin of the face, abdomen, or foot can estimate the temperature, but measuring the



temperature is more exact. Measuring temperature in the armpit (axilla) is safer than measuring a rectal temperature. Measure temperature in all babies within 90 minutes after birth. A thermometer used with babies must measure temperatures below 35.5 °C.

2.11 NEONATAL RESUSCITATION

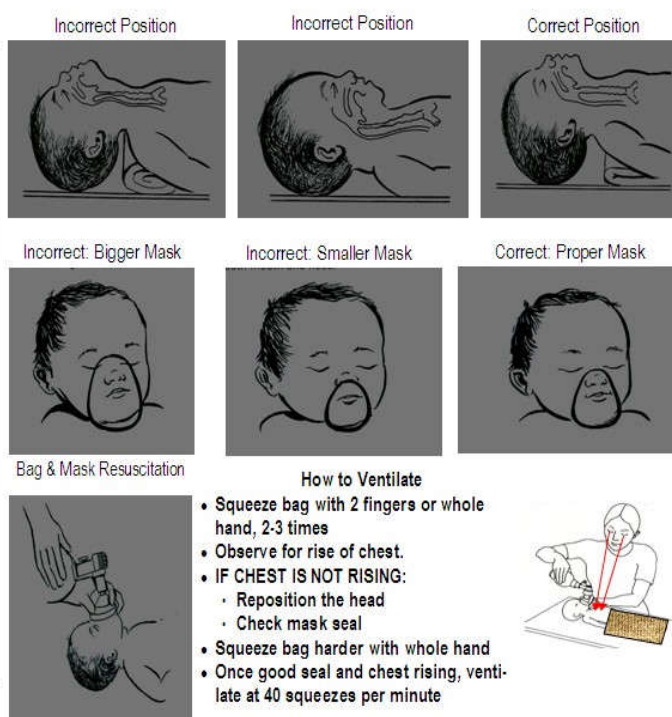
The great majority of infants with asphyxia can be successfully managed by appropriate ventilation without drugs, volume expanders or other interventions. Applying the basic principles of resuscitation to all infants at all levels of care will substantially improve newborn health and decrease deaths. Timely and correct resuscitation will not only revive them but will enable them to develop normally. Most will need no further special care after resuscitation.

Every birth attendant should know the **basic principles of resuscitation** have basic skills in neonatal resuscitation and have access to **appropriate resuscitation equipment**. Whenever possible, a person skilled in resuscitation, and who can devote full attention to the infant, should attend deliveries when complications are anticipated. Resuscitation equipment should not only be available in every delivery room, but its presence and proper working order should be verified before every delivery.

Figure 5.14: Steps in new born resuscitation

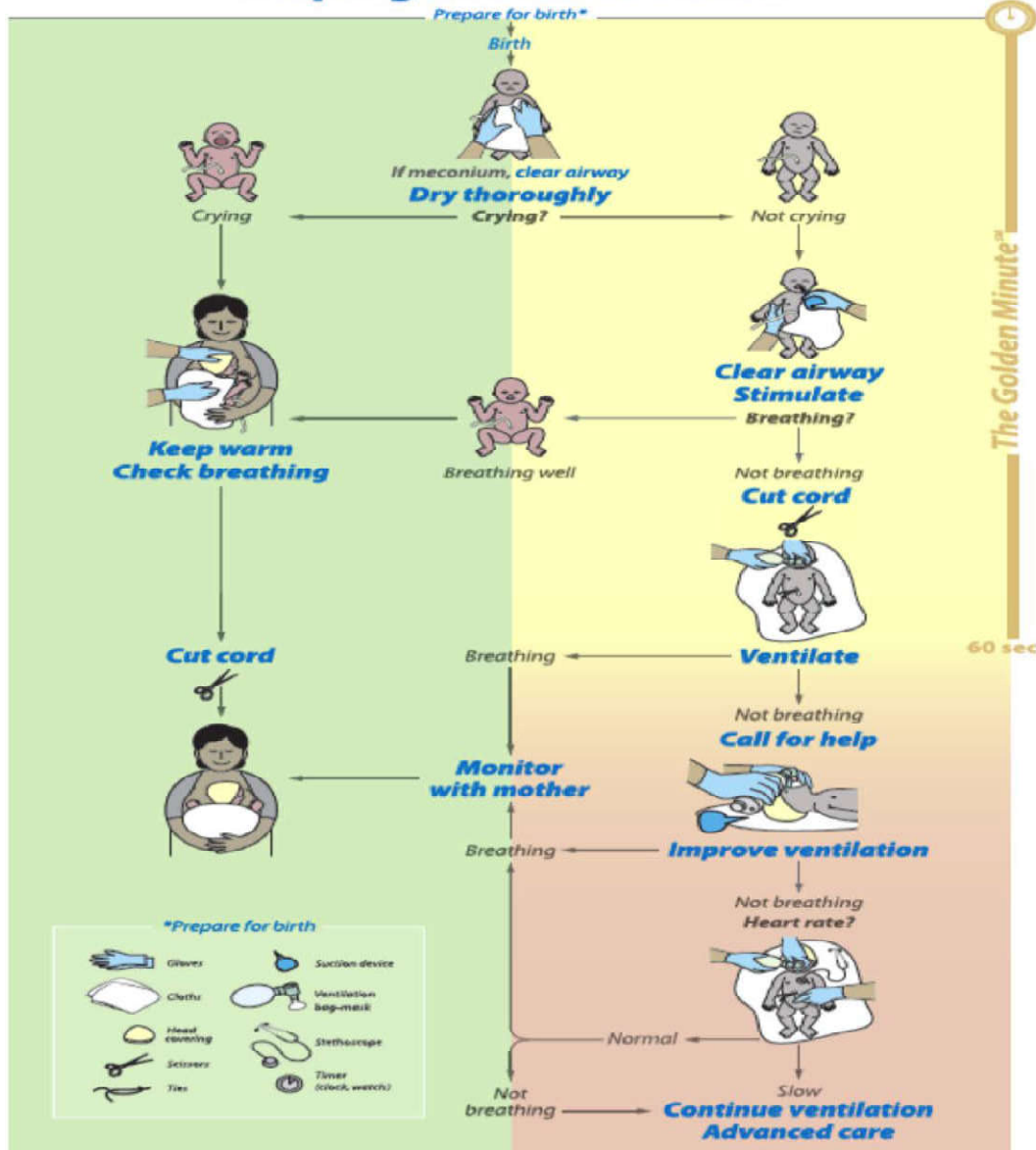
ESSENTIAL NEWBORN CARE: NEWBORN RESUSCITATION

Position	- Place the baby on his back with the neck slightly extended. - Put a towel or cloth behind the shoulder to facilitate positioning
Clear airway	- Clear the airway by wiping out the mouth with gauze - Suction the baby's nose and mouth gently - Reassess the baby's breathing
Ventilate	- Use baby bag and mask to ventilate at 40 breaths per minute - Continue to ventilate until the baby breathes independently - Stop after 20 minutes if the baby has not responded
Monitor	- Keep the baby warm (skin-to-skin) - Defer bathing for 24 hours after the baby is stable - Breastfeed as soon as possible - Watch for signs of a breathing problem: rapid, labored, or noisy breathing, blue color of the tongue, trunk - If a breathing problem occurs, stimulate, give oxygen [if available], and refer



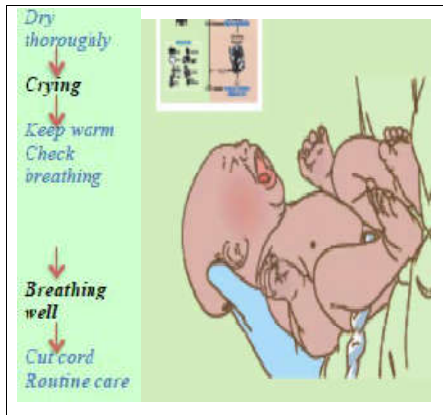
ACTION PLAN Helping Babies Breathe

Prepare for birth*



Routine Care

1. Baby is crying: Provide routine care



About 1 in 10 babies needs help to breathe. Rapid assessment at the moment of birth is the best way to know if a baby needs help to breathe.

If the baby is crying continue with routine essential newborn care discussed in the module 2 above. Most babies cry at birth. Crying means a baby is breathing well. Crying baby usually moves his/her arms and legs and has good muscle tone. After crying for some seconds, a baby may stop crying and begin to breathe quietly and regularly. A baby also may continue to cry for some time.

Dry, Stimulate and Clear Airway

2. Baby is not crying: Dry, stimulate breathing and Clear the

airway

A baby who does not cry needs help to breathe. Babies who do not cry may not be breathing at birth. A baby who is not breathing is limp and may be pale or bluish. A baby who is breathing shallowly, gasping, or not breathing at all needs help to breathe. Prompt attention will increase the chance of good response. If not help is given to baby who is not breathing, that baby may die or experience serious brain damage.

If the baby is not crying or breathing well after drying and stimulation, you will need to help the baby breathe in “The Golden Minute”.

Position the baby with the neck slightly extended to help keep the airway open. When the baby’s head is in the correct position, the nose will be as far forward as possible. If the neck is flexed or extended too far, air may not enter freely.

Clear the airway: Clear the mouth and then the nose with a clean suction device or wipe. Clear the mouth first to remove the largest amount of secretions before the baby gasps or cries. Suctioning the nose first may cause gasping and aspiration of secretions.





When using bulb suction, squeeze the bulb before inserting the tip in the mouth or nose and release before withdrawing the bulb. Stop suctioning when secretions are cleared, even if the baby does not breathe. Suctioning too long, too vigorously, or too deeply can cause injury, slow heart rate and prevent breathing.

When using a suction device with a tube and reservoir, insert the tube into the side of the baby's mouth no more than 5 cm beyond the lips. Apply suction while withdrawing the tube. Insert the suction tube 1 to 2 cm into each nostril and apply suction while withdrawing the tube.

Stimulating breathing.

Gently rub the back once or twice. Do not delay or stimulate longer. Move quickly to evaluate breathing and decide if ventilation is needed. Drying, stimulating, and clearing the airway breathing should take less than 1 minute. Your actions in The Golden Minute helps many babies begin to breathe. If the baby breathes well after you cleared the airway and with simple stimulation, keep him/her skin-to-skin with the mother and continue with routine essential newborn care.

Check if the baby is breathing well

A baby who is breathing well will be: Crying or Breathing quietly and regularly

A baby who is not breathing well will be: Gasping (taking a single deep breath followed by a long pause) or not breathing at all.

Some babies will have shallow, irregular, slow, or noisy breathing immediately after birth. Others may have chest indrawing (retractions). These babies will require close monitoring of their breathing, heart rate, and color to decide if they need more help to breathe.

Decide what care the baby needs after stimulation and clearing the airway: If the baby is breathing well, no further intervention is required. Continue to check the breathing. Clamp or tie and cut the umbilical cord. Encourage breastfeeding within the first hour.

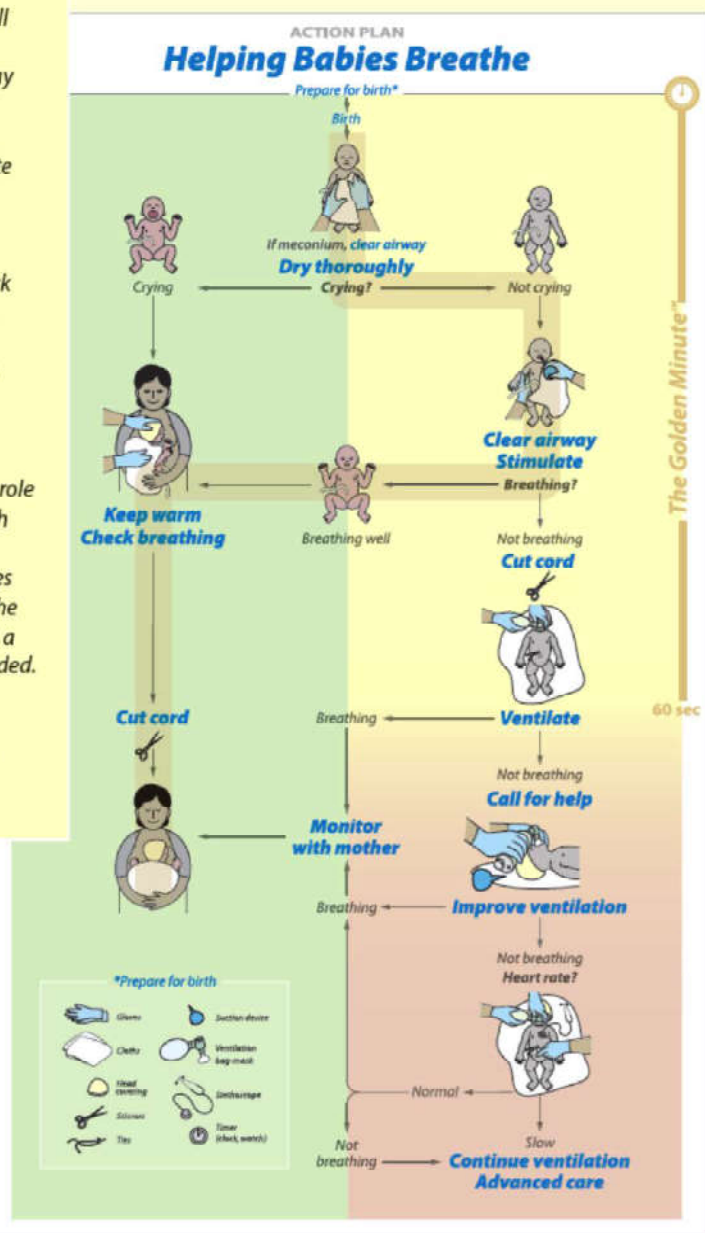
If the baby is not breathing well (gasping or not breathing at all) quickly clamp or tie and cut the umbilical cord before moving the baby to the area for ventilation, then begin ventilation with bag and mask. Delay in ventilation may result in preventable death or brain damage.

Exercise: The Golden MinuteSM – clear the airway and stimulate breathing

The facilitators will demonstrate clearing the airway and stimulating breathing during The Golden Minute and the baby's responses.

Learners will work in pairs with the mannequin to practice clearing the airway and stimulating breathing. One person takes the role of the skilled birth attendant. The other person gives the response of the baby and acts as a helper when needed.

Learners switch roles and repeat the exercise.



Bag and Mask Ventilation

3. *Baby is not crying: Ventilate with bag and mask*

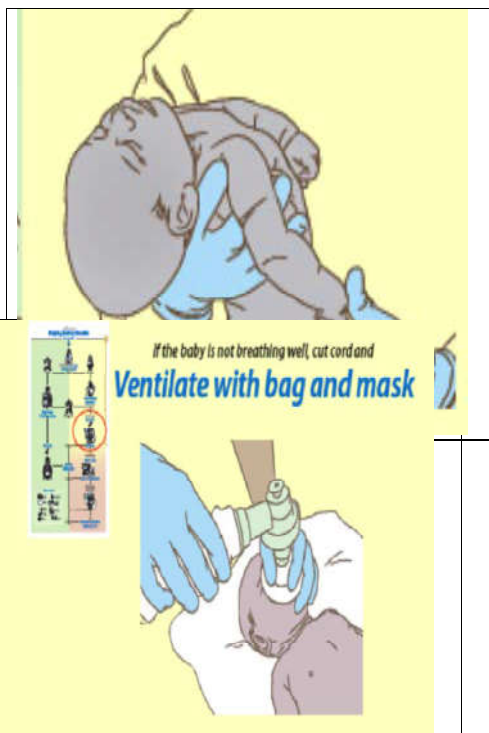
If the baby does not breathe well after clearing the airways and stimulating with gentle rubbing of the back, then clamp or tie and cut the cord and ventilate the baby with bag and mask.

Ventilation with bag and mask is the most important and effective way to help the baby who is not breathing or is gasping. Ventilation opens the lungs with air.

How to ventilate with bag and mask



Initiate ventilation: Place the baby on a flat clean, warm and dry area with a good light to assess the baby. You should have prepared this area prior to the birth.





Stand at the baby's head: You will need to control the position of the head and look for movement of the chest

Select the correct mask: The mask should cover the chin, mouth, and nose, but not the eyes. The mask should make a tight seal on the face so that air will enter the baby's lungs. A mask that is too large will not seal well on the face. Air will escape under the mask. A mask that is too small will not cover both the mouth and nose and may block the nose. Air will not enter the lungs freely.

Position the head slightly extended: Help keep the baby's airway open by positioning the head slightly extended and supporting the chin. Avoid over extension or flexion.

Position the mask on the face: Position the rim of the mask on the tip of the chin, then place the mask over mouth and nose.

Make a firm seal between the mask and the face while squeezing the bag to produce a gentle movement of the chest: Hold the mask on the face with the thumb and index finger on top of the mask. Use the middle finger to hold the chin up toward the mask. Use the 4th and 5th fingers along the jaw to lift it forward and help keep the airway open.

Form a tight seal by pressing lightly on the top of the mask and gently holding the chin up toward the mask. If the seal is not tight, you will not move air into the lungs as you squeeze the bag. The air will escape under the rim of the mask. Do not push the mask down onto the face. This may change the head position and interfere with air entering the lungs.

Squeeze the bag to produce a gentle movement of the chest, as if the baby were taking an easy breath. Make sure there is no leak between the mask and the baby's face. Squeeze the bag harder if you need to deliver more air with each breath.

Give 40 to 60 breaths per minute: Count aloud "Breath.... two.... three". If you squeeze the bag as you say, "Breath," and release while you say, "two.... three," you will ventilate at a rate that helps air move into and out of the lungs.

Check if the baby is breathing well

After one minute of ventilation, some babies improve quickly and begin

breathing well after brief ventilation Whereas\ some babies require continued ventilation with bag and mask.

Check if the baby is breathing well

A baby who is breathing well will be: Crying or Breathing quietly and regularly

A baby who is not breathing well will be: Gasping (taking a single deep breath followed by a long pause) or not breathing at all.

Decide what next care baby needs after starting spontaneous breathing.

- If spontaneous breathing established, stop ventilation and provide essential newborn care.
- Look for one of the danger signs (fast breathing, grunting and chest in-drawing)

A baby who is not breathing well (gasping or not breathing at all) needs continued ventilation with bag and mask.

Exercise: The Golden MinuteSM - ventilation

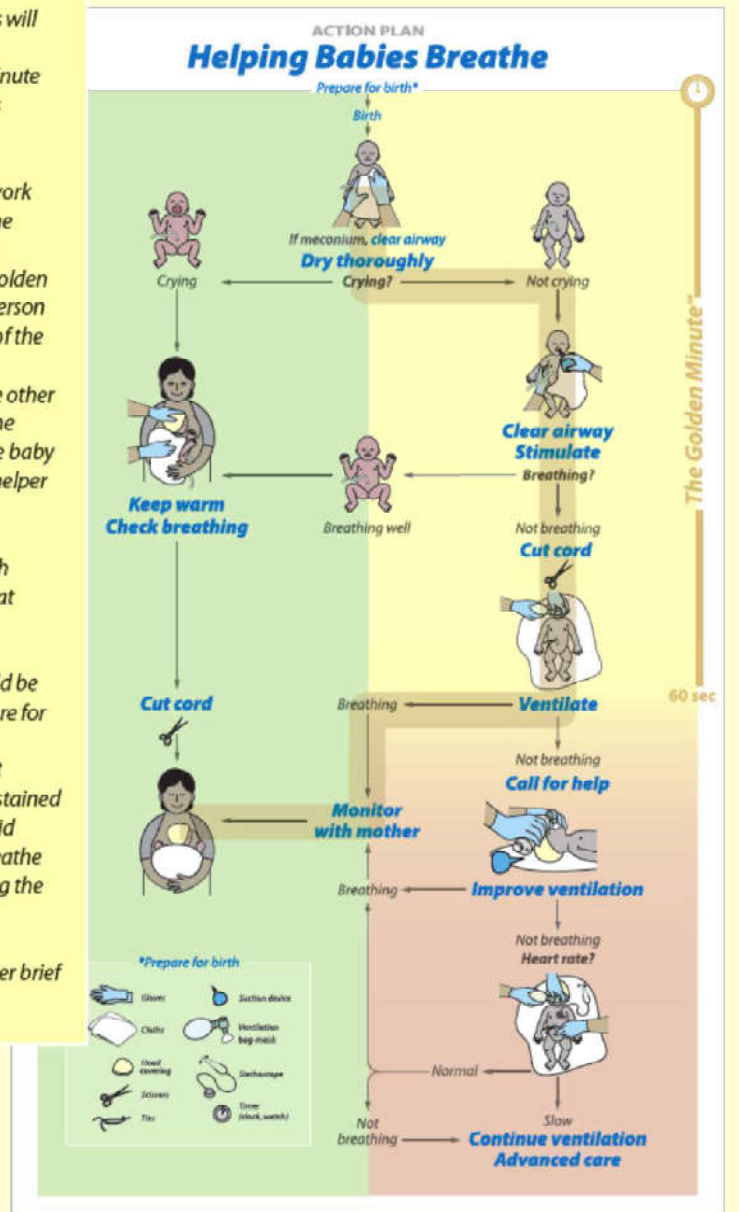
The facilitators will demonstrate The Golden Minute and the baby's responses.

Learners will work in pairs with the mannequin to practice The Golden Minute. One person takes the role of the skilled birth attendant. The other person gives the response of the baby and acts as a helper when needed.

Learners switch roles and repeat the exercise.

Learners should be prepared to care for a baby who

- has clear OR meconium-stained amniotic fluid
- does not breathe after clearing the airway and stimulating
- breathes after brief ventilation

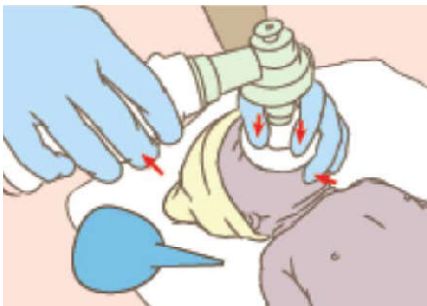


Improved Ventilation

4. Baby is not breathing with beginning ventilation: Continue/improve ventilation

If the baby doesn't breathe well after beginning ventilation call for help and continue resuscitation. Ask for more skilled professional available. Check that ventilation breaths produce movement of the chest as if the baby were breathing normally. Take steps to improve ventilation if the chest is not moving. Check the baby's mouth, head and the bag.

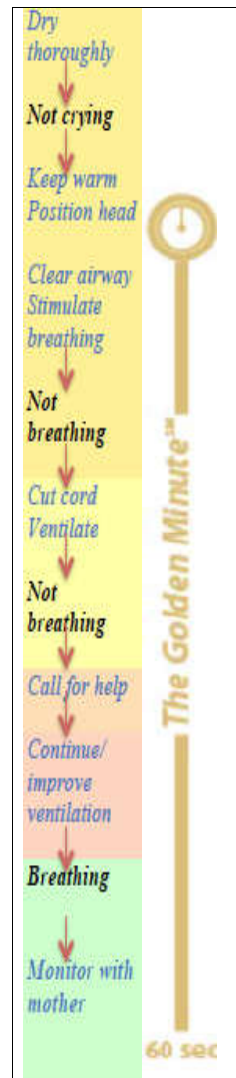
- Put the baby in lateral position
- Check the mouth and nose and do fast suction, first the mouth and then the nose.



- Re-position the head properly
- Check fast for mal-functioning of the bag.
- Open the baby's mouth slightly before reapplying the mask
- Reapply the mask to the face to form a better seal.
- Squeeze the bag harder to give a larger breath

An air leak under the mask or incorrect position of the head is a common reason for poor chest movement. If you still do not see gentle movement of the chest, try to find the problem and repeat the necessary steps to improve ventilation. Recheck the function of the ventilation bag. Replace it if another bag is available.

If the baby does not begin to breathe after 1 minute of ventilation with chest movement, evaluate heart rate to decide if ventilation is adequate. Checking the heart rate is easier and faster with the help of another skilled person. A skilled helper can count the umbilical cord pulsation while you are giving the first minute of ventilation. If you have no skilled helper or the cord pulse cannot be felt, you will need to rely on movement of the chest as an indicator of adequate ventilation. Continue ventilation for 1 minute before stopping to listen to the heartbeat.



Decide if the heart rate is normal or slow: Evaluate the heart rate by feeling the umbilical cord pulse or listening to the heartbeat with a stethoscope. Feel the pulse in the umbilical cord. If pulse cannot be felt in the cord, you or your helper must listen over the left chest with a stethoscope and count the heartbeat. Pause ventilation every minute in order to hear the heartbeat for 6 seconds and multiply by 10.

- A heart rate of 100 beats per minute or more is normal.



- A heart rate of less than 100 beats per minute is abnormal.

Minimize the time without ventilation. Listen to the heart rate just long enough to recognize if it is normal or slow. If the heart rate sounds faster than your own, it is probably normal. If the heart rate sounds slower than your pulse, it is slow.

Normal Heart Rate and Ventilation

5. Baby is not breathing with ventilation, has normal heart rate: Continue ventilation



If the heart rate is normal, continue to ventilate until the baby is breathing well. Gradually reduce the rate of ventilation and look for the baby's breathing. If the heart rate stays normal as the baby begins to breathe, stop ventilation. Ventilation can stop when the baby is breathing and the heart rate stays normal (more than 100 beats per minute). Monitor the baby who is breathing after ventilation.

Monitor the baby with the mother. Extended skin-to-skin care may be of special value to the small or sick baby who required ventilation. Monitor vital signs including breathing, heart rate, temperature, and color. A baby who received ventilation with bag and mask may need assistance with feeding, consider also NEC. Talk with mother and the birth companion about the baby and the plan of care.

Continue ventilation and seek advanced care if the baby is not breathing or not breathing well

A baby who has a normal heart rate and pink color but does not breathe needs continued ventilation. A slow decrease in the rate of ventilation over several minutes may allow return of spontaneous breathing. If the baby still does not breathe, continue ventilation and consider specialty consultation and/or referral. It is always good to refer the baby to Neonatal Intensive Care Unit (NICU) if the health facility has one but if the facility does not have Neonatal Intensive Care Unit (NICU), continue ventilation.

The baby who begins to breathe, but has difficulty breathing and a slow heart rate without ventilation needs continued ventilation and specialty care

The baby who is blue, pale, or breathing fast may be helped by supplemental oxygen through nasal prongs or

catheter. Severe chest in-drawing, grunting or frequent apnea (pauses in breathing longer than 15 to 20 seconds) may require mechanical support of breathing. A baby who has received continued ventilation (longer than 5 minutes) needs close monitoring and specialty consultation or referral. Warmth and assistance with feeding will be necessary.

Exercise: Continued ventilation with normal heart rate

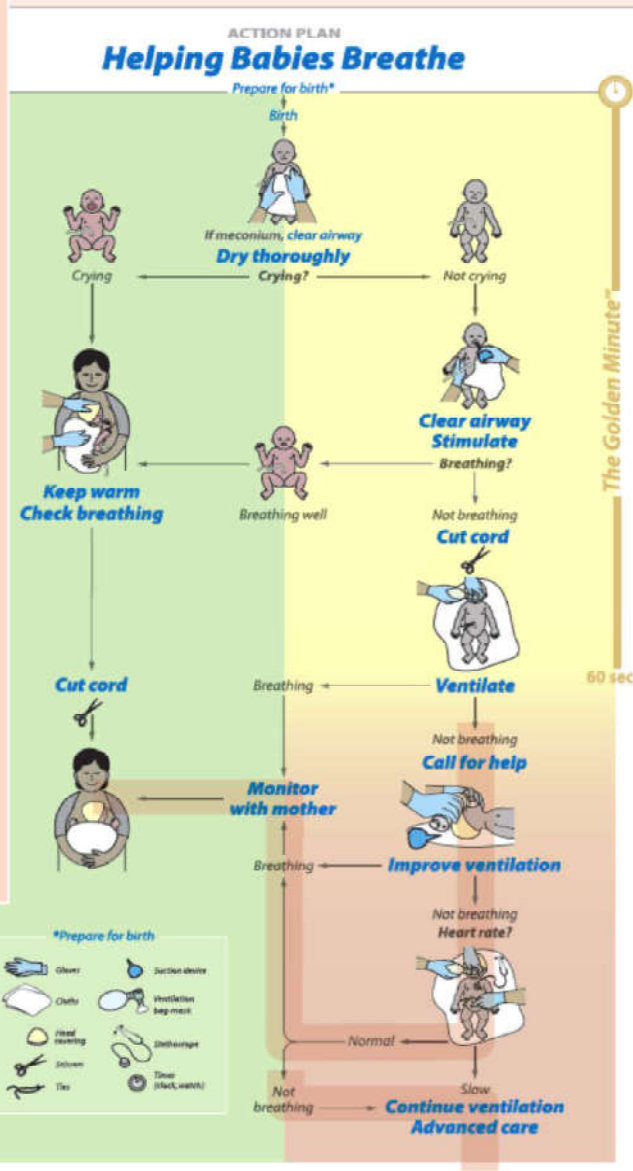
The facilitators will demonstrate continued ventilation with normal heart rate.

Learners will work in pairs with the mannequin to practice continued ventilation with normal heart rate. One person takes the role of the skilled birth attendant. The other person gives the response of the baby and also acts as a helper when needed.

Learners switch roles and repeat the exercise.

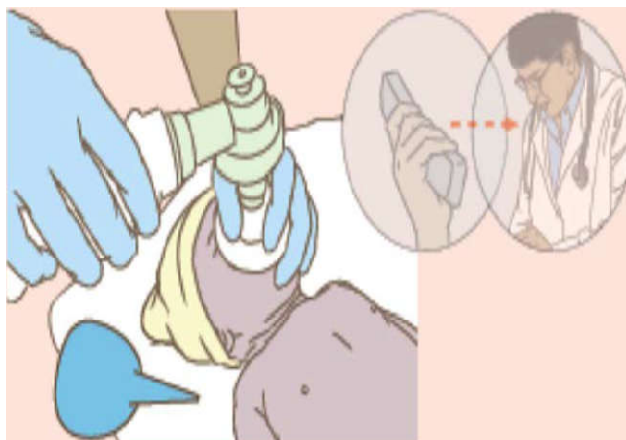
Learners should be prepared to care for a baby who

- has poor chest movement
- has a normal heart rate and breathes well OR does not breathe well



Seeking for Advanced Care

6. Baby is not breathing with ventilation, heart rate is slow: Continue ventilation and seek advanced care



If the heart rate is slow, make sure that you have taken all the steps to improve ventilation. There may be a serious problem.

Such problems include pneumonia, meconium aspiration, immature lungs (prematurity), or a congenital malformation. The baby may need endotracheal intubation and supplemental oxygen or chest compression and medications.

Activate the emergency plan to access advanced care at a specialty facility.

Continue ventilation during transport if the baby must be moved for advanced care.

If the baby has no heart rate and no breathing after giving ventilation for 10 minutes, the baby is dead. Stop ventilation.

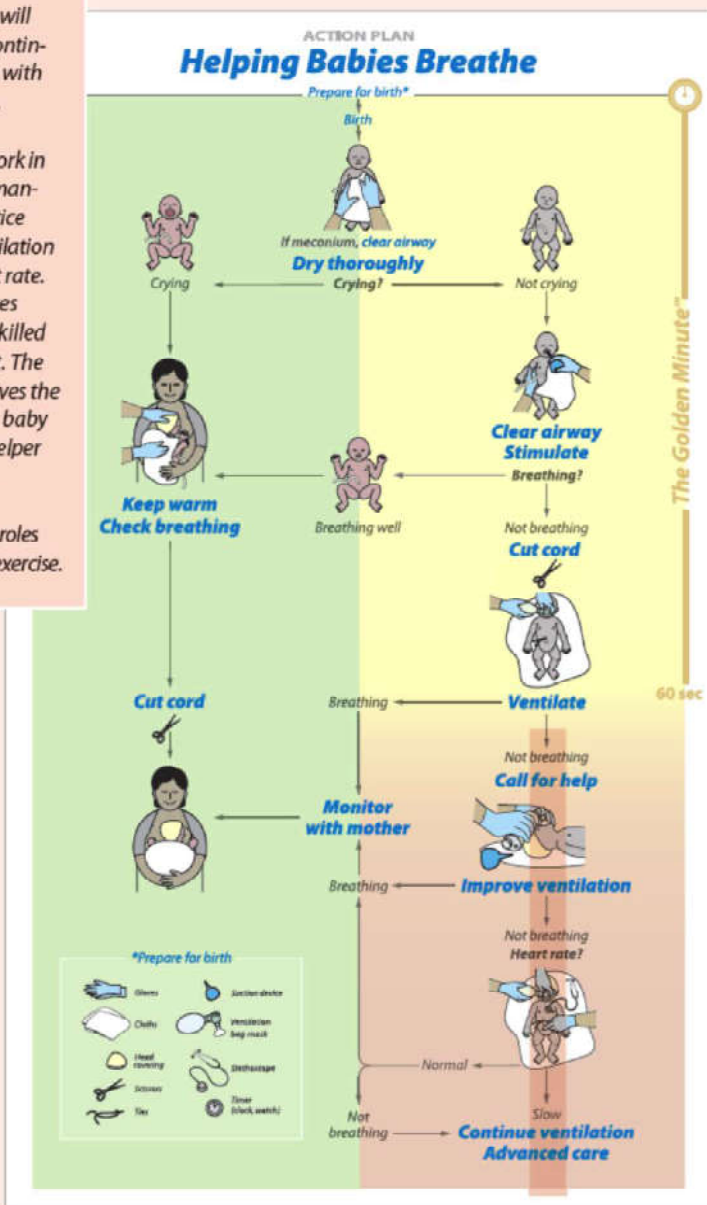
Skin that is purple-white or peeling (maceration) suggests that a baby died long before delivery. If recognized at delivery, ventilation should not begin. Ventilation can be stopped whenever maceration is recognized. No intervention is indicated. A baby who never had a heart rate and never breathed after birth is stillborn.

Exercise: Continued ventilation with slow heart rate

The facilitators will demonstrate continued ventilation with slow heart rate.

Learners will work in pairs with the mannequin to practice continued ventilation with slow heart rate. One person takes the role of the skilled birth attendant. The other person gives the response of the baby and acts as a helper when needed.

Learners switch roles and repeat the exercise.



Matrix for Helping Babies Breathe: Proposed actions for the six possible scenarios

Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
<i>If meconium, clear the airway</i>					
Dry thoroughly on abdomen ↓	Dry thoroughly on abdomen ↓	Dry thoroughly on abdomen ↓	Dry thoroughly on abdomen ↓	Dry thoroughly on abdomen ↓	Dry thoroughly on abdomen ↓
Crying ↓	Not crying ↓	Not crying ↓	Not crying ↓	Not crying ↓	Not crying ↓
Keep warm Check breathing ↓	Keep warm Position head Clear airway Stimulate breathing ↓	Keep warm Position head Clear airway Stimulate breathing ↓	Keep warm Position head Clear airway Stimulate breathing ↓	Keep warm Position head Clear airway Stimulate breathing ↓	Keep warm Position head Clear airway Stimulate breathing ↓
Breathing well ↓	Breathing well ↓	Not breathing ↓	Not breathing ↓	Not breathing ↓	Not breathing ↓
Cut cord Routine care	Cut cord Routine care	Cut cord Ventilate ↓	Cut cord Ventilate ↓	Cut cord Ventilate ↓	Cut cord Ventilate ↓
		Breathing well ↓	Not breathing ↓	Not breathing ↓	Not breathing ↓
		Monitor with mother	Call for help ↓	Call for help ↓	Call for help ↓
			Continue/improve ventilation ↓	Continue/improve ventilation ↓	Continue/improve ventilation ↓
			Breathing ↓	Not breathing ↓	Not breathing ↓
			Monitor with mother	Continue ventilation ↓	Continue ventilation ↓
				Normal heart rate ↓	Slow heart rate OR normal heart rate ↓
				Breathing ↓	Not breathing ↓
				Monitor with mother	Continue ventilation Advanced care

Oxygen use in newborns:

Indications for Oxygen Therapy in Neonates

Oxygen therapy in newborn infants, particularly when they are born preterm, should reflect the fact that in the first hours of life they have lower normal oxygen saturation than older newborns. **See the Table below.** Pulse oximetry should be used to monitor SpO₂, which should be maintained $\geq 88\%$, but in pre-term babies no higher than 95% to prevent eye damage.

Table xx: Normal Oxygen Saturation in Newborns

Age	Normal oxygen saturation in SpO ₂
1min	60 to 65
2min	65 to 70
3 min	70 to 75
4 min	75 to 80
5min	80 to 85
10min	85 to 95

A number of conditions that may lead to hypoxemia occur more frequently in neonates, notably

- Birth asphyxia,
- Respiratory distress syndrome,
- Transient tachypnea of the neonate, and
- Pneumonia is also common.

Neonates who are very unwell for reasons such as prematurity, sepsis, seizures or hypoglycemia are also prone to apnea. Apnea and hypoventilation also occur in otherwise well infants of very low birth weight (< 1.5 kg or gestational age < 32 weeks) because of immature respiratory drive (apnea of prematurity). Apnea can lead to hypoxemia and slowing of the heart rate (bradycardia), further reducing oxygen delivery to tissues.

Recommended indication for oxygen in newborns also include;

- Bradycardia,
- Cyanosis,
- Floppy infant,
- Increased respiratory effort, and
- On post resuscitation care.

Pulse Oximetry in Neonatal Care

Prolonged severe hypoxemia, especially if combined with low cardiac output, can acutely lead to:

- Hypoxic brain injury,
- Renal failure,
- Pulmonary hypertension, and
- Predispose to necrotizing enterocolitis and long-term cognitive and intellectual impairment.

On the other hand, hyperoxia in premature neonates causes

- **Retinopathy** of prematurity, and
- Increases the risk of bronchopulmonary dysplasia and brain injury.

The implementation of a package of measures, including intermittent monitoring with pulse oximetry, and apnea monitoring, to improve the quality of neonatal care in a health facility can significantly reduce neonatal mortality from very low birth weight, pneumonia and sepsis.

When using oxygen, remember:

- Supplemental oxygen should only be used for difficulty in breathing or cyanosis;
- If the newborn is having **severe in-drawing of the chest**, is **gasping for breath** or is **persistently cyanotic**, increase the concentration of oxygen by nasal catheter, nasal prongs or oxygen hood.

Note: Indiscriminate use of supplemental oxygen for premature infants has been associated with the risk of blindness.

Flow rates for oxygen administration:

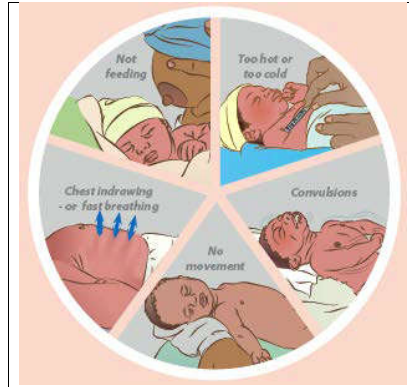
- **Nasal prongs and Nasal catheter:** -
 - **Low** = 0.5 L /minute, **Moderate** = 0.5 to 1 L/minute, **High**= more than 1L/minute.
- **Head box:** -
 - **Low** = 3 L/minute, **Moderate** = 3 to 5 L/minute, **High** = more than 5L/minute.
- **Facemask:** -
 - **Low** = 1 L/minute, **Moderate** = 1 to 2 L/minute, **High** = more than 2L/minute.
- **Incubator:** - If using a head box inside the incubator, see above

- If connecting oxygen directly to the incubator, follow the manufacturer's instructions

Classify the Baby

Assessing newborns for Danger Signs

At any time if you suspect a problem assess for danger signs to detect problems early and reduce the risk of death.



Danger Signs are caused by infection or other serious conditions and indicate that a baby may die. All babies should be assessed for Danger

Signs every 15 minutes within one hour after birth and every 30 minutes the next two hours and every hourly the following hours until the baby is discharged. A baby with any of the Danger Signs needs urgent antibiotic treatment and advanced care.

The following are the Danger Signs: Fast breathing and chest in-drawing can be caused by pneumonia or sepsis. Chest in-drawing means the spaces between, above or below the ribs indent with each breath. Fast breathing is breathing rate more than 60 breaths per minute. Babies with breathing problems may also have a blue color of the skin and inside the mouth. Temperature that is too low ($<35.5^{\circ}\text{C}$) or high ($\geq 37.5^{\circ}\text{C}$) is a sign of infection. A temperature that is $35.5\text{--}36.4^{\circ}\text{C}$ and does not rise with re-warming is also a Danger Sign.

Not feeding may be a sign of infection, prematurity, or other serious problems.

No movement, or very little movement, even when stimulated, may be a sign of infection or other serious problems.

Convulsions are repeated back-and-forth movements of the arms and legs that cannot be stopped by holding the arm or leg. Jitteriness of the arms and legs may look like convulsions but is a less serious problem. Unlike convulsions, jitteriness can be caused by a stimulus such as a loud noise or sudden movement. Jitteriness can be stopped by holding the arms and legs.

A baby with a Danger Sign needs urgent antibiotic treatment and advanced care.

By 90 minutes after birth classify the baby to determine further care



At about 90 minutes following birth, babies should be classified as normal and well, having a problem or needing advanced care. Classification is based on the baby's weight, temperature and exam.

- Well babies breathe at a normal rate (40-60 per minute) without effort, have a temperature of 36.5-37.5 °C, and weigh >2000 grams.
- Babies who have a problem may have a temperature of 35.5-36.5 °C, birth weight of 1500-2000 grams, or may feed poorly.
- Babies needing advanced care may have a Danger Sign, severe jaundice or a birth weight <1500 grams.

Some babies do not attach to the breast during the first 90 minutes after birth and therefore do not feed. If these babies are normal in all other ways, feeding should be attempted again. Babies who do not feed after several attempts should be classified as having a Danger Sign.

All babies should be classified by one hour of age.

Newborn danger signs; refer newborn urgently if any of these is present:

- Breathing less than 30 or more than 60 breaths per minute, grunting, severe chest in-drawing, blue tongue & lips, or gasping
- Unable to suck or sucking poorly
- Feels cold to touch or axillary temperature < 35°C
- Feels hot to touch or axillary temperature > 37.5°C
- Red swollen eyelids and pus discharge from the eyes
- Convulsion
- Jaundice / yellow skin — at age < 24 hours or > 2 weeks; involving soles and palms
- Pallor, bleeding
- Repeated vomiting, swollen abdomen, no stool after 24 hours

Avoid separating mother from newborn whenever possible. Do not leave mother and newborn unattended at any time.

Figure 5.14: Chart to assess the newborn for birth asphyxia.

ASK THE MOTHER WHAT THE YOUNG INFANT'S PROBLEMS ARE

- Determine if this is an initial visit for this problem.
- initial visit, assess the young infant as follows.

**IF YOU ARE ATTENDING DELIVERY or
BABY BROUGHT TO YOU IMMEDIATELY AFTER BIRTH**

Assess and check for Birth Asphyxia after drying and wrapping with dry cloth

**USE ALL BOXES THAT MATCH INFANT'S
SYMPTOMS AND PROBLEMS TO CLASSIFY
THE ILLNESS**

Assess, Look, Listen

- Is baby not breathing?
- Is baby gasping?
- Count breaths in one minute

Classify

ALL

Newborn

SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-treatments are in bold print)
If any of the following sign: ☐ Not breathing ☐ Gasping ☐ Is breathing poorly (less than 30 per minute)	BIRTH ASPHYXIA	Start resuscitation ☐ Position the newborn supine with neck slightly extended ☐ Clear the mouth and nose with gauze or clean cloth ☐ Ventilate with appropriate size mask and self-inflating bag ☐ If the resuscitation is successful continue giving essential newborn care ☐ If the newborn remains weak or is having irregular breathing after 20 minutes of resuscitation; refer urgently to hospital ☐ Stop resuscitation after 20 minutes if no response (no spontaneous breathing) ☐ Monitor continuously for 6 hours ☐ Follow after, 12 hrs, 24 hrs, 3 days and 6 weeks
• Strong cry • Breathing more than 30 per minute	NO BIRTH ASPHYXIA	☐ Cord care ☐ Eye care ☐ Vitamin K ☐ Initiate skin-to-skin contact ☐ Initiate exclusive breastfeeding ☐ Advise mother when to return immediately ☐ Follow after, 6 hrs, 3 days and 6 weeks

2.12 IMMEDIATE POSTPARTUM MATERNAL & NEWBORN CARE

2.12.1 INTRODUCTION TO CARE OF THE MOTHER AND NEWBORN WITHIN FIRST HOUR OF DELIVERY OF PLACENTA

The first hour after delivery of placenta is one of the critical moments for both the mother and newborn. Use Figure 5-14 to monitor & provide care.

Table 5-5:- Use this chart for woman and newborn during the first hour after complete delivery of placenta.

MONITOR MOTHER EVERY 15 MINUTES :	MONITOR NEWBORN EVERY 15 MINUTES:
▪ For emergency signs, using rapid assessment (RAM). ▪ Feel if uterus is hard and round. ▪ Record findings, treatments and procedures in <i>Labor record and Partograph</i>. ▪ Keep mother and newborn in delivery room - do not separate them. ▪ Never leave the woman and newborn alone.	■ Breathing: listen for grunting, look for chest in-drawing and fast breathing. ■ Warmth: check to see if feet are cold to touch.
CARE OF MOTHER AND NEWBORN WOMAN ▪ Assess the amount of vaginal bleeding. ▪ Encourage the woman to eat and drink. ▪ Ask the companion to stay with the mother. ▪ Encourage the woman to pass urine. NEWBORN ▪ Wipe the eyes. ▪ Apply an antimicrobial eye drop within 1 hour of birth. → either 1% silver nitrate drops or 2.5% povidone iodine drops or 1% tetracycline ointment. ▪ DO NOT wash away the eye antimicrobial. ▪ If blood or meconium, wipe off with wet cloth and dry. ▪ DO NOT remove vernix or bathe the newborn. ▪ Continue keeping the newborn warm & in skin-to-skin contact with the mother. ▪ Encourage the mother to initiate breastfeeding when newborn shows signs of readiness. Offer her help. ▪ DO NOT give artificial teats or pre-lacteal feeds to the newborn: no water, sugar water, or local feeds. ▪	INTERVENTIONS, IF REQUIRED ▪ If pad soaked in less than 5 minutes, or constant trickle of blood, manage as for PPH management. ▪ If uterus is soft, manage as PPH ▪ If bleeding is from a perineal tear, repair if required or consult/refer to hospital. ▪ If breathing with difficulty — grunting, chest in-drawing or fast breathing, examine the newborn and manage as for asphyxia management ▪ If feet are cold to touch or mother and newborn are separated: → Ensure the room is warm. Cover mother and newborn with a blanket → Reassess in 1 hour. If still cold, measure temperature. If less than 36.5°C, manage as for hypothermia management. ▪ If unable to initiate breastfeeding (mother has complications): → Plan for alternative feeding method. → If mother HIV+ and chooses replacement feeding, feed accordingly. ▪ If baby is stillborn or dead, give supportive care to mother and her family. Refer to hospital/consult if woman had serious complications at admission, during delivery or post-partum.

2.12.2 CARE OF THE MOTHER ONE HOUR AFTER DELIVERY OF PLACENTA

A. MONITOR MOTHER AT 2, 3 AND 4 HOURS, THEN EVERY 4 HOURS:

- ☐ For emergency signs, using rapid assessment.
- ☐ Feel uterus if hard and round.
- ☐ Record findings, treatments and procedures in *Labor record and Partograph*.
- ☐ Keep the mother and newborn together.
- ☐ **Never leave the woman and newborn alone.**
- ☐ **DO NOT** discharge before 24hrs

Table 5-6:-Use this chart for continuous care of the mother until discharge.

CARE OF MOTHER	INTERVENTIONS, IF REQUIRED
▪ Accompany the mother and newborn toward. ▪ Advise on <i>Postpartum care and hygiene</i>. ▪ Ensure the mother has sanitary napkins or clean material to collect vaginal blood. ▪ Encourage the mother to eat, drink and rest. ▪ Ensure the room is warm (25°C).	▪ Make sure the woman has someone with her and they know when to call for help.
▪ Ask the mother's companion to watch her and call for help if bleeding or pain increases, if mother feels dizzy or has severe headaches, visual disturbance or epigastric distress.	▪ If heavy vaginal bleeding, palpate the uterus. → If uterus not firm, massage the fundus to make it contract and expel any clots. → If pad is soaked in less than 5 minutes, manage as bleeding after child birth. → If bleeding is from perineal tear, repair or refer to hospital.
▪ Encourage the mother to empty her bladder and ensure that she has passed urine.	▪ If the mother cannot pass urine or the bladder is full (swelling over lower abdomen) and she is uncomfortable, help her by gently pouring water on vulva. ▪ DO NOT catheterize unless you have to.

▪ Check record and give any treatment or prophylaxis which is due. ▪ Advise the mother on postpartum care and nutrition. ▪ Advise when to seek care. ▪ Counsel on birth spacing and other family planning methods. ▪ Repeat examination of the mother before discharge using <i>Assess the mother after delivery</i>. For newborn, see module-5.	▪ If tubal ligation or IUD desired, make plans before discharge. ▪ If mother is on antibiotics because of rupture of membranes >18 hours but shows no signs of infection now, discontinue antibiotics.
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B. GIVE PREVENTIVE MEASURES

Table 5-7: -Ensure that all are given before discharge.

ASSESS, CHECK RECORDS	TREAT AND ADVIS
- Check RPR status in records. - If no RPR during this pregnancy, do the RPR test.	- If RPR positive: → Treat woman and the partner for syphilis according to the national STI management guideline. → Treat the newborn.
- Check tetanus toxoid (TT) immunization status. - Check when last dose of mebendazole was given.	- Give tetanus toxoid if due. - Give mebendazole once in 6 months.
Check woman's supply of prescribed dose of iron/folate.	Give 3 month's supply of iron and counsel on compliance.
- Ask whether woman and newborn are sleeping under insecticide treated bednet. - Counsel and advise all women.	- Encourage sleeping under insecticide treated bednet. - Advise on postpartum care. - Counsel on nutrition. - Counsel on birth spacing and family planning. - Counsel on breastfeeding. - Counsel on correct and consistent use of condoms. - Advise on routine and follow-up postpartum visits. Advise on danger signs and discuss how to prepare for an emergency in postpartum
Record all treatments given	

C. ADVISE ON POSTPARTUM CARE:

Advise on postpartum care and hygiene:

Advise and explain to the woman:

- ☐ To always have someone near her for the first 24 hours to respond to any change in her condition.
- ☐ Not to insert anything into the vagina.
- ☐ To have enough rest and sleep.
- ☐ The importance of washing to prevent infection of the mother and her baby:
 - o wash hands before handling and feeding newborn
 - o wash perineum daily and after faecal excretion

- change perineal pads every 4 to 6 hours, or more frequently if heavy lochia
- wash used pads or dispose of them safely
- wash the body daily.

To avoid sexual intercourse until the perineal wound heals and mother feels comfortable.

Counsel on nutrition:

- Advise the woman to eat a greater amount and variety of healthy foods, such as meat, fish, oils, nuts, seeds, cereals, beans, vegetables, cheese, milk, to help her feel well and strong (give examples of types of food and how much to eat).
- Reassure the mother that she can eat any normal food – these will not harm the breastfeeding newborn.
- Spend more time on nutrition counseling with very thin women and adolescents.
- Determine if there are important taboos about foods which are nutritionally healthy. Advise the woman against these taboos.
- Talk to family members such as partner and mother-in-law, to encourage them to help ensure the woman eats enough and avoids hard physical work.

D. COUNSEL ON BIRTH SPACING AND FAMILY PLANNING:

Counsel on the importance of family planning

Refer to module III for details

Lactational amenorrhoea method (LAM)

- ☐ A breastfeeding woman is protected from pregnancy only if:
 - she is no more than 6 months postpartum, and
 - she is breastfeeding exclusively (8 or more times a day, including at least once at night: no daytime feedings more than 4 hours apart and no night feedings more than 6 hours apart; no complementary foods or fluids), and
 - her menstrual cycle has not returned.
- ☐ A breastfeeding woman can also choose any other family planning method, either to use alone or together with LAM (refer module 3, section on family planning)

E. ADVISE ON WHEN TO RETURN:

- ☐ All mothers in the postpartum period should have at least 3 routine postpartum care visits. Use **table 5.8** for advising on postpartum care visit for problems. Encourage woman to bring her partner or family member to at least one visit. **Provide four postnatal visits for all mothers in the postnatal period.**

Table 5.8:-Routine postpartum care visits (WHO 2015):

Visit	Time
FIRST VISIT	48 -72hours
SECOND VISIT	7-14 days,
THIRD VISIT	6wks

UNIT-3: MANAGEMENT OF ABNORMAL LABOR

3.1 – LEARNING OBJECTIVES

At the end of this unit, participants will be able to:

- Describe common diseases and complications during labor, delivery & the immediate postpartum.
- Describe additional care required for diseases and complications during labor, delivery and the immediate postpartum
- Describe the pre-referral management of severe complications during labor, delivery and immediate postpartum.

3.2 - UNSATISFACTORY (POOR) PROGRESS OF LABOR

PROBLEMS:

- * Latent phase: cervical dilation below 4 cm after 8 hours of *adequate contractions*.
- * Active Phase: Cervical dilatation is to the right of the alert line on the partograph.
- * The woman has been experiencing true labor pains for 12 hours or more without delivery (prolonged labor).
- * Second Stage Expulsive phase: The cervix is fully dilated and the woman has the urge to push, but there is no descent for 2hrs.

3.2.1. GENERAL MANAGEMENT:

- Carefully explain the situation to the woman and talk to her about how you will evaluate her. Listen to the woman and be sensitive to her feelings.
- Perform a rapid evaluation of the woman's general condition, vital signs (pulse, blood pressure, respiration), level of consciousness, presence of anxiety and/or confusion, blood loss, and skin color and temperature.
- Check the fetal heart rate:
 - If there are fetal heart rate abnormalities (less than 100 or more than 180 beats per minute), suspect fetal distress.
 - If the fetal heart cannot be heard, ask several other persons to listen or use a Doppler stethoscope, if available.
 - If the fetal heart cannot be heard, suspect fetal death.
- Evaluate the cause of unsatisfactory progress of labor (Table 5.10):

- Review the partograph, especially to rule out inadequate uterine activity or prolonged/obstructed labor.
- Test urine for ketones. If ketotic, encourage the woman to eat or drink; otherwise, treat with IV fluids.
- Assess if the woman is anxious, fearful or distressed by pain:
- If the woman is distressed by pain, encourage breathing techniques and a warm bath or shower; if necessary, give analgesics.
- If the woman is anxious or fearful, provide supportive care.
- Check for malposition, malpresentation and macrosomia.
- Talk to the woman and any support person she would like present about findings and the proposed plan, and obtain consent before initiating treatment.
- Provide specific care for the identified cause.

3.2.2. DIAGNOSIS:

See table 5.10 below to diagnose unsatisfactory progress of labor.

Table 5.10:- Diagnosis of unsatisfactory progress of labor

Findings	Diagnosis
• Cervix not dilated • No palpable contractions/infrequent contractions	• False labor,
• Cervix not dilated beyond 4 cm after 8 hours of regular contractions	• Prolonged latent phase,
• Cervical dilatation to the right of the alert line on the partograph	• Prolonged active phase
• Secondary arrest of cervical dilatation and descent of presenting part in presence of good contractions	• Cephalopelvic disproportion,
• Secondary arrest of cervical dilatation and descent of presenting part with large caput, third degree moulding, cervix poorly applied to presenting part, oedematous cervix, ballooning of lower uterine segment, formation of retraction band, maternal and fetal distress	• Obstruction
• Less than three contractions in 10 minutes, each lasting less than 40 seconds	• Inadequate uterine activity,
• Presentation other than vertex with occiput anterior	• Mal-presentation or mal-position,
• Cervix fully dilated and woman has urge to push, but there is no descent in 1hr or baby not delivered in 2hrs (varies with parity)	• Prolonged expulsive phase,

Table 5.11 causes of prolonged labor categorized into the 5 “P” s of dystocia

Passenger	Passage
<i>Head:</i> ● Large fetal head (big for that pelvis) ● Hydrocephalus (brain surrounded by fluid, which makes the skull swell) <i>Presentation and position:</i> ● Brow, face, shoulder ● Persistent malposition <i>Twin pregnancy:</i> ● Locked twins (locked at the neck) ● Conjoined twins (fused together with some shared organs)	<i>Bony pelvis:</i> ● Contracted (due to malnutrition) ● Deformed (due to trauma, polio) <i>Soft tissue:</i> ● Tumour in the pelvis ● Viral infection in the uterus or abdomen ● Scars (from female circumcision) Powers: Inadequate contractions Psyche: psychological make up fearful Physician

3.2.3 MANAGEMENT:

A. FALSE LABOR

Examine for urinary tract or other infection or ruptured membranes and treat accordingly. If none of these are present, discharge the woman and encourage her to return if signs of labor recur or she has danger signs.

B. PROLONGED LATENT PHASE

The diagnosis of prolonged latent phase is made retrospectively. When contractions cease, the woman is said to have had false labor. When contractions become regular and dilatation progresses beyond 4 cm, the woman is said to have been in the latent phase.

Note: Misdiagnosing false labor or prolonged latent phase leads to unnecessary induction or augmentation, which may fail. This may lead to unnecessary caesarean section and amonites.

If a woman has been in the latent phase for more than 8 hours and there is little sign of progress, reassess the situation by assessing the cervix:

- If there has been **no change in cervical effacement or dilatation** and there is no fetal distress, review the diagnosis. The woman may not be in labor.
- If contraction persist for 8 hours and cervical dilatation remains less than 4 cm, the diagnosis is prolonged latent phase. Refer her to a higher facility.

First explore less invasive interventions that could lead to vaginal birth.

- If there has been a **change in cervical effacement or dilatation**, augment labor using oxytocin.

Note: Do not use oral misoprostol for labor augmentation.

Do not perform amniotomy as a sole intervention for treatment of confirmed delay in labor, especially in settings with high HIV prevalence.

- Continue to monitor the woman and fetus.
- Reassess cervical dilatation every four hours.
- If the **woman has not entered the active phase after eight hours of oxytocin** infusion, refer for possible caesarean section
- If there are **signs of infection** (fever, foul-smelling vaginal discharge):
 - Augment labor immediately with oxytocin.
 - Give a combination of antibiotics until the baby is born:
ampicillin 2 g IV every six hours PLUS gentamicin 5 mg/kg body weight IV every 24 hours.

Once labor augmentation has begun, the woman should never be left alone.

C. PROLONGED ACTIVE PHASE:

In the active phase of labor, plotting of the cervical dilatation usually remains to the right of the alert line warning that the labor may be prolonged. A thorough assessment of the mother, fetus and progress using the partogram may indicate the cause.

- **Inadequate/Inefficient uterine contractions:**

- If **contractions are inefficient** (less than three contractions in 10 minutes, each lasting less than 40 seconds), suspect inadequate uterine activity;
- ☐ General methods of labor support may improve contractions and accelerate progress. The support includes: rehydration, emptying the bladder and encouraging the woman to be more active and move around or adopt an upright position will speed up the progress of labor and normal progress will continue.
- ☐ Augmentation with oxytocin is also recommended in higher centers.

If **contractions are efficient** (three contractions in 10 minutes, each lasting more than 40 seconds) suspect cephalopelvic disproportion, obstruction, malpositions or malpresentation **& refer to a higher facility.**

- **Cephalo Pelvic Disproportion:**

Cephalopelvic disproportion occurs because the fetus is relatively too large or the maternal pelvis is too small. If **labor persists with Cephalopelvic disproportion**, it may become arrested or obstructed. The best test to determine if a pelvis is adequate is a trial of labor. Clinical pelvimetry is of limited value.

- **Obstruction:**

It is always caused by mechanical factors (either contracted pelvis and/or malpositions, malpresentations, pelvic tumors..)

Patient should be referred to a higher center with CS facility after appropriate prereferral care (Iv fluids, antibiotics, catheterization, analgesics accompanied with a midwife)

Note: Rupture of an unscarred uterus is usually caused by obstructed labor.

D. PROLONGED SECOND STAGE/EXPULSIVE PHASE

Maternal expulsive efforts increase fetal risk by reducing the delivery of oxygen to the placenta. Allow spontaneous maternal “pushing”, but do not encourage prolonged effort and holding the breath. The management of protracted second stage depends on the identified cause and complications.

- Assist the woman in assuming a position of her choice, including upright positions.
- If the presentation is vertex with live fetus and station at or below O (O/5) delivery can be assisted/ with forceps or vacuum.
 - o In tight perineum : vaginal delivery may be achieved with vaginal procedure such as episiotomy
- All other conditions require immediate referral to a higher facility.

3.3. FETAL DISTRESS INLABOR

PROBLEMS

- Abnormal fetal heart rate (less than 100 or more than 180 beats per minute).
- Thick meconium-stained amniotic fluid.

GENERAL MANAGEMENT

- Prop up the woman or place her on her leftside.
- Stop oxytocin if it is being administered.
- Open IV line to hydrate.
- Look for possible causes like cord prolapse.
- Start intranasal oxygen

ABNORMAL FETAL HEART RATE

Box 5.3: Abnormal fetal heartrate

- A **normal fetal heart rate** may slow during a contraction but usually recovers to normal as soon as the uterus relaxes.
 - A **very slow fetal heart rate** in the absence of contractions or persisting after contractions is suggestive of fetal distress.
 - A **rapid fetal heart rate** may be a response to maternal fever, drugs causing rapid maternal heart rate (e.g. tocolytic drugs), hypertension or amnionitis. In the absence of a rapid maternal heart rate, a rapid fetal heart rate should be considered a sign of fetal distress.
-
- If a **maternal cause is identified** (e.g. maternal fever, drugs), initiate appropriate management.
 - If a **maternal cause is not identified** and the **fetal heart rate remains abnormal** throughout at least three contractions, perform a vaginal examination to check for explanatory signs of distress:
 - If there is **bleeding with intermittent or constant pain**, suspect abruptio placentae;
 - If there are **signs of infection** (fever, foul-smelling vaginal discharge) give antibiotics as for amnionitis;
 - If the **cord is below the presenting part** or **in the vagina**, manage as prolapsed cord.
 - If **fetal heart rate abnormalities persist** or there are **additional signs of distress** (thick meconium-stained fluid):
 - Plan delivery if the **cervix is fully dilated** and the **fetal head is not more than 1/5 above** the symphysis pubis or the leading bony edge of the **head is at or below 0 station**, deliver by vacuum extraction or forceps;
 - If the **cervix is not fully dilated** or the **fetal head is more than 1/5 above** the symphysis pubis or the leading bony edge of the **head is above 0 station**, consult/refer for delivery by caesarean section.

MECONIUM

- Meconium staining of amniotic fluid is seen frequently as the fetus matures and by itself is not an indicator of fetal distress. A slight degree of meconium without fetal heart rate abnormalities is a warning of the need for vigilance.
- **Thick meconium** suggests passage of meconium in reduced amniotic fluid and may indicate the need for expedited delivery and meconium management of the neonatal upper airway at birth to prevent meconium aspiration.
- In **breech presentation**, meconium is passed in labor because of compression of the fetal abdomen during delivery. This is not a sign of distress unless it occurs in early labor.

3.4 CORD PROLAPSE

PROBLEMS

- The umbilical cord lies in the birth canal below the fetal presenting part.
- The umbilical cord is visible at the vagina following rupture of the membranes.

GENERAL MANAGEMENT

When managing the woman's problem, apply basic principles of providing care .

- Give oxygen at 4-6 L per minute by mask or nasal cannulae.

SPECIFIC MANAGEMENT

PULSATING CORD

If the **cord is pulsating**, the fetus is alive.

- Diagnose the stage of labor by an immediate vaginal examination.
- If the woman is in the **first stage of labor**, perform the following in all cases:
 - Wearing sterile gloves, insert a hand into the vagina. Push the presenting part up to decrease pressure on the cord and dislodge the presenting part from the pelvis.
 - Place the other hand on the abdomen in the suprapubic region to keep the presenting part out of the pelvis.
 - Once the presenting part is firmly held above the pelvic brim, remove the other hand from the vagina. Keep the hand on the abdomen until a caesarean can be performed.
 - If available, give tocolytics (drugs to arrest labor e.g nifedipine 20mg po stat) to reduce contractions.
 - Perform immediate caesarean (in referral center).
- If the woman is in the **second stage of labor**:
 - Expedite birth with obstetric vacuum or forceps in vertex presentation .
 - For **breech presentation**, perform breech extraction and apply Piper or long forceps to the after-coming head.
 - Prepare for resuscitation of the newborn.

CORD NOT PULSATING

- If the **cord is not pulsating**, the fetus is dead. Proceed with birth of the baby in the manner that is safest for the woman.

3.5 SHOULDER DYSTOCIA

PROBLEM

- The fetal head has been born but the shoulders are stuck and cannot be delivered.

GENERAL MANAGEMENT

- Be prepared for shoulder dystocia at all births, especially if a large baby is anticipated.
- Have several persons available to help.

Shoulder dystocia cannot be predicted.

DIAGNOSIS

- The fetal head is born but remains tightly applied to the vulva.
- The chin retracts and depresses the perineum.
- Traction on the head fails to deliver the shoulder, which is caught behind the symphysis pubis.

MANAGEMENT

When managing the woman's problem, apply basic principles when providing care.

- **SHOUT FOR HELP.** Urgently mobilize all available personnel.
- Make an adequate episiotomy to reduce soft tissue obstruction and to allow space for manipulation.
- With the woman on her back, ask her to flex both thighs, bringing her knees as far up as possible towards her chest. Ask two assistants to push her flexed knees firmly up onto her chest.
- Wearing sterile gloves:
 - Apply firm, continuous traction downwards on the fetal head to move the **shoulder that is anterior under the symphysis pubis**.
Note: Avoid excessive traction on the fetal head as this may result in brachial plexus injury.
 - Have an assistant simultaneously apply suprapubic pressure downwards to assist delivery of the shoulder.
Note: Do not apply fundal pressure. This will further impact the shoulder and can result in uterine rupture.
- If the **shoulder has still not been born**:
 - Insert a hand into the vagina along the baby's back.
 - Apply pressure to the shoulder that is anterior in the direction of the baby's sternum to rotate the shoulder and decrease the diameter of the shoulders.

- If needed, apply pressure to the shoulder that is posterior in the direction of the sternum.
- If the **shoulder still is not born despite the above measures**: - Insert a hand into the vagina.

FIGURE 5.17 Assistant pushing flexed knees firmly towards chest



- Grasp the humerus of the arm that is posterior and, keeping the arm flexed at the elbow, sweep the arm across the chest. This will provide room for the shoulder that is anterior to move under the symphysis pubis.

FIGURE 5.18 Grasping the humerus of the arm that is posterior and sweeping the arm across the chest



If all of the **above measures fail to deliver the shoulder**, other options include:

- Fracture the clavicle to decrease the width of the shoulders and free the shoulder that is anterior.
- Apply traction with a hook in the axilla to extract the arm that is posterior.

3.6 BREECH DELIVERY

3.6.1 GENERAL

Breech presentation is a longitudinal lie of the fetus with the caudal pole (buttock or lower extremity) occupying the lower part of the uterus and cephalic pole in the uterine fundus. The breech of the fetus is palpated at the pelvic brim. Breech presentation may be caused by an underlying fetal or maternal abnormality, or may be an apparently chance occurrence, or related to an otherwise benign variant such as cornual placental position.

Diagnosis:

The history may reveal:

- discomfort under the rib (due to the hard head)
- feeling of more fetal movement in the lower segment

On abdominal palpation (Leopold's maneuvers):

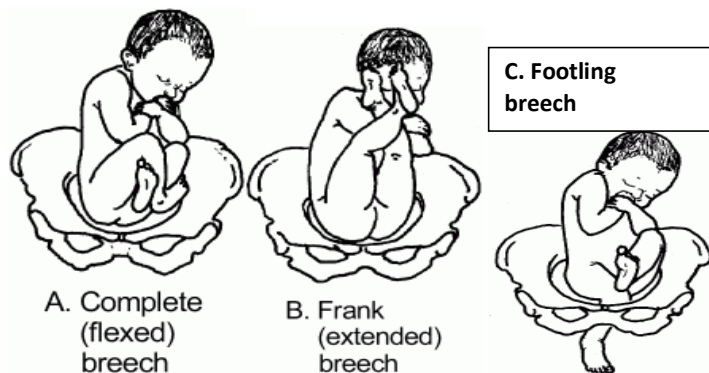
- A round hard and smooth mass (head) occupying the fundus
- A soft, broad, indefinite and non ballotable mass (the breech) occupying the lower pole of the uterus
- FHB loudest just above the umbilicus (may be lower with engagement)

Vaginal examination:

- Ischial tuberosities, sacrum and its spines, genitals and the anus (are usually palpable in frank breech)
- The foot distinguished by the heel should be searched for round the breech to determine the type:
 - Complete breech: feet are felt alongside the buttock
 - Frank breech
 - Footling breech: one or both feet are inferior to the buttock
- Ischial tuberosity and anus are in straight line (where as the malar eminence & mouth form a triangle in face presentation)
- Suckling in live fetus (but not by the anus in breech)

The examination should also include pelvic assessment and cord presentation or prolapse.

Figure 5.19 Breech presentation



Management:

- Review for indications. Ensure that all conditions for safe vaginal breech delivery are met.
- Refer urgently if the following specific breech-related conditions are identified:
 - Footling breech
 - Estimated fetal weight of > 3500gm (By clinical or ultrasound)

- Extended or deflexed neck, or
- The presence of compounding factors such as
 - Previous CS
 - Elderly primigravida
 - Infertility
 - Bad obstetric history
 - Severe IUGR
 - Rh-immunization
 - Any degree of CPD
 - Uterine dysfunction, prolonged labor or failure to progress in labor
- Review general care principles and start an IV infusion.
- Provide emotional support and encouragement. If necessary, use a pudendal block.
- Perform all manoeuvres gently without undue force.

3.6.2 COMPLETE OR FRANK BREECH

Delivery of the buttocks and legs:

- Once the buttocks have entered the vagina and the cervix is fully dilated, tell the woman she can bear down with the contractions.
- If the perineum is very tight, perform an episiotomy.
- Let the buttocks deliver until the lower back and then the shoulder blades are seen.
- Gently hold the buttocks in one hand, but do not pull.
- If the legs do not deliver spontaneously, deliver one leg at a time:
 - Push behind the knee to bend the leg;
 - Grasp the ankle and deliver the foot and leg;
 - Repeat for the other leg.

Do not pull the baby while the legs are being delivered.

- Hold the fetus by the hips, as shown in Fig 5.20. Do not hold the baby by the flanks or abdomen as this may cause kidney or liver damage.

Delivery of the arms:

Arms are felt on chest:-

- Allow the arms to disengage spontaneously one by one. Only assist if necessary.
- After spontaneous delivery of the first arm, lift the buttocks towards the mother's abdomen to enable the second arm to deliver spontaneously.
- If the arm does not spontaneously deliver, place one or two fingers in the elbow and bend

the arm, bringing the hand down over the fetus's face.

Figure 5.20 Hold the fetus at the hips, but do not pull



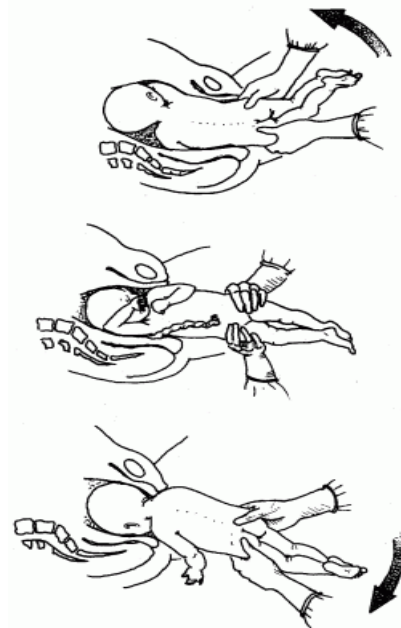
Arms are stretched above the head or folded around the neck:

Use the Lovset's manoeuvre (Fig 5-21):

- Hold the fetus by the hips and turn half a circle, keeping the back uppermost and applying downward traction at the same time, so that the arm that was posterior becomes anterior and can be delivered under the pubic arch.

- Assist delivery of the arm by placing one or two fingers on the upper part of the arm. Draw the arm down over the chest as the elbow is flexed, with the hand sweeping over the face.
- To deliver the second arm, turn the baby back half a circle, keeping the back uppermost and applying downward traction, and deliver the second arm in the same way under the pubic arch.

Figure 5.21- Lovset's manoeuvre

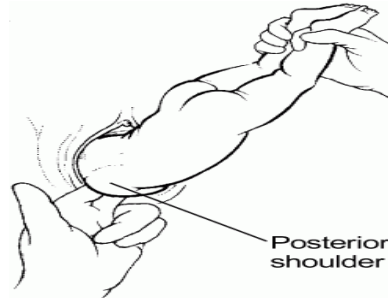


BABY'S BODY CANNOT BE TURNED:

If the baby's body cannot be turned to deliver the arm that is anterior first, deliver the shoulder that is posterior (Fig 5.22):

- Hold and lift the baby up by the ankles.
- Move the baby's chest towards the woman's inner leg. The shoulder that is posterior should deliver.
- Deliver the arm and hand.
- Lay the baby back down by the ankles. The shoulder that is anterior should now deliver.
- Deliver the arm and hand.

Figure 5.22- Delivery of the shoulder that is posterior



DELIVERY OF THE HEAD

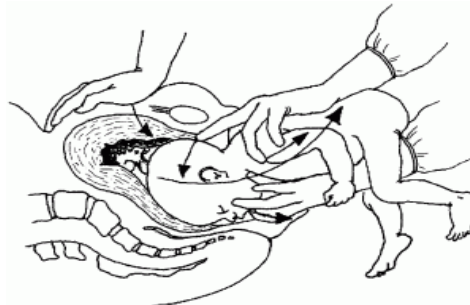
Deliver the head by the Mauriceau-Smellie-Veit manoeuvre (Fig 4-31) as follows:

- Lay the baby face down with the length of its body over your hand and arm.
- Place the first and third fingers of this hand on the baby's cheekbones and place the second finger in the baby's mouth to pull the jaw down and flex the head.
- Use the other hand to grasp the baby's shoulders.
- With two fingers of this hand, gently flex the baby's head towards the chest, while applying downward pressure on the jaw to bring the baby's head down until the hairline is visible.
- Pull gently to deliver the head.

Note: Ask an assistant to push above the mother's pubic bone as the head delivers. This helps to keep the baby's head flexed.

- Raise the baby, still astride the arm, until the mouth and nose are free.

Figure 5.23 The Mauriceau-Smellie-Veit manoeuvre



ENTRAPPED (STUCK) HEAD:

- Catheterize the bladder.
- Be sure the cervix is fully dilated.

Apply firm pressure above the mother's pubic bone to flex the baby's head and push it through the pelvis.

3.7 EPISIOTOMY AND PERINEAL TEARS

3.7.1 INTRODUCTION

Episiotomy is no longer recommended as a routine procedure. There is no evidence that routine episiotomy decreases perineal damage, future vaginal prolapse or urinary incontinence. In fact, routine episiotomy is associated with an increase of third and fourth degree tears and subsequent anal sphincter muscle dysfunction.

- Review for indications.

Episiotomy should be considered only in the case of:

- Complicated vaginal delivery (breech, shoulder dystocia, forceps, vacuum);
- Scarring from female genital cutting or poorly healed third or fourth degree tears;
- Fetal distress.

- Review general care principles and apply antiseptic solution to the perineal area.
- Provide emotional support and encouragement. Use local infiltration with lignocaine or a pudendal block.
- Make sure there are no known allergies to lignocaine or related drugs.
- Infiltrate beneath the vaginal mucosa, beneath the skin of the perineum and deeply into the perineal muscle using about 10 mL 0.5% lidocaine solution (**Figure 4-16**).

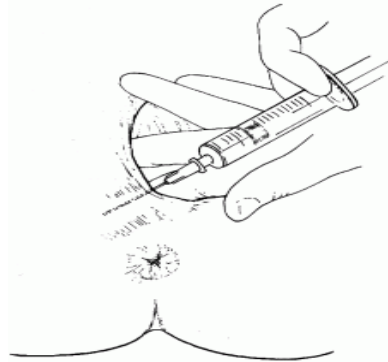
Note: Aspirate (pull back on the plunger) to be sure that no vessel has been penetrated. If **blood is returned in the syringe with aspiration**, remove the needle. Recheck the position carefully and try again. Never inject if blood is aspirated. **The woman can suffer seizures and death if IV injection of lidocaine occurs.**

- At the conclusion of the set of injections, wait 2 minutes and then pinch the incision site with forceps. If the **woman feels the pinch**, wait 2 more minutes and then retest.

Anaesthetize early to provide sufficient time for effect.

- Wait to perform episiotomy until:
 - The perineum is thinned out; and
 - The head crowns i.e. 3–4 cm of the fetus's head is visible during a contraction.

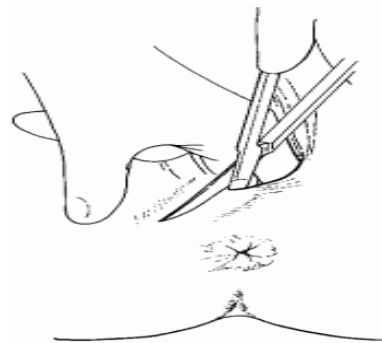
Figure 5.24 Infiltration of perineal tissue with local anaesthetic



Performing an episiotomy will cause bleeding. It should not, therefore, be done too early.

- Wearing high-level disinfected gloves, place two fingers between the baby's head and the perineum (**Figure 5.24**)
- Use scissors to cut the perineum about 3–4 cm in the medio-lateral direction.
- Use scissors to cut 2–3 cm up the middle of the posterior vagina (**Fig 5.25**)
- Control the fetal head and shoulders as they deliver, ensuring that the shoulders have rotated to the midline to prevent an extension of the episiotomy.
- Carefully examine for extensions and other tears and repair (see below).

Figure 5.25 Making the incision while inserting two fingers to protect the fetal head

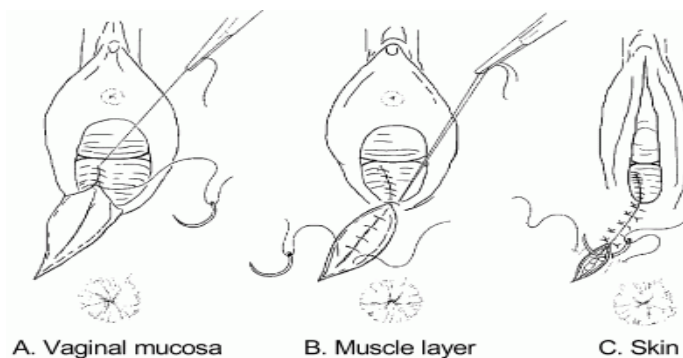


3.7.2 REPAIR OF EPISIOTOMY:

Note: It is important that absorbable sutures be used for closure. Polyglycolic sutures are preferred over chromic catgut for their tensile strength, non-allergenic properties and lower probability of infectious complications and episiotomy breakdown. Chromic catgut is an acceptable alternative, but is not ideal.

- Clean with gauze the area around the episiotomy.
- If the **episiotomy is extended** through the anal sphincter or rectal mucosa, manage as third or fourth degree tears, respectively (i.e. refer if consultation not possible)
- Close the vaginal mucosa using continuous 2-0 suture: (see **Figure 5.26**)
 - Start the repair about 1 cm above the apex (top) of the episiotomy. Continue the suture to the level of the vaginal opening;
 - At the opening of the vagina, bring together the cut edges of the vaginal opening;
 - Bring the needle under the vaginal opening and out through the incision and tie.
- Close the perineal muscle using interrupted 2-0 sutures.
- Close the skin using interrupted (or subcuticular) 2-0 sutures.

Figure 5.26- Repair of episiotomy



3.7.3. COMPLICATIONS:

- If a **haematoma occurs**, open and drain. If there are **no signs of infection** and **bleeding has stopped**, reclose the episiotomy.
- If there are **signs of infection**, open and drain the wound. Remove infected sutures and debride the wound:

- If the **infection is mild**, antibiotics are not required;
- If the **infection is severe but does not involve deep tissues**, give a combination of antibiotics:
 - Ampicillin 500 mg by mouth four times per day for 5 days;
 - PLUS metronidazole 500 mg by mouth three times per day for 5 days.
- If the **infection is deep, involves muscles and is causing necrosis** (necrotizing fasciitis), give a combination of antibiotics until necrotic tissue has been removed and the woman is fever-free for 48 hours:
 - Penicillin G 2 million units IV every 6 hours;
 - PLUS gentamicin 5 mg/kg body weight IV every 24 hours;
 - PLUS metronidazole 500 mg IV every 8 hours;
 - Once the **woman is fever-free for 48 hours**, give:
 - Ampicillin 500 mg by mouth four times per day for 5 days;
 - PLUS metronidazole 400 mg by mouth three times per day for 5 days;

Note: Necrotizing fasciitis requires wide surgical debridement. Perform secondary closure in 2–4 weeks (depending on resolution of the infection).

3.7.3 REPAIR OF VAGINAL AND PERINEAL TEARS

There are four degrees of perineal tears that can occur during delivery:

- **First degree tears** involve the vaginal mucosa and connective tissue.
- **Second degree tears** involve the vaginal mucosa, connective tissue and underlying muscles.
- **Third degree tears** involve complete transection of the anal sphincter.
- **Fourth degree tears** involve the rectal mucosa.

Note: It is important that absorbable sutures be used for closure. Polyglycolic sutures are preferred over chromic catgut for their tensile strength, non-allergenic properties and lower probability of infectious complications. Chromic catgut is an acceptable alternative, but is not ideal.

REPAIR OF FIRST AND SECOND DEGREE TEARS:

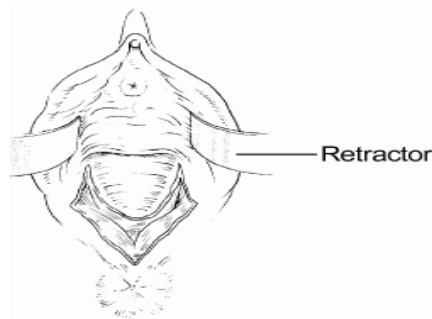
Most first degree tears close spontaneously without sutures.

- Review general care principles.
- Provide emotional support and encouragement. Use local infiltration with lignocaine. If

necessary, use a pudendal block.

- Ask an assistant to massage the uterus and provide fundal pressure.
- Carefully examine the vagina, perineum and cervix.
- If the **tear is long and deep through the perineum**, inspect to be sure there is no third or fourth degree tear:
 - Place a gloved finger in the anus;
 - Gently lift the finger and identify the sphincter;
 - Feel for the tone or tightness of the sphincter.
- Change to clean, high-level disinfected gloves.
- If the **sphincter is not injured**, proceed with repair.

Figure 5.27 Exposing a perineal tear



- Clean the area around the tear.
- Make sure there are no known allergies to lignocaine or related drugs.

Note: If **more than 40 mL of lidocaine** solution will be needed for the repair, add adrenaline to the solution (the resulting vasoconstriction caused by adrenalin will prevent intravascular complications associated with administration of large doses of lidocaine)

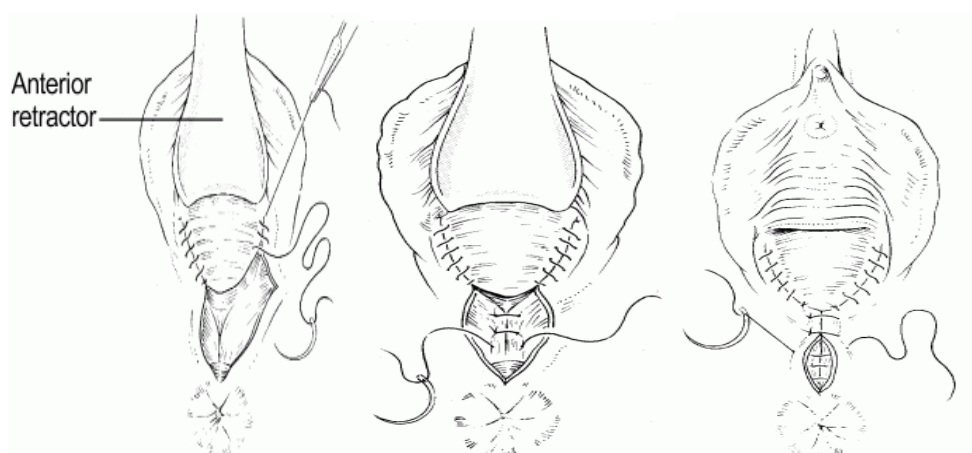
- Infiltrate beneath the vaginal mucosa, beneath the skin of the perineum and deeply into the perineal muscle using about 10 mL 0.5% lignocaine solution.

Note: Aspirate (pull back on the plunger) to be sure that no vessel has been penetrated. If **blood is returned in the syringe with aspiration**, remove the needle. Recheck the position carefully and try again. Never inject if blood is aspirated.

- At the conclusion of the set of injections, wait 2 minutes and then pinch the area with forceps. If the **woman feels the pinch**, wait 2 more minutes and then retest.

- Repair the vaginal mucosa using a continuous 2-0 suture:
 - Start the repair about 1 cm above the apex (top) of the vaginal tear. Continue the suture to the level of the vaginal opening;
 - At the opening of the vagina, bring together the cut edges of the vaginal opening;
 - Bring the needle under the vaginal opening and out through the perineal tear and tie.
- Repair the perineal muscles using interrupted 2-0 suture. If the **tear is deep**, place a second layer of the same stitch to close the space.
- Repair the skin using interrupted (or subcuticular) 2-0 sutures starting at the vaginal opening.
- **If the tear was deep, perform a rectal examination. Make sure no stitches are in the rectum**

Figure 5-28:- Repairing the vaginal mucosa(A), perineal muscles(B) and the skin(C)



Post Procedure counseling (both episiotomy and perineal tear repairs)

- Change pad/cloths frequently to keep wound dry
- warm soapy baths 3–4 times per day
- Do not insert anything in the vagina
- Get rest and good nutrition
- Delay intercourse to avoid breaking sutures
- Do not return for suture removal as they are absorbable

THIRD AND FOURTH DEGREE PERINEAL TEARS

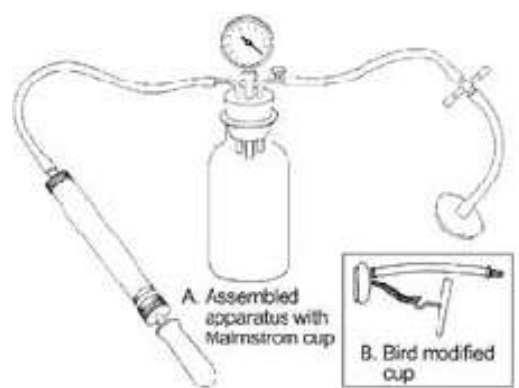
The woman may suffer loss of control over bowel movements and gas if a torn anal sphincter is not repaired correctly. If a **tear in the rectum is not repaired**, the woman can suffer from infection

and recto-vaginal fistula (passage of stool through the vagina). Repair of such tears should be done in the operating room, therefore refer her immediately

3.8 VACUUM EXTRACTION

In general, vacuum devices are easier to apply, place less force on the fetal head, require less maternal anesthesia, result in less maternal soft tissue trauma, and do not affect the diameter of the fetal head compared to forceps. Vacuum delivery is effected using the ventouse (vacuum extractor). The main action of the ventouse is traction with an additional effect of rotation. Its main components are the suction cup, vacuum pump and traction devices.

Figure 5.29 shows the essential components of the obstetric vacuum.



Indications

1. Prolonged second stage of labor
2. Fetal distress and cord prolapse in second stage of labor.
3. Shortening of 2nd stage in some maternal illnesses (heart disease, severe anemia, severe preeclampsia/eclampsia)

Contraindications:

- CPD,
- Non-vertex presentation such as face, breech (after-coming head)

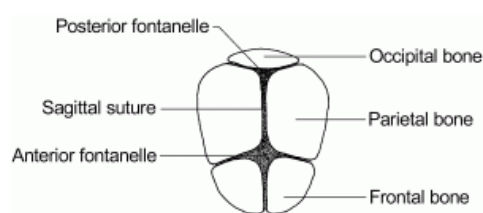
Prerequisites

- Vertex presentation with fetal position identified
- Fully dilated cervix
- Engaged head: station at 0 or not more than 2/5 above symphysis pubis
- Ruptured membranes
- Live fetus
- Term fetus

Preparation

- Empty bladder
- Local anesthesia infiltration for episiotomy
- Assembled and tested vacuum extractor
- ☐ - Check all connections and test the vacuum on a gloved hand.
- ☐ Assess the position of the fetal head by feeling the sagittal suture line and the fontanelles.
- ☐ Identify the posterior fontanel.

Figure 5.30 Landmarks of the fetal skull



Identify the flexion point (Fig 5.31)

- it is located 3cm anterior to the posterior fontanelle

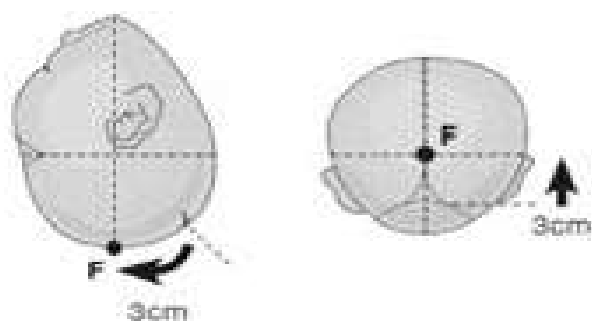
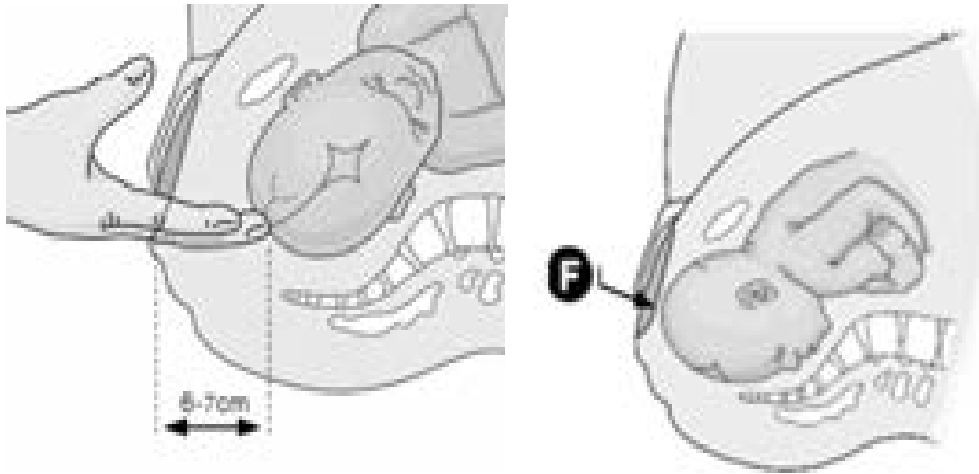
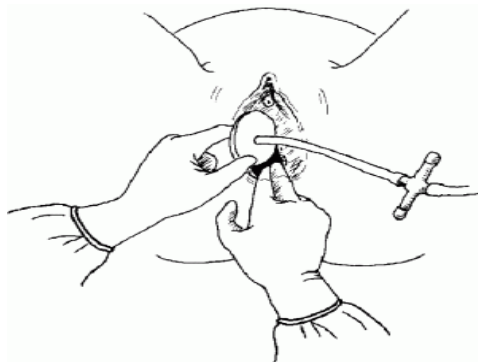


Fig 5.31: Locating the flexion point (F), 3 cm anterior to the posterior fontanelle, and calculating cup insertion distance



- Apply the largest cup that will fit, with the center of the cup over the flexion point, where the edge of the cup will be about 1 cm anterior to the posterior fontanel. This placement will promote flexion, descent and autorotation with traction (Fig 5.32).
- An episiotomy may be needed for proper placement at this time. If an episiotomy is not necessary for placement, delay the episiotomy until the head stretches the perineum or the perineum interferes with the axis of traction. This will avoid unnecessary bloodloss.
- Check the application. Ensure there is no maternal soft tissue (cervix or vagina) within the rim.
- With the pump, create a vacuum of 0.2 kg/cm² negative pressure and check the application.
- Increase the vacuum to 0.8 kg/cm² and check the application.

Figure 5.32- Applying the vacuum cup

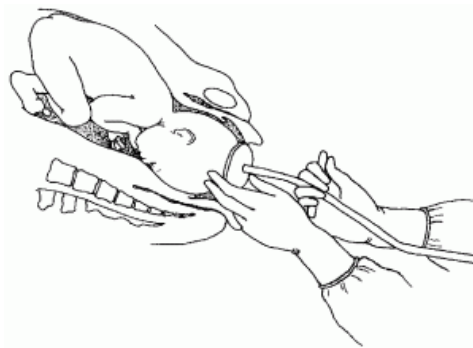


- After maximum negative pressure, start traction in the line of the pelvic axis and perpendicular to the cup. If the fetal head is tilted to one side or not flexed well, traction

should be directed in a line that will try to correct the tilt or deflexion of the head (i.e. to one side or the other, not necessarily in the midline).

- With each contraction, apply traction in a line perpendicular to the plane of the cup rim (Fig 5.33). Wearing high-level disinfected gloves, place a finger on the scalp next to the cup during traction to assess potential slippage and descent of the vertex.
- Between contractions check:
 - Fetal heart rate;
 - Application of the cup.

Figure 5-33:-Applying traction



TIPS

- Never use the cup to actively rotate the baby's head. Rotation of the baby's head will occur with traction.
- The first pulls help to find the proper direction for pulling.
- Do not continue to pull between contractions and expulsive efforts.
- With progress, and in the absence of fetal distress, continue the "guiding" pulls for a maximum of 30 minutes.

Support the perineum to assist the delivery of the head.

Removal of cup. As soon as the head is crowned or after delivery of the head, release the vacuum, remove the cup, and complete delivery of the baby.

FAILURE

- Vacuum extraction failed if:
 - The head does not advance with each pull;
 - The fetus is undelivered after three pulls with no descent.
 - The cup slips off the head twice at the proper direction of pull with a maximum

negative pressure.

- Every application should be considered a trial of vacuum extraction. Do not persist if there is no descent with every pull.
- If vacuum extraction fails, refer her urgently providing basic support.

COMPLICATIONS

Complications usually result from not observing the conditions of application or from continuing efforts beyond the time limits stated above.

Fetal complications

- Localized scalp oedema (artificial caput or chignon) under the vacuum cup is harmless and disappears in a few hours.
- Cephalohaematoma requires observation and usually will clear in 3–4 weeks.
- Scalp abrasions (common and harmless) and lacerations may occur. Clean and examine lacerations to determine if sutures are necessary. Necrosis is extremely rare.
- Intracranial bleeding is extremely rare and requires immediate intensive neonatal care.

Maternal complications

Tears of the genital tract may occur. Examine the woman carefully and repair any tears to the cervix or vagina or repair episiotomy.

MODULE 6 : MANAGEMENT OF OBSTETRICHEMORRHAGE

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UNIT I - INTRODUCTION

Obstetric haemorrhage is the single most significant cause of maternal mortality worldwide accounting for 25– 30% of all maternal deaths. It encompasses antepartum, intrapartum, and postpartum bleeding.

Vaginal bleeding in early pregnancy which occurs during the first 28 weeks of pregnancy affects one fourth of pregnancy. Possible causes of bleeding in early pregnancy include threatened abortion, incomplete abortion, inevitable abortion, ectopic pregnancy, and gestational trophoblastic disease.

Antepartum haemorrhage (APH) is defined as bleeding from or in to the genital tract, occurring from 28 weeks of pregnancy and prior to the birth of the baby. The most important causes of APH are placenta praevia and placental abruption, although these are not the most common. APH complicates 3–5% of pregnancies and is a leading cause of perinatal and maternal mortality worldwide.

Postpartum hemorrhage (PPH) is the single largest cause of maternal death worldwide, accounting for an estimated 132,000 deaths each year. In developing countries, where most births occur in homes or local clinics, the interventions needed to treat PPH—emergency referrals, obstetric care, blood transfusion, and surgery— are simply out of reach for the majority of women. Fortunately, the effectiveness of a feasible and inexpensive intervention that prevents PPH has already been proven. Active management of the third stage of labor (AMTSL) can prevent postpartum hemorrhage and eliminate at least half of PPH cases.

Although specific interventions will be required depending on the underlying cause of the haemorrhage, several basic steps are essential in the initial and continued management of any patient with obstetric haemorrhage. The general principles in the management of obstetric hemorrhage includes (i) organization of the multidisciplinary team; (ii) restoration of blood volume via large bore access using fluid and blood; (iii) correction of defective coagulation with blood products and factors; (iv) evaluation of response to treatment by hemodynamic and laboratory assessment; and (v) remedying of the underlying cause of the bleeding.

This module introduces to the management of causes of obstetric hemorrhage and is structured as bleeding in three parts; Vaginal bleeding in early pregnancy, vaginal bleeding in later pregnancy and vaginal bleeding after child birth.

Participant learning objective: After completing this module, participants will be able to detect the causes of obstetric hemorrhage early in pregnancy, late in pregnancy and after child birth.

Enabling objectives:

- Detect and provide care for women with bleeding early in pregnancy
- Describe the components of post abortion care

- Detect and provide care for women with bleeding late in pregnancy
- Detect and provide care for women with bleeding after child birth

UNIT 2 -VAGINAL BLEEDING IN EARLY PREGNANCY

2.1. INTRODUCTION

OVERVIEW

Vaginal bleeding in early pregnancy encompasses bleeding which occurs during the first 28 weeks of pregnancy. Vaginal bleeding in early pregnancy occurs in about one fourth of pregnancies. About one half of those who bleed will miscarry. Possible causes of bleeding in early pregnancy include threatened abortion, incomplete abortion, inevitable abortion, ectopic pregnancy, and gestational trophoblastic disease. In threatened abortion guarded reassurance and watchful waiting are appropriate if fetal heart beat is detected, if the patient is medically stable, and if there is no adnexal mass or clinical sign of intraperitoneal bleeding. For all other causes prompt management is warranted which will depend on the amount of bleeding, gestational age and cause of the bleeding.

GENERAL MANAGEMENT

- Make a **rapid evaluation** of the general condition of the woman including vital signs (pulse, blood pressure, respiration, temperature).
- If **shock is suspected**, immediately begin treatment. Even if signs of shock are not present, keep shock in mind as you evaluate the woman further because her status may worsen rapidly. If **shock develops**, it is important to begin treatment immediately.
- If the **woman is in shock**, consider ruptured ectopic pregnancy.
- Start an IV infusion and infuse IV fluids.
- Definitive management of abortion depends on a number of factors, including: the type of abortion; the stage of abortion; and the size of the uterus as identified by a pelvic exam.

DIAGNOSIS (table 6.1.)

- **Consider ectopic pregnancy** in any woman with anemia, pelvic inflammatory disease (PID), threatened abortion or unusual complaints about abdominal pain.
 - **Note:** If **ectopic pregnancy is suspected**, perform bimanual examination gently because an early ectopic pregnancy is easily ruptured.
- **Consider abortion** in any woman of reproductive age who has a missed period (delayed menstrual bleeding with more than a month having passed since her last menstrual period) and has one or more of the following: bleeding, cramping, partial expulsion of products of conception, dilated cervix or smaller uterus than expected.
- If **abortion is a possible diagnosis**, identify and treat any complications immediately

(table 6.2.).

- Other causes include: Molar pregnancy, coincidental local genital lesions of the cervix, vagina or vulva (e.g. cervical cancer).

TABLE 6.1.:-Diagnosis of vaginal bleeding in early pregnancy

Presenting Symptoms and Signs Present	Other Symptoms and Signs Typically Present	Sometimes Probable Diagnosis
• Light^a bleeding • Closed cervix • Uterus corresponds to dates	• Cramping/lower abdominal pain • Uterus softer than normal	Threatened abortion
• Light bleeding • Abdominal pain • Closed cervix • Uterus slightly larger than normal • Uterus softer than normal	• Fainting • Tender adnexal mass • Amenorrhoea • Cervical motion tenderness	Ectopic pregnancy
• Light bleeding • Closed cervix • Uterus smaller than dates • Uterus softer than normal	• Light cramping/lower abdominal pain • History of expulsion of products of conception	Complete abortion,
• Heavy^b bleeding • Dilated cervix • Uterus corresponds to dates	• Cramping/lower abdominal pain • Tender uterus • No expulsion of products of conception	Inevitable abortion
• Heavy bleeding • Dilated cervix • Uterus smaller than dates	• Cramping/lower abdominal pain • Partial expulsion of products of conception	Incomplete abortion
• Heavy bleeding • Dilated cervix • Uterus larger than dates • Uterus softer than normal • Partial expulsion of products of conception which resemble grapes	• Nausea/vomiting • Spontaneous abortion • Cramping/lower abdominal pain • Ovarian cysts (easily ruptured) • Early onset pre-eclampsia • No evidence of a fetus	Molar pregnancy

^a Light bleeding: takes longer than 5 minutes for a clean pad or cloth to be soaked.

^b Heavy bleeding: takes less than 5 minutes for a clean pad or cloth to be soaked.

2.2. THREATENED ABORTION

- Medical treatment is usually not necessary.
- Advise the woman to avoid strenuous activity and sexual intercourse.
- If **bleeding stops**, follow up in antenatal clinic. Reassess if bleeding recurs.
- If **bleeding persists**, assess for fetal viability (pregnancy test/ultrasound) or ectopic pregnancy (ultrasound). Persistent bleeding, particularly in the presence of a uterus larger than expected, may indicate twins or molar pregnancy.

2.3. INEVITABLE ABORTION

- If **pregnancy is less than 12 weeks**, plan for evacuation of uterine contents. If **evacuation is not immediately possible**:
 - Give ergometrine 0.2 mg IM (repeated after 15 minutes if necessary) OR misoprostol 400 mcg by mouth (repeated once after 4 hours if necessary);
 - Arrange for evacuation of uterus as soon as possible.
- If **pregnancy is greater than 12 weeks**:
 - Await spontaneous expulsion of products of conception and then evacuate the uterus to remove any remaining products of conception;
 - If necessary, infuse oxytocin 40 units in 1 L IV fluids (normal saline or Ringer's lactate) at 40 drops per minute to help achieve expulsion of products of conception.
- Ensure follow-up of the woman after treatment.

2.4. INCOMPLETE ABORTION

- If **bleeding is light to moderate** and **pregnancy is less than 12 weeks**, use fingers or ring (or sponge) forceps to remove products of conception protruding through the cervix.
- If **bleeding is heavy** and **pregnancy is less than 12 weeks**, evacuate the uterus:
 - Manual vacuum aspiration is the preferred method of evacuation. Evacuation by sharp curettage should only be done if manual vacuum aspiration is not available;
 - If **evacuation is not immediately possible**, give ergometrine 0.2 mg IM (repeated after 15 minutes if necessary) OR misoprostol 400 mcg orally (repeated once after 4 hours if necessary).
- If **pregnancy is greater than 12 weeks**:
 - Infuse oxytocin 40 units in 1 L IV fluids (normal saline or Ringer's lactate) at 40 drops per minute until expulsion of products of conception occurs;
 - If necessary, give misoprostol 200 mcg vaginally every 4 hours until expulsion, but do not administer more than 800 mcg;
 - Evacuate any remaining products of conception from the uterus.
- Ensure follow-up of the woman after treatment (see below).

Table 6.2. :- Diagnosis and management of complications of abortion

Symptoms and Signs	Complication	Management
• Lower abdominal pain • Rebound tenderness • Tender uterus • Prolonged bleeding • Malaise • Fever • Foul-smelling vaginal discharge • Purulent cervical discharge • Cervical motion tenderness	Infection/sepsis	Begin antibiotics ^a as soon as possible before attempting manual vacuum aspiration.
• Cramping/abdominal pain • Rebound tenderness • Abdominal distension • Rigid (tense and hard) abdomen • Shoulder pain • Nausea/vomiting • Fever	Uterine, vaginal or bowel injuries	Consult or refer to a higher facility urgently.

^a Give ampicillin 2 g IV every 6 hours PLUS gentamicin 5 mg/kg body weight IV every 24 hours PLUS metronidazole 500 mg IV every 8 hours until the woman is fever-free for 48 hours.

2.5. COMPLETE ABORTION

- Evacuation of the uterus is usually not necessary.
- Observe for heavy bleeding.
- Ensure follow-up of the woman after treatment (see below).

2.6.FOLLOW-UP OF WOMEN WHO HAVE HAD ANABORTION

Before discharge, tell a woman who has had a spontaneous abortion that spontaneous abortion is common and occurs in at least 15% (one in every seven) of clinically recognized pregnancies. Also reassure the woman that the chances for a subsequent successful pregnancy are good unless there has been sepsis or a cause of the abortion is identified that may have an adverse effect on future pregnancies (this is rare).

Some women may want to become pregnant soon after having an incomplete abortion. The woman should be encouraged to delay the next pregnancy until she is completely recovered.

It is important to counsel women who have had an unsafe abortion. If **pregnancy is not desired**, certain methods of family planning can be started immediately (within 7 days) provided:

- There are no severe complications requiring further treatment;
- The woman receives adequate counseling and help in selecting the most appropriate family planning method.

Table 6.3. :-Family planning methods

Type of Contraceptive	Advise to Start
Hormonal (pills, injections, implants)	• Immediately
Condoms	• Immediately
Intrauterine device (IUD)	• Immediately • If infection is present or suspected, delay insertion until it is cleared • If Hb is less than 7 g/dL, delay until anemia improves • Provide an interim method (e.g. condom)
Voluntary tubal ligation	• Immediately • If infection is present or suspected, delay surgery until it is cleared • If Hb is less than 7 g/dL, delay until anemia improves • Provide an interim method (e.g. condom)

Also identify any other reproductive health services that a woman may need. For example some women may need:

- Tetanus prophylaxis or tetanus booster;
- Treatment for sexually transmitted diseases (STDs);
- Cervical cancer screening.
- HIV testing and counseling

2.7. ECTOPIC PREGNANCY

An ectopic pregnancy is one in which implantation occurs outside the uterine cavity. The fallopian tube is the most common site of ectopic implantation (greater than 90%). Symptoms and signs are extremely variable depending on whether or not the pregnancy has ruptured (table 2-10). The classic triad of symptoms ascribed to ectopic pregnancy is: amenorrhea (symptoms of early pregnancy) followed by vaginal bleeding and abdominal/ pelvic pain. Culdocentesis (cul-de-sac puncture) is an important tool for the diagnosis of ruptured ectopic pregnancy, but is less useful than a serum pregnancy test combined with ultrasonography. If **non-clotting blood is obtained**, begin immediate management.

IMMEDIATE MANAGEMENT

In ruptured ectopic pregnancy, resuscitation is immediately instituted (see section on shock) and Laparotomy is performed as soon as possible. Therefore if ectopic pregnancy is diagnosed/ suspected immediately start resuscitation and consult or refer for immediate laparotomy.

Table 6.4. :-Symptoms and signs of ruptured and unruptured ectopic pregnancy

Unruptured Ectopic Pregnancy	Ruptured Ectopic Pregnancy
• Symptoms of early pregnancy (irregular spotting or bleeding, nausea, swelling of breasts, bluish discoloration of vagina and cervix, softening of cervix, slight uterine enlargement, increased urinary frequency) • Abdominal and pelvic pain	• Collapse and weakness • Fast, weak pulse (110 per minute or more) • Hypotension • Hypovolaemia • Acute abdominal and pelvic pain • Abdominal distension^a • Rebound tenderness • Pallor

^a Distended abdomen with shifting dullness may indicate free blood.

2.8. MOLAR PREGNANCY

Molar pregnancy is characterized by an abnormal proliferation of chorionic villi. The diagnosis is certain if vesicles were passed. If the **diagnosis of molar pregnancy is certain**, refer immediately after resuscitation and providing supportive care.

UNIT 3 - VAGINAL BLEEDING IN LATER PREGNANCY AND LABOR

PROBLEMS:

- Vaginal bleeding after 28 weeks of pregnancy, Antepartum Hemorrhage (APH).
- Vaginal bleeding in labor before delivery.

CAUSES:

- Placental causes
 - Abruptio placentae
 - Placenta praevia
 - Rare causes: vasa praevia and other placental abnormalities
- Uterine rupture
- Local lesions of the cervix, vagina and vulva
- Indeterminate: cause not identified even after delivery and examining the placenta

DIAGNOSIS: (see table below)

Table 6.5.:—Diagnosis of antepartum haemorrhage

Presenting Symptom and Other Symptoms and Signs Typically Present	Symptoms and Signs Sometimes Present	Probable Diagnosis
• Bleeding after 28 weeks gestation (may be retained in the uterus) • Intermittent or constant abdominal pain	• Shock • Tense/tender uterus • Decreased/absent fetal movements • Fetal distress or absent fetal heart sounds	Abruptio placentae,
• Bleeding (intra-abdominal and/or vaginal) • Severe abdominal pain (may decrease after rupture)	• Shock • Abdominal distension/ free fluid • Abnormal uterine contour • Tender abdomen • Easily palpable fetal parts • Absent fetal movements and fetal heart sounds • Rapid maternal pulse	Ruptured uterus,
• Bleeding after 28 weeks gestation	• Shock • Bleeding may be precipitated by intercourse • Relaxed uterus • Fetal presentation not in pelvis/ lower uterine pole feels empty • Normal fetal condition	Placenta praevia,

GENERAL MANAGEMENT:

- **SHOUT FOR HELP.** Urgently mobilize all available personnel.
- Make a rapid evaluation of the general condition of the woman including vital signs (pulse, blood pressure, respiration, temperature).

Note:- Vaginal or rectal examination should not be done as it can aggravate bleeding.

- If **shock is suspected**, immediately begin treatment. Even if signs of shock are not present, keep shock in mind as you evaluate the woman further because her status may worsen rapidly.
- Start an IV infusion and infuse IV fluids.
- All cases of antepartum hemorrhage should be managed in a facility with set up for cesarean delivery; therefore **refer urgently** while continuing resuscitation and supportive care.
- If the diagnosis is ruptured uterus (in addition to the above care and before referral):-
 - Catheterize bladder
 - Give loading dose of triple antibiotic
 - Ampicillin 2 gm IV stat
 - Metronidazole 500mg or Chloramphenicol 2gm IV stat, and
 - Gentamycin 80mg IV stat

UNIT 4 - VAGINAL BLEEDING AFTER CHILDBIRTH (POSTPARTUM HEMORRHAGE)

4.1. INTRODUCTION

Postpartum hemorrhage (PPH) is defined as vaginal bleeding in excess of 500 ml after childbirth (Or in excess of 1000 ml after Cesarean section. There are, however, some problems with this definition:

- Estimates of blood loss are notoriously low, often half the actual loss. Blood is mixed with amniotic fluid and sometimes with urine. It is dispersed on sponges, towels and linens, in buckets and on the floor.
- The importance of a given volume of blood loss varies with the woman's hemoglobin level. A woman with a normal hemoglobin level will tolerate blood loss that would be fatal for an anaemic woman.

Even healthy, non-anaemic women can have catastrophic blood loss.

- Bleeding may occur at a slow rate over several hours and the condition may not be recognized until the woman suddenly enters shock.

Therefore, PPH can also be defined as any amount of vaginal bleeding after child birth that is sufficient enough to make the patient symptomatic &/or results in signs of hypovolemia.

Risk assessment in the antenatal period does not effectively predict those women who will have PPH. Active management of the third stage should be practiced on all women in labor since it reduces the incidence of PPH due to uterine atony. All postpartum women must be closely monitored to determine those that have PPH.

- Increased vaginal bleeding within the first 24 hours after childbirth is called immediate or primary PPH.
- Increased vaginal bleeding following the first 24 hours after childbirth is called delayed PPH.

More than half of all maternal deaths occur within 24 hours of childbirth and immediate/primary PPH from uterine atony is the major cause contributing for more than one-quarter of all maternal deaths worldwide.

CAUSES:

- Atonic uterus
- Genital trauma
- Retained placenta
- Coagulation failure
- Acute inversion of the uterus

See table 6.6.

GENERAL MANAGEMENT:

- SHOUT FOR HELP. Urgently mobilize all available personnel.
- Make a rapid evaluation of the general condition of the woman including vital signs (pulse, blood pressure, respiration, temperature).
- **If shock is suspected**, immediately begin treatment. Even if signs of shock are not present, keep shock in mind as you evaluate the woman further because her status may worsen rapidly. If shock develops, it is important to begin treatment immediately.
- Massage the uterus to expel blood and blood clots. Blood clots trapped in the uterus will inhibit effective uterine contractions.
- Give oxytocin 10 units IM.
- Start an IV infusion and infuse IV fluids. Establish two IV lines if necessary.
- Take blood (5 mL) for hemoglobin (Hg)/ hematocrit (Hct) and crossmatching.
- Catheterize the bladder.
- Check to see if the placenta has been expelled and examine the placenta to be certain it is complete.
- Examine the cervix, vagina and perineum for tears.
- Provide specific treatment for the specific cause identified (see below).
- After bleeding is controlled (24 hours after bleeding stops), determine haemoglobin or hematocrit to check for anemia:
 - If haemoglobin is below 7 g/dL or hematocrit is below 20% (severe anemia):
 - Treat with iron if clinically stable.
 - Refer immediately for transfusion if signs and symptoms of de-compensation develop.
 - If haemoglobin is between 7–11 g/dL, treat with iron.
 - Where hookworm is endemic (prevalence of 20% or more), give one of the following anthelmintics:
 - Albendazole 400 mg PO stat
 - Mebendazole 500 mg PO stat or 100mg PO bid for three days.
 - If hookworm is highly endemic (prevalence of 50% or more), repeat the anthelmintic treatment 12 weeks after the first dose.

DIAGNOSIS:

For diagnosis of vaginal bleeding after childbirth use Table 7.6. below.

Table 6.6.: -Diagnosis of vaginal bleeding after childbirth

Presenting Symptom and Other Symptoms and Signs Typically Present	Symptoms and Signs Sometimes Present	Probable Diagnosis
• Immediate PPH^a • Uterus soft and not contracted	• Shock	Atonic uterus,
• Immediate PPH^a	• Complete placenta • Uterus contracted	Tears of cervix, vagina or perineum,
• Placenta not delivered within 30 minutes after delivery	• Immediate PPH^a • Uterus contracted	Retained placenta,
• Portion of maternal surface of placenta missing or torn membranes with vessels	• Immediate PPH^a • Uterus contracted	Retained placental fragments,
• Uterine fundus not felt on abdominal palpation • Slight or intense pain	• Inverted uterus apparent at vulva • Immediate PPH^b	Inverted uterus,
• Bleeding occurs more than 24 hours after delivery • Uterus softer and larger than expected for elapsed time since delivery	• Bleeding is variable (light or heavy, continuous or irregular) and foul-smelling • Anemia	Delayed PPH,
• Immediate PPH^a (bleeding is intra-abdominal and/or vaginal) • Severe abdominal pain (may decrease after rupture)	• Shock • Tender abdomen • Rapid maternal pulse	Ruptured uterus,

^a Bleeding may be light if a clot blocks the cervix or if the woman is lying on her back.

^b There may be no bleeding with complete inversion.

4.2. ATONIC UTERUS

An atonic uterus fails to contract after delivery. Atonic uterus is the most common cause of primary PPH.

- Continue to massage the uterus.
- Use oxytocic drugs which can be given together or sequentially (Table 6.7.).

Table 6.7. Use of medicines in the management of PPH

	Dose and route	Continuing dose	Maximum dose	Precautions/ Contra- indications
Oxytocin	IV: Infuse 20 units in 1 L IV fluids at 60 drops per minute IM: 10 units	IV: Infuse 20 units in 1 L IV fluids at 40 drops per minute	Not more than 3 L of IV fluids containing oxytocin	Do not give as an IV bolus
Ergometrine/ Methyl- ergometrine	IM or IV (slowly): 0.2 mg	Repeat 0.2 mg IM after 15 minutes If required, give 0.2 mg IM or IV (slowly) every 4 hours	5 doses (Total 1.0 mg)	Pre-eclampsia, hypertension, heart disease, retained placenta
MISOPROSTOL (CYTOTECH)	ORAL/SL INTRAVAG RECTAL 800mcg	200 -800mcg Every 4 hours	Not more than 1600 mcg	Asthma Heart Dis*
Tranexamic acid:	IV (slowly) 1 g	Repeat after 30 minutes if bleeding continues	Not more than 10 mg per kg body weight, three to four times daily	History of coagulopathy or active intravascular clotting, convulsions

- If bleeding continues:
 - o Check placenta again for completeness;
 - o If there are signs of retained placental fragments (absence of a portion of maternal surface or torn membranes with vessels), remove remaining placental tissue;
 - o Assess clotting status using a bedside clotting test. Failure of a clot to form after 7 minutes or a soft clot that breaks down easily suggests coagulopathy.
- If bleeding continues in spite of management above:
 - o Perform bimanual compression of the uterus (Fig 6.1.):
 - Wearing high-level disinfected gloves, insert a hand into the vagina and form a fist;
 - Place the fist into the anterior fornix and apply pressure against the anterior wall of the uterus;
 - With the other hand, press deeply into the abdomen behind the uterus, applying pressure against the posterior wall of the uterus;

- Maintain compression until bleeding is controlled and the uterus contracts.
- o Alternatively, compress the aorta (fig6.2.):
 - Apply downward pressure with a closed fist over the abdominal aorta directly through the abdominal wall:

→ The point of compression is just above the umbilicus and slightly to the left;

→ Aortic pulsations can be felt easily through the anterior abdominal wall in the immediate postpartum period.

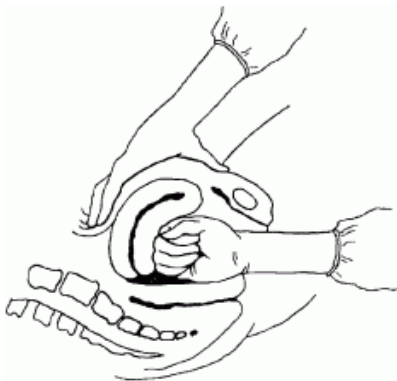
- With the other hand, palpate the femoral pulse to check the adequacy of compression:

→ If the pulse is palpable during compression, the pressure exerted by the fist is inadequate;

→ If the femoral pulse is not palpable, the pressure exerted is adequate;

- Maintain compression until bleeding is controlled.

Figure 6.1.:-Bimanual compression of the uterus **Figure 6.2.:-**Compression of Abdominal aorta and palpation of femoral pulse



Packing the uterus is ineffective and wastes precious time.

- If **bleeding continues** in spite of compression: refer immediately accompanying her and continuing providing the above measures or perform intrauterine balloon tamponade (See below)

INTRAUTERINE BALLOON TAMPONADE

• If **bleeding continues** in spite of bimanual and aortic compression, perform intrauterine balloon tamponade (**Fig. 6.3. and 6.4.**)

- When a balloon catheter designed specifically for treatment of PPH is not available, low-cost adaptations such as a condom balloon tamponade can function as a substitute.

- Before performing the procedure for condom balloon tamponade:

- Provide emotional support and encouragement.
- Insert an indwelling Foley catheter into the bladder.

• Procedure for condom balloon tamponade:

- Apply antiseptic solution to the perineal area and vagina.
- Prior to commencing procedure, secure a condom to the end of a Foley catheter.
- Perform a vaginal examination to identify the cervix. - Gently insert a high-level disinfected or sterile speculum into the vagina.
- Gently grasp the anterior lip of the cervix with a ring or sponge forceps.

Note: A ring or sponge forceps is preferable, as it is less likely than the tenaculum to tear the cervix.

- Hold the catheter with a high-level disinfected or sterile forceps and gently introduce it through the cervix. Ensure that the inflatable bulb or catheter is beyond the internal cervical os.

- Once the balloon end has been placed in the uterine cavity, inflate the condom with 300–500 mL of warm saline solution.

- Inflate the condom until it is visible in the cervix. Beware of overfilling the balloon as this might cause the balloon to bulge out of the cervix and be expelled.

- If there is **no bleeding through the cervix**, the tamponade test is positive. No further fluid is added and further surgery is not required at this stage.

- If **bleeding continues**, consider surgical interventions.
- Pack the upper vagina with roller gauze to prevent expulsion of the balloon.
- Palpate the uterine fundus abdominally and mark with a pen.

Note: This mark is used as a reference line from which any uterine enlargement or distension would be noted during the period of observation.

- Give oxytocin 20 units in 1 L IV fluids (normal saline or Ringer's lactate) at 60 drops per minute.

- A single dose of antibiotics (ampicillin or first-generation cephalosporin) is recommended:
 - ampicillin 2 g IV;
 - OR cefazolin 1 g IV.

Post-procedure care:

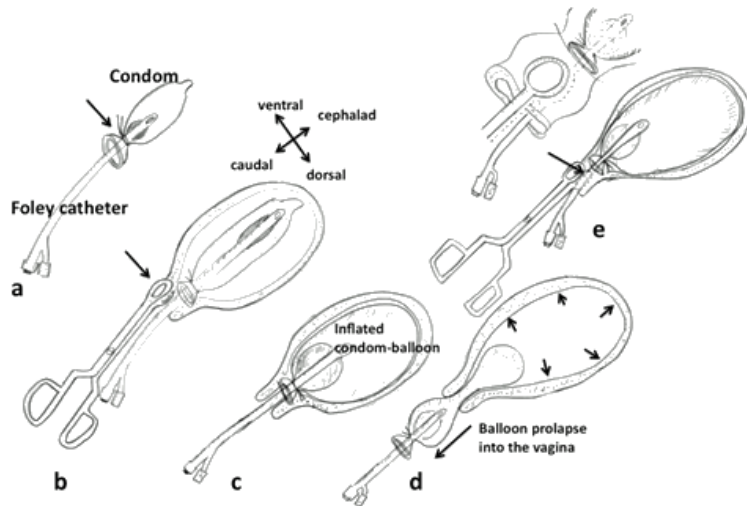
- Monitor vital signs, uterine fundal height and vaginal bleeding.
- Monitor urine output every hour.

- After 6–24 hours, if the uterus fundus remains at the same level and there is no active vaginal bleeding, deflate the balloon 50–100 mL every hour as long as there is no further bleeding at each interval.
- If **there is no further vaginal bleeding** 30 minutes after the balloon is totally deflated, remove the balloon and stop the oxytocin.
- If the **woman starts to bleed when the balloon is deflated or the oxytocin has stopped**, reinflate the balloon and recommence the oxytocin infusion; prepare for surgical interventions when her condition becomes stable.
- Remove the urinary catheter once the woman is stable.

Figure 6.3.:-Materials needed for uterine condom balloon tamponade



Figure 6.4.: - Steps of uterine condom balloon tamponade



4.3. TEARS OF CERVIX, VAGINA OR PERINEUM

Tears of the birth canal are the second most frequent cause of PPH. Tears may coexist with atonic uterus. Postpartum bleeding with a contracted uterus is usually due to a cervical or vaginal tear.

- Examine the woman carefully and repair tears to the cervix or vagina and perineum.
- If **bleeding continues**, assess clotting status using a bedside clotting test. Failure of a clot to form after 7 minutes or a soft clot that breaks down easily suggests coagulopathy.
 - If coagulopathy is diagnosed refer urgently accompanying her and providing the above basic support.

4.4. RETAINED PLACENTA

There may be no bleeding with retained placenta.

- If you can see the placenta, ask the woman to push it out. If you can feel the placenta in the vagina, remove it.
- Ensure that the bladder is empty. Catheterize the bladder, if necessary.
- If the **placenta is not expelled and no active bleeding**, give oxytocin 10 units IM if not already done for active management of the third stage, attempt controlled cord traction.

Note: Avoid forceful cord traction and fundal pressure as they may cause uterine inversion.

Do not give ergometrine because it causes tonic uterine contraction, which may delay expulsion.

- If there is active bleeding or the **placenta is undelivered after 30 minutes of oxytocin stimulation** and **controlled cord traction**, attempt manual removal of placenta.
 - If placenta is delivered and uterus well contracted, closely monitor and continue with frequent rubbing of the uterus
 - In case the placenta is delivered but the uterus remains hypotonic and bleeding continues, manage as atonic uterus.
- If **there are signs of infection** (fever, foul-smelling vaginal discharge), give antibiotics as for metritis.

4.5. RETAINED PLACENTAL FRAGMENTS

There may be no bleeding with retained placental fragments.

When a portion of the placenta—one or more lobes—is retained, it prevents the uterus from contracting effectively.

- Feel inside the uterus for placental fragments. Manual exploration of the uterus is similar to the technique described for removal of the retained placenta.
- Remove placental fragments by hand, ovum forceps or large curette.

Note: Very adherent tissue may be placenta accreta. Efforts to extract fragments that do not separate easily may result in heavy bleeding or uterine perforation, therefore stop the procedure and refer urgently providing supportive care.

4.6. INVERTED UTERUS

The uterus is said to be inverted if it turns inside-out during delivery of the placenta. Repositioning the uterus should be performed immediately. With the passage of time the constricting ring around the inverted uterus becomes more rigid and the uterus more engorged with blood.

- If the **woman is in severe pain**, give pethidine 1 mg/kg body weight (but not more than 100 mg) IM or IV slowly or give morphine 0.1 mg/kg body weight IM.

Note: Do not give oxytocic drugs until the inversion is corrected.

- Immediately after diagnosis, it is often possible to replace the uterus by applying gentle trans-vaginal pressure. The Johnson technique calls for lifting the uterus and the cervix into the abdominal cavity with the fingers in the fornix and the inverted uterine fundus on the palm. The fundus is then gently pushed back through the cervix. The operator's hand should be kept in the uterus until the fundus begins to climb up. If the placenta is still attached, it should not be removed until after the uterus is replaced through the cervix.

- If vaginal replacement is unsuccessful (as may occur in delayed recognition and treatment), refer urgently providing basic support.

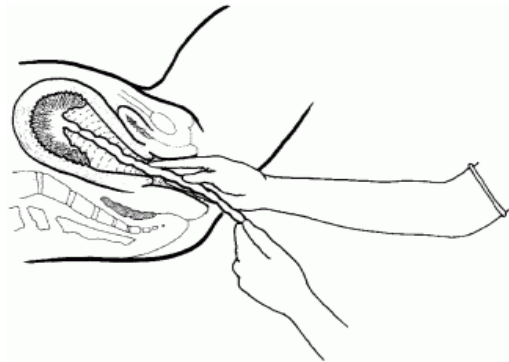
- Give a single dose of prophylactic antibiotics after correcting the inverted uterus:

- Ampicillin 2 g IV;
- OR cefazolin 1 g IV.
- If **there are signs of infection** (fever, foul-smelling vaginal discharge), give antibiotics as for metritis.
- If **necrosis is suspected**, referral to a tertiary care center.

4.7. MANUAL REMOVAL OF PLACENTA

- Review for indications.
- Review general care principles and start an IV infusion.
- Provide emotional support and encouragement. Give pethidine and diazepam IV slowly (do not mix in the same syringe) or use Ketamine.
- Give a single dose of prophylactic antibiotics:
 - Ampicillin 2 g
 - OR cefazolin 1 g IV.
- Hold the umbilical cord with a clamp. Pull the cord gently until it is parallel to the floor.
- Wearing high-level disinfected long sleeve gloves or similar modification, insert a hand into the vagina and up into the uterus. (Figure 6.6.)

Figure 6.6.: -Introducing one hand into the vagina along cord

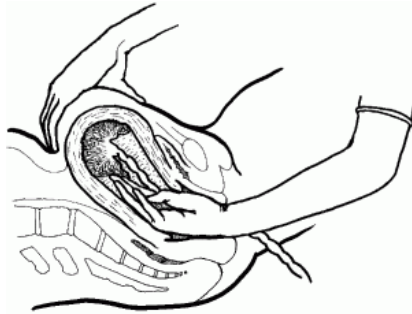


- Let go of the cord and move the hand up over the abdomen in order to support the fundus of the uterus and to provide counter-traction during removal to prevent inversion of the uterus.

Note: If **uterine inversion occurs**, reposition the uterus.

- Move the fingers of the hand laterally until the edge of the placenta is located.
- If the **cord has been detached previously**, insert a hand into the uterine cavity. Explore the entire cavity until a line of cleavage is identified between the placenta and the uterine wall.

Figure 6.7.: Supporting the fundus while detaching the placenta

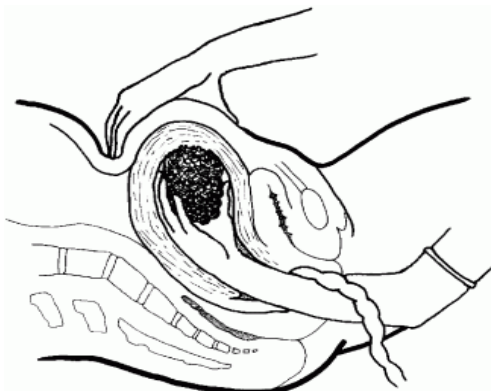


- Detach the placenta from the implantation site by keeping the fingers tightly together and using the edge of the hand to gradually make a space between the placenta and the uterine wall.
- Proceed slowly all around the placental bed until the whole placenta is detached from the uterine wall.

NOTE: If the **placenta does not separate from the uterine surface** by gentle lateral movement of the fingertips at the line of cleavage, suspect adherent placenta, stop the procedure and urgently refer providing supportive care.

- Hold the placenta and slowly withdraw the hand from the uterus, bringing the placenta with it.
- With the other hand, continue to provide counter-traction to the fundus by pushing it in the opposite direction of the hand that is being withdrawn.

Figure 6.8.: -Withdrawing the hand from the uterus



- Palpate the inside of the uterine cavity to ensure that all placental tissue has been removed.
- Give oxytocin 20 units in 1 L IV fluids (normal saline or Ringer's lactate) at 60 drops per minute.
- Have an assistant massage the fundus of the uterus to encourage a tonic uterine contraction.

- If there is **continued heavy bleeding**, give ergometrine 0.2 mg IM or prostaglandins.
- Examine the uterine surface of the placenta to ensure that it is complete. If any **placental lobe or tissue is missing**, explore the uterine cavity to remove it.
- Examine the woman carefully and repair any tears to the cervix or vagina, or repair episiotomy.

POST-PROCEDURE CARE

- Observe the woman closely in labor ward or where she can be monitored closely for at least 6 hours or until stable.
- Check and rub the uterus every 15 minute for the next two hours.
- Monitor the vital signs (pulse, blood pressure, respiration) every 30 minutes for the next 6 hours
- Continue with IV fluid and oxytocin drip for next 4-6 hours.
- Continue infusion of IV fluids.
- If patient is stabilized, assist her to initiate breast feeding if appropriate

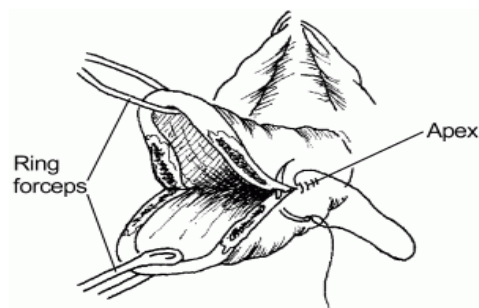
4.8. REPAIR OF GENITAL TEARS

4.8.1. REPAIR OF CERVICAL TEARS

- Review general care principles and apply antiseptic solution to the vagina and cervix.
- Provide emotional support and encouragement. Anesthesia is not required for most cervical tears. For tears that are high and extensive, give pethidine and diazepam IV slowly (do not mix in the same syringe).
- Ask an assistant to massage the uterus and provide fundal pressure.
- Gently grasp the cervix with ring or sponge forceps. Apply the forceps on both sides of the tear and gently pull in various directions to see the entire cervix. There may be several tears.
- Close the cervical tears with continuous 0 chromic catgut (or polyglycolic) suture starting at the apex (upper edge of tear), which is often the source of bleeding.
- If a **long section of the rim of the cervix is tattered**, under-run it with continuous 0 chromic catgut (or polyglycolic) suture.
- If the **apex is difficult to reach and ligate**, it may be possible to grasp it with artery or ring forceps. Leave the forceps in place for 4 hours. Do not persist in attempts to ligate the bleeding points as such attempts may increase the bleeding. Then:
 - After 4 hours, open the forceps partially but do not remove;
 - After another 4 hours, remove the forceps completely.

If the apex of the cervical tear is not visible, the tear may have involved the uterus. A laparotomy may be required to repair a cervical tear that has extended deep beyond the vaginal vault. Hence, in such condition, refer urgently or consult as appropriate.

Figure 6.9.:-Repair of a cervical tear



4.8.2. REPAIR OF VAGINAL AND PERINEAL TEARS

Refer to Module 5 (section: repair of vaginal & perineal tears)

4.9. DELAYED (“SECONDARY”) POSTPARTUM HAEMORRHAGE

Secondary PPH (2^o PPH) is abnormal bleeding occurring from the second day to the sixth week postpartum.

CAUSES:

Bleeding in the first 3 weeks after the first day of delivery is mainly due to:

- Sub-involution of the uterus
- Infection
- Retained pieces of placental tissue or clot
- Breakdown of the uterine wound after Cesarean delivery or ruptured uterus
- After obstructed labor, bleeding may occur due to sloughing of dead vaginal tissue (cervix, vagina, bladder, and rectum).

- Bleeding from the third week to sixth week is mainly due to:

- Choriocarcinoma (rare)
- Local causes (vaginal or cervical) such as severe infection, malignancies, trauma
- Early onset menstruation

MANAGEMENT:

- Usually sub-involution of the uterus, intrauterine infection and retained pieces of placental tissue are considered as a triad of causes for bleeding in the first two weeks especially from

the 5th to 10th day postpartum. All the three conditions are treated together:

- Oxitocics in drip or ergometrine (PO 1 tablet twice a day for 2-3day)
- Semi-sitting position to enhance drainage from the uterus
- Antibiotics against common organisms of the vagina
- If the **cervix is dilated**, explore by hand to remove large clots and placental fragments. *Manual exploration of the uterus is similar to the technique described for removal of the retained placenta.*
- If the **cervix is not dilated**, evacuate the uterus to remove placental fragments. using ‘finger curettage’, vacuum aspiration or large blunt curette
- Treat anemia and shock as appropriate

NOTE: Very rarely, bleeding persists and needs referral to a higher facility.

- When bleeding occurs late in the postpartum period, (3rd - 6th week), pregnancy test has to be performed to rule out choriocarcinoma. Specimen of uterine evacuation must be submitted for histological examination.
- In non-breast feeding woman, it may be the first menstrual bleeding postpartum. Local lesion of the vagina or cervix are excluded by speculum examination.
- If **anemia is severe** (haemoglobin less than 7 g/dL or hematocrit less than 20%), provide oral iron and folic acid and consider referral for transfusion.
- If **there are signs of infection** (fever, foul-smelling vaginal discharge), give antibiotics as for metritis.

Prolonged or delayed PPH may be a sign of metritis.

4.10 . NON-PNEUMATIC ANTI-SHOCK GARMENT (NASG)

The Non-Pneumatic Anti-Shock Garment (NASG) is a lightweight, relatively inexpensive, washable neoprene suit composed of articulated horizontal segments with three segments on each leg, one segment over the pelvis and another, over the abdomen, which includes a foam compression ball (Figure 1). Using the three-way elasticity of neoprene and the tight closure of the Velcro, the garment applies 20–40 mm Hg circumferential counter-pressure to the lower body to reverse hypovolemic shock by shunting blood to the vital core organs.

The (NASG) is a new invention and first-aid equipment which helps to stabilize the woman circulatory system by shunting blood from the peripheral parts for the body to the central part and helps to buy time while she is being transported to emergency obstetric care center for definitive treatment.

A key component of managing hypovolemic shock is ensuring that the available blood in the body is directed mostly to the upper body so that the vital organs (heart, lungs, kidneys, and brain) continue to receive oxygen and the woman is protected from vital organ damage and death. One way to achieve this is to place the woman in Trendelenburg position, where the head is lower than the feet and hips. A more effective way of ensuring this is to apply the NASG.

- The NASG is a lightweight (1.5 kg) compression suit made of neoprene with 6 segments that Close around the legs at the ankle, calf, thigh, pelvis, and abdomen.
- Velcro fastenings help keep the garment on tightly.
- The abdominal segment (#5) incorporates a small foam ball that applies pressure to the uterus to decrease bleeding.
- Markings on each section show how to apply it on a woman.
- When applied tightly by one person, the garment supplies enough circumferential pressure from the ankles to the diaphragm to reverse hypovolemic shock by shunting blood from the capacitance vessels of the abdomen and lower extremities to the vital core organs- heart, lungs, kidneys, and brain.
- It is washable and reusable (more than 100 times).
- The NASG can be easily packed into a bag for storage.

If available, apply a non-pneumatic anti-shock garment (NASG) as a temporizing measure until appropriate care is available. An NASG applies pressure to the lower body and abdomen, thereby stabilizing vital signs and resolving hypovolemic shock. Follow the manufacturer’s instructions below to apply and remove the NASG.

NASG’s Unique Role in Obstetric Hemorrhage and Hypovolemic Shock

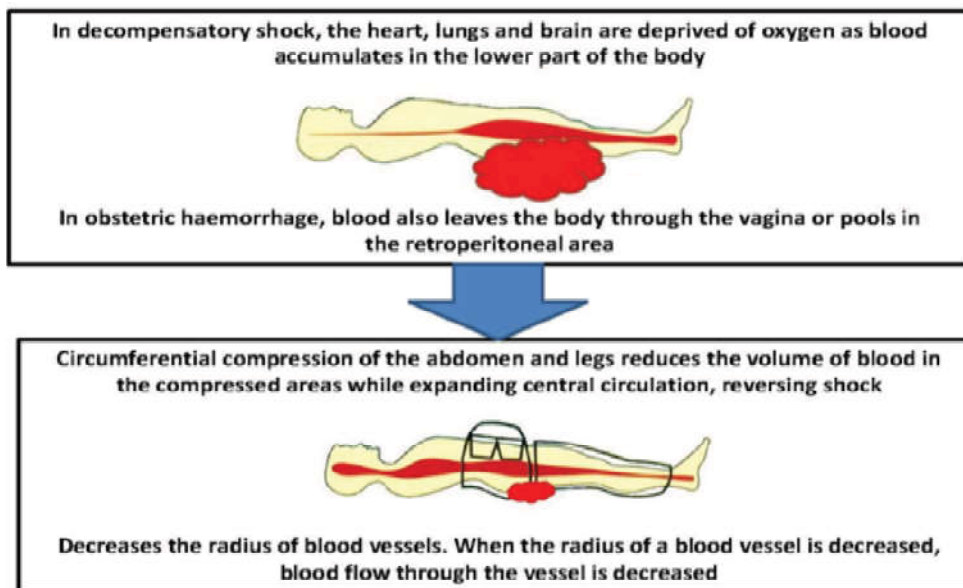
- Does **not** compete with other approaches: Not an either/or situation
- Used with hemorrhage therapies, uterotonics, massage, vaginal procedures, even surgeries,
- A technology that can be used when patient does not respond to uterotonics
- A woman can safely and comfortably wear the NASG for up to 48 hours.

- The NASG remains on the woman during treatments and until she has been hemodynamically stable for 2 hours.
- May help avoid unnecessary emergency hysterectomy for intractable uterine atony
- May decrease need for or number of units of blood transfusions

Mechanism of Action

NASG is a first-aid device that can reverse shock, decrease blood loss and stabilize women suffering hypovolemic shock secondary to severe obstetric hemorrhage

Figure 6.10.: -Mechanism of action of NASG



Effects of the NASG

- The NASG provides efficient, simple, and safe circumferential counter pressure
- Reduces hemorrhage in lower body
- Does not act like a tourniquet, it does not completely cut off blood supply to lower limbs
- Decreases arterial perfusion pressure to uterus, comparable to ligation of the internal iliac arteries
- Overcomes pressure in capillary and venous system (15-25 mmHg)
- Reduces trans mural pressure, vessel radius, and blood flow

Contraindication: Do not use the NASG with:

- A viable fetus (unless there is no other way to save the mother's life and both mother and fetus will die)
- Bleeding above the diaphragm
- Open thoracic wounds
- Pulmonary edema
- Congestive heart failure due to mitral stenosis
- A major cause of morbidity and mortality when using the NASG is the premature and inappropriate release of external counter pressure.

Benefits of NASG

- It stabilizes the patient while trying to look for the cause of bleeding.
- It keeps patient alive while transporting to a referral centres
- It may stop bleeding and thus prevent surgery.
- It may reduce or prevent blood transfusion and its associated problems.
- It can be applied by any trained health care personnel

When to Apply the NASG

- INITIAL CONTROL OF HEMORRHAGE
- SHOCK (When a woman shows signs of hypovolemic shock from obstetric hemorrhage)
 - Postpartum hemorrhage - any cause
 - Ectopic Pregnancy
 - Post cesarean hemorrhage
- Trauma with injury/hemorrhage below the diaphragm
- Applying the NASG before inserting an IV may improve access to veins
- Use the NASG along with standard treatment protocols (the NASG does not replace them)

RECOMMEND: Apply the ASG after 750 ml of blood loss, before signs of shock occur!

Figure 6.11. Non-pneumatic anti-shock garment



KEY:Components of NASG

- 1- Segment 1-Is part of NASG applied to the distal part of the lower leg starting from ankle
- 2- Segment 2- Is part of NASG applied to the proximal part of the lower leg around calf
- 3- Segment 3- Is part of NASG applied to thigh (below pubic bone and above knee)
- 4- Segment4- Is part of NASG applied to the area of pubic bone
- 5- Segment5- Is part of NASG applied over to umbilical area
- 6- Segment 6- Is part of NASG which is used to close segment 5

Application can be for

- General application
- Application for shorter women
- Application if the woman is unconscious

Steps of Application

General Application

Step 1

- To apply the NASG, place it under the woman; the top of the NASG should be at the level of her lowest rib.
- Starting at the ankles, close segment #1 tightly around each ankle.
- Make sure it is tight enough so that you can snap it and hear a sharp sound!

Step 2

- Next, close segment #2 on each leg as tightly as possible.
- Try to leave the woman's knee free in the space between segments so that she can bend her leg. She may be in the NASG for a long time.

Step 3

- Apply segments #3, the thigh segments, in the same way as segments #1 and #2.
- Remember: close segments tightly enough so that you can snap it and hear a sharp sound!

Step 4

- Segment #4, the pelvic segment, goes all the way around the woman at the level of the pubic bone.

Step 5

- Place segment #5 with the pressure ball directly over her umbilicus.
- Then close the NASG using segment #6. If there are two people present they can rapidly apply the three leg segments together, each working on one leg, starting at the ankle. However, only one person using as much strength as possible should close the pelvic and abdominal segments. If two people close the pelvic and abdominal segments they can apply too much pressure and compromise the patient's breathing. Do not close the segment so tightly that it restricts the woman's breathing. One person can sufficiently manage the whole application if necessary.

When Finished

- Make sure the patient can breathe normally with the NASG segment #6 in place.
- If the source of bleeding appears to be uterine atony, administer uterotonic drugs and massage the uterus. The NASG stretches, allowing room for your hand to fit between the woman's abdomen and the NASG.

Application for Short Women (if the very small garment is not available)

If a woman is shorter than those for whom the NASGs available are designed, a simple adjustment can be made so that the larger garment will still fit. To apply the NASG to a short woman:

Step 1

- If the woman is short, fold segment #1 to the inside of segment #2
- Begin segment #2 at her ankles.

Steps 2-4

- Apply segment #3 to the thighs, as usual. Continue with the rest of the segments as with all women.

Application if the Woman is Unconscious

You will need 2 people!

Applying the garment to an unconscious woman will require 2 people to position the garment beneath her. The final segment, however, should still be closed using only the strength of one person.

Step 1:

- Open the NASG on a flat surface and only open segments #4, #5, #6, keeping segments #1,#2, and #3, closed (but not fastened with the Velcro).

Step 2:

- Fold segments #4 and #6 (the sides that do NOT contain the ball) once towards the yellow midline dots (in toward the black side of the NASG). This will prevent the Velcro from sticking to other parts of the NASG and to the patient or bed linen.

Step 3

Turn segments #4 and #6 once more toward the yellow midline so that the folded edge lies along the yellow midline.

Step 4

- Take the folded segments #4 and #6 and turn them over towards the colored (maroon or blue) outside of the NASG placed approximately where the yellow midline is on the outside. This will divide the upper portion of the NASG in half along the dotted line.

Step 5

- Turn the woman on her side with her back facing you and place the folded NASG on the bed with the dotted line along the woman's spine and the top edge of the NASG at the level of her lowest rib. Push the folded/rolled sections #4 and #6 under her body.

Step 6

- Roll her towards you, turning her to her other side, over the rolled portions of the NASG. She is now facing you.
- The second person pulls the folded segments #4 and #6 out from under the woman.

Step 7

- Turn the woman on her back. She is now lying in the middle of the NASG, with the yellow dots along her spine, with the top edge of the NASG at the level of her lowest rib on the side. Check positioning by placing, but not closing, the #5 segment with the ball over her navel.

Step 8

- Starting at the ankles, close segments #1 tightly around each ankle. Remember: Make sure it is tight enough so that you can place a finger under the segment and snap the leg segments to hear a sharp sound.

Flowchart for Applying the NASG

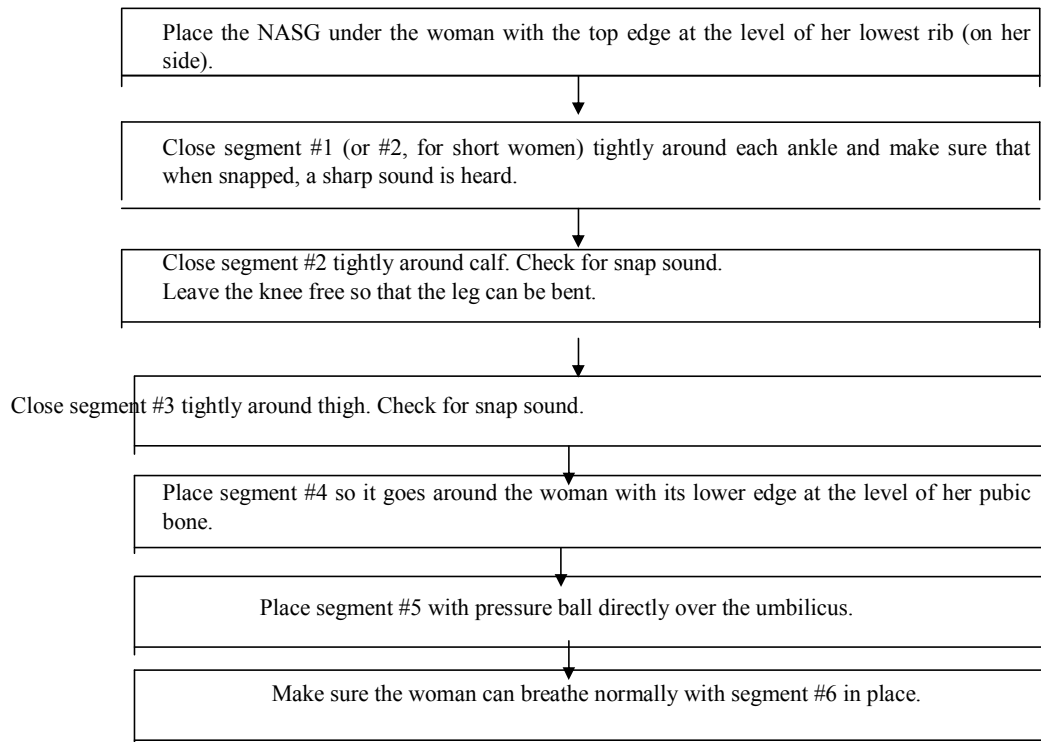


Figure 6.12. Non-pneumatic anti-shock garment after application



Removal of the NASG

Indications for Removal of NASG/When to remove the NASG

- Intravascular volume re-expanded with blood and saline
- Vital signs stable and systolic >100/pulse <100
- Hb>7.0 or Hct (PCV) >20
- Remove one segment every 15 minutes starting below with #1 - check pulse/BP

The NASG should only be removed:

- Under medical supervision,
- When the woman is stable, and
- According to the time line outlined below. Rapid removal of the NASG or removal of the segments in the wrong order can result in death.
- At Hospital level (where definitive management of obstetric heamorrhage is available)

Step 1

Begin removing the NASG only when the woman's condition has been stable for two hours:

- Bleeding has decreased to <50 ml/hour,
- Hemoglobin level is >7 or the hematocrit 20% (unless the woman's usual hemoglobin is less than 7% and hematocrit is less than 20%),
- Pulse <100 and systolic BP 90 mm/Hg or greater, and
- The woman is conscious and aware.

Step 2

- Wait 15 minutes for redistribution of blood to occur between removing each segment.
- Always wear gloves when handling a soiled garment.
- Removal of the NASG begins with the lowest segment (typically #1, #2 if the woman is short and segment #2 is at her ankles) and proceeds upwards.
- 15 minutes after removing the first segment and before proceeding to the next step, take her pulse and blood pressure to verify that she is ready for removal of the next segment.

Step 3

- If pulse and blood pressure stable 15 minutes later, remove the next section.

Steps 4 and 5

- After 15 minutes, take pulse and blood pressure. If stable, remove the next segment.
- Continue following this procedure—remove a segment, wait 15 minutes, take pulse and blood pressure—until all parts of the NASG are removed.

CAUTION: Rule of 20

If the blood pressure falls by 20 mmHg OR the pulse increases by 20 beats/min after a segment is removed:

- Rapidly replace ALL segments, and consider the need for more saline or blood transfusions.
- If there is recurrent bleeding, replace all segments of the NASG and determine the source of bleeding.

Avoiding Adverse Events when Using the NASG

- Only one person should apply the pelvic and abdominal sections of the NASG (even if the woman is unconscious and two people were required to begin apply the NASG).
- Monitor urine output.
- Ensure airway protection and aspiration prevention as required.
- Ensure one-on-one nursing care throughout.
- Ensure presence of a relative/support person with the unconscious patient, ready to explain the purpose of the garment when patient returns to consciousness and call for help. Surprised, confused, or frightened patients may attempt to remove the garment prematurely, resulting in death.
- Never open the abdominal panel first.

Performing Vaginal Procedures with the NASG On

Pelvic exams with the NASG On:

The design of the NASG permits complete perineal access. Thus, the source of most obstetric hemorrhages can be located and treated while the garment maintains the woman's vital signs.

The following vaginal procedures can be performed on a woman in an NASG:

- Repair of episiotomy or vaginal and cervical lacerations,
- Manual removal of the placenta,
- Bimanual compression (external or internal),
- Dilatation and curettage (D&C),
- Dilatation and evacuation (D&E), and

- Manual vacuum aspiration (MVA).

Abdominal Surgery with the NASG On

Surgery to obtain hemostasis can also be performed with the NASG in place. The abdominal and pelvic panels must be opened, but only immediately before the first incision. Such procedures may include:

- Cesarean section,
- Laparotomy,
- Repair of ruptured uterus,
- Laparoscopy,
- Hysterectomy,
- Removal of placenta accreta,
- Salpingectomy/salpingostomy,
- Repair of broad ligament hematoma, and
- Ligation of arteries, N
- B-Lynch or other uterine compression sutures.

Prepare the operating theatre for surgery, have all members of the surgical team scrubbed, gowned, gloved, in place, and ready to operate immediately prior to surgery. The woman should be catheterized; the anesthesiologist must be prepared to administer IV fluids to manage drop in blood pressure.

Step 1

- ✓ Remove ONLY segments #4, #5, and #6. With the abdominal portion removed, much of the benefit of the NASG is lost, and the patient may go back into shock.

Step 2

- ✓ Place the patient in steep Trendelenberg position.
- ✓ Operate as quickly as possible.

Step 3

- ✓ Replace segments #4, #5, and #6 as soon as the vaginal procedure is complete.

How to transport with NASG:

NASG the potential to keep women alive during long transports from lower level facilities or home to the CEmOC facilities for definitive therapies

When to remove the NASG

- The NASG should only be removed:
- Under medical supervision,
- When the woman is stable, and

- According to the time line given. Rapid removal of the
- NASG or removal of the segments in the wrong order can result in death.
- At Hospital level (where definitive management of obstetric hemorrhage is available)

N: B. With training, NASG application and removal is easy, but removal needs precaution as the shock could return.



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MODULE 7

POSTPARTUM MATERNAL (UP-TO 6 WEEKS) AND NEWBORN CARE

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UNIT-1 INTRODUCTION

The days and weeks following childbirth – the postnatal period – is a critical phase in the lives of mothers and newborn babies. It is an important transitional time for a woman, her newborn and her family, on a physiological, emotional and social level. Major changes occur during this period which determine the well-being of mothers and newborns.

Most maternal and infant deaths occur during this time. And almost half of all postnatal maternal deaths occur within the first 24 hours, and 66% occur during the first week. In 2016, of the 2.8 million newborns died in their first month of life – 1 million died on the first day.

Yet, this is the most neglected period for the provision of quality care. In both developing and developed countries rates of provision of skilled care are lower after childbirth when compared to rates before and during childbirth. Lack of appropriate care during this period could result in significant ill health and even death.

Skilled care and early identification of problems in both the mother and the newborn in the postnatal period could reduce the incidence of death and disability. Increased awareness of warning signals and appropriate intervention is needed at all levels. Poor quality care reduces opportunities for health promotion and for the early detection and adequate management of problems and disease.

This module describes the aims and standards of postpartum care for both the mothers and their newborns, based on the needs, evidences and challenges. It offers guidance on the way postpartum care could be organized. With respect to clinical problems, attention is focused on primary care, directed at the prevention, early diagnosis and treatment of common selected maternal and newborn diseases and complications, and at referral to hospital if necessary.

Enabling objectives

By the end of this module, the participants will be able to:

1. Provide basic care to the woman in post-partum period.
2. Explain the basic care of neonates presenting to the health facility in the postnatal period
3. Describe steps contained in the initial rapid assessment and emergency management of a sick neonate presenting to the health facility
4. Detect and provide care for sick neonates presenting to the health facility with illnesses and complications during the postnatal period.

UNIT-2: BASIC POST-PARTUM MATERNAL CARE

2.1. LEARNING OBJECTIVES

At the end of this unit, participants will be able to:

- Define post-partum period.
- Understand the value/significance of proper postnatal follow-up.
- Describe the basic components of post-partum care for all mothers and immediate post-partum care of their newborns.
- Describe problems and complications of the post-partum period for the mother and newborn.

2.2. BASICS OF POSTPARTUM PERIOD

2.2.1. THE POSTPARTUM PERIOD

The words "postpartum" and "postnatal" are sometimes used interchangeably. The postpartum period (also called the puerperium) is defined as the period from one hour after delivery of the placenta up to 6 weeks (42 days). The one hour interval is considered to be part of childbirth; during that time the immediate care of the mother (e.g. assessment of her condition, suturing, control of blood loss) and the infant (assessment of its condition, maintaining body temperature, initiating breastfeeding, etc.) take place. The period of 6 weeks fits very well into cultural traditions in many countries, where often the first 40 days after birth are considered a time of convalescence for the mother and her newborn infant.

In many countries at that time a routine postnatal visit and examination are planned. Six weeks after delivery, the body of the woman has largely returned to the non-pregnant state. The uterus and vagina have regained their pre-pregnant proportions. Physiological changes during pregnancy, such as increased cardiac output and blood volume, increased extracellular fluid (edema) and changes in the composition of the blood have subsided. The sudden disappearance of placental hormones after delivery, and the start of lactation have caused drastic endocrinological changes in the first weeks, but after six weeks a steady state has been reached. The psychosocial adaptation of the mother, the newborn and the family to the new situation usually has attained a new balance.

However, this does not mean that the pre-pregnant state has completely returned: lactation usually continues, often the menstrual cycle has not yet normalized, and sexual activity may not have been resumed yet. Contraception, though an important need, may be problematic for many couples at this time. For the infant the age of six weeks is not a decisive turning point in his or

her life, but the continuation or discontinuation of breastfeeding is directly related to the social and economic activities of the mother and her choice of contraceptive method. Although the main focus is on the first six weeks postpartum, it is fully recognized that the life of the woman and her baby is a continuum, and discussions will be extended to the following weeks and months where appropriate.

2.2.2. AIMS AND TIMING OF POSTPARTUM CARE

The postpartum period is a very special phase in the life of a woman and her newborn. For women experiencing childbirth for the first time, it marks probably the most significant and life-changing event they have yet lived. It is marked by strong emotions, dramatic physical changes, new and altered relationships and the assumption of and adjustment to new roles. It is a time of profound transition, making great demands on the woman's resilience and capacity to adapt. Postpartum care all too often does not incorporate all the essential elements required for the health of a woman or her newborn in a comprehensive package. Quality postpartum services are a long-term investment in the future health of women and their newborn.

The aims of care in the postpartum period are:

- Support of the mother and her family in the transition to a new family constellation, and response to their needs
- Prevention, early diagnosis and treatment of complications of mother and infant, including the prevention of vertical transmission of diseases from mother to infant
- Referral of mother and infant for specialist care when necessary
- Counseling on baby care
- Support of breastfeeding
- Counseling on maternal nutrition, and supplementation if necessary
- Counseling and service provision for contraception and the resumption of sexual activity
- Immunization of the infant.

Components of postpartum care:

1. Early detection and management of complications
2. Promoting health and preventing disease
3. Providing woman-centered education and counseling

1. Early detection and management of complications

The postpartum evaluation starts by reviewing the parturient clinical document including antepartum and intrapartum records. Complications such as cardiac disease, preeclampsia, obstructed labor, cesarean delivery that require close monitoring and treatment are identified in

the patient's medical document. Besides reviewing the clinical records, the parturient should be evaluated thoroughly during the immediate postpartum period, 3rd -6th day and 6th week postpartum visits. Generally, the postpartum evaluation of the *woman* during hospital discharge, or subsequent postpartum visits includes:

- How she feels; ask for pain, bleeding, difficulty in urination, breast feeding, any other concern she may have
- General physical
- Depression /psychosis.mood
- Malnutrition: general health, night blindness,goiter
- BP, pulse,temperature
- Anemia: conjunctiva/tongue/palms, hemoglobin (if necessary)
- Condition of the breast and nipple; establishment of proper breastfeeding
- Checking for bladder distension and urine passed, incontinence/fistula
- Fundal height and uterine consistency
 - The uterus involutes progressively:
 - After 5 to 7 days, it is firm and no longer tender, extending midway between the symphysis and umbilicus.
 - By the second week, it is no longer palpable abdominally.
- Inspecting the vaginal, perineum: episiotomy, lochia, bleeding, discharge, hematoma
- Thrombophlebitis: Homan's sign, inspection of legs

Usually, no specific tests are required in the postpartum period if all the basic investigations are done in the ANC and intrapartum care. In case these tests were not undertaken, they have to be done before discharge. Besides, tests required in the management of any complication have to be undertaken as required. If the HIV status of the parturient is unknown, offer her testing. After thorough evaluation and review of the parturient's antepartum and intrapartum records, parturients with complication are provided with emergency care and then specific treatment or referral for specialized care to higher centers. Women with normal conditions are managed in the labor ward or postpartum clinic and get the basic postpartum care.

The following education regarding complication and preparedness should be provided.

Danger signals for woman

- Sudden and profuse blood loss, persistent or increased blood loss
- Fainting, dizziness, palpitations
- Fever, shivering, abdominal pain, and/or offensive vaginal discharge
- Painful or hot breast(s)

- Abdominal pain
- Calf pain, redness or swelling
- Shortness of breath or chest pain
- Excessive tiredness
- Severe headaches accompanied, visual disturbances
- Edema in hands and face

Danger signals for newborn

- Cord red or draining pus
- Suckling poorly
- Eyes swollen, sticky or draining pus
- Cold to touch in spite of re-warming
- Hot to touch in spite of undressing
- Difficulty breathing
- Lethargy
- Convulsions

Complication readiness

- Establish savings plan/scheme
- Make plan for decision-making
- Arrange system of transport
- Establish plan for blood donation

2. Promoting health and preventing disease

- Iron/ folate: 1 tablet to be taken by mouth once a day for 3 months postpartum.
- Six monthly presumptive treatments with broad-spectrum anti-helminthics in areas of significant prevalence
- Sleeping under a bed net (ITN) in malarious areas
- VDRL/RPR
- HIV testing (opt-out)
- Continue TT vaccination started at ANC based on the schedule
- Continue TT vaccination started at ANC based on the schedule below

Tetanus Toxoid Immunization Schedule	
TT Injection	Due
TT 1	At first contact with woman of child bearing age or as early as possible in pregnancy (at 1st ANC visit)
TT 2	At least 4 weeks after TT 1
TT 3	At least 6 months after TT 2
TT 4	At least 1 year after TT 3
TT 5	At least 1 year after TT 4

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3. Providing woman-centered education and counseling

The education and counseling should address postpartum needs such as nutrition, breastfeeding, family planning, sexual activity, early symptoms of complications and preparations for possible complications. The counseling should take place at a private area to allow women to ask questions and express their concerns freely. It is advisable to involve husbands (after the permission of the woman) in this counseling and in receiving instructions before discharge.

Specific postpartum care and treatment

Nutrition:

- A regular diet should be offered as soon as the woman requests food and is conscious.
- Intake should be increased by 10% (not physically active) to 20% (moderately or very active) to cover energy cost of lactation.
- Women should be advised to eat a diet that is rich in proteins and fluids.
- Eating more of staple food (cereal or tuber)
- Greater consumption of non-saturated fats
- Encourage foods rich in iron (e.g., liver, dark green leafy vegetables, etc.)
- Avoid all dietary restrictions

Breastfeeding

- Early skin to skin contact of mother and newborn and immediate initiation of breast

feeding

- In case breast feeding can't be started due to either maternal or newborn illness, feeding the newborn has to be initiated if possible by milk expressed from the mother herself.
- Rooming in (keeping the mother and the baby together) throughout the facility stay of mother and newborn

- Women should be encouraged to maintain exclusive breast feeding for six months and should be educated about effective breastfeeding practices, as well as common breastfeeding problems, how to continue breast feeding for two years and to start complementary feeding after six months. See PMTCT section on breast feeding advice regarding HIV infected women. Postpartum education and counseling includes:
 - Correct positioning of the newborn at the breast
 - Exclusive breast feeding.: No other fluids e.g. herbs, glucose, or sugar water should be given
 - Encouraging breast feeding on demand
- If there is a medical contraindication to breastfeeding, firm support of the breasts can suppress lactation. For many women, tight binding of the breasts, cold packs, and analgesics followed by firm support effectively control temporary symptoms while lactation is being suppressed

Breast care:

- To prevent engorgement, breastfeed every 2-3 hours
- Wear supportive (but not tight) bra or binder
- Keep nipples clean and dry
- Wash nipples with water only once per day – no soap
- After breastfeeding, leave milk on nipples and allow to air dry

Postpartum family planning

- All postpartum women should receive family planning education and counseling before discharge.
- Ideally, counseling for postpartum contraception should start during the antenatal period, and should be an integral part of antenatal care.
- Women who had no antenatal care and those who did not receive counseling during the ante natal period, should be counseled for family planning in the immediate post partum period, after their own and their newborn's condition have stabilized.
- Women should be informed about the advantages of birth spacing for at least two (2-5 years) years before getting pregnant again and about different family planning options.
- Women should also be given a choice of receiving a family planning method in the labor ward before discharge from hospital or at a family planning clinic within the first 40 days postpartum.
- Facilitate free informed choice for all women: The provider should make sure that the mother is not in pain and that her other concerns have been addressed. It is preferable to offer family planning counseling some time before discharge from hospital so as to give the woman time to make a free decision and to consider different contraceptive options.
- Family planning services should be provided by the attending doctor and nurse in the

ward as well as FPworkers.

- In settings where family planning methods are not available on the ward, the health provider should provide family planning counseling and refer the woman either to the hospital family planning clinic (if the woman is interested in immediate initiation) or to a family planning clinic near her residence.
- The counseling should focus on initiation of long-term family planning methods such as intrauterine contraceptive device (IUCD) and implants. Women who are interested in initiating such family planning method should receive the appropriate method of their choice before discharge.
- If the woman was adequately counseled during ANC and has decided to initiate IUCD, it can be inserted during delivery. It should be noted that insertion of IUCD is contraindicated between 48 hrs and 42 days of the postpartum period, thus every effort should be made to insert the IUCD soon after delivery, if that is the method of choice for the mother.
- Breastfeeding women who are less than 6 weeks postpartum can generally use progestogen-only pills (POPs), and levonorgestrel (LNG) and etonogestrel (ETG) implants (2015 WHO medical eligibility criteria category 2).
- Advise against use of combined oral contraceptives in breastfeeding women in the first 6 months after childbirth or until weaning, whichever comes first.
- Women who are interested in immediate initiation of contraception should be offered a family planning method before discharge.
- Women who were counseled during antenatal care and who had indicated a desire for postpartum IUD insertion or tubal ligation could have an IUD inserted within 48 hours after delivery (postplacental IUD insertion) or referred for minilap for tubal ligation.

Exercise

- Normal activities may be resumed as soon as the woman feels ready.
- When to start an exercise routine depends on the woman; its safety depends on whether complications or disorders are present. Usually, exercises to strengthen abdominal muscles can be started once the discomfort of delivery (vaginal or cesarean) has subsided, typically within one day for women who deliver vaginally and later for those who deliver by cesarean section.
- Sit-ups or curl-ups, (rising from supine to semi-sitting position), done in bed with the hips and knees flexed, tighten abdominal muscles, usually without causing backache.
- Kegel's exercise are also recommended to strengthen the pelvic floor.

Rest and activity:

- Increase rest time
- All postpartum need additional rest to speed recovery

- Breastfeeding women need even more rest
- Wait at least 4 to 5 weeks to resume normal activity; start back gradually

Personal hygiene and perineal care

- If delivery was uncomplicated, showering and bathing are allowed.
- Vaginal douching is avoided in early puerperium, till after bleeding stops completely and all wounds are healed.
- The vulva should be cleaned from front to back.
- Women are encouraged to defecate before leaving the hospital, although with early discharge, this recommendation is often impractical.
- Maintaining good bowel function can prevent or help relieve existing hemorrhoids, which can be treated with warm sitz baths.

Emotional support

- Transient depression (baby blues) is common during the first week after delivery.
 - Symptoms are typically mild and usually subside by 7 to 10 days.
 - Treatment is supportive care and reassurance.
- Persistent depression, lack of interest in the infant, suicidal or homicidal thoughts, hallucinations, delusions, or psychotic behavior may require intensive counseling and antidepressants or antipsychotics.
- Women with a preexisting mental disorder are at high risk of recurrence or exacerbation during the puerperium and should be monitored closely.

Sexual activity

Intercourse may be resumed after cessation of bleeding and discharge, and as soon as desired and comfortable to the woman. However, a delay in sexual activity should be considered for women who need to heal from lacerations or episiotomy repairs. Sexual activity after childbirth may be affected due to decreased sexual desire (due to fatigue and disturbed sleep patterns, genital lacerations/episiotomy), hypo-estrogenization of the vagina, and power issues in marriage.

Bladder care:

- Avoid distention & encourage urination: voiding must be encouraged and monitored to prevent asymptomatic bladder overfilling.
- Do not routinely catheterize unless retention necessitates catheterization (e.g. retention of urine due to pain from peri urethral laceration at vaginal delivery)
- Rapid diuresis may occur, especially when oxytocin is stopped.

Pain management

Common causes: after-pain and episiotomy

- Episiotomy pain: immediately after delivery, ice packs may help reduce pain and edema at the site of an episiotomy or repaired laceration; later, warm sitz baths several times a day can be used. Analgesics are used if not relieved.
- Contractions of the involuting uterus, if painful (after-pains), may require analgesics. Commonly used analgesics include:
 - Aspirin 600mg,
 - Acetaminophen 650mg
 - Ibuprofen 400 mg orally every 4 to 6 hours

Support for Mother-Baby-Family Relationships

● **Bonding**

- Encourage touching, holding, exploring
- Encourage Rooming-in

● **Challenges**

- Discuss woman's increased need for rest and (if breastfeeding) intake of food/fluids
- Discuss woman's increased workload

● **Support**

- Encourage sharing in care of newborn
- Assist in devising strategies for overcoming challenges

● **Information**

- Discuss key aspects of postpartum and newborn care
- Encourage questions

Rh-negative blood group

Women with Rh-negative blood group, who have an infant with Rh-positive blood and are not sensitized, should be given Rh0(D) immune globulin 300 µg IM, as soon as possible (preferably within 72 hours of delivery) to prevent sensitization.

Hospital stay and follow up

Hospital stay:

Hospital stay after delivery varies depending on the parturients' condition. Women with complications stay longer period. Most women have normal pregnancy, labor, delivery and postpartum period; and usually they are discharged within a day of delivery. The minimum in-hospital stay should be 24 hours after which the risk of life endangering conditions such as PPH due to atonic uterus or genital trauma, postpartum eclampsia etc. are less likely. The use of this hospital stay should be well organized so that all parturients and their babies get the basic postpartum care. At the time of discharge, the evaluation should be thorough; the parturient and her baby should get all the basic postpartum care; and appointment be

given. Providing women with a summary of their condition, especially for those with complication, is essential.

Follow-up visit:

Women should be informed that they should make a follow up visit to the health facility or to a health unit on day 3 (48–72 hours), between days 7–14 and at six weeks postpartum (WHO PNC recommendation 2013). The schedule should not be rigid. The summary of care to be provided during each visit is shown in table 7-1. It should incorporate maternal (family) convenience and medical condition. They should also be informed to come back to the health facility if they feel any symptoms that worry them or have the above danger signs.

The vast majority of women and newborns needing care are in the community, whether urban or rural, throughout the postpartum period, and many will not access the formal health system for care even if it is available. Complex patterns of traditional support exist in many societies to provide protection and nurture for around seven to forty days. Formal care provision can build on this pattern.

Table 7-1: Summary of care that can be offered at each point of contact during the puerperium.

Postpartum Care of All Women and their Babies	
First 24 hours after birth	
Mother Assess maternal well-being • All postpartum women should have regular assessment of vaginal bleeding, uterine contraction, fundal height, temperature and heart rate (pulse) routinely during the first 24 hours starting from the first hour after birth. • Blood pressure should be measured shortly after birth. If normal, the second blood pressure measurement should be taken within six hours. • Urine void should be documented within six hours. • Counseling on safe disposal of potentially infectious soiled pads or other materials. • Advice on where to call for help in case of emergency (for home based delivery) • Immunize with Tetanus Toxoid if it was not done during pregnancy • Support initiation of breastfeeding • Offering HIV testing if not done already	Baby • Warm baby by keeping mother and baby together (Bedding in, Rooming in) • Frequent and exclusive breastfeeding • Keeping baby clean and clean cord care • Weighing the baby • Examination of the newborn for health as per standards • Frequent observation of baby by the mother for danger signs • Immunization with BCG, and OPV₀ • Schedule return visit
Schedule return visit At day 3 and 7 – 14 days	
Mother • At each subsequent postnatal contact, enquiries should continue to be made about general well-being and assessments made regarding the following: micturition and urinary incontinence, bowel function, healing of any perineal wound, headache, fatigue, back pain, perineal pain and perineal hygiene, breast pain, uterine tenderness and lochia. • Breastfeeding progress should be assessed at each postnatal contact. • At each postnatal contact, women should be asked about their emotional wellbeing, what family and social support they have and their usual coping strategies for dealing with day-to-day matters. • All women and their families/partners should be encouraged to tell their health care professional about any changes in mood, emotional state and behaviour that are outside of the woman's normal pattern.	Baby Concerns about breastfeeding and wellbeing of baby as mother perceives it • Assess general condition of baby: Active, feeding well, frequently • Observe how baby is breastfeeding* • Observe skin for signs of pallor and jaundice • Assess vital sign if baby is not active • Immunization with BCG, and OPV₁ if not provided already • Advise on direct sunlight exposure

Postpartum Care of All Women and their Babies

At 6 weeks

- Routine postpartum physical examination
- Assessment for signs of postpartum complications
- Counseling on appropriate nutrition, and micronutrient supplementation
- Counseling on family planning and safe sex practices
- Counseling on breastfeeding and support as needed
- Counseling on personal hygiene and disposal of soiled pads.
- Micronutrient supplementation as appropriate
- Encourage on continued use of ITN for women living in malaria endemic areas
- Routine offering of HIV testing if not already done
- Plan for revisit and immunization of baby
- Identify warning signs of complications
- Routine examination of the baby
- Advise on direct sunlight exposure
- Immunization: OPV₁, Pentavalent₁

2.3. ASSESSMENT (HISTORY AND PHYSICAL EXAMINATION) AND CARE OF A POSTPARTUM WOMAN

Postpartum care is a continuum of care which should start immediately after expulsion of placenta. The health care provider should be very efficient in assessing the woman to identify signs and symptoms that indicate complications/problems and provide basic care to all mothers. Use **Table 7-2** for examining the mother who presented after discharge from a facility or after home delivery.

Table 7-2:-POSTPARTUM EXAMINATION OF THE MOTHER (UP TO 6 WEEKS)

ASK, CHECK RECORD	LOOK, LISTEN, FEEL	SIGNS	CLASSIFY	TREAT AND ADVISE
■ When and where did you deliver? ■ How are you feeling? ■ Have you had any pain or fever or bleeding since delivery? ■ Do you have any problem with passing urine? ■ Have you decided on any contraception? ■ How do your breasts feel? ■ Do you have any other concerns? ■ Check records: →Any complications during delivery? →Receiving any treatments? →HIV status.	■ Measure blood pressure and temperature. ■ Feel uterus. Is it hard and round? ■ Look at vulva and perineum for: →tear →swelling →pus. ■ Look at pad for bleeding and lochia. →Does it smell? →Is it profuse? ■ Look for pallor.	■ Mother feeling well. ■ Did not bleed >250ml. ■ Uterus well contracted and hard. ■ No perineal swelling. ■ Blood pressure, pulse and temperature normal. ■ No pallor. ■ No breast problem, is breastfeeding well. ■ No fever or pain or concern. ■ No problem with urination	NORMAL POSTPARTUM	■ Make sure woman and family know what to watch for and when to seek care. ■ Advise on Postpartum care and hygiene, and counsel on nutrition ■ Counsel on the importance of birth spacing and family planning - Refer for family planning counselling. ■ Dispense 3 months iron supply and counsel on compliance. ■ Give any treatment or prophylaxis due: →tetanus immunization if she has not had full course. ■ Promote use of impregnated bednet for the mother and baby. ■ Record on the mother's home-based maternal record. ■ Advise to return to health centre within 4-6 weeks.

Table 7-2:-POSTPARTUM EXAMINATION OF THE MOTHER (UP TO 6 WEEKS) contd.

ASK, CHECK RECORD	LOOK, LISTEN, FEEL	SIGNS	CLASSIFY	TREAT AND ADVISE
IF PALLOR ON SCREENING, CHECK FOR ANEMIA				
■ Check record for bleeding in pregnancy, delivery or postpartum. ■ Have you had heavy bleeding since delivery? ■ Do you tire easily? ■ Are you breathless (short of breath) during routine housework?	■ Measure haemoglobin, if history of bleeding. ■ Look for conjunctival pallor. ■ Look for palmar pallor. If pallor: → Is it severe pallor? → Some pallor? → Count number of breaths in 1 minute	■ Haemoglobin <7 g/dl. AND/OR ■ Severe palmar and conjunctival pallor or ■ Any pallor and any of: → >30 breaths per minute → tires easily → <u>breathlessness at rest.</u>	SEVERE ANEMIA	■ Give double dose of iron (1 tablet 60 mg twice daily for 3 months) . ■ Consult or refer urgently to hospital. ■ Follow up in 2 weeks to check clinical progress and compliance with treatment.
		■ Haemoglobin 7-11 g/dl. OR ■ Palmar or conjunctival pallor.	MODERATE ANEMIA	■ Give double dose of iron for 3 months. ■ Reassess at next postnatal visit (in 4 weeks). If anemia persists, refer to hospital.
		■ Haemoglobin >11 g/dl ■ No pallor.	NO ANEMIA	■ Continue treatment with iron for 3 months altogether Give iron/folate for 3 months
IF HEAVY VAGINAL BLEEDING				
		More than 1 pad soaked in 5 minutes.	POSTPARTUM BLEEDING	■ Give 0.2 mg ergometrine IM. ■ Give appropriate IM/IV antibiotics. ■ Manage as in <i>Rapid assessment and management.</i> ■ Consult or refer urgently to hospital.
IF ELEVATED DIASTOLIC BLOOD PRESSURE				
History of pre-eclampsia or eclampsia in pregnancy, delivery or after delivery?	If diastolic blood pressure is ≥90 mmHg , repeat after a 1 hour rest.	■ Diastolic blood pressure ≥110 mmHg.	SEVERE HYPERTENSION	■ Give appropriate antihypertensive. ■ Manage as in module 3, section 3-5. ■ Consult or refer urgently to hospital.
		■ Diastolic blood pressure ≥90 mmHg on 2 readings.	MODERATE HYPERTENSION	■ Reassess in 1 week. If hypertension persists, refer to hospital.
		■ Diastolic blood pressure <90 mmHg after 2 readings	BLOOD PRESSURE NORMAL	■ No additional treatment

Table 7-2:-POSTPARTUM EXAMINATION OF THE MOTHER (UP TO 6 WEEKS) contd.

ASK,CHECK RECORD	LOOK,LISTEN, FEEL	SIGNS	CLASSIFY	TREAT ANDADVISE
IF FEVER OR FOUL-SMELLING LOCHIA				
■ Have you had: →heavy bleeding? →foul-smelling lochia? →burning on urination?	■ Feel lower abdomen and flanks for tenderness. ■ Look for abnormal lochia. ■ Measure temperature. ■ Look or feel for stiff neck. ■ Look for lethargy.	■ Temperature >38°C and any of: →very weak →abdominal tenderness →foul-smelling lochia →profuse lochia →uterus not well contracted →lower abdominal pain →history of heavy vaginal bleeding.	UTERINE INFECTION	■ Insert an IV line and give fluids rapidly. ■ Give appropriate IM/IV antibiotics. ■ Consult or refer urgently to hospital.
		■ Fever >38°C and any of: →burning on urination →flank pain.	UPPER URINARY TRACT INFECTION	■ Give appropriate IM/IV antibiotics. ■ Consult or refer urgently to hospital.
		■ Burning on urination.	LOWER URINARY TRACT INFECTION	■ Give appropriate oral antibiotic. ■ Encourage her to drink more fluids. ■ Follow up in 2 days. If no improvement, refer to hospital.
		■ Temperature >38°C and any of: →stiff neck →lethargy.	VERY SEVERE FEBRILE DISEASE	■ Insert an IV line. ■ Give appropriate IM/IV antibiotics. ■ Give artemether IM (or quinine IM if artemether not available) and glucose or ■ Consult or refer urgently to hospital.
		■ Fever >38°C	MALARIA	■ Give oral antimalarial. ■ Follow up in 2 days. If no improvement, refer to hospital.
IF DRIBBLING URINE				
		■ Dribbling or leaking urine.	URINARY INCONTINENCE	■ Check perineal trauma. ■ Give appropriate oral antibiotics for lower urinary tract infection. ■ If conditions persist more than 1 week, refer the woman to hospital
IF PUS OR PERINEAL PAIN				
		■ Excessive swelling of vulva or perineum.	PERINEAL TRAUMA	■ Refer the woman to hospital.
		■ Pus in perineum. ■ Pain in perineum.	PERINEAL INFECTION OR PAIN	■ Remove sutures, if present. ■ Clean wound. Counsel on care and hygiene. ■ Give paracetamol for pain. ■ Follow up in 2 days. If no improvement, refer to hospital

2.4. POSTPARTUM CARE FOR PMTCT

A comprehensive and integrated PMTCT intervention does not end at administering anti-retroviral drugs during pregnancy and delivery but must extend beyond these periods. The postpartum period is a critical bridge for the continuum of care that needs to be provided for HIV positive women, their babies and families. In the postpartum period, a mother with HIV infection should receive care that supports her health, prevents complications, and improves the family's ability to live positively with HIV infection.

In addition to the minimum package which should be provided for all mothers and their newborns irrespective of the HIV status, HIV infected mothers and their babies need specific additional attentions. The additional care HIV positive mothers and their newborns should get in the postpartum period are related to HIV infection and its progress. These include

- Prevention and treatment of opportunistic infection
- Family planning methods
- Antiretroviral therapy
- Positive living
- Nutritional supplements and
- Social and psychological support.

Prevention and Treatment of Opportunistic Infections:

Helping the HIV-positive mother stay healthier by providing appropriate preventive management of opportunistic infection is a vital component of a quality postpartum care. Pulmonary tuberculosis (TB), *Pneumocystis Jirovecii* Pneumonia (PCP), Toxoplasmosis, and Cryptococcus meningitis are the common opportunistic infections in Ethiopia. Most of the opportunistic infections occur when the CD4 count is below 200/mm³.

Family planning:

Preventing unintended pregnancies in HIV infected women has to take into considerations different cultural, religious and social factors. HIV infected women and their partners decision to have babies depend on many factors. In addition to considering the different factors for and against use of family planning for HIV-positive women, health care providers should be familiar with the different potential drug- drug interactions that could occur with use of ARV drugs and contraceptives.

- **Dual protection should be recommended for all women on HAART.** Effective and appropriate contraception should be available, specifically for women who are on HAART with Nevirapine (NVP), Lopinavir/Ritonavir (LPV/r), Nelfinavir (NLF) and Ritonavir (RTV). It is also recommended that women on COCs need to be monitored closely.

- **Considerations for HIV positive women on Rifampicin and COC.**

Rifampicin, which is often used to treat tuberculosis in HIV+ clients, also decreases the effectiveness of COCs by reducing circulating estrogen. Any woman on Rifampicin and COCs should use dual protection.

In all circumstances the health care provider has to provide up-to-date and unbiased information to the client and must follow confidentiality. All the time the client's choice has to be respected unless there is clear evidence based medical contraindications. The health care provider should encourage dual protection and/or dual method use for all HIV positive clients.

Anti-Retroviral Therapy:

A health care provider should offer HIV testing and counseling for a woman who comes for postpartum care with unknown HIV status. If she is found to be HIV positive, assessment for ART eligibility must be done using immunological and clinical criteria. For eligible women, the health care provider has to facilitate access to ART either in the facility or through referral.

Neonatal ARV prophylaxis

Administer NVP once daily for six weeks for the newborn; check adherence to infant prophylaxis during follow-up.

NVP syrup dosing beyond 6 weeks of age in special situations in which prolonged dosing of up to 12 weeks should be considered (such as the mother having had less than one month coverage with ART before giving birth and not being likely to be virally suppressed; the infant is identified as HIV exposed after birth and is breastfeeding)

Enhanced postnatal prophylaxis for HEI (NVP + AZT for 6 weeks and continued NVP prophylaxis for another 6 weeks) for high risk HEI is suggested whenever essential inputs are adequate.

The different scenarios of administering neonatal ARV prophylaxis is shown in the table below

Scenarios	New born ARV prophylaxis regimen		
	Mother	Exclusive Breastfeeding	Replacement feeding
HIV positive mothers on ART for more than one	Continue ART	NVP daily until 6 weeks starting immediately at	NVP daily until 6 weeks starting immediately at birth

month during pregnancy		birth	
mother is diagnosed with HIV during labor or immediately post partum	Initiate ART	NVP daily for 6 weeks and consider extending for 12 weeks	NVP daily for 6 weeks

Scenarios		New born ARV prophylaxis regimen	
	Mother	Exclusive Breastfeeding	Replacement feeding
HIV positive mothers on ART for more than one month during pregnancy	Continue ART	NVP daily until 6 weeks starting immediately at birth	NVP daily until 6 weeks starting immediately at birth
mother is diagnosed with HIV during labor or immediately post partum	Initiate ART	NVP daily for 6 weeks and consider extending for 12 weeks	NVP daily for 6 weeks

Positive Living:

Most women diagnosed to be HIV-positive through the PMTCT programs are quite healthy and have no, or few, symptoms.

There are things that mothers and community can do to maintain or improve health. These include:

- Eating variety diet.
- Cook meat and eggs thoroughly and wash fruits and raw vegetables with potable water or treated water before eating (1 teaspoon of bleach per liter of water).
- Maintaining good personal hygiene with particular attention to oral hygiene, fingernails, toenails, and private areas.
- Getting immediate and proper treatment for every infection or illness.
- Avoiding drinking alcohol and smoking cigarettes.
- Using birth control to avoid unintended pregnancies
- Practicing safer sex. Using condoms consistently and regularly to avoid re-infection or infecting a partner, and avoiding sex during menses or when either partner has symptoms of a sexually transmitted infection (STI). Getting STIs treated promptly.
- Preventing malaria infection through the use of impregnated bed nets, clearing bushes around the house, etc.
- Exercise and rest (most women's regular daily work schedule is exercise enough, but they need to rest and at least 8 hours' sleep a night).
- Accepting her condition and understanding the implications.

- Obtaining medical care promptly when she gets sick.
- Working toward disclosing her status to her partner and family.
- Planning for the future of her children and family (succession planning) including will-making.
- Taking care of herself mentally and spiritually by obtaining regular counseling, from religious counselors and participating in a support group.
- Continuing to participate in community activities.
- Developing or consolidating her support network and participating in a support group for HIV-positive women.
- Challenging stigma and discrimination against persons living with HIV/AIDS.

Nutrition:

Eating variety of food is very important for HIV-positive mothers and especially for those who are breast-feeding. Breast-feeding mothers should continue taking micronutrients such as vitamins, folic acid, iron and iodized salt, and should have two extra meals a day. Nutrition counseling is especially important for persons living with HIV/AIDS, caregivers, family members, and HIV-exposed or -infected infants and children. HIV/AIDS and malnutrition are closely interrelated. HIV infection causes malnutrition through multiple mechanisms:

- ☐ Negative impact on food intake
- ☐ Losses through diarrhea and vomiting
- ☐ Reduction of nutrient absorption
- ☐ Increase of energy requirements

Social and psychological support:

The follow-up treatment, care and support for women who are HIV-infected and the care of their children and families, can be strengthened if linkages are made with comprehensive community health services. It is important that treatment and care extend beyond PMTCT prophylaxis for women, infants, and family members at risk for or infected with HIV. In order to offer comprehensive care, health facilities must function in partnership with governmental, non-governmental, community-based, private organizations that provide care, treatment, and support services for mothers who are HIV-infected and their family members. Linkages provide clients with comprehensive care and enable healthcare workers to feel part of the broader strategy to improve the health of the community and, as such, increase staff motivation and satisfaction.

Peer/ Mother Support Groups:

Support groups provide opportunities for ongoing support, information/education, skills building, peer support and encouragement, and follow-up medical care for HIV-positive mothers and their babies. Mothers who have not joined support groups should have the opportunity to do so at any time during follow-up.

Peer support group such as mothers-to-mothers groups help HIV-positive women to demand their rights and to access appropriate treatment and care. Moreover, they empower the women's negotiation skills which help them to disclose their status to their partners, family members and to the community at large. In addition to the above mentioned benefits these support are found to be effective in ensuring initiation of ARVs and adherence to treatment. Peer/mother support group can also play the major role in sharing experiences that include problem solving, palliative care (home-based care). It can also empower women on self-care.

UNIT 3: ONGOING SUPPORT FOR NEWBORN

3.1.KEEP THE NEWBORN WARM

- **Warm the room** with the temperature of 25-28°C, no air current.
- **Always keep the newborn covered with dry and warm clothing.**
- **Skin-to-skin contact:** bed in the newborn with the mother. If the mother cannot keep the newborn skin-to-skin because of problems, wrap the newborn in a clean, dry, warm cloth and place in a cot covered with a blanket.
- Explain to the mother that keeping newborn warm is important for the newborn to remain healthy before discharge.
- In areas where prevalence of malaria is high, advice to mother-newborn pair to sleep under insecticide treated bed net.

Do not put the newborn on any cold or wet surface.

Do not bath the newborn at birth. Delay bathing for 24 hours.

3.2.Counselling the mother and families on cord care:

Provide the following messages to the mother

- Don't to cover the cord with the diaper
- Don't use bandages as it may delay healing and introduce infection
- Don't use alcohol for cleansing as it may delay healing.
- Don't apply traditional remedies to the cord as it may cause tetanus and other infections.
- Apply chlorhexidine on the cord after washing hands with soap and water based on the guidance given to you
- Watch out for pus discharge from the cord stump, redness around the cord especially if there is swelling, or fever (temperature more than 38°C) or other signs of infection. Visit health worker if you see any of these signs.

3.3.COUNSEL AND SUPPORT SAFE INFANT FEEDING

I. BREASTFEEDING:

Continuously emphasize on the importance of exclusive breast feeding after birth to lactating mothers including partner or care takers within the family.

Explain to the mother that:

- Breast milk contains most optimal level of nutrients to satisfy the newborn needs. It is easily digestible and protects the newborn against infection.
- Exclusive breastfeeding should be initiated within 1 hour of birth. This means babies should not have any other food or drink up to the age of six months.
- Breastfeeding helps newborn's growth and development and promotes mother-newborn bonding.
- Encourage breastfeeding on demand, day and night, as long as the newborn wants, 8 or more times in 24 hours
- **Note:** Exclusive breastfeeding helps delay a new pregnancy (see for breastfeeding and family planning).
- Check that position and attachment are correct at the first feed. Offer assistance any time when the mother needs.
- Offer the breasts alternatively until each breast empties
- If the mother is ill and unable to breastfeed, help her to express breast milk and feed the baby by cup.
- Advise the mother on medication and breastfeeding:
 - Commonly given drugs to lactating mothers are generally considered safe; thus the baby can continue to breastfed. However, if mother is taking some drugs, cotrimoxazole, fansidar or the like, monitor newborn for jaundice and any other abnormal signs.
- If mother is not around, let her express breast milk and let somebody else feed the expressed breast milk to the newborn by cup.

Reminders:

DO NOT force the newborn to take the breast.

DO NOT interrupt newborn's feeding. **DO NOT** give any other feeds or water. **DO NOT** use artificial teats or pacifiers.

II. FEEDING METHOD FOR LOW BIRTH WEIGHT NEONATE:

Help a mother breastfeed her low birth weight newborn:

- ☐ Express a few drops of milk on the baby's lip to help the newborn start nursing.
- Give the baby short rests during a breastfeed; feeding is hard work for LBW baby.
- If the baby coughs, gags, or spits up when starting to breastfeed, the milk may be letting down too fast for the little baby. Teach the mother to take the baby off the breast if this happens.
- Hold the baby against her chest until the baby can breathe well again then put it back to the

breast after the let-down of milk has passed.

- If the LBW baby does not have enough energy to suck for long or a strong enough sucking reflex: Teach the mother to express her breast milk and feed the baby with cup.

Steps for expressing breast milk (can take 20-30 minutes or longer in the beginning)

- Wash hands with soap and water
- Prepare a cleaned and boiled cup or container with a wide opening.
- ☐ Sit comfortably and lean slightly toward the container. Hold the breast in a “C-hold”.
- Gently massage and pat the breast from all directions.
- Press thumb and fingers toward the chest wall, roll thumb forward as if taking a thumb print so that milk is expressed from all areas of the breast.
- Express the milk from one breast for at least 3-4 minutes until the flow slows and shift to the other breast.

How to feed a newborn with a cup:

- Hold the newborn sitting upright or semi-upright on your lap
- Hold a small cup of milk to the newborn’s lips
 - Tip the cup so the milk just touches the newborn’s lips
 - The cup rests gently on the newborn’s lower lip and the edges of the cup and touch the outer part of the newborn’s upper lip
 - The newborn becomes alert and opens his mouth and eyes
- Do not pour the milk into the newborn’s mouth. A young infant starts to take the milk with the tongue. An older/ bigger newborn sucks the milk, spilling some of it
- When the newborn has had enough he closes his mouth and will not take any more. If the newborn has not taken the required amount, wait and then offer the cup again or feed more frequently

Storing and using stored breast milk:

Fresh breast milk has the highest quality. If the breast milk must be saved, advise the mother and family to:

- Use either a glass or hard plastic container with a large opening and a tight lid to store breastmilk.
- Use a container and lid which have been boiled for 10minutes.
- If the mother is literate, teach her to write the time and date the milk was expressed(or morning, afternoon, evening) on the container before storing.
- Empty the breast and store the milk in the coolest place possible.

III: GIVE SPECIAL SUPPORT TO BREASTFED TWINS:

COUNSEL THE MOTHER:

- Reassure the mother that she has enough breast milk for two babies.
- Encourage her that twins may take longer to establish breastfeeding since they are frequently born preterm and with low birthweight.

HELP THE MOTHER:

- Start feeding one newborn at a time until breastfeeding is well established.
- Help the mother find the best method to feed the twins:
- If one is weaker, encourage her to make sure that the weaker twin gets enough milk.
- If necessary, she can express milk for her/him and feed her/him by cup after initial breastfeeding.
- Daily alternate the side each newborn is offered.

IV: MOTHERS WITH SPECIAL CONDITIONS

Babies born to mothers who may be unable to breast feed due to special conditions (loss of consciousness, severe psychiatric problems, or HIV positive mothers who chose replacement feeding, death, and others) may require replacement feeding. (Refer to IMNCI and PMTCT guidelines for further detail in preparation and feeding)

Immunize the newborn

Advise the mother to seek immediate care for the newborn for the following danger signs:

- difficulty breathing
- convulsions
- fever or feels cold
- bleeding
- diarrhea
- very small, just born
- not feeding at all
- difficulty feeding
- pus discharging from the eyes
- many skin pustules
- yellow skin
- a cord stump which is red or draining pus
- Movement less than normal(lethargy)

UNIT 4: POST NATAL NEWBORN CARE (VISIT FROM/AT HOME)

4.1. LEARNING OBJECTIVES

At the end of this unit, participants will be able to:

- Describe how to organize care of the sick or small newborn
- Describe the steps and principles of rapid initial assessment and immediate management.
- Describe further basic assessment (history and physical examination) and management of the newborn.

4.2. ORGANIZING CARE OF THE SICK OR SMALL NEWBORN

Whether babies who need care are brought to the health facility from home or transferred from another institution, the care management involves a cycle of planning (preparedness), monitoring (assessing status and/or progress), implementing (decision for action), and re-evaluating the condition before discharge. The care that the newborn receives at the health facility follows several steps described below:

Note: Before undertaking any intervention, make sure to follow the principles and practices of infection prevention; especially hand washing before and after examining clients at all times, when examining and treating babies. (Example: if the newborn has diarrhea or a possible infection of the skin, eye, or umbilicus)

ARRIVAL, RAPID ASSESSMENT, AND IMMEDIATE MANAGEMENT:

- There must be Health provider to attend the sick and small newborns at all time (from arrival to the site of care) and prioritize for immediate care.
- Ensure that the admission and reception area is organized so that every baby can be seen without undue delay
- Assessment and care of the newborn should be initiated even before any administrative procedures
- Start by assessing the baby for danger signs to curb further damage and preventable deaths.

While looking for danger signs introduce yourself to the mother (or caretaker): and ask her

- *What is wrong with the baby?*
- *When did the problem(s) first start?*
- *What are the names of the mother and baby?*
- *How old is the baby?*
- *Was the baby brought in from outside the health care facility?*
- *Keep the baby with the mother, if possible, and allow her to be present during the assessment and for any procedures, if appropriate.*

4.3. RAPID ASSESSMENT AND IMMEDIATE MANAGEMENT

Few babies may have emergency signs that indicate a problem that is life threatening. Therefore this section of the unit will assist you to rapidly assess the baby and provide immediate lifesaving care.

Note: Examine the newborn immediately for the following emergency signs, and provide immediate management (refer to Table 5.a):

- *Not breathing at all; gasping; or if the heart beat is less than 100 bpm; OR*
- *Bleeding; OR*
- *Shock (pallor/cyanotic, cold to the touch, heart rate less than 100 beats per minute, extremely lethargic /floppy or unconscious).*

4.3.1. RAPID ASSESSMENT

a) If newborn has Breathing difficulty

- Place the baby on a warm surface and with good light. For example under a radiant heat warmer, or use pre-warmed towel/linen (using hot water bottle).
- Assess immediately if newborn:
 - o *is not breathing at all, after cleaning the air ways and stimulation ; OR*
 - o *is gasping; OR*
 - o *has a heart rate less than 100 beats per minute (bpm)*

- Start ventilating the newborn using bag and mask until the heart rate is >100bpm. Always check for the heart rate or umbilical pulse rate every 30 seconds and decide on to the next action.
- If there is no response after resuscitation through ventilation, i.e. heart rate is <60 per minute and did not start spontaneous breathing, then call for assistance
- If there is no change; continue the procedure for 20 minutes until the heart rate started to rise
- If no further response refer immediately.

Refer to module five for further details of resuscitation

b) If newborn has Bleeding

- Assess & check for signs of bleeding including weighing the baby
- Stop visible bleeding, if possible; (e.g. if there is **bleeding from the umbilicus**, re-clamp or retie the umbilical stump; if the **bleeding is from a cut or male circumcision site**, press on the bleeding site by applying a sterile dressing....);
- Administer vitamin K 1 mg IM on the anterior thigh if it had not been given
- Monitor the vital signs and if there is sign of shock (pallor, cyanotic, cold to touch, heart rate <100 bpm or > than 160 bpm and skin is clammy)
- Establish intravenous line with Normal Saline or Ringers lactate 10ml/kg over 10 minutes and repeat once after 10 minutes if your facility is able to do so;
- Take a blood sample from the heel prick to type and cross-match, and measure hemoglobin
- **Urgently refer** the newborn if no improvement

c) If newborn has Convulsion

If the **baby is having a convulsion or spasm**, or twitching after the above emergency management, check for hypoglycemia (blood glucose level < 45mg/dl) and administer bolus dose of 2mg/kg of 10% glucose iv slowly over 5 minutes. If these are not possible REFER urgently.

4.3.2. FURTHER ASSESSMENT AND MANAGEMENT

After examining and managing the emergencies signs (i.e. not breathing, gasping, heart rate less than 100 beats per minute, bleeding, or shock) and if the newborn is not being referred, continue to assess the newborn and proceed to the next action.

- Obtain complete history of the newborn and the mother;
- Use the findings from the history and examination to decide on the most appropriate intervention;
- Complete additional examinations, if necessary, and determine the required laboratory investigations;
- Perform appropriate laboratory investigations, and treat the newborn (and/or the mother or her partner(s), as necessary);
- Record all information, including:
 - o *The findings of the history, examination, and laboratory investigations;*
 - o *Treatment given;*
 - o *Changes in the newborn's condition.*

BABY

Ask the mother or caretaker the following:

- What is the problem? Why did you bring the baby?
- When did the problem first start?
- Has the baby's condition changed since the problem was first noted? Is the problem getting worse? If so, how rapidly and in what way?
- What kind of care, including specific treatment, has the baby already received?
- How old is the baby?
- How much did the baby weigh at birth?
- Was the baby born at term? If not, at how many weeks gestation was the baby born?
- Where was the baby born? Did anybody assist the birth?
- How was the baby immediately after birth?
 - *Did the baby spontaneously breathe at birth?*
 - *Did the baby require resuscitation? If so, what was the length of time before spontaneous breathing was established?*
 - *Did the baby move and cry normally?*

- Is the baby having problems in feeding, including any of the following?
 - *Poor or no feeding since birth or after a period of feeding normally;*
 - *coughing or choking during feeding;*
 - *vomiting after a feeding*
- Ask the mother if she has any questions or concerns (e.g. special concerns or anxiety about breastfeeding)
- If the **mother is not present**, determine where she is, what her condition is, and whether she will be able to care for the baby, including breastfeeding or expressing breastmilk.

MOTHER

- Review the mother's medical, obstetric, and social history.
 - Duration pregnancy; any complication during your pregnancy and treatment taken
 - Chronic illnesses: including hepatitis B, Tuberculosis, diabetes, or syphilis, symptomatic or seropositive HIV
 - Any complication during labor and delivery (uterine infection or fever at onset of labor and 3 days after giving birth. Ask about the rupture of membrane or fetal distress; prolonged labor; caesarean section; instrumental vaginal delivery (e.g. forceps or vacuum extraction); mal-position or mal-presentation of the baby (e.g. breech); any other complications after birth.

4.3.3 IMMEDIATE ACTIONS ONCE ASSESSMENT IS COMPLETE

Determine appropriate Classification and management according to the tables (Table 5.4a and Table 5.4b) below:

Figure 5-a: Chart to assess, classify and treat for possible bacterial infection and jaundice

ASSESS CLASSIFY

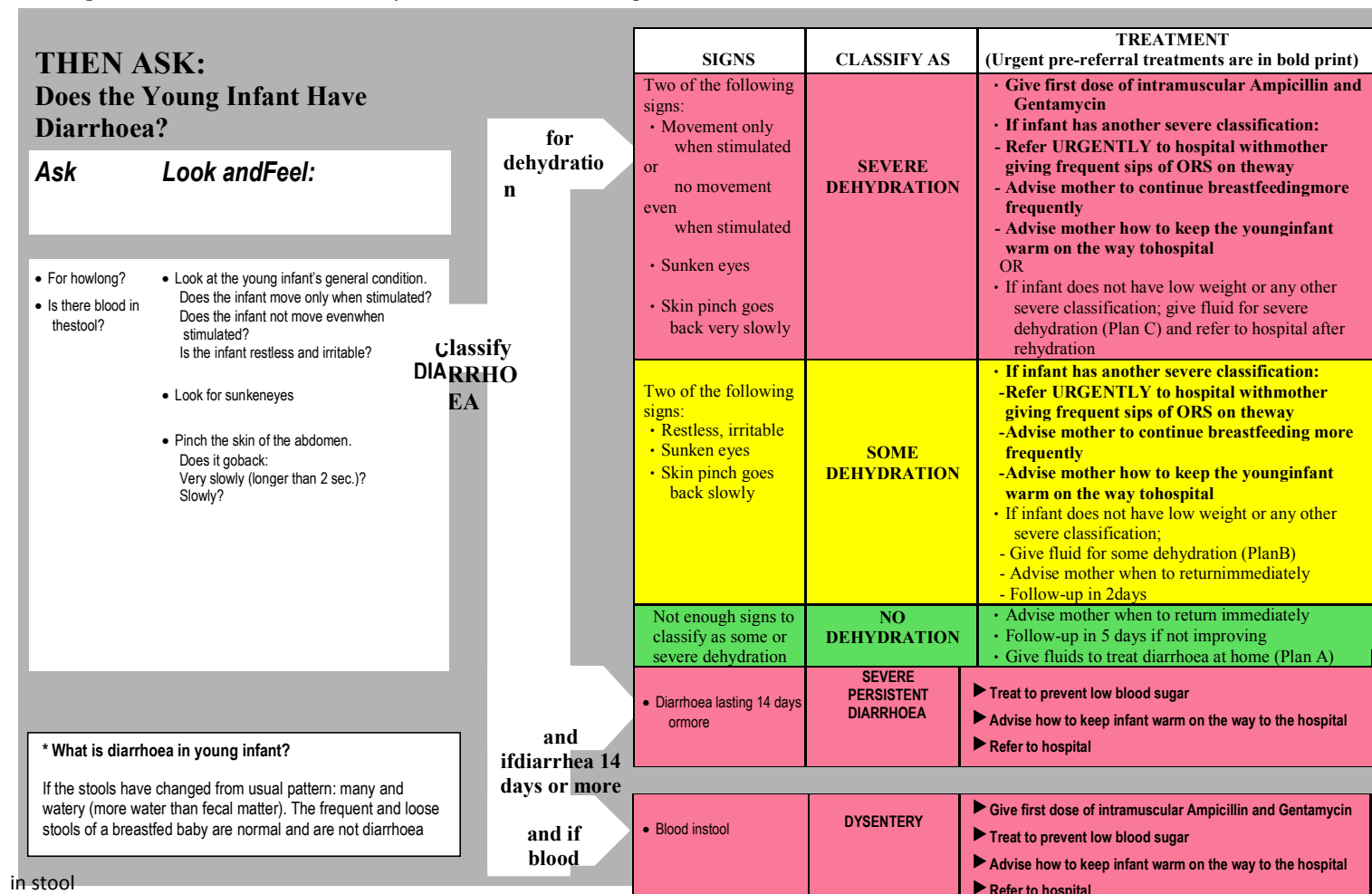
IDENTIFY TREATMENT

USE ALL BOXES THAT MATCH INFANT'S SYMPTOMS AND PROBLEMS TO CLASSIFY THE ILLNESS

CHECK FOR POSSIBLE BACTERIAL INFECTION AND JAUNDICE		Classify all young infants	SIGNS	CLASSIFY AS	TREATMENT (Urgent pre-referral treatments are in bold print)
Ask ☐ Has the infant had convulsions? ☐ Is there difficulty feeding?	Look, Listen, Feel: ☐ Count the breaths in one minute. Repeat the count if elevated. ☐ Look for severe chest indrawing. ☐ Look and listen for grunting. ☐ See if the young infant is not feeding. ☐ See if the infant is convulsing now. ☐ Look at the umbilicus. Is it red or draining pus? ☐ Measure temperature (or feel for fever or low body temperature) ☐ Look for skin pustules. ☐ Look at the young infant's movements. Does the infant move only when stimulated? Does the infant not move even when stimulated? ☐ Look for jaundice: Are the palms and soles yellow? Are skin on the face or eyes yellow?		YOUNG INFANT MUST BE CALM And if temp. is between 35.5-36.4°C	• Not feeding well or • Convulsions/convulsing now or • Fast breathing (60 breaths per minute or more) or • Severe chest indrawing or • Grunting or • Fever (37.5°C* or above or feels hot) or • Low body temperature (less than 35.5°C* or feels cold) or • Movement only when stimulated or no movement even when stimulated.	POSSIBLE SERIOUS BACTERIAL INFECTION Or VERY SEVERE DISEASE
			• Red umbilicus or draining pus or • Skin pustules	LOCAL BACTERIAL INFECTION	• Give Cotrimoxazole or Amoxycillin for 5 days • Teach the mother to treat local infections at home • Advise mother when to return immediately • Follow-up in 2 days
			• None of the signs of possible serious bacterial infection or local bacterial infection	SEVERE DISEASE, or LOCAL INFECTION UNLIKELY	• Advise mother to give home care for the young infant
			• Temperature between 35.5°C – 36.4°C (both values inclusive)	LOW BODY TEMPERATURE	• Treat to prevent low blood sugar • Warm the young infant using skin-to-skin contact for one hour and re-assess. If temperature remains same or worse, refer. • Advise mother when to return immediately • Follow-up in 2 days
			• Palms and/or soles yellow OR • Age < 24 hrs OR • Age 14 days or more	SEVERE JAUNDICE	• Treat to prevent low blood sugar • Warm the young infant by skin-to-skin contact if temperature is less than 36.5°C (or feels cold to touch) while arranging referral • Advise mother how to keep the young infant warm on the way to the hospital • Refer URGENTLY to hospital
			• Only skin or eyes yellow	JAUNDICE	• Advise mother to give home care for the young infant • Advise mother when to return immediately • Follow-up in 2 days

Thresholds are based on axillary temperature. The threshold rectal temperature readings are approximately 0.5°C higher.

Figure 5-b: Chart to assess, classify and treat newborn who presented with diarrhea.



4.4. Care for preterm and low birth weight babies

A preterm baby is a baby born before 37 completed weeks of gestation. The following are common problems associated with prematurity:

- Feeding
- Respiratory distress, apnea
- Jaundice
- Intracranial bleed
- Hypoglycemia
- Temperature instability
- Infection

PREVENTION OF HYPOTHERMIA

All small babies need attention to basic thermal care to prevent them from becoming cold.

In general, newborns are at risk of hypothermia because of their large surface area for small body mass and premature and LBW babies in particular for the following reasons.

- Highly permeable skin which increases epidermal water loss
- Deficient subcutaneous fat with less insulation
- Deficient stores of brown fat
- Immature central thermoregulation
- Poor caloric intake
- Poor oxygen consumption because of associated pulmonary problems

Newborns may lose heat by the following mechanisms

- Convection – where heat is lost from the skin to moving air.
- Radiation – where heat is dissipated from the baby to a colder object in the surrounding like to the floor, wall or window.
- Conduction – where the baby loses heat to the surface on which he or she lies.
- Evaporation – major cause of heat loss immediately after birth where water is evaporated from wet infants skin like evaporation from boiling water.

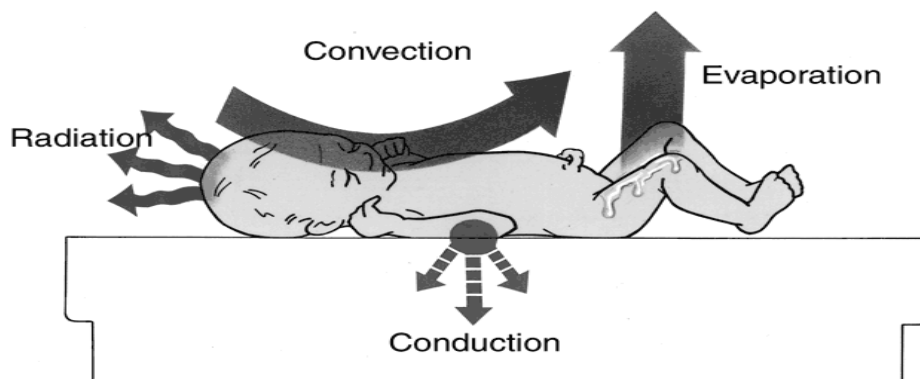


Figure above illustrates physical mechanisms of heat loss

Hypothermia

Classification: Based on its severity hypothermia could be:

- Mild (cold stress) = 36°C - 36.4°C
- Moderate = 32°C - 35.9°C and
- Severe (neonatal cold injury) $< 32^{\circ}\text{C}$

Prevention and management of hypothermia: There are several actions that providers can take to prevent small babies from going into hypothermia.

Care point	Type of care
Before delivery	• Warm delivery room • Organize newborn corner with adequate heat source
At delivery	• Deliver the baby on mother's abdomen • Dry the baby thoroughly immediately after birth and remove wet clothes. • Use cap to prevent significant heat loss through the scalp • Keep the newborn in skin to skin contact with the mother • Keep the newborn under radiant warmer – if there is a need for resuscitation • Cover weighing scales with warm towel when weighing the baby • Initiate early breastfeeding
Subsequent care	• Arrange appropriate transportation if needed: continue provision of KMC during transportation • Postpone bathing (after 24 hours) • Warm hands and stethoscope before touching the baby • Do examination/resuscitation of the infant under the radiant warmer • Practice rooming in wards/postnatal rooms • Keep the newborn away from windows and drafts • Continue breastfeeding

General management	• Identify and treat cause of hypothermia (disease process and environmental conditions) • Put hypothermic infants on KMC, in incubators or under radiant warmer. • Warm the new born slowly • Monitor axillary temperature every 30 minutes till newborn temperature becomes stable • Monitor environmental temperature
Management of newborns with severe hypothermia	• Warm the baby using a radiant warmer. • Remove cold or wet cloths. • Cover the baby with warm clothes and hat. • Treat for sepsis, if present • Measure blood glucose and treat if hypoglycemic. • Keep IV line under the radiant warmer to warm the fluid. • Measure the baby's temperature every hour. • If the baby's temperature is increasing at least 0.5 °C per hour in the 1st three hours, re warming is successful. • Then measure the baby's temperature every two hours. • If the baby's temperature does not rise or is rising more slowly than 0.5 °C per hour, check and reset temperature of the warmer. • Once the baby's temperature is normal, measure the temperature every three hours for 12 hours and then 12 hourly. • Monitor for complications and manage accordingly - o Look for respiratory problems - o Monitor vital signs - o Monitor urine output - o Monitor blood sugars - o Look for signs multi organ failure

Dangers of warmers: Providers should not use radiant warmer unless it is necessary. There are some dangers associated with using warmers.

- Hyperthermia
- Burn
- Dehydration
- Mask serious infections

Kangaroo mother care (KMC):

Kangaroo mother care consists of skin-to-skin care of babies (usually low birth weight or very low birth weight). KMC also promotes early and exclusive breastfeeding, but may be used even when babies are formula fed.

The cornerstones of KMC

Kangaroo Position

Dress the baby in a nappy and cap and place in an upright position against the mother's bare chest, between her breasts and inside her blouse. One may use a special garment, or one can tuck the mother's blouse under the baby or into her waistband. Cover both mother and baby with a Gabi, blanket or jacket if it is cold. You too can be innovative.

Kangaroo Nutrition

Babies who are unable to suckle should be fed expressed breast milk via a nasogastric tube or cup if they can swallow. Keep babies in the KMC position whilst being tube fed. Allow them to try to suckle during the tube feed.

In the KMC position, babies will declare themselves ready to suckle, as their rooting and suckling reflexes become manifest. Once the baby is able to suckle, allow the baby to breastfeed on demand but at least every three hours.

Kangaroo Support

It is very important to explain and demonstrate to the mother until she is motivated and confident to try the kangaroo position. Assist the mother with positioning and feeding, and give emotional support. The concept should be explained to other family members (especially the maternal grandmother), and they can also practice KMC (especially the father).

When to discharge from the hospital

Discharge when the baby has a sustained weight gain of at least 15 grams /kg /day. Bring the baby back for follow up in the next few days to ensure that baby is well and growing. It is advised practice to follow up KMC babies in a designated place.

Types of Kangaroo Mother Care

Intermittent KMC

This type of KMC is not done on a 24-hour basis but only for certain periods of the day. The mother stays at home or within the hospital but comes to the neonatal unit to do KMC at specified times; the newborn is left in an incubator for the remainder of the time. Intermittent KMC is mostly used for very small and sick babies, and/or for mothers who do not want or are not yet ready or able to practice continuous KMC. Examples include very LBW infants or mothers who are recovering from surgery (e. g., C-section). Intermittent KMC can be practiced while the baby is still in neonatal unit or delivery room. It is possible even with babies on oxygen and IV therapy. Frequency is determined by how stable baby is. A common sense approach is best.

Continuous KMC:

This is when KMC is practiced 24 hours every day (except for very short periods when the mother has to bathe or attend to other personal needs) and requires support from family members, including the husband. It is the ideal type of KMC for LBW babies. Continuous KMC can be instituted once the baby is stable, suckling well and needs no additional care. The baby can then be transferred to an adjoining KMC ward. Smaller babies may be able to go onto continuous KMC if they are stable and do not require oxygen.

Where do we do continuous KMC?

The KMC ward should be in close proximity to the neonatal unit and under the supervision of the neonatal staff, with 24-hour nursing coverage. The ward should be comfortable and warm but not heated.

Who can provide Kangaroo Mother Care?

- Everyone can provide KMC as long as they understand the method and are motivated to practice it.
- All those who want to assist the mother can practice KMC, such as grandmothers, sisters, aunts, husbands, and even friends.

Duration of KMC

Both **intermittent KMC** and **continuous KMC** are practiced as long as possible until the baby no longer tolerates the method. Babies who outgrow KMC become restless and will usually try to get out of the skin-to-skin position. Local KMC protocols may vary regarding the weight when babies are discharged from KMC follow-up. It is important to note, however, that babies should still be breastfed and kept warm even when KMC is no longer practiced.

How to practice kangaroo mother care

When to start KMC

KMC should be started when the small preterm or LBW baby is stable; otherwise it will have to be delayed. Exactly when KMC can begin depends on the condition and status of the baby and the mother. It is important, however, to encourage the mother to adopt KMC very early on.

Eligibility criteria for KMC

The following criteria should be used to decide whether a mother should begin KMC:

- The willingness of the mother to do KMC
- The baby should be in a stable condition:
 - No major illness present such as sepsis, pneumonia, meningitis, respiratory distress and convulsions.
 - Babies who have been started on antibiotics for suspected infection can start KMC as soon as they are stable.
 - Intermittent KMC can be used until the baby is fully stable.
- Babies under phototherapy may be evaluated to receive intermittent KMC.

Start KMC at your health facility or refer all LBW babies with a weight below 2,000 grams to the nearest health facility with KMC services or to a higher level of care.

Positioning of the mother and baby

In KMC the baby, wearing only a nappy, socks and a hat, is held upright between the mother's breasts in continuous contact with her skin (skin-to-skin contact). The position of the baby against the mother's chest underneath the cloth should secure the position of the baby's head and neck. The mother covers her baby with her own clothes and an additional blanket or shawl to cover the baby. While resting, the mother should be in a comfortable, moderately inclined position at about a 30-degree angle, supported with pillows to keep her comfortable.

When the mother walks around, the baby is still kept upright by a cloth. It is important that the nappy is changed soon after wetting or soiling, not only for the comfort of mother and baby but to reduce the body's heat loss.

Keeping the baby in the KMC position can be demanding for the mother, as continuous KMC practice is a tiring job. To assist the mother when she is tired or is attending to personal needs such as bathing, other family members (such as husbands, grandmothers, mothers-in-law, or older siblings) can be taught how to care for the baby in the kangaroo position so they can give the mother relief when necessary.

Steps in positioning the baby for KMC:

1. Dress the baby in socks, a nappy, and a cap.
2. Place the baby between the mother's breasts.
3. Secure the baby on to the mother's chest with a cloth
4. Put a blanket or a shawl on top for additional warmth.
5. Instruct the mother to put on a front-opened top: a top that opens at the front to allow the face, chest, abdomen, arms and legs of the baby to remain in continuous skin-to-skin contact with the mother's chest and abdomen.
6. Instruct the mother to keep the baby upright when walking or sitting.
7. Advise the mother to have the baby in continuous skin-to-skin contact 24 hours a day (or less in the case of intermittent KMC).
8. Advise the mother to sleep in a half-sitting position in order to maintain the baby in a vertical position.



4.5. FOLLOW-UP CARE FOR THE SICK NEWBORN

General consideration for follow-up care:

Advise mother to give home care for the neonate. Advise her to breastfeed frequently, as often and for as long as the infant wants, day or night, during sickness and health; but more frequently during sickness.

When to Return:

See table 5-5 and box 5-1 for follow up schedules and advise.

Table 5-5: Schedule for follow-up visit in specific conditions:

If the infant has:	Return for follow-up in:
LOCAL BACTERIAL INFECTION ANY FEEDING PROBLEM THRUSH LBW/PRETERM JAUNDICE SOME DEHYDRATION LOW BODY TEMPERATURE	2 days
BIRTH ASPHYXIA ALL NEWBORNS	3 days
LOW WEIGHT FOR AGE HIV INFECTION /EXPOSED	14 days

Box 5-1: When to return immediately

Advise the mother to return immediately if the young infant has any of these signs:

- Breastfeeding or drinking poorly
- Becomes sicker
- Develops a fever or feels cold to touch
- Fast breathing
- Difficult breathing
- Blood in stool
- Deepening of yellow colour of the skin
- Redness, swollen discharging eyes
- Redness, pus or foul smell around the cord

Note: after providing emergency management for the sick newborn the follow-up of the health care shall be undertaken as per the IMNCI strategy manual in the under-five clinic

MODULE 8

MANAGEMENT & PREVENTION INFECTIONS IN OBSTETRICS

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UNIT 1: INTRODUCTION

Infection in pregnancy is associated with numerous complications including preterm labor, PROM, Abruption placentae, IUGR and IUFD. Today puerperal infections are still a major cause of preventable maternal mortality in developing countries and, to a lesser degree, in developed countries. Predisposing factors for puerperal genital infections are prolonged labor, pre-labor rupture of the membranes, frequent vaginal examination, internal (vaginal) electronic fetal monitoring and caesarean section. The most important causative agents of genital infections are *E.coli*, streptococci, anaerobic microorganisms like *Bacteroides*, and gonococci. The clinical picture of genital infections is fairly uniform. Fever (temperature >38.0°C) is the main clinical symptom. Worldwide Infection accounts for 11% of all maternal deaths. Infection/pneumonia accounts for 26% of newborn deaths. In the 2016 Ethiopian EmONC assessment, maternal deaths due to maternal sepsis was 2% and it was the 2nd highest contributor of cause specific case fatality rate (1.2%)

Participant learning objective: After completing this module, participants will be able to recognize the common of causes of fever during pregnancy and after childbirth in developing countries including their usual clinical manifestations. They will also be able to describe the evidence based strategies for the prevention of infections in Obstetrics.

Enabling objectives

1. Detect and manage causes of fever during pregnancy
2. Detect and manage causes of fever after childbirth.
3. Describe the strategies for the prevention of infection in Obstetrics
4. Recognize serious causes of fever that need referral to higher centers.

UNIT 2: INFECTION IN PREGNANCY AND CHILDBIRTH

2.1 learning objectives

- describe the symptom signs of infection during pregnancy
- Detect causes of fever during pregnancy
- Detect causes of fever after childbirth.

2.2 The diagnosis of infection in pregnancy & childbirth

PROBLEM

- A woman has fever (temperature 38°C or more) during pregnancy/labor or 24hrs after childbirth.

Rapid evaluation & emergency Management

- Perform a rapid evaluation of the woman's general condition, vital signs (pulse, blood pressure, and respiration), level of consciousness, presence of anxiety and/or confusion, blood loss, and skin color and temperature.
- If shock is suspected, immediately begin treatment. Even if signs of shock are not present, keep shock in mind as you evaluate the woman further, because her status may worsen rapidly. If shock develops, it is important to begin treatment immediately.

DIAGNOSIS: see table 8.1 below

* Carefully assessing and developing a differential diagnosis of fever and other signs of infectious illness in pregnant and postpartum women is important.

Table 8.1:- Symptom/signs (differential diagnosis) of common causes of fever during pregnancy and postpartum

Presenting Symptom and Other Signs Typically Present	Symptoms and Signs Sometimes Present	Probable Diagnosis
during pregnancy		
• Foul-smelling vaginal discharge in first 28weeks • Fever • Tender uterus	• Lower abdominal pain • Rebound tenderness • Prolonged bleeding • Purulent cervical discharge	Septic abortion,
• Fever/chills • Foul-smelling watery discharge after 28weeks • Abdominal pain	• History of loss of fluid • Tender uterus • Rapid fetal heartrate • Light vaginal bleeding	Amnionitis,
Both During pregnancy and Postpartum		

• Dysuria • Increased frequency and urgency of urination	• Retropubic/suprapubic pain • Abdominal pain	Cystitis,
• Dysuria • Spiking fever/chills • Increased frequency and urgency of urination • Abdominal pain	• Retropubic/suprapubic pain • Looin pain/tenderness • Tenderness in ribcage • Anorexia • Nausea/vomiting	Acute pyelonephritis,
• Fever • Chills/rigors • Headache • Muscle/joint pain	• Enlarged spleen	Uncomplicated malaria,
• Symptoms and signs of uncomplicated malaria • Coma • Anemia	• Convulsions • Jaundice	Severe/complicated malaria,
• Cough with expectoration • Chest pain	• Consolidation • Rapid/difficulty breathing • Rhonchi/rales (wheezing)	Pneumonia
• Fever, Headache • Dry cough • Malaise, Anorexia • Enlarged spleen	• Confusion • Stupor	Typhoid
• Fever, Malaise, Anorexia, Nausea • Dark urine and pale stool • Jaundice • Enlarged liver	• Muscle/joint pain • Urticaria • Enlarged spleen	Hepatitis b
after childbirth		
• Fever/chills • Lower abdominal pain • Purulent, foul-smelling lochia • Tender uterus	• Light vaginal bleeding^a • Shock	Postpartum endometritis,
• Persistent spiking fever/chills • Lower abdominal pain and distension • Tender uterus	• Poor response to antibiotics • Swelling in adnexa or pouch of Douglas • Pus obtained upon	Pelvic abscess,
• Low-grade fever/chills • Lower abdominal pain • Absent bowel sounds	• Rebound tenderness • Abdominal distension • Anorexia • Nausea/vomiting	Peritonitis,
• Breast pain and tenderness three to six days after giving birth	• Hard, enlarged breasts • Both breasts affected	Breast engorgement, age
• Breast pain and tenderness • Reddened, wedge-shaped area on breast	• Inflammation preceded by engorgement • Usually only one breast affected	Mastitis,
• Firm, very tender breast • Overlying erythema	• Fluctuant swelling in breast • Draining pus	Breast abscess,

• Unusually tender wound with bloody or serous discharge		Slight erythema (extending beyond edge of incision)	Wound abscess, wound seroma or wound haematoma,
• Painful and tender wound • Erythema and oedema beyond edge of incision	•	• Hardened edges of wound • Purulent discharge • Reddened area around wound	Wound cellulitis,
• Spiking fever despite • Swelling in the affected leg • Calf muscle tenderness	• antibiotics	• Warmth and redness of the affected leg	Deep vein thrombosis^b
• Abrupt onset of pleuritic chest pain • Shortness of breath • Tachypnea • Hypoxia	•	• Dry cough • Cough with bloody sputum • Swollen leg or arm • Dizziness or syncope	Pulmonary embolism^c

^b Provide supportive therapy and observe.

UNIT 3: THE MANAGEMENT OF INFECTIONS DURING PREGNANCY & AFTER CHILDBIRTH

3.1 unit learning objectives

- manage causes of fever during pregnancy & after childbirth
- Recognize serious causes of fever that need referral to higher centers

3.2 GENERAL MANAGEMENT

- Encourage rest.
- Ensure adequate hydration by mouth or IV.
- Use a fan or tepid sponge and, if necessary, open a window to help decrease temperature.
- Consider paracetamol 500–1000 mg (oral tablets or rectal suppositories) every six to eight hours to help decrease temperature (maximum 4000 mg in 24 hours).

3.3 SPECIFIC MANAGEMENT

URINARY TRACT INFECTIONS

TESTS

Urine dipstick, microscopy and culture tests can be used to evaluate for a urinary tract infection, but they will not differentiate between cystitis and acute pyelonephritis.

- A dipstick leukocyte esterase test can be used to detect white blood cells, protein or blood; a nitrate reductase test can be used to detect nitrites.
- Microscopy of a urine specimen may show white cells in clumps, bacteria and, sometimes, red cells.
- Urine culture and sensitivity tests should be done, if available, to identify the organism and its antibiotic sensitivity.

Note: Urine examination requires a clean-catch midstream specimen to minimize the possibility of contamination. Contamination may be indicated by presence of many epithelial cells noted on microscopy.

CYSTITIS

Cystitis is infection of the bladder (lower urinary tract).

- Treat with an antibiotic
 - amoxicillin 500 mg by mouth every eight hours for three days;
 - nitrofurantoin 100 mg by mouth every eight hours for three days.
- **Note:** Avoid nitrofurantoin at term as it can cause neonatal haemolysis.
- If treatment fails or if infection recurs two or more times during pregnancy,

check urine culture and sensitivity, if available, and treat with an antibiotic appropriate for the organism.

ACUTE PYELONEPHRITIS

Acute pyelonephritis is an infection of the upper urinary tract, mainly of the renal pelvis, which may also involve the renal parenchyma. Acute pyelonephritis can cause significant illness in pregnant women and should be promptly investigated and treated in every pregnant woman with fever, urinary symptoms and flank pain.

- If **shock is present or suspected**, initiate immediate treatment.
- Start an IV infusion and infuse IV fluids at 150 mL per hour (.
- Check urine dipstick and urine culture if possible and begin empiric antibiotic treatment promptly (pending results of urine culture, if available).
- Treat with an IV antibiotic until the woman is fever-free for 48 hours:
 - ampicillin 2 g IV every six hours;
 - PLUS gentamicin 5 mg/kg body weight IV every 24 hours.
- Once the **woman is fever-free for 48 hours, is drinking well and is not vomiting**, switch from an IV to an oral antibiotic, for a total of 14 days treatment. Oral antibiotic options include, among others:
 - amoxicillin 1 g by mouth every eight hours to complete 14 days of treatment.

Note: Clinical response is expected within 48 hours. If there is **no clinical response in 48 hours**, re-evaluate urine culture results (if available) and antibiotic coverage. Also re-evaluate diagnosis to be sure that there is not another source of infection.

- Ensure adequate hydration by mouth or IV.
- Give paracetamol 500–1000 mg by mouth three to four times daily as needed for pain and to lower temperature (4000 mg maximum in 24 hours).

Note: If there are **palpable contractions and blood-stained mucus discharge**, suspect preterm labor.

MALARIA

Uncomplicated Malaria

The symptoms and complications of malaria in pregnancy vary according to transmission intensity and the level of acquired immunity.

Pregnant women living in areas of low or unstable malaria transmission (like many malarious areas in Ethiopia) have little or no immunity to malaria, and are at higher risk of developing severe malaria than are non-pregnant adults living in the same area. In these areas, malaria is a major cause of maternal anaemia, spontaneous abortion, stillbirth, premature

delivery, low birth weight (birth weight < 2.5kg), neonatal death and maternal death. In non-immune women, severe malaria symptoms (hypoglycaemia, cerebral malaria, and pulmonary oedema being particular problems) are more common in pregnancy.

In stable transmission settings, the deleterious impact of malaria is particularly apparent in first and second pregnancies. Partial clinical immunity acquired during years of exposure to the malaria parasite prior to pregnancy does not prevent infection, but does reduce the risk of severe disease. Clinical malaria is not, therefore, a prominent feature of infection during pregnancy, and the major detrimental effects of infection are low birth weight (LBW) and maternal anaemia.

HIV infection impairs pregnant women's ability to control *P. falciparum* infection. Women with HIV infection are more likely to have symptomatic malaria infections and to have an increased risk of an adverse birth outcome due to malaria. In the presence of HIV infection, placental malaria appears to be independent of the number of pregnancies, so that the risk of placental malaria is similar in HIV-infected multigravidae and HIV-negative primagravidae.

Severe anaemia, exacerbated by malaria, is an important complication of pregnancy in many tropical countries. Especially in communities where chronic hookworm anaemia is prevalent, high output anaemic cardiac failure may develop in late pregnancy.

There are five important malaria parasites: *Plasmodium (P.) falciparum*, *P. vivax*, *P. ovale*, *P. malariae* and *P. knowlesi*. Two species of these parasites, *P. falciparum* and *P. vivax*, account for the majority of cases. Symptomatic falciparum malaria in pregnant women may cause severe disease and death if not recognized and treated early. When malaria presents as an acute illness with fever, it cannot be reliably distinguished from many other causes of fever on clinical grounds. Malaria should be considered the most likely diagnosis in a pregnant woman with fever who has been exposed to malaria.

Tests

- If facilities for testing are not available, begin therapy with antimalarial drugs based on clinical suspicion (e.g. headache, fever, joint pain).
- Where available, the following tests will confirm the diagnosis:
 - microscopy of a thick and thin blood film:
 - thick blood film is more sensitive at detecting parasites (absence of parasites does not rule out malaria);
 - thin blood film helps to identify the parasite species.
 - rapid antigen detection tests.

Management:

Pregnant women with symptomatic acute malaria are a high-risk group, and require effective antimalarial medication. There is insufficient information on the safety and

efficacy of most antimalarial medicines in pregnancy, particularly for exposure in the first trimester, and treatment recommendations differ from those for non-pregnant adults. Therefore as a standard practice for the administration of any medicine pregnant women, all women of child-bearing age should be asked whether they are, or could possibly be, pregnant before an antimalarial medicine is prescribed. The following are the antimalarial medicines recommended for the treatment of uncomplicated falciparum malaria during pregnancy:

FALCIPARUM MALARIA

First Trimester

- Give **quinine salt** (dihydrochloride or sulfate) 10 mg/kg body weight by mouth three times daily **PLUS clindamycin** 300 mg every six hours for seven days.
- If **clindamycin is not available**, treat with quinine monotherapy: quinine salt (dihydrochloride or sulfate) 10 mg/kg body weight by mouth three times daily for seven days (The FMOH training manual on malaria, 2017).
- An artemisinin-based combination therapy (ACT) can be used if quinine is not available, or if quinine plus clindamycin fails, or if adherence to seven-day treatment with quinine cannot be guaranteed.

Second and Third Trimesters

- Treat orally based on national policy with any of the ACTs (assuming a body weight of 50 kg or more):
e.g artemether (80 mg) PLUS lumefantrine (480 mg) twice daily for three days; (MOH training manual on malaria)

NB: For the treatment of severe falciparum malaria, intramuscular or intravenous quinine should be administered according to the national malaria management guidelines.

Quinine is associated with an increased risk of hypoglycaemia in late pregnancy. It should be used, in combination with clindamycin, only if effective alternatives are not available.

P. VIVAX, OVALE, MALARIAE, KNOWLESI MALARIA

First Trimester

Areas with Chloroquine-Sensitive *P. Vivax* Parasites

Chloroquine is the treatment of choice in areas with chloroquine-sensitive vivax malaria.

- Give **chloroquine** 10 mg/kg body weight by mouth once daily for two days followed by 5 mg/kg body weight by mouth on day three (FMOH recommendation).

Areas with Chloroquine-Resistant *P. Vivax* Parasites

Chloroquine-resistant *P. vivax* has been reported in several countries. Before considering second-line drugs for treatment failure with chloroquine, clinicians should exclude poor patient compliance and a new infection with *P. falciparum*. If **diagnostic testing is not**

available, treat as for falciparum malaria. The treatment option for confirmed chloroquine resistant vivax malaria is quinine salt (dihydrochloride or sulfate) 10 mg/kg body weight by mouth three times a day for seven days.

Note: The dose of quinine is the same for all species of malaria.

Second and Third Trimesters

Areas with Chloroquine-Sensitive *P. Vivax* Parasites

ACT and chloroquine alone are the two treatment options in areas with chloroquine-sensitive vivax malaria

Areas with Chloroquine-Resistant *P. Vivax* Parasites

Treat with ACT (see dose above)

AREAS OF MIXED FALCIPARUM-VIVAX MALARIA

In areas of mixed transmission, the proportions of malaria species and their drug sensitivity patterns vary. If **microscopic diagnosis is available**, specific treatment can be prescribed. Where unavailable, assume the infection is due to *P. falciparum* and treat accordingly.

SEVERE MALARIA

- A pregnant woman with severe malaria should be given a parenteral antimalarial medicine in full doses without delay. Parenteral artesunate is more effective than parenteral quinine in reducing the risk of death from severe malaria.

Although safety data on the use of artemisinins in the first trimester are limited, saving the mother's life is the primary objective. Therefore artesunate (IV or IM) is the preferred drug for all severe forms **in all trimesters**:

- Begin treatment with IV or IM route for at least 24 hours and until the woman can tolerate oral medications.
- Then give a complete oral treatment with ACT **for three days**.
- Parenteral Quinine dihydrochloride can be given in areas where artemether is unavailable
 - Loading: 20 mg/kg body in IV fluids (5% dextrose) over four hours
 - Maintenance: 10 mg/kg body weight over four hours. Repeat every eight hours from the start of the last dose (i.e. quinine infusion for four hours, no quinine for four hours, quinine infusion for four hours, etc.).

PREVENTION OF MALARIA DURING PREGNANCY

WHO recommends the following package of interventions for the prevention and treatment of malaria during pregnancy:

- use of long-lasting insecticidal nets (LLINs);

An insecticide-treated net (ITN) is a net (usually a bed net), designed to block mosquitoes physically, that has been treated with safe, residual insecticide for the purpose of killing and repelling mosquitoes, which carry malaria. A long-lasting insecticide-treated net (LLIN) is an ITN designed to remain effective for multiple years

without retreatment

- intermittent preventive treatment in pregnancy (IPTp) with sulfadoxine-pyrimethamine (SP), in all areas with moderate to high malaria transmission in Africa, as part of antenatal care services;
- Prompt diagnosis and effective treatment of malaria infections.

AMNIONITIS

Is infection of the fetal membranes (chorio amniotic) usually following prelabor rupture of membranes.

- Give a combination of antibiotics until the woman gives birth
 - ampicillin 2 g IV every six hours;
 - PLUS gentamicin 5 mg/kg body weight IV every 24 hours.
- Pregnancy should be terminated irrespective of GA in women with chorio-
amnionitis
- If the **woman gives birth vaginally**, continue treatment for 24–48 hours after the symptoms and signs of infection (e.g. fever, uterine tenderness) have subsided.
- If the **woman has a caesarean**, cleanse the vagina with povidone-iodine before the procedure.

POSTPARTUM ENDOMETRITIS

Postpartum endometritis is a major cause of maternal death. Delayed or inadequate treatment of postpartum endometritis may result in pelvic abscess, peritonitis, septic shock, deep vein thrombosis, pulmonary embolism, chronic pelvic infection with recurrent pelvic pain and dyspareunia, tubal blockage, or infertility.

- Transfuse as necessary. Use packed cells, if available.
- Give the woman a combination of antibiotics for 24–48 hours after complete resolution of clinical signs and symptoms (fever, uterine tenderness, purulent lochia, leukocytosis:
 - Clindamycin phosphate 600 mg IV every eight hours; PLUS gentamicin 5 mg/kg body weight IV every 24 hours.

If clindamycin is not available administer:

- Ampicillin 2 g IV every 6 hours;
- PLUS gentamicin 5 mg/kg body weight IV every 24 hours + metronidazole 500mg iv TID

When available, clindamycin (in combination with gentamycin) is more effective than ampicillin or a penicillin antibiotic for the treatment of postpartum endometritis.

- If fever is still present 72 hours after starting antibiotics, re-evaluate and revise diagnosis.

Causes of antibiotic treatment failure might include:

- RPC, pelvic abscess, mis-diagnosis (e.g malaria), drug resistance

Note: Oral antibiotics are not necessary after stopping IV antibiotics.

- If retained placental fragments are suspected, perform a digital exploration of the uterus to remove clots and large pieces. Use ovum forceps or a wide curette if required.
- If there is no improvement with conservative measures and there are signs of general peritonitis (fever, rebound tenderness, abdominal pain), perform a laparotomy to drain the pus.

PELVIC ABSCESS

- Give a combination of antibiotics before draining the abscess; continue antibiotics until the woman is fever-free for 48 hours ():
 - Ampicillin 2 g IV every six hours;
 - PLUS gentamicin 5 mg/kg body weight IV every 24 hours; PLUS metronidazole 500 mg IV every eight hours.

N;B; Generally patients are managed at higher centers & management may include drainage of abscess

PERITONITIS

- Provide nasogastric suction.
- Start an IV infusion and infuse IV fluids.
 - Referral to a higher center and further management include:
- Give the woman a combination of antibiotics until she is fever-free for 48 hours :
 - Ampicillin 2 g IV every six hours;
 - PLUS gentamicin 5 mg/kg body weight IV every 24 hours PLUS metronidazole 500 mg IV every eight hours.
- Identify and treat the underlying cause of the peritonitis.
- Perform additional diagnostics such as X-ray or ultrasound to assist in identifying the underlying cause.
- The type of surgical intervention needed depends on the diagnosis of the cause of the peritonitis. For example, closure may need to be performed for an intestinal or uterine perforation, whereas an abscess may need to be drained.

BREAST ENGORGEMENT

Breast engorgement is an exaggeration of the lymphatic and venous engorgement that occurs before lactation. It is not the result of overdistension of the breast with milk.

BREASTFEEDING

- If the woman is breastfeeding and the baby is able to suckle:
 - Encourage the woman to breastfeed more frequently, without restrictions, using both breasts at each feeding.
 - Show the woman how to hold the baby and help the baby attach.
- If the woman is breastfeeding and the baby is not able to suckle, encourage the woman to express milk by hand or with a pump.
- Relief measures before feeding or expression may include:
 - Applying warm compresses to the breasts just before breastfeeding, or encouraging the woman to take a warm shower;
 - Massaging the woman's neck and back; and
 - Having the woman express some milk manually before breastfeeding, and wetting the nipple area to soften the areola to help the baby latch on properly and easily.
- Relief measures after feeding or expression may include:
 - Supporting breasts with a binder or bra;
 - Applying cold compresses to the breasts between feedings to reduce swelling and pain.
- Give ibuprofen 200–400 mg every six to eight hours orally (maximum dose of 1200 mg in 24 hours);
 - OR paracetamol 500–1000 mg every six to eight hours orally as an appropriate alternative (maximum 4000 mg in 24 hours).
- Follow up in three days to ensure response.

NOT BREASTFEEDING

- If the woman is not breastfeeding:
 - Encourage her to support breasts with a binder or bra.
 - Apply cold compresses to the breasts to reduce swelling and pain.
 - Avoid massaging or applying heat to the breasts.
 - Avoid stimulating the nipples.
 - Give ibuprofen 200–400 mg every six to eight hours (maximum dose 1200 mg in 24 hours);
 - OR paracetamol 500–1000 mg every six to eight hours orally as an appropriate alternative (maximum dose 4000 mg in 24 hours).
 - Follow up in three days to ensure response.

BREAST INFECTION

MASTITIS

- Treat with antibiotics:
 - Cloxacillin 500 mg by mouth every six hours for 10 days; OR erythromycin 250 mg every eight hours for 10 days.
- Encourage the woman to:
 - continue breastfeeding;
 - support the breasts with a binder or bra; and
 - apply cold compresses to the breasts between feedings to reduce swelling and pain.
- Give the woman ibuprofen 200–400 mg every six to eight hours (maximum dose 1200 mg in 24 hours);
 - OR paracetamol 500–1000 mg every six to eight hours as an appropriate alternative (maximum dose 4000 mg in 24 hours).
- Follow up in three days to ensure response.

BREAST ABSCESS

Antibiotic Treatment

- Treat with antibiotics:
 - cloxacillin 500 mg by mouth every six hours for 10 days; - OR erythromycin 250 mg every eight hours for 10 days.

Surgical Treatment (at a higher center or by experienced professional at health center)

Pus must be drained either by incision and drainage or ultrasound-guided needle aspiration (which may need to be repeated).

- Drain the abscess:
 - General anaesthesia (e.g. ketamine, is usually required.
 - Make the incision radially, extending from near the areolar margin toward the periphery of the breast, to avoid injury to the milk ducts.
 - Wearing sterile gloves, use a finger or tissue forceps to break up the pockets of pus.
 - Loosely pack the cavity with gauze.
 - Remove the gauze pack after 24 hours and replace with a smaller gauze pack.
- If there is still pus in the cavity:
 - Place a gauze pack in the cavity and bring the edge out through the wound as a wick, to facilitate drainage of any remaining pus;
 - OR perform ultrasound-guided aspiration for abscesses in which overlying skin is intact and the abscess is less than 5 cm in diameter.

- Local anaesthesia is generally sufficient.
- This can often be done as an outpatient procedure.
- If laboratory capacity is available, send drained or aspirated pus for culture and sensitivity testing.

Note: A large surgical incision should be avoided because it could damage the areola and milk ducts and interfere with subsequent breastfeeding.

Supportive Treatment

- Encourage the woman to:
 - continue breastfeeding even when there is collection of pus;
 - support breasts with a binder or bra;
 - apply cold compresses to the breasts between feedings to reduce swelling and pain.
- Give ibuprofen 200–400 mg by mouth every six to eight hours as needed (maximum dose 1200 mg in 24 hours);
- OR paracetamol 500–1000 mg by mouth every six to eight hours (maximum dose 4000 mg in 24 hours).
- Reassess in two to three days to determine the woman's clinical status.
- Review danger signs with the woman (fever, etc.).
- For women with a breast abscess, more frequent assessment and wound dressing is necessary.

INFECTION OF PERINEAL AND ABDOMINAL WOUNDS

- If there is pus or fluid, open and drain the wound.
- Remove infected skin or subcutaneous sutures and debride the wound. Do not remove fascial sutures.
- If there is an abscess without cellulitis, antibiotics are not required.
- Place a damp dressing on the wound and have the woman return to change the dressing every 24 hours.
- Advise the woman to practice good hygiene and to wear clean pads or cloths that she changes often.

WOUND CELLULITIS AND NECROTIZING FASCIITIS

- If there is fluid or pus, open and drain the wound.
- Remove infected skin or subcutaneous sutures and debride the wound. Do not remove fascial sutures.

- If infection is superficial and does not involve deep tissues, monitor for development of an abscess and give a combination of antibiotics (:

- ampicillin 500 mg by mouth every six hours for five days.

- If the infection is deep, involves muscles and is causing necrosis (necrotizing fasciitis), (referral to a higher center or management by experienced physician or an ESO at Health center)

- * give a combination of antibiotics until necrotic tissue has been removed and the woman is fever-free for 48 hours - penicillin G 2 million units IV every six hours (or Ampicillin 2gm IV qid);

- PLUS gentamicin 5 mg/kg body weight IV every 24 hours;

- PLUS metronidazole 500 mg IV every eight hours;

Note: Necrotizing fasciitis requires wide surgical debridement. Perform delayed primary closure two to four weeks later, depending on resolution of infection.

- If the woman has a severe infection or necrotizing fasciitis, admit her to the hospital for management, and change the wound dressing twice daily.

UNIT 4: PREVENTION OF INFECTION IN OBSTETRICS (DURING PREGNANCY AND POSTPARTUM)

4.1 Unit Learning Objectives

- describe strategies for the prevention of infections in Obstetrics

4.2 Infection prevention practices (See Module 1, Unit 4)

4.3 Prophylactic antibiotic administration for prevention of infection in obstetrics

4.3.1 Maternal

- it is the administration of antibiotics in a woman with no clinical evidence of infection. It is indicated for (**Table 8.2**)

- i) prevention of infection in the setting of established risk factors
(e.g. vaginal colonization with Group B streptococcus in pregnant women

- ii) prophylaxis for medical procedures

Performing certain obstetrical procedures (e.g. caesarean birth, manual removal of placenta) increases a woman's risk of infectious morbidity. This risk can be reduced by providing prophylactic antibiotics at the time of the procedure in addition to following recommended infection prevention practices as discussed above

Whenever possible, give prophylactic intravenous antibiotics 15–60 minutes before the start of a procedure to achieve adequate blood levels of the antibiotic at the time of the procedure. One dose of prophylactic antibiotics is sufficient and is no less effective than three doses or 24 hours of antibiotics for preventing infection after an obstetrical procedure. If the **procedure lasts longer than six hours** or

blood loss is 1500 mL or more, give a second dose of prophylactic antibiotics to maintain adequate blood levels during the procedure.

Table 8.2 Administration of prophylactic antibiotics

Obstetrical procedure or condition	Recommended antibiotic(s) and dosage
☐ • Caesarean birth (elective and emergency)—administer prophylactic antibiotics before the procedure, not <i>after</i> clamping and cutting the cord; also, now recommend cleansing the vagina with povidone-iodine before the procedure ☐ • Manual removal of the placenta ☐ • Placement of uterine balloon tamponade	Single dose of antibiotics (ampicillin or first-generation cephalosporin): • Ampicillin 2 g intravenously OR • Cefazolin 1 g intravenously
☐ • Repair of third and fourth degree lacerations	Single dose of antibiotics: ☐ Ampicillin 500 mg
☐ • Preterm prelabor rupture of membranes	• Oral erythromycin 250 mg every six hours for 10 days (or until birth) OR • Ampicillin 2 g intravenously every six hours

N;B Administration of prophylactic antibiotics **is not recommended** for the following conditions: uncomplicated vaginal birth; operative vaginal birth; episiotomy; and first or second degree lacerations.

4.3.2 Neonatal

newborns delivered under the following circumstances are at increased risk of infection and should be treated with prophylactic antibiotics—ampicillin (intravenously or intramuscularly) and gentamycin—for at least two days, until infection is ruled out:

- ☐ - Preterm prelabor rupture of membranes occurred.
- ☐ - The membranes ruptured more than 12 hours before birth.
- ☐ - The mother is being treated with antibiotics for chorioamnionitis.
- ☐ - The mother had a fever greater than 38 °C before childbirth or during labor.
- ☐ - The amniotic fluid was foul smelling or purulent.
- ☐ - The mother has confirmed maternal colonization with Group B streptococcus without adequate antibiotic therapy during labor.

4.4 . Evidence based obstetrical clinical practices

Avoiding/modifying traditional Clinical practices associated with increased risk of infections and applying evidence based recommendations is important

- Improving nutritional status (micronutrients) & hygiene
- Control of diabetes
- Treatment of asymptomatic bacteriuria
- Screening for genital GBS colonization and Treatment
- Health education on safe sex practice & Personal hygiene
- Screening for HIV and care provision
- Prevention/treatment of Anemia,
- prevent prolonged labor (Partographic management of labor)
- avoid multiple vaginal exams in labor;
- avoid early artificial ROM
- avoid unnecessary Episiotomies
- perform manual removal of placenta only if indicated
- repair genital tears appropriately

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